

**A Capstone Project presented to the College of Information
Technology Education PHINMA University of Iloilo
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**Design and Development of an Integrated Study Hub Management
System for WS Students & Professionals Lounge**

Submitted in partial fulfillment of the requirements for
the degree of
Bachelor of Science in Information Technology
System Development

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Abstract

The rapid expansion of shared workspaces has increased the operational demand for efficient facility management systems. The WS Students & Professionals Lounge - La Paz Branch previously managed its daily transactions through manual workflows, paper logbooks, decentralized spreadsheets, and third-party social media platforms. This fragmented operational setup resulted in structural vulnerabilities, including resource double-bookings, prolonged customer waiting times, data entry errors, and financial tracking discrepancies. To mitigate these administrative bottlenecks, this study developed the Integrated Study Hub Management System, a centralized, cross-platform web application designed to optimize back-office utilities and enhance user service quality. Utilizing an agile development methodology, the system was engineered with six primary modules: Reservation Management, Membership and Time Tracking, Payment Billing, Sales Recording and Reporting, Admin and Staff Dashboards, and Role-Based Access Control (RBAC).

The technical architecture was built on a web-based framework supported by a relational MySQL database engine, ensuring secure data synchronization and eliminating data fragmentation. Evaluation metrics indicated that the automation of elapsed-time calculations and live space allocation rules effectively eradicated transcription errors, secured historical financial ledgers, and minimized context-switching for frontline desk personnel. Ultimately, the integration of this system converted manual front-desk friction into actionable business

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intelligence, providing a secure, transparent, and frictionless facility environment for both administrative personnel and clients.

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CHAPTER 1

1.1 Introduction to the Study

In the modern digital era, the operational efficiency of service-oriented enterprises relies heavily on the integration of centralized software architectures that mitigate manual friction and eliminate data fragmentation. Shared workspace facilities, such as study hubs and co-working lounges, process complex daily transaction streams that demand high computational precision. These transactions include real-time space scheduling, user subscription authentication, and dynamic, time-based financial billing. When an enterprise manages these highly volatile parameters using non-integrated, manual procedures, its operational framework becomes fundamentally vulnerable to processing delays, human error, and structural data silos. Consequently, modern organizational management dictates that transitioning to an automated, centralized platform is an operational necessity to preserve data integrity and optimize service delivery.

The WS Students & Professionals Lounge - La Paz Branch previously conducted its business functions through a fragmented combination of manual, paper-reliant processes, localized spreadsheet applications, and third-party social media utilities. Specifically, the establishment utilized Facebook and Facebook Messenger as its primary communication interfaces for

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coordinating customer reservations and handling client availability inquiries. Frontline desk personnel had to manually parse incoming chat logs, verify physical availability, and retrospectively transcribe these transactional details into decentralized paper ledgers or detached spreadsheets. This disconnected manual workflow demanded significant administrative labor and created a highly volatile operational environment where data synchronization could not be guaranteed in real time.

Additionally, the lounge's customer registration and long-term membership management followed an identical manual methodology. New clients were registered via handwritten forms, and the facility issued physical membership cards while administrative staff manually logged subscription dates into individual spreadsheets. This management model created extreme operational friction when tracking, updating, or verifying expiration timelines as the lounge's active user base expanded. Because there was no centralized database engine to link these operations, front-desk personnel were constantly forced to context-switch between digital chat windows, physical logbooks, and desktop spreadsheet rows simply to execute a single check-in or validate a membership tier. This procedural friction rendered daily front-office operations heavily time-consuming and highly susceptible to data entry errors.

As a result, the La Paz branch faced persistent operational liabilities that threatened its business performance and customer satisfaction metrics. These liabilities manifested as accidental room double-bookings, overlooked or missed client

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reservations, asynchronous data updates, and prolonged queues at the service counter. Managing overlapping transaction records across disconnected media significantly inflated the administrative workload of staff, leading to severe service bottlenecks during high-volume operating hours. These systemic inefficiencies collectively highlighted an urgent, immediate requirement for a structured, unified, and permanent technological intervention.

To address these operational bottlenecks, this study developed the Integrated Study Hub Management System for WS Students & Professionals Lounge to centralize and automate key operations such as user registration, room reservations, membership tracking, billing, time-duration tracking, and automated sales reporting. By replacing manual processes with a digital solution, the system successfully reduced administrative errors, streamlined facility workflows, and improved overall service efficiency. Ultimately, the system was designed and implemented to simplify administrative procedures for staff while providing a faster, seamless, and more convenient experience for customers.

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1.2 Background of the Study

Study hubs and co-working spaces have experienced an unprecedented surge in popularity. This growth is largely driven by the increasing demand from students, freelancers, and working professionals for structured, quiet, and highly productive environments that foster deep focus and collaboration. These spaces fulfill a critical societal and infrastructure need by providing essential facilities, including high-speed internet connectivity, ergonomically sound work areas, proper lighting, and specialized meeting rooms. However, as customer volume expands, the operational complexity of managing these shared facilities scales accordingly. To sustain high service quality and maintain customer satisfaction, modern study hubs are increasingly required to manage their day-to-day administrative frameworks with a high degree of operational efficiency.

The WS Students & Professionals Lounge - La Paz Branch is one such study hub that offers dedicated spaces for studying, meetings, and collaboration. Previously, however, the establishment relied almost exclusively on a combination of fragmented, manual procedures, handwritten logbooks, and legacy spreadsheet software to administer its daily operations. Furthermore, the lounge utilized external social media channels—primarily Facebook and Facebook Messenger—as its primary interfaces for handling client inquiries and booking requests. This operational framework spanned across multiple administrative silos, including customer registration, workspace

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reservations, membership subscription tracking, elapsed time logging, invoicing, and retrospective financial reporting. Because these workflows were fundamentally decentralized and detached from one another, data retrieval and verification processes remained heavily disorganized and administratively cumbersome.

This reliance on non-integrated platforms exposed the establishment to significant operational vulnerabilities. Frontline personnel had to constantly navigate and manually cross-reference communication threads on social media, physical logbooks, and individual digital spreadsheets simply to verify a single customer transaction or room availability status. This structural fragmentation frequently resulted in serious service bottlenecks, such as accidental resource double-bookings, delayed response times to online clients, prolonged front-desk waiting lines, and chronic inaccuracies in tracking real-time room usage and membership status. The logistical strain was particularly magnified during peak operating hours when the high influx of clients drastically increased the administrative workload, rendering the previous workflow acutely susceptible to human calculation and transcription errors.

To resolve these documented administrative bottlenecks, there existed a critical and immediate imperative to transition from a manual, paper-reliant architecture to a secure, digitized infrastructure. The Integrated Study Hub Management System for WS Students & Professionals Lounge was directly engineered to

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address these operational vulnerabilities. By consolidating and automating the core business mechanisms of client registration, interactive reservations, membership management, time-duration tracking, instant billing computation, and real-time sales reporting into a centralized web-based platform, the system established an integrated, database-driven workflow. Ultimately, the system was designed and successfully implemented to improve operational efficiency, reduce human error, and enhance overall service quality for both staff and customers.

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1.3 Statement of the Problem

As the demand for study hubs and co-working spaces continues to grow, businesses like WS Students & Professionals Lounge faced significant operational challenges due to their previous reliance on manual and paper-based management systems. The lack of a centralized platform for managing reservations, memberships, payments, and sales often led to double bookings, long waiting times, and inaccuracies in tracking room usage and sales reports. Because most records were maintained in physical logs or scattered spreadsheets, the business was highly prone to data loss, human error, and delays in service delivery.

Without an integrated system, staff were forced to spend excessive time manually verifying records and updating information, which resulted in operational inefficiency and customer dissatisfaction, especially during peak hours. Furthermore, the absence of real-time reporting made it difficult for management to monitor sales and occupancy accurately, hindering their ability to make data-driven decisions. To maintain its reputation and provide quality service, it was essential for the lounge to transition into an automated and secure management framework.

Specifically, this study sought to address and evaluate the following key objectives and questions:

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1. How was the real-time reservation module designed and implemented to effectively prevent double bookings, optimize table allocation, and minimize client waiting times?
2. In what ways was the membership tracking system developed to automatically monitor user activity, track usage duration, and manage subscription validity accurately?
3. What methods were applied to integrate billing and sales reporting so that every transaction was automatically recorded and updated to minimize manual errors?
4. What approaches were employed to implement a centralized dashboard that displayed real-time information on time usage, occupancy, and sales to assist management in decision-making?
5. How did the web-based management system improve the overall service quality and customer satisfaction of WS Students & Professionals Lounge compared to its previous manual processes?
6. What technical features were implemented to ensure that data and transaction records were securely stored and accessible only to authorized personnel?
7. In what ways did the system improve the customer experience in terms of faster transactions, reliable booking, and efficient service delivery?

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8. To what extent did the Integrated Study Hub Management System enhance the overall management of user registrations and memberships?
9. How effective was the system in handling automated billing, time-based room usage, and subscription expiration tracking?
10. What impact did the system have on the accuracy and efficiency of automated sales report generation?

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1.4 Objectives of the Study

1.4.1 General Objective

The primary objective of this study was to design, develop, and evaluate an Integrated Study Hub Management System for WS Students & Professionals Lounge - La Paz Branch. The system aimed to enhance operational efficiency by automating and streamlining core business processes, improving the quality and speed of service delivery, and ensuring accurate, reliable, and centralized management of user registration, membership records, billing transactions, time-based room usage, and sales report generation. Ultimately, the system was created to provide an organized and efficient platform supporting both administrative functions and customer needs.

1.4.2 Specific Objectives

Specifically, this study aimed to:

- Design and develop a centralized web-based management system that served as a single platform for all lounge operations and data storage.
- Automate the reservation and booking process to effectively eliminate double bookings and optimize table allocation for customers.

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- Implement an automated membership and time usage tracking feature to monitor customer subscriptions and facility usage duration accurately.
- Develop an integrated billing and sales recording system that minimized human errors and ensured all financial transactions were logged correctly.
- Generate comprehensive real-time reports for sales, room occupancy, and time tracking to assist management in data-driven decision-making.
- Conduct system testing and evaluation to ensure that the developed application met all functional requirements and provided a user-friendly experience for staff and administrators.

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1.5 Scope and Limitations

This study focused primarily on the design, development, and implementation of an Integrated Study Hub Management System engineered specifically to optimize the administrative framework and customer-facing workflows of the WS Students & Professionals Lounge - La Paz Branch. Designed as a centralized web platform, the project directly addressed the systemic operational vulnerabilities inherent in manual, legacy business setups.

The system automated traditionally labor-intensive workflows—such as handwritten reservation books, manual elapsed-time calculations, and paper-based invoicing—thereby drastically minimizing human calculation errors, mitigating data fragmentation, and eliminating accidental resource double-bookings. By modernizing these core daily transactions into real-time digital processes, the developed platform enhanced overall data accuracy, secured historical transaction records, and optimized resource allocation. Ultimately, this integration elevated operational efficiency for frontline staff while providing management with the transparent, data-driven oversight necessary to maintain superior service quality and business integrity.

The scope of the developed system included the following features and modules:

1. Reservation Management - This feature delivered an interactive, functional module allowing both frontend lounge

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staff and registered clients to schedule and secure specific study spaces, study pods, or conference rooms in advance. It incorporated programmatic validation logic to prevent double-bookings, handle reservation overlaps, and optimize physical table allocation in real time.

2. Membership and Time Tracking - This module securely recorded detailed customer profiles and automatically monitored physical usage durations via server-side check-in and check-out timestamps. It programmatically tracked subscription validity periods and membership tier benefits to ensure workspace privileges were validated automatically before a session began.

3. Payment Billing - This component handled the internal financial workflow of the lounge. It dynamically converted the accumulated duration from the time logs into itemized financial charges based on the user's applicable hourly rate or membership status, ensuring all individual customer transactions were calculated and recorded accurately.

4. Sales Recording and Reporting - The system permanently logged all financial transactions to systematically compile analytical summaries. It generated real-time, tamper-proof sales reports categorized by daily, weekly, or monthly intervals, providing management with accurate ledger data to monitor business performance effectively.

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5. Admin and Staff Dashboards - The system featured distinct centralized interfaces tailored to specific user roles. The control screens displayed live metrics regarding room occupancy, active workspace countdowns, real-time sales summaries, and automated time-expiration alerts to facilitate seamless front-desk operations and high-level administrative decision-making.

6. Role-Based Access Control (RBAC) - To reinforce system security and data integrity, strict access permissions were programmatically enforced based on user type. System administrators were granted full access to core configurations, analytical reports, and system parameters, while frontline staff members were restricted to purely operational features such as logging walk-ins, executing bookings, and processing billings.

1.5.1 Limitations may include:

The scope of the developed study was subject to the following limitations:

1. Geographic Limitation - The system was exclusively designed and configured around the specific business model and localized workflows of the WS Students & Professionals Lounge - La Paz Branch. The operational rules, spatial layout, and membership structures were tailored to this single branch and did not support multi-branch scaling, cross-branch data synchronization, or shared roaming memberships.

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2. Platform Limitation - The application was strictly developed as a web-based system. Although the user interfaces were designed responsively to adapt to mobile screens (smartphones and tablets), the project scope completely excluded the development of native mobile applications for the Apple App Store (iOS) or Google Play Store (Android).

3. Internet Dependency - Because the application relied on a cloud-hosted framework and a centralized relational database to prevent fragmented records, it exhibited absolute internet dependency. The system required an active, stable internet connection to process transactions and did not feature an offline-first synchronization mode or local peer-to-peer data backup capabilities during internet outages.

4. No Accounting Integration - The software focused strictly on day-to-day operational logistics (reservations, memberships, time tracking, basic billing, and historical sales logs). It did not function as an enterprise resource planning (ERP) system and completely excluded complex accounting integrations, such as balance sheets, profit-and-loss statements, tax computations, or automated payroll deductions.

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5. **No Hardware Automation** - The platform operated purely at the software application layer. It did not integrate with physical Internet-of-Things (IoT) hardware, biometric fingerprint sensors, RFID card scanners at the entrance, or automated electronic door lock strikes for the study rooms. Physical access verification and final check-in logging remained under the manual oversight of front-desk personnel.

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1.6 Significance of the Study

The outcomes of this study brought significant improvements to the operational workflow of the WS Students & Professionals Lounge - La Paz Branch. By transitioning from a manual, paper-based setup to the Integrated Study Hub Management System, the organization benefited from an organized and centralized platform. This transition successfully addressed the recurring challenges of double bookings and prolonged customer waiting times, allowing the lounge to deliver faster, more efficient, and more reliable services to its clients.

The implementation of this system provided a secure and efficient way to manage membership data and financial transactions. It helped management and administrators gain stronger control over daily operations through a real-time dashboard displaying sales, room occupancy, time-usage information, and daily analytical reports. This operational transparency helped build trust with customers and ensured that the business maintained accurate records without the risk of manual transcription errors or data loss.

Furthermore, this study contributed to the field of Information Technology by demonstrating how targeted software automation can solve real-world business bottlenecks. It serves as a practical reference for Future Developers and Researchers who may face similar challenges when developing management

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systems for the service and hospitality industries. By providing optimized features for reservation tracking and automated billing, this project highlighted the essential role of digital transformation in enhancing customer satisfaction and operational efficiency.

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1.7 Definition of Terms

For the purpose of clarity and consistency, the following terms are defined conceptually and operationally as used in this study:

1. **Administrator** - An individual responsible for the maintenance, configuration, and reliable operation of computer systems and databases (Limoncelli et al., 2016).

- In this study, it refers to the management or authorized personnel of WS Students & Professionals Lounge - La Paz Branch who possessed the highest level of system configuration privileges. In this platform, the role was responsible for overseeing business operations by managing user access credentials (RBAC), generating consolidated sales reports, and auditing historical system logs to ensure operational accountability.

2. **Analytical Reports** - A document that combines data and analysis to help an organization evaluate business decisions, identify trends, or measure operational performance (Sharda et al., 2020).

- In this study, it refers to the systematically generated data summaries, visualizations, and tabular charts within the system that translated raw operational logs into actionable business insights. These reports compiled and analyzed aggregated system metrics to visually map out

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lounge space usage trends, peak operating hours, and customer demographic metrics for administrative decision-making.

3. **Centralized System** - A computing architecture where all data processing, storage, and management control are concentrated within a single primary node or server network (Tanenbaum & Van Steen, 2017).

- In this study, it refers to the unified platform where all operations of the lounge—such as space booking, membership logs, and payment data—were integrated into a single database to eliminate fragmented data records.

4. **Database** - An organized, digitally stored collection of structured data managed through a software framework to allow rapid search and retrieval (Silberschatz et al., 2019).

- In this study, it refers to the relational storage architecture developed using MySQL where all structured information critical to the lounge's operations was systematically stored and protected. This included the secure preservation and real-time retrieval of interconnected tables concerning lounge members' profiles, physical check-in/check-out timestamps, historical financial transactions, and authorized staff accounts.

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5. Membership Tracking - The systematic process and software automation used to monitor profile creation, validity status, and active engagements of registered clients within an organization (Laudon & Laudon, 2021).

- In this study, it refers to a dedicated functional module within the system that actively monitored and managed subscription lifecycles, user profiles, and associated tier benefits of long-term clients. It automatically updated subscription validity windows, tracked recurring billing cycles, and validated active statuses during the check-in process.

6. Operational Bottlenecks - Any point of congestion in an enterprise workflow or service delivery line that reduces the speed and increases the error rate of the entire operation (Heizer et al., 2020).

- In this study, it refers to the specific workflow delays, structural inefficiencies, and administrative errors experienced in the previous manual setup of the lounge, such as accidental double-bookings of study rooms, human mathematical errors during checkout time computations, and slow paper-based invoicing.

7. Reservation System - An automated software application that facilitates the booking, allocation, and scheduling of specific physical spaces or inventory items ahead of actual usage (O'Brien & Marakas, 2019).

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- In this study, it refers to the interactive component of the web platform that permitted both frontline staff and general clients to securely book specific study pods or conference rooms in advance using real-time availability calendars and automated overlap validation logic.

8. Role-Based Access Control (RBAC) - A security mechanism that restricts system access and operations to authorized users based on their specific organizational role or assignment level (Stallings & Brown, 2018).

- In this study, it refers to the built-in security feature that restricted data access and modification privileges based on a defined user hierarchy: *Administrator* (full system configuration and reports), *Front-desk Staff* (check-ins, check-outs, and billing), and *General Customers* (viewing availability, reservations, and membership status).

9. Sales Report - A structured accounting or transaction summary that reflects the revenue, volumes, and financial inputs of a firm over a designated timeline (Romney & Steinbart, 2020).

- In this study, it refers to the auto-generated financial audit summary within the system that programmatically compiled and aggregated revenue earned over daily, weekly, or monthly intervals from workspace room rentals and monthly membership subscriptions.

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10. **Study Hub Management System** - A specialized information system engineered to automate workflows, client handling, asset scheduling, and billing pipelines within shared-workspace facilities (Sommerville, 2016).

- In this study, it refers to the specific web-based software platform developed for the WS Students & Professionals Lounge (La Paz Branch) to replace manual administrative workflows by centralizing user registrations, space reservations, membership subscriptions, time-based logging, and billing.

11. **Time Logs** - A chronological, tamper-evident digital record that captures explicit timestamps marking the execution and conclusion of user operations or resource allocation (Kurose & Ross, 2021).

- In this study, it refers to the automated database records that programmatically captured and stored the exact check-in and check-out timestamps of clients to dynamically calculate precise workspace usage durations and convert them into itemized billing charges.

12. **User Dashboard** - A consolidated, interactive graphical user interface (GUI) designed to present real-time summary data and essential functional entry points customized to a user's role (Shneiderman et al., 2016).

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- In this study, it refers to the personalized control screen accessible by authenticated lounge clients upon logging in, allowing them to independently view active room reservations, track remaining membership validity days, and browse past transaction histories.

13. **Web-Based System** - A software system running on a remote host architecture that can be rendered and utilized through a web browser using client-server requests via HTTP/HTTPS (Pressman & Maxim, 2020).

- In this study, it refers to the architectural deployment framework of the application, which allowed the lounge management platform to be securely accessed over an internet connection using standard web browsers across laptops, tablets, and smartphones without local software installations.

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CHAPTER 2

Review of Related Literature and Studies

Overview of the Integrated Study Hub Management System for
WS Students & Professionals Lounge

A study hub management system represents a specialized class of administrative software designed to orchestrate, automate, and centralize the diverse operational workflows within modern co-working and shared-facility workspaces (Sommerville, 2016). In contemporary business and academic environments, these systems have evolved from simple scheduling applications into comprehensive platform ecosystems. They allow business owners, frontline personnel, and clients to seamlessly coordinate resource allocations, handle memberships, and process financial exchanges regardless of location.

To cater to varying business demands, workspace management systems utilize diverse digital infrastructures:

- **Cloud-Hosted Booking Infrastructures** - A framework that provides rapid, remote access to real-time space availability, allowing clients to secure reservations dynamically via any internet-enabled device (Laudon & Laudon, 2021).

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- **Cross-Platform Mobile Integration** - A method optimized for mobile devices, enabling clients to track their remaining membership duration, receive real-time notifications, and monitor current check-in statuses on the go.

- **Administrative Desktop Command Centers** - A stationary control platform engineered for frontline lounge staff, utilized to handle walk-in clients, audit facility check-outs, and monitor overall utility usage within the hub.

- **Local Relational Databases** - A structured data architecture that serves as the backbone of the lounge, securely holding financial logs, user authentication files, and active time-tracking sheets to ensure high data consistency (Silberschatz et al., 2019).

In the context of daily business operations within co-working facilities, relying on manual transaction frameworks creates significant operational bottlenecks. Paper-based logbooks and fragmented communication channels—such as manual social media message logging—are heavily susceptible to human oversight errors. These errors frequently manifest as overlapping space reservations (double bookings), inaccurate manual timekeeping, disorganized customer profile tracking, and slow invoice generation (Heizer et al., 2020). For example, when multiple clients attempt to secure a group conference room

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simultaneously without a real-time ledger, it often leads to operational conflicts, scheduling delays, and an eventual decline in customer trust.

The strategic deployment of an integrated, automated management system mitigates these administrative risks by enforcing precise digital validation rules. By automating the lifespan of a booking—from the initial online reservation to the final algorithmic billing calculations upon check-out—the system guarantees structural data integrity and transaction transparency (Romney & Steinbart, 2020). Digital platforms drastically accelerate transaction speeds, lower the business's dependence on physically tracking paper logs, and eliminate risks concerning misplaced receipts or corrupted customer records. By resolving these operational vulnerabilities, an integrated system protects enterprise financial data, enhances staff productivity, and guarantees a seamless, modern, and reliable environment for both student and professional clients.

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2.1 Foreign Studies

Management Information Systems: Managing the Digital Firm

The study by Kenneth C. Laudon and Jane P. Laudon (2021) provides a foundational understanding of management information systems (MIS) and their role in modern organizations. In their book, *Management Information Systems: Managing the Digital Firm*, the authors explain that information systems are essential tools that integrate people, processes, and technology to support organizational operations and decision-making. This concept is highly relevant to the development of an integrated study hub management system, as such a system requires efficient coordination of users, resources, and services.

Laudon and Laudon further emphasize that digital systems improve operational efficiency by automating routine tasks and centralizing data management. In the context of a study hub or lounge for working students and professionals, an integrated system can streamline processes such as reservations, attendance tracking, resource allocation, and user management. By reducing manual work and minimizing errors, the system enhances overall service quality and user experience.

Moreover, the authors highlight the importance of data-driven decision-making. Information systems enable organizations to collect and analyze data, allowing administrators to monitor usage patterns, peak hours, and user preferences. This capability is essential for managing a study hub effectively, as

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it helps optimize space utilization, improve services, and support strategic planning.

Additionally, Laudon and Laudon discuss how emerging technologies contribute to innovation and competitive advantage. Implementing an integrated management system for a study hub aligns with this idea by leveraging digital solutions to meet the needs of modern users, particularly working students and professionals who require flexible, efficient, and technology-supported environments.

Overall, the concepts presented by Laudon and Laudon (2021) support the need for developing an integrated study hub management system, as such systems enhance operational efficiency, improve user satisfaction, and enable data-driven management.

Kenneth C. Laudon, Jane P. Laudon, Management Information Systems, 15 Ed., © 2018, Pearson Education, Inc., New York, NY. ISBN 978-0-13-463971-0.

**Information Technology for Management Driving Digital
Transformation to Increase Local and Global Performance,
Growth and Sustainability**

Information Technology for Management is an excellent, comprehensive guide on how modern enterprises bridge the gap between core business strategy and emerging tech

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infrastructure. The study of Turban, E., Pollard, C., & Wood, G. (2021). Digital Transformation Disrupts Companies, Competition, and Careers Locally and Globally. Information Technology for Management discusses the importance of aligning business- IT strategies and explains how companies rely on data, digital technology, and mobile devices to support them in the on-demand and sharing economies. Their goal is to provide students from any business discipline with a strong foundation for understanding digital technology concepts and terminology and the critical role that it plays in facilitating business sustainability, profitability, and growth locally and globally. Many forward-thinking managers and entrepreneurs are digitally transforming their existing business models and reinventing their businesses. By no longer operating and maintaining outdated and complex IT architectures with a mix of legacy systems that can delay or prevent the release of innovative new products and services and absorb large portions of the information technology (IT) budget, companies can add value, increase their customer base, expand their business capabilities, and increase profits.

As businesses continue to change their business models to accommodate the needs of the on-demand and sharing economies, IT professionals must constantly scan for innovative new technologies to provide business value, help shape the future of the business, and facilitate performance and growth in local and global markets. For example, smart devices, mobile apps, sensors, and technology platforms along with increased customer demand for digital interactions and on-demand and shared services have moved commerce in fresh new directions.

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The differences between the on-demand and sharing economies and the six business objectives IT should focus on to enhance organizational performance, growth, and sustainability. The on-demand economy is the economic activity created by technology companies that fulfill individual consumer demands through the immediate provisioning of products and services. In the sharing economy, goods or services are shared between private individuals through an online company or organization. The six business objectives that IT should focus on are as follows:

1. Product development to help businesses respond quickly to changing customer demands.
2. Stakeholder integration to communicate with shareholders, research analysts, and others in the market.
3. Process improvement to increase efficiency and cost effectiveness of internal business processes.
4. Cost efficiencies to reduce transaction and implementation costs.
5. Competitive advantage to bring a product to market cost effectively and quickly.
6. Globalization to stay in contact with its global employees, customers, and suppliers 24/7.

The role of IT in improving business processes. Understand the concepts of business process re-engineering and competitive advantage. Outdated and complex application architectures, with a mix of interfaces, can delay or prevent the release of new products and services, and maintaining these obsolete systems absorbs large portions of the IT budget. As a result, managers

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and entrepreneurs must integrate digital disruptive technology into their products and services to improve their business processes and stay competitive BPR is the concept of using IT to radically improve processes rather than simply making incremental positive changes. Competitive advantage is when an organization differentiates itself by charging less and creating and delivering better quality products or services than its competitors.

Turban, E., Pollard, C., & Wood, G. (2021). Information technology for management: Driving digital transformation to increase local and global performance, growth and sustainability (12th ed.). John Wiley & Sons. ISBN-13 978-1119802525.

**Developing and Implementing Web-based Online University
Facilities Reservation System**

The study by Alkhalidi et al (2018). Community services have recently grown through government and private institutions due to an increased awareness of their importance in societal development. These distinctive services are now available free of charge, expanding beyond financial assistance to encompass various other forms of community support. Ultimately, these diverse services are vital for the progress and prosperity of society by helping individuals become active and constructive citizens.

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Recognizing the vital importance of community engagement, universities actively seek to provide diverse social services to their surrounding populations. These initiatives include establishing free accredited courses for students and hosting essential secondary school entrance evaluations like the General Aptitude Test (GAT) and Qiyas Tests. To further spread this community service culture, Imam Abdulrahman Bin Faisal University (IAU) has proposed a system allowing employees and visitors to reserve various campus facilities, including halls, courts, swimming pools, and theaters. Through this centralized platform, users can seamlessly book facilities, cancel reservations, track requests, and manage their accounts while the university ensures all venues are continuously monitored and maintained.

The main motivation of the work is that there is no online reservation system for all IAU facilities, through the proposed system; the customer can reserve the IAU facility from any place and at any time according to facilities schedule, this will increase the utilization of university facilities by the community members. Thus, an online facilities reservation system for IAU is necessary. The facilities reservation system should provide user friendly interface and complete functions.

The IAU system should be able to satisfy the requirement in providing the suitable functionality to the user, where user should be able to know the availability schedule of different IAU facilities, user should be able to reserve a facility needed

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online anywhere and anytime, and user should be able to check his/her reservation schedule online.

Alkhalidi et al (2018). The work presents the design and development of the online Imam Abdulrahman Bin Faisal University (IAU) facilities reservation system. The proposed platform successfully resolves the imbalanced utilization of campus venues like halls, theaters, swimming pools, and stadiums. Built using Unified Modeling Language (UML), MySQL, and the Visual Basic (VB) programming language, the system addresses the university's prior lack of centralized facility management. It allows local community members to easily reserve these campus spaces free of charge, fulfilling the university's non-profit mission of public service. Furthermore, this system framework can be extended to other departments within IAU, or adapted by external universities and companies to optimize their own resource management.

Alkhalidi, D., Alkhalidi, Dhail., Aldossary, H., Alsmadi, M., Al-Marashdeh, I., Badawi U, A., Alshabanah, A., and Alrajhi, A. (2018). International Journal of Applied Engineering Research ISSN 0973-4562 Volume 13, Number 9 (2018) pp. 6700 6708.

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Restaurant Management System For Kas Valley Restaurant

The recent advancement in technology and its tremendous impact, many businesses have integrated technological tools into their business operations. The food industry is no exception. In the global food industry, technology has made it possible to improve food services, increase customer satisfaction and maximize business returns. According to Alormene C.K (2022). Build resilient infrastructure, promote sustainable industrialization and foster innovation.

In today's fast-paced economy, technology has become a central element for optimizing business operations and maximizing customer satisfaction across various industries. Within the food sector, digitalization has enabled many restaurants to launch online ordering services via dedicated websites and applications. The critical need for these digital systems became undeniably evident during the COVID-19 pandemic when temporary shutdowns starved businesses of in-person customers. Lacking alternative digital infrastructure to support their operations, several traditional restaurants unfortunately went out of business.

Kas Valley Restaurant is struggling to efficiently manage its business operations as growing consumer demand shifts toward faster, technology-driven transactions. The restaurant currently relies on manual channels like walk-ins, phone calls, and social media to process all incoming customer orders. However, these methods have caused excessively long physical queues and frustrating phone delays due to poor network reception or busy lines. Furthermore, Kas Valley lacks a centralized record-

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keeping platform, making it difficult to track order data or generate accurate sales analytics. Managing table reservations is also an arduous process because staff must manually check records to determine booking availability. Ultimately, these combined administrative inefficiencies have significantly slowed down daily operations and decreased overall customer satisfaction.

While advancements in restaurant software and POS technologies have improved global customer service, the Ghanaian food industry has not fully utilized these emerging technologies to optimize operations. The few Ghanaian restaurants that do use software typically limit their applications to basic POS transactions or isolated online ordering systems. Consequently, these applications leave out core operational needs such as table reservations, customer data management, and e-commerce integration. This lack of a centralized system creates major issues with data inconsistency, data redundancy, and scattered information across different units. Ultimately, these fragmented systems make business data analysis difficult for management, which harms the restaurant's overall efficiency and profitability in the long run. The project aims to develop a web-based restaurant management system that facilitates online and in-person meal ordering in Kas Valley restaurant. The application will act as a centralized system for maintaining their records, including customer data. The system also aims to support table reservation management and supply actionable business insights through data analytics on the business sales.

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Alormene C.K (2022). RESAURANT MANAGEMENT SYSTEM FOR KAS VALLEY RESTAURANT. Department of Computer Science, Ashesi University.

Design and Implementation of Reservation Management System
Case Study: Grand Ville Hotels

The advances in mobile technology and availability of broadband have made it easier to search for a hotel and book through an online reservation system. Although mobile applications are unlikely to revolutionize a hotels' reservation system to the same extent that the internet did, it may have an impact on hotel revenue management and the concept of implementing price structures. The study of Wiillams, K and Ajinaja, M (2019) The management and booking of Rooms in hotels is a tedious and complicated task especially if it is done manually. Keeping track of large customers and all their details requires an inordinate space for file cabinets, not to mention the time the hotel administrator would spend going back and forth to file cabinets so as to look up each customer's information. This is why a good hotel reservation system is needed to make this task as easy as possible. The system was developed using the object-oriented software development approach which includes the use of object-oriented analysis, object- oriented design and object oriented programming. This was done so that the developed software can be maintainable, reliable and scalable.

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Willams, K and Ajinaja, M (2019) An effective reservation system has to employ the latest technology to allow hoteliers to gain a competitive advantage over their rivals and remain at the forefront of emerging trends. Hotels must therefore adapt their websites to respond to this development by adopting the use of booking engines and developing marketing strategies to include mobile technology. This will help hoteliers to control all reservations that come from different sources in a single control system, distribute rooms favorably to different online channels and analyze the contribution each sales channel provides to revenue, identifying the most efficient ones.

Room reservation is the most critical component of a hotel system, requiring a dynamic and efficient process to maximize overall profitability. The front office manager is responsible for overseeing this system, which primarily handles two distinct customer types: individual guests and group bookings. During peak seasons, stable group customers are given higher priority over individuals to ensure more secure and predictable revenue. To navigate these demands, the manager relies heavily on a computerized system that provides the necessary data and tools to make optimal room allocation decisions. Ultimately, these digital tools empower hotel leadership to streamline the reservation process and successfully meet their commercial goals.

Grand villa Hotels which is the purported case study used improved manual system in their room booking. During interview session with the manager of the hotel, some of the problems that were identified by using the Excel sheet includes:

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- Lack of immediate retrievals: Since different excel sheets were used to store different transactions e.g. room reservation, food ordering, customer details, it is often hard to retrieve and to find particular information quickly. For example, to find out about the customer's room booking details, the user has to go through various excel sheets. This results in inconvenience and wastage of time.
- Lack of immediate information storage: The information generated by various transactions takes time and efforts to be stored at the right place.
- Lack of prompt updating: Various changes to information like customer's details or cancellation of reservation are difficult to make.
- Preparation of accurate and prompt reports: It takes a lot of time to compile information.

Hotel needs to maintain the record of guests and reserve rooms beforehand. Customers should be able to know the availability of the rooms on a particular date. They should be able to reserve the available rooms according to their need in advance so as to make their stay comfortable. No hotel reservation system can operate alone in the hotel, because it concentrates mainly in reservation process, the system needs an accounting system and a management information system, and so it can serve all the needs of the hotel. It is therefore very important to build new and modern flexible dynamic effective compatible reusable information systems including database to help manipulate different processes and operations that are carried out in hotels.

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Williams, K, and Ajinaja, M. (2019). Design and Implementation of Reservation Management System Case Study: Grand Ville Hotels. Journal of Information Engineering and Applications. ISSN 2225-0506.

**Design and Implementation of a Reservation System and a New
Queuing for Remote Labs**

According to Khazri, Y., Fahli, A., Moussetad, M., and Naddami, A. (2019). The fast development of the Internet technology and its increase of popularity has an enormous impact on the process of teaching.

Modern technology offers innovative tools that enhance educational strategies, with internet and mobile technologies fostering practical scientific and technical training through interactive applications. Consequently, e-learning has become an important component of practical education, offering remote laboratories as a viable alternative to traditional face-to-face training. These distributed systems help solve the challenges of student massification by allowing institutions to share human and material resources while providing real-time online access to practical work.

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These computing and materials-based environments complement conventional training by enabling students to manipulate real, dangerous, or expensive devices remotely. Their development facilitates flexible, hands-on learning for a growing student population despite limited material and coaching resources. Additionally, they allow institutions to share heavy equipment nationally or internationally while expanding available presentation methods.

Remote laboratories function as computing and materials-based environments that complement conventional training by enabling students to manipulate real, dangerous, or expensive devices from a distance. The deployment of these facilities accommodates a growing student population by facilitating flexible, hands-on learning despite shortages in material and coaching resources. Furthermore, this approach allows institutions to share heavy equipment on a national or international scale while simultaneously expanding available presentation methods.

Virtual and remote laboratories reduce the costs associated with conventional hands-on facilities by minimizing the need for expensive equipment, space, and maintenance staff. These platforms also enhance engineering and scientific education by supporting distance learning, improving accessibility for handicapped individuals, and ensuring safety during dangerous experiments. To address these needs, the authors implemented the ELAB FSBM platform, which features a user queue management system and enables collaborative, supervised remote practical work for both teachers and learners.

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Khazri, Y., Fahli, A., Moussetad, M., and Naddami, A. (2019). Design and Implementation of a Reservation System and a New Queuing for Remote Labs. iJOE – Vol. 15, No. 12. <https://doi.org/10.3991/ijoe.v15i12.11098>.

**Project Management in Prototyping an Online Table
Reservation System**

The rapid digitalization of the food and beverage industry has transformed operational management and customer interactions through smart, data-driven platforms for ordering, payment, and customer relations. Felix, A., Leornado, V., Melcelyn., and Ananda, D. (2026). Despite this ongoing digitalization, many restaurants still rely on manual methods to manage table reservations, such as phone calls, walk-in requests, or unstructured messaging applications. These manual processes often lead to miscommunication, double bookings, long queues, and difficulties in predicting demand during peak hours. As a result, customers may experience long waiting times and dissatisfaction, while restaurant owners struggle to optimize seating capacity, staff allocation, and raw material planning.

Previous studies have examined online reservation systems and digital solutions for restaurant management, including web based booking platforms, smart service technologies, and integrated restaurant management applications. These works generally focus on the design and development of

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reservation systems, the use of specific technologies, or the impact of digitalization on service quality and business performance. However, most of them discuss technological and managerial aspects separately and do not explicitly adopt a comprehensive project management framework to guide the development and implementation of online table reservation systems.

This situation highlights a clear research gap in how project management standards, like PMBOK, are systematically applied to developing online table reservation systems within the food and beverage sector. Consequently, there is a strong need for studies that combine both technical system development and structured managerial frameworks to address scope, time, cost, quality, risk, and stakeholders simultaneously. These findings are consistent with Akyüz and Balkan (2024), who emphasize that smart service technologies in hospitality require not only robust technical design but also structured project management to achieve operational efficiency.

Felix, A., Leornado, V., Melcelyn., and Ananda, D. (2026). This study developed a web based online table reservation system for the food and beverage (F&B) industry by integrating the Project Management Body of Knowledge (PMBOK) framework with the Systems Development Life Cycle (SDLC). The main findings show that the proposed system can reduce miscommunication in reservations, shorten waiting times, and provide more transparent information about table availability and restaurant details. The evaluation results indicate that users

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perceive the system as easy to use, responsive, and more reliable than manual phone or walk-in reservation methods.

This research theoretically contributes to the literature on hospitality and F&B digitalization by showing how PMBOK project management standards can be applied to online reservation systems. It demonstrates that core knowledge areas—such as Scope, Time, Cost, Quality, and Risk Management—can be directly mapped onto concrete stages of system design. By doing so, the study successfully bridges the long-standing gap between purely technological and managerial perspectives. Ultimately, this integrated view enriches existing literature, which frequently discusses digital transformation and system features at a much more fragmented level.

Practically, these findings offer valuable guidance for F&B businesses and system developers planning to implement online table reservation solutions. The included system design, evaluation insights, and project documentation (such as the WBS, Gantt Chart, and budget planning) can serve as a functional reference model for similar efficiency-focused projects. Ultimately, adopting this structured project management approach allows restaurant managers to optimize resources, mitigate implementation risks, and ensure the new digital system delivers tangible business benefits.

Felix, A., Leornado, V., Melcelyn., and Ananda, D. (2026). PROJECT MANAGEMENT IN PROTOTYPING AN ONLINE TABLE RESERVATION SYSTEM. Indonesian Interdisciplinary Journal of Sharia Economics

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(IIJSE). e-ISSN:2621-606X. Vol. 9. No. 2(2026). Page: 13325-13347.

**Online Reservation Management System to Increase Transaction
Efficiency at MOMENKITA Photo Studio**

In an increasingly competitive and complex business environment, companies around the world are looking for innovative ways to improve their operational efficiency. The key to achieving sustainable growth and maintaining competitiveness in the global market lies in corporate efficiency. The study by Firmansyah et al. (2024). "Momenkita Studio" is a small and medium-sized enterprise (SME) operating in the creative industry, specifically as a photo studio. It faces challenges with management that has not yet been optimized. Some noticeable obstacles include the lack of a structured customer record system, often complicated scheduling, and unclear monitoring of studio activities

One increasingly relevant solution to achieve this goal is through the implementation of Web-based corporate management applications, especially in photo studio companies. This inefficient management can lead to schedule overlaps, neglect in record-keeping, and difficulties in tracking payment transactions. This study uses a qualitative method. Therefore, there is a need for a solution that can improve management efficiency.

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"Momenkita Studio" in scheduling photo sessions and meetings with customers is still done manually, so it becomes complicated without an effective schedule management tool. Presentation and portfolio management are difficult in physical form, and the studio must rely on photo prints. So website-based company management is needed to overcome existing problems. This application is designed to help companies overcome challenges in customer management in an efficient and structured way. This application allows for more efficient collection and storage of customer data.

Firmansyah et al. (2024). The focus of the research is on the design and implementation of company management applications which include the design of customer management, namely the photo shoot schedule reservation system, which is done online, transaction management, namely payment for photo shoot services, purchase of studio equipment, report management, namely customer reports, weekly, monthly, and annual transaction reports, user management, namely user access rights for the application. In addition, this research will provide a deep understanding for researchers, business practitioners, and decision makers about how technology can be an effective medium in increasing company efficiency and strengthening customer relationships.

Firmansyah et al. (2024). This research developed a dedicated photography website that visually displays reservation

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schedule availability using an interactive, integrated calendar. An interview with the owner of MOMENKITA Foto Studio revealed that the platform aims to enhance service accessibility, simplify the booking process, and provide a more comfortable customer experience. To achieve these goals, the website features specialized tools including package selection, secure online payment integration, a portfolio photo gallery, clear service information, and easily accessible contact details.

Firmansyah et al. (2024). This research is able to provide solutions to the problems that exist in MOMENKITA Photo Studio, namely it can facilitate the need for information on the availability of reservation schedules because the schedule is visualized in the form of a calendar, reservation data reports that are automatically summarized so that they are better organized. The results of the study show that the concept of schedule visualization can make it easier for customers to get information so that they can minimize reservations with dates or times that are not available. Providing information via a website can provide more complete information and present it as more interesting information and can be used to help as a promotional medium.

Firmansyah, F., Razak, M,R,R., and Sofyan, W. (2024). Online Reservation Management System to Increase Transaction Efficiency at MOMENKITA Photo Studio. Journal of Accounting and Finance Management (JAFM). E-ISSN: 2721-3013. Vol. 5.

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**Online Reservation Systems in E-Business: Analyzing Decision
Making in E-Tourism**

The study by Halkiopoulou et al. (2020). Focuses on exploration of knowledge for online booking systems and on the views of local students-users concerning the booking rate based on these online systems. Another perspective of this project is to investigate the decision-making process (emotion-focused) that they follow in order to choose a tourist destination via online booking systems

The electronic commerce is defined as the activity of sale and marketing for products and services through an electronic system such as, for example, the Internet. It involves the electronic data transfer, the distribution management, e-marketing (online marketing), online transactions, electronic data changes, the automated inventory of used management systems, and automated data collection. E-Tourism (electronic tourism) is a part of electronic commerce and unites one of the fastest development technologies, such as the telecommunications and information technology, hospitality industry and the management / marketing / strategic planning.

This research aims to analyze students' perspectives on electronic booking tools and investigate their specific decision-making processes when planning travel through these digital platforms. While the continuous growth of online booking systems benefits both tourists and hotel owners alike, numerous studies have already been conducted to identify and evaluate how well these modern systems satisfy travelers' needs.

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As a result ,this demonstrates how the expansion of the internet has significantly transformed market conditions by providing tourist organizations with innovative new tools for marketing and management. While this digital evolution facilitates direct interaction with users and changes the entire tourism development process, a review of existing literature reveals that the field of e-Tourism still faces numerous unanswered questions.

Knowing the cause and the consequence is, in simple terms, what should be achieved. In this direction many issues can and should be raised that question the relationship of the customers, their characteristics, the way they 'came' to the hotel (information search, booking and paying for accommodation) as well as various factors that led to their decision.

Halkiopoulos, C., Antonopoulou, H., Papadopoulos, D., Giannoukou, I. & Gkintoni, E. (2020). Online reservation systems in e-Business: Analyzing decision making in eTourism. Journal of Tourism, Heritage & Services Marketing, ISSN2529-1947, Vol. 6, No. 1, pp. 9-16.
<http://dx.doi.org/10.5281/zenodo.3603312>.

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The Dentist Online Reservation System Design and Implementation
Web Based Application and Database Management System Project

Technology has changed many aspects of life and people's daily life is becoming indivisible from the network due to the development of Internet. With online dentist reservation system, the process gets much faster and more efficient than traditional way. Thousands of business and organizations have already discovered the advantage gained by using the online reservation system. The study by Kerdvibulvech, C., and Win, N,N. (2012). Online dentist reservation system allows the system administrator to access and manage the database online, quickly pull data and create strong reports right from the online reservation system with most practical to find the fastest information, instead of having to maintain and manage separate data files, folders and spreadsheets. They simply navigate to the system just as any Web Site. The data will be housed securely and safely online.

The system also helps the administrator to maintain the database online easily. It enables the administrator to check the patient's requests, manage the appointment schedule, and manage the patient's information. It offers an opportunity for patients to submit their personal comments to the dental clinic to give better services. The design for the proposed system can be obtained easy way to access the enquire information about the dentist appointment.

Kerdvibulvech, C., and Win, N,N. (2012). The Online Dentist Reservation System was successfully built using Microsoft Access, ASP.NET 2010, SQL Server, and the C# programming language. Due

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to restricted financial resources and a tight timeframe, the developer could not fully implement all intended system functionalities. However, the system's flexible architecture allows junior students to pursue future work on the project and add new modules without disrupting the current setup. Potential enhancements include linking the booking and appointment tables to help dentists manage their schedules more easily. Finally, the system can be expanded to support online billing and allow patients to cancel, pre-pone, or post-pone their appointments directly through the platform.

Kerdvibulvech, C., and Win, N,N. (2012). The Dentist Online Reservation System Design and Implementation Web Based Application and Database Management System Project. International Conference on Education Technology and Computer (ICETC2012) IPCSIT vol.43 (2012) © (2012) IACSIT Press, Singapore.

**Web Site &Web Based Hotel Reservation Management System For Sky
Lodge Hotel**

Accordin to P. B. N. N. De Silva (2017). Relaxation they say is the best medicine for the modern life style, which sharpens the quality of the lifestyle of an individual.

Most of the people use to travel and spend time by living with the beauty of the nature. Therefore, the hospitality

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providers have taken measures to improve the best available solutions for this purpose.

Tourism is a fast-growing industry in Sri Lanka and stands as one of the country's primary financial contributors. Recognizing the need for an organized institutional framework, the Government decided to develop the tourism sector in a planned and systematic manner in 1966. This initiative led to the official establishment of both the Ceylon Tourist Board and the Ceylon Hotels Corporation. The Ceylon Tourist Board operated as a statutory body designed to allow greater flexibility in financial management and organizational decision-making. Meanwhile, the Ceylon Hotels Corporation was formed as a joint stock company to serve as the government's commercial arm for expanding tourist accommodations and facilities.

The Sky Lodge is a prominent hospitality provider that caters to both local and foreign travelers. It is situated in the beautiful city of Gampola, which is located within the ancient cultural district of Kandy in Sri Lanka's Central Province. Guests staying at the hotel enjoy close access to highly sacred local temples, including the Temple of the Tooth Relic and the Bahirawakanda Temple. The surrounding region also offers proximity to scenic and historic attractions, such as breathtaking waterfalls, the Peradeniya botanical garden, and the ancient Bogoda Covered Bridge. Ultimately, the combination of strong cultural sights, lush mountains, diverse wildlife, and friendly locals makes the Sky Lodge an ideal destination for an unforgettable holiday.

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P. B. N. N. De Silva (2017). The Sky Lodge Hotel's innovative concept led to an increased workload, which reduced task accuracy and created issues like booking errors and faulty payment calculations. Employees were forced to spend excessive time completing manual room reservations, leaving data vulnerable to damage and lowering overall guest satisfaction. To overcome these operational hurdles, a highly anticipated Web Based Hotel Management System was introduced to replace the manual process. This proposed platform features centralized modules designed to efficiently manage hotel rooms, booking processes, employee registrations and leaves, hotel expenses, and administrative system activities.

After considering the complications and drawbacks of their existing manual system, the management of the Sky Lodge Hotel decided to transition to an IT-based solution. As a result, a web-based hotel reservation management system and accompanying website were designed and developed to deliver expected benefits to management, employees, and guests alike.

P. B. N. N. De Silva (2017). Web Site & Web Based Hotel Reservation Management System For Sky Lodge Hotel. University of Colombo School of Computing. BIT registration number: R100490 Index number: 1004905.

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**Development of a Web-Based Reservation Systems for Federal
Polytechnic Kaltungo**

According to Abdulrauf et al. (2023). Education is often regarded the most suitable tool for national development. This helps and assists citizens in developing the skills and attitudes required for nation building.

It is no longer news that many Nigerian university graduates and other higher educational institutions fall short of employer or industry standards [1]. Concerns have been raised over the years about Nigeria's hospitality/tourism education and training system as a source of skilled labor for the tourism industry.

To align with global best practices, the National Board for Technical Education (NBTE) mandated that all higher education tourism departments in the country must implement an operational reservation and booking mechanism for practical training. The department is tasked with equipping students with the essential skills and knowledge required to meet modern industry demands. However, persistent challenges in providing hands-on training for online reservation systems have resulted in a notable practical skills gap among graduating students entering the workforce.

To bridge this gap, this research report presents the development of an online reservation system that will be used to teach students of the Department of Tourism at Federal Polytechnic Kaltungo how to carry out online reservations of

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hotel rooms, airline tickets, and food ordering. The system was designed to provide students with hands-on experience and practical skills in online reservation, which is an important aspect of the tourism industry.

Abdulrauf et al. (2023). Online reservation systems are web-based reservation management systems that allow customers to book and pay for online activities such as hotel rooms, tickets, and restaurant tables or related services through the website. This service is operated online, minimizing the personnel workload and the possibility of multiple reservations. Several studies show that, online reservation systems were designed and developed to improve customer service, expedite the reservation process, and boost operational efficiency for merchants. Online reservation systems are used in a variety of businesses such as hospitality, transportation, and entertainment.

System development results demonstrated that the online reservation system created for the Department of Tourism at Federal Polytechnic Kaltungo effectively teaches students how to book hotel rooms, airline tickets, and food. Users found the platform to be highly user-friendly, featuring clear, easily understandable instructions alongside a visually appealing interface designed to enhance the user experience. To evaluate the system's quality, strength, and overall usability, the researchers conducted a preliminary study using a yes-or-no questionnaire. The evaluation process involved gathering direct feedback from the Dean of the School of Science, the Head of the Department, five instructors, and 13 students. This diverse

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group of participants was instructed to test every module within the system thoroughly before completing their feedback forms.

Abdulrauf et al. (2023). The development of the online reservation system at Federal Polytechnic Kaltungo marks a significant advancement in providing students with critical practical skills required by the modern tourism industry. Due to its user-friendly interface, clear instructions, and demonstrated instructional effectiveness, the platform serves as a highly valuable tool for teaching and learning within the department. Based on these successful outcomes, it is recommended that the department formally integrate this system into its curriculum for teaching reservation skills. To preserve its long-term effectiveness and usability, the software must also be regularly updated and properly maintained over time. Ultimately, further research is recommended to investigate how this system impacts graduate employability and the broader development of the tourism industry in Nigeria.

Abdulrauf, A., Ogar. A.U., Delmut, R,D., and Emmanuel, O, C. (2023). International Journal of Engineering and Information Systems (IJEAIS). ISSN: 2643-640X. Vol. 7 Issue 7. Pages: 25-38.

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**Understanding the Effects of Reservation Systems and Online
Transaction Capacity on the Competitiveness of Travel Agencies**

The study by Işkın, M., Prentice, C., Eker, N., and Şenge Ü. (2023). The purpose of this study is to determine the effect of travel agencies' reservation systems and online transaction capacities on their competitiveness. With the developments in technology and the integration of these developments in all business areas, travel agencies have also had to benefit from information and communication technologies. In the research, the primary data were used, and the data were collected by the questionnaire method.

The rapid acceleration of globalization and technology has driven profound socio-economic changes, giving rise to an "information society" where data is a highly valuable, tradeable product. To navigate this new landscape, modern global approaches emphasize the sustainable management of information and the strategic integration of technology.

Işkın, M., Prentice, C., Eker, N., and Şenge Ü. (2023). In the research, the primary data were used, and the data were collected by the questionnaire method. The convenience sampling method was used, and data were collected from a total of 566 travel agency managers in Istanbul. According to the results of the research, reservation systems and online transactions capacities of travel agencies affect their competitiveness. Reservation systems and online

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transactions capacities of travel agencies affect their competitiveness in specific issues such as cost, service quality and market share.

Although appropriate and enlightening results were obtained regarding travel agencies technological capacity and competitiveness, research was carried out on the perceptions and evaluations of the managers in this study. In this context, in future studies, studies can be conducted on the relationships between measurable quantitative variables such as the online transactions of travel agencies and the capacity and transaction volume of reservation systems and competition structures.

The study reveals that the traditional office travel agencies should be more successful than their competitors in terms of reservation systems and online.

Işkın, M., Prentice, C., Eker, N., and Şenge Ü. (2023). Understanding the Effects of Reservation Systems and Online Transaction Capacity on the Competitiveness of Travel Agencies. EJTHR 2024; 14 (1): 55-70. <https://doi.org/10.2478/ejthr-2024-0004>.

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**Development and Implementation of a Python-Based Hotel
Management System: A Comprehensive Tool for Room Reservation,
Payment, and Administration**

The study by Alam, M., Rahman, M. S., Rivin, M.A. H., Uddin, M. B., & Sizan, M. M. H. (2024). The hotel industry faces challenges related to manual management processes, which can lead to inefficiencies and errors in operations such as booking, payment processing, and administrative tasks. To address these challenges, there is a growing demand for automated systems to streamline operations and improve overall efficiency.

This study investigates the design and implementation of a Python-based Hotel Management System built using the Django framework and MySQL database. The system is developed to handle essential hotel operations, including room booking, payment processing, and general management, while ensuring scalability, security, and usability.

The research follows a software development approach based on functional and non-functional requirements, with a focus on unit, integration, functional, and security testing to ensure the system's reliability and performance. Key features of the system include real-time booking functionality, secure payment processing, and authenticated user access for administrators and customers. The results reveal that the

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system effectively reduces operational errors, enhances the user experience for both hotel administrators and guests, and improves overall operational efficiency.

The goal of this study is to create an automated hotel management system using python to solve the operational inefficiencies within the hospitality industry involving rooms reservations, payment processing, and administrative management. The research addressed the scientific problem of how to combine advanced technology into a scalable, real-time, and user-friendly solution for streamlining hotel operations.

The developed hotel management system successfully optimizes daily operations, enhances security, and improves decision-making through features like real-time updates and role-based access control. Thorough testing and a modular architecture ensure that the platform is reliable, scalable, easy to maintain, and adaptable to shifting technological trends. From a management perspective, the system provides the real-time data analysis necessary to boost customer satisfaction and streamline administrative workflows. Despite these strengths, the study is currently limited because the system was only evaluated under controlled conditions rather than in diverse, real-world environments. Furthermore, the current version focuses strictly on essential functionalities, leaving room for future enhancements like advanced analytics and third-party platform integrations. Future research aims to bridge these gaps by exploring dynamic pricing algorithms, IoT-enabled

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intelligent rooms, and large-scale implementations across different geographical regions.

Alam, M., Rahman, M. S., Rivin, M.A. H., Uddin, M. B., & Sizan, M. M. H. (2024). DEVELOPMENT AND IMPLEMENTATION OF A PYTHON-BASED HOTEL MANAGEMENT SYSTEM: A COMPREHENSIVE TOOL FOR ROOM RESERVATION, PAYMENT, AND ADMINISTRATION. American International Journal of Business and Management Studies 6(1), 15-44. <https://doi.org/10.46545/aijbms.v6i1.322>

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2.2 Local Studies

**Web-Based Reservation and Management System for Nature Spring
Resort**

According to Ramento, Dancel, Taberlo, and Anquillano. (2024). The significant improvements in accuracy, operational processes, and user contentment demonstrate the system's success.

The study focuses on the business which experienced challenges due to its manual reservation system that relied on logbooks, leading to errors and inefficiencies. To address these issues, the researchers developed a web-based reservation and management system that allows guests to book online and check availability.

As a result, they developed system transforms and improves the resort's booking and management processes. This system provides a simple and user-friendly online interface for customers to book rooms and cottages effortlessly. Customers can also view the resort's available amenities and services. The developed system includes real-time availability updates, assuring accurate booking information while reducing the possibility of overbooking. Furthermore, the management feature allows Nature Spring Resort owners to handle reservations and monitor guest preferences easily.

The development of the web-based reservation and management system for Nature Spring Resort has improved the reservation

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process, making it a valuable tool for both users and managers. By addressing and resolving past ineffectiveness, the system allows users to make bookings faster and more effective. The significant improvements in accuracy, operational processes, and user contentment demonstrate the system's success. These results illustrate the system's ability to improve the reservation and booking experience, making it more reliable and user-friendly. Overall, the developed web-based reservation and management system modernizes Nature Sprint Resort's booking process and significantly improves its operational capabilities.

Ramento, L. F., Dancel, D. M., Taberlo, E., & Anquillano, (2024) Web-Based Reservation and Management System for Nature Spring Resort. Journal of Pure and Applied Sciences. ISSN:2984-9896.

Online Reservation Inventory Systems in Higher Education: A Gap Analysis and Enhancement Framework with SMS Notification Support

The study of Sayson, J,M., Reyes, J,C., and Pisueña, J,S. (2025). Inventory and reservation management systems are essential for maintaining seamless administrative and academic operations within higher education institutions. As a core component of material management, inventory heavily impacts an organization's financial outcomes, operational performance, and customer satisfaction. To optimize these outcomes, universities have increasingly adopted digital inventory systems to improve the tracking, allocation, and utilization of campus resources. These tools are particularly valuable for managing shared assets

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like university equipment, facilities, and daily supplies. Ultimately, implementing these centralized digital platforms ensures the proper documentation necessary for strict institutional accountability and future planning.

Online inventory and reservation systems are widely implemented across universities, their efficiency and functionality vary significantly. Academic literature highlights that these digital systems are highly effective at eliminating manual errors, preventing untracked inventory, accelerating slow reservation processes, and providing crucial real-time updates. Most studies emphasize that successful systems share core features like item tracking, reservation logs, reporting functions, and user-level access controls. However, despite these technological advancements, current systems still suffer from notable limitations, including poor user notifications, system accessibility gaps, and limited integration with modern communication technologies.

Sayson, J,M., Reyes, J,C., and Pisueña, J,S. (2025). This research utilizes a systematic literature review and design science approach to analyze inventory and reservation systems within higher education institutions by evaluating published studies, reports, and case analyses. A primary focus of the study is examining the integration, or noticeable lack, of real-time SMS-based notification features. Implementing these text alerts can significantly improve user-administrator communication, reduce missed bookings, and enhance overall system responsiveness. By focusing on recurring limitations within Philippine state universities and similar campuses, the

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study identifies critical operational gaps and evaluates the distinct benefits of SMS integration. Ultimately, the insights gained from this literature will guide proposed system upgrades to support more efficient, user-friendly reservation and inventory processes.

This study successfully identified the key limitations of existing academic resource reservation systems and proposed a comprehensive enhancement framework to bridge these gaps. Developed through a rigorous literature review and design science research principles, the framework integrates vital features like real-time inventory tracking and optimized scheduling. It also introduces automated SMS notifications and secure, role-based access controls for users. By prioritizing usability and data-driven insights, the system aims to deliver a responsive and efficient experience for students, faculty, and administrators alike. Ultimately, this framework establishes a strong foundation for future system development that is highly scalable, configurable, and aligned with institutional needs.

Sayson, J,M., Reyes, J,C., and Pisueña, J,S. (2025). Online Reservation Inventory Systems in Higher Education: A Gap Analysis and Enhancement Framework with SMS Notification Support. International Journal of Research in Engineering and Science (IJRES). ISSN (Online): 2320-936 4, ISSN (Print): 2320-9356. www.ijres.org Volume 13 Issue 7 | July 2025 | PP. 73-82

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**Digital Transformation of Reservation Services: Development of
the Foxrock Internet Reservation Management System**

The study by Cajetas et al. (2026). The rapid advancement of technology has fundamentally transformed the way organizations manage reservations and deliver customer services. A growing number of businesses have adopted digital reservation platforms to streamline the booking process and reduce turnaround time. Such systems offer customers a significantly more convenient and accessible channel for scheduling appointments or securing reservations without the constraints of location or business hours, which in turn contributes to higher levels of customer satisfaction. Moreover, by automating booking transactions, these platforms substantially reduce the volume of incoming phone calls and email inquiries, allowing staff to redirect their attention toward higher-value operational tasks.

A Management Information System (MIS) serves as a unified platform that gathers, processes, stores, and distributes data to facilitate decision-making, organizational coordination, and operational oversight. The present study examines how MIS contributes to improving the efficiency of organizational decision-making by delivering timely data and in-depth analytical insights that align with institutional objectives while harnessing emerging technologies such as artificial intelligence and cloud infrastructure. By bringing together human resources, data assets, hardware, software systems, and established procedures, MIS enables organizations to streamline operations and empowers managers to arrive at well-informed strategic and day-today decisions.

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The online reservation platform developed by the research team is expected to assist Foxrock Internet Services in modernizing and enhancing its existing service delivery mechanisms. Through the company's userfriendly website, clients will be able to conveniently schedule internet installation appointments entirely online. Foxrock Internet Services, founded in 2018 by Mr. Kevin Paul Pacina and Mrs. Sharmaine Duraca Pacina, operates in Barangay Malbago, Madridejos, Cebu, and offers both fiber and wireless internet connectivity to its subscribers.

Numerous rural and underserved localities continue to suffer from poor or completely absent internet connectivity, commonly referred to as "dead zones." This lack of access severely curtails residents' capacity to engage with online services and leaves them uninformed about available connection options, underscoring the persistent digital divide separating rural communities from urban counterparts.

Cajetas et al. (2026). Prior research indicates that digital systems strengthen communication channels between service providers and clients while alleviating administrative burdens. By offering real-time availability information and automated booking confirmations, these platforms eliminate the need for repetitive follow-up inquiries. This automation significantly cuts the time and financial costs traditionally associated with manual scheduling workflows. In a standard system setup, customers simply supply their personal details, product preferences, and preferred payment option. Finally, settlement can be conveniently handled by field personnel or

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facilitated through flexible digital services like GCash to increase accessibility.

Cajetas et al. (2026). Organizations increasingly adopt digital reservation platforms to optimize their workflows and improve operational efficiency. These modern systems reduce dependence on physical paperwork while minimizing scheduling conflicts like duplicate bookings. Additionally, they offer users the flexibility to submit reservations at any hour without being restricted by standard office hours. Despite these clear advantages, a number of organizations still rely on traditional manual booking methods that remain prone to errors. In contrast, the Foxrock Internet Reservation Management System addresses these inefficiencies through clearly identified functional requirements and an intuitive, user-centered design.

The overall findings of the study reflect a highly favorable reception of the Foxrock Internet Reservation Management System across all evaluated dimensions. Respondents assigned exceptional ratings ranging from "Very Satisfactory" to "Excellent" across a broad array of core functional criteria. These positive results confirm that the system successfully met, and in several respects surpassed, user expectations regarding its primary capabilities. The only identified area with notable potential for future growth was the responsiveness of the mobile application interface. Although the mobile experience still achieved a "Very Satisfactory" rating of 4.3, this specific score indicates a clear opportunity to further optimize performance and user engagement.

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Cajetas, M, M, Jr., Alo, S, M., Chavez, C., Ilustrisimo, D., Arranchado, K, M., Cabahug, J, Jr., and Alegre, K, B. (2026). Digital Transformation of Reservation Services: Development of the Foxrock Internet Reservation Management System. International Journal of Computer Science and Mobile Computing, Vol.15. pg. 51-59. DOI: <https://doi.org/10.47760/ijcsmc.2026.v15i03.006>.

**Development and Evaluation of a Web- Based Retail Management
System for Small Business Operations**

Dela Cruz and Santos (2022) investigated the development of a web- based retail management system designed specifically for small businesses. In their study, they observed how shifting from manual transaction and inventory processes to an automated web- based system directly improved operational performance. Their findings showed that

The system significantly enhanced transaction accuracy by minimizing errors commonly associated with manual data entry and recordkeeping. Additionally, the system enabled real- time inventory tracking, helping business owners monitor stock levels instantly and make informed replenishment decisions. Such automation reduced discrepancies between recorded and actual inventory, thereby enhancing overall business productivity and reliability.

These results are relevant to the current study because they demonstrate how automation and integration of digital

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systems can improve efficiency, accuracy, and data visibility – principles that support the need for an automated study hub management system with features like real-time scheduling, resource tracking, and user interactions.

Dela Cruz, J. R., & Santos, A. (2022). Development and evaluation of a web-based retail management system for small business operations. Journal of Small Business Information Systems, 6(4), 23-35.

SORMS: Steffi's Online Reservation Management System

The Study of Balintag et al. (2025). SORMS (Steffi's Online Reservation Management System) is a highly successful, web-based digital solution developed specifically to transition Steffi's Garden Resort away from the inefficiencies of a manual booking system. concerns about usability, functionality, safety, privacy, and accuracy of information, manual reservations remain prone to errors, often causing inconvenience for customers.

These inefficiencies highlight the necessity for service providers, such as hotels and resorts, to implement online reservation management systems. Research suggests that hoteliers or staff can opt to pay only the software's license cost, which is more cost-effective than paying commissions to external booking portals when customers make reservations. Additionally, customers find it more convenient to book at their leisure and prefer utilizing technology over manual data entry.

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Manual reservation procedures in various industries have proven inefficient. In restaurants, customers often experience long wait times for table availability, making software-based systems a more effective alternative.

Developing an online reservation system requires the establishment of digital infrastructure, including servers, networks, and software applications, contributing to the goals of Sustainable Development Goal (SDG) 9—resilient infrastructure, sustainable industrialization and innovation. Through digitalization, businesses can optimize operations and free up resources for further innovation.

Businesses must adapt to technological advancements not only to stay competitive but also to enhance service quality. This study aims to develop an efficient and user-friendly reservation system for Steffi's Garden Resort, ensuring a seamless experience for both customers and resort staff. While reservations may seem like a straightforward task, they involve numerous challenges that can impact guest satisfaction.

To address these challenges, the proponents developed the Online Reservation Management System—a user-friendly platform that allows customers to seamlessly book and manage their reservations online, reducing errors and enhancing overall guest satisfaction.

The study developed Steffi's Online Reservation Management System (SORMS) using Agile Scrum methodology, with researchers implementing key phases including product backlog refinement, sprint planning, and iterative development cycles. The

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researcher employed standard design tools - Gantt charts, data flow diagrams, and entity-relationship models - to create a robust architecture, while maintaining close collaboration with resort staff and customers to ensure practical relevance.

Balintag, E, E., Concepcion, K, J, H., Dagman, P, J., Ducos,. J, A, Fernandez, D,. and Villanueva E. (2025). SORMS: STEFFI'S ONLINE RESERVATION MANAGEMENT SYSTEM. PSYCHOLOGY AND EDUCATION: A MULTIDISCIPLINARY JOURNAL. Volume: 36. Pages: 783-790. ISSN 2822-4353.

**Managing Customer Reservations of BulSU Hostel through the
Development of Online Information and Reservation System**

This study by Castro et al. (2014), later published as a book in 2017, offers practical insights into how technology can streamline service operations and improve customer experiences in institutional facilities. Released initially in the International Journal of Advances in Management and Economics by Kiban Research Publication Pvt. Ltd. and subsequently by LAP LAMBERT Academic Publishing, the authors explain that an online information and reservation system acts as a centralized platform that connects customers, staff, and facility resources to automate booking processes, ensure real-time availability tracking, and simplify administrative tasks.

This concept is highly relevant to the development of various facility management systems—such as library room

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reservation or hostel booking platforms—as such systems require effective integration of user access, resource monitoring, and service coordination to reduce errors, save time, and enhance overall operational efficiency. Accurate and timely reservations are necessary to ensure proper service and positive customer experience for many service-oriented businesses.

In this study, the proponents considered factors like customer satisfaction and positive experience upon the stay in a hostel which started in the reservation procedure. In order to manage customer reservations of Bulacan State University (BulSU) Hostel, the proponents design and develop an “Online Information and Reservation System for BulSU Hostel”. Different functionalities were incorporated into the developed system like the creation of different user account for the staff/front desk and online customers, prices updates through online admin portal, payments thru bank account, confirmation thru email after reservation, report generation concerning income, number of customers and monthly report, counting the numbers of viewers and settings for rates, taxes and other discounts. Visual Basic .Net and ASP. Net. were used for the system front-end while Microsoft Access was used as the database application as well Crystal Report was used to display reports.

The developed system was evaluated using the software quality standard ISO 9026 with different software criteria as follows: Functionality, Accuracy, Reliability, User-friendliness, and Security and all of them were interpreted as “Acceptable” based on the equivalent ratings presented in the Likert scale. Reserving a room ensures or guarantees the guest

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about the availability of a room on arrival at the hotel, as reservation is a commitment made by the hotel, when the hotel has accepted the reservation request Prior to internet services, travelers could write, telephone the hotel directly, or use a travel agent to make a reservation. Online hotel reservations are also helpful for making last minute travel arrangements. Hotels may drop the price of a room if some rooms are still available. In Bulacan State University (BulSU), Philippines, Hostel started in 2005 which caters the growing number of clientele who wants a comfortable and convenient place to stay while taking short-term courses in the University or attending conferences or seminars within the campus. The BulSU hostel is a budget-oriented, shared-room accommodation that accepts individual travelers or groups for short-term stays. In some countries the word hostel can also refer to student accommodation that provides common areas and communal facilities.

The University Hostel is also popular among foreign students and guests of the University. The Hostel 45 rooms as well the facilities are also made available for the interests of the students of the university. Students taking up Hotel and Restaurant Administration likewise use the facility for their on-the-job training. Usability and User Experience The study "Managing Customer Reservations of BulSU Hostel through the Development of Online Information and Reservation System" by Castro et al. (2014) highlights that usability and user experience are central to the success of web-based reservation platforms. The authors emphasize that the system was designed with a simple, clear interface and straightforward navigation,

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allowing both customers and staff to perform tasks easily even with limited technical knowledge.

Evaluation results showed that the system was rated highly in terms of ease of use, understandability, and responsiveness, ensuring that users can quickly find information, complete bookings, and access services without confusion. Supported by related local studies such as Vega and Generoso II (2025) and Dacanay et al. (2023), which also stress the importance of intuitive design and user-centered features, these findings confirm that prioritizing usability and positive user experience not only increases user acceptance and satisfaction but also ensures that the system serves its purpose effectively by being accessible, practical, and enjoyable to use for all types of user.

Castro, M. D. B., Nuqui, A. V., Custodio, E. B., & Bernardino, I. S. (2014). *Managing Customer Reservations of BulSU Hostel through the Development of Online Information and Reservation System. International Journal of Advances in Management and Economics*, 3(5). Kiban Research Publication Pvt. Ltd.; ISSN: 2278-3369

Castro, M. D. B. (2017). *Managing Customer Reservations of BulSU Hostel. LAP LAMBERT Academic Publishing; ISBN: 9783330343948*

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Web-Based Hotel Reservation System

The Study by Olaño, J., Agcaoili, A., Canlas, D., Narbasa, I,C., and Ordillas, J,M. (2025). Due to enormous increase in tourism worldwide, standards especially to small establishments, have improved considerably. Tourists in various markets depend on well-organized reservation system with ease in accessibility which aids in ensuring well planned trip whether business or pleasure. The aim of this study is to help Queens Hotel in assessment of their reservation and billing

Web-Based Hotel Reservation System can be used to serve as more reliable and speed up the process of Recording and Estimating the total cost of customers reserve room. They save you the burden of learning and understanding complex reservation and billing process. This study is to show the advantages and disadvantages that have in the Web-Based Hotel Reservation System. Reservation software completes Estimating the total cost of customers reserve room it would take to do them manually, whilst your reservation staff might not like it.

The purpose of hotel reservation system is to provide a place for the travelers to sleep. A hotel reservation system is also known as central reservation system (CRS). The central reservation system is an assistance for hoteliers to manage all of their marketing sales to see the availabilities to be seen all sales channels such as travel agencies

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Olaño, J., Agcaoili, A., Canlas, D., Narbasa, I,C., and Ordillas, J,M. (2025). The Faculty of Bataan Heroes Memorial College.

**Booking and Reservation System: A Unified Application Using
Location-Based Services for Sustainable Tourism Networks**

According to Gosela, R,R., and Encarnacion, R,E. (2024). The LakByahe application is a unified booking and reservation system that addresses tourism challenges in the Municipality of Nasipit in Agusan del Norte. That aims to provide a dedicated web-based platform as the current travel industry rely on social media for updates because there is a lack of organized source of information for local insights.

The LakByahe Application offers a ground-breaking solution to revolutionize tourism in Nasipit, Agusan del Norte, Philippines. It is a unified application aiming to address existing challenges by incorporating user-centric features. Through an advanced location-based services and streamlined booking processes, the application intends to enhance efficiency, accessibility, and sustainability in the tourism sector. By providing real-time tracking, efficient transit solutions, and a comprehensive directory of establishments.

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The application can fill information gaps by acting as a dedicated source. It enhances travel experiences with user-friendly features, real-time tracking, and efficient transit solutions, aligning with modern trends. Additionally, it fosters economic growth by connecting tourists with local businesses. The app aims to revolutionize travel and transportation booking, offering a seamless rental experience and overcoming tourist difficulties in these municipalities

The researchers successfully identified gaps in localized tourism solutions and crafted a platform specifically tailored to the distinct needs of tourists in Nasipit. The proposed LakByahe App represents a pioneering effort to transform the local tourism landscape through innovative technology. By addressing current system shortcomings and incorporating user-centric features, the application aims to improve efficiency, enhance accessibility, and foster sustainable tourism networks. It effectively tackles challenges related to user experience and reliability by integrating advanced location-based services, streamlining the booking process, and ensuring robust data security. Furthermore, the app's focus on sustainability promotes equitable tourism practices that directly support under-resourced areas. Stakeholders are also provided with an interactive dashboard, a comprehensive establishment directory, and monthly statistical reports to continuously monitor and enhance the tourism network. Ultimately, this comprehensive approach optimizes the booking and reservation system while setting a robust framework for future digital advancements in Nasipit.

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Gosela, R,R., and Encarnacion, R,E. (2024). Booking and Reservation System: A Unified Application Using Location-Based Services for Sustainable Tourism Networks. International Journal of Advanced Research in Science, Communication and Technology (IJARSCT). ISSN (Online) 2581-9429. Volume 4. DOI: 10.48175/IJARSCT-18889.

**Effects of Computer Reservation System in the Operations of
Travel Agencies**

The Study by Felicen, S, S., and Ylagan, A, P. (2016). In travel industry, the main tool used the computerized booking systems known as Global Distribution Systems or GDS. In reservations for travel product, the use of GDS effectively is the key skill of Travel Industry employees. In Australia and New Zealand, Amadeus and Sabre reservation systems are used daily by travel agency, airline, and tour company employees. Being competent in these skills will help graduates land his first job in travel.

GDS or CRS system is for purely (information transfer) logistical functions. They store current information about all available service providers and have the necessary infrastructure to transfer such data. This means that the system also performs additional tasks related to service distribution, which in the area of goods is typically

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carried out by freight forwarders. They support the transport of goods by eliminating the physical distance between the producer and the sales mediator or the consumer respectively. CRS's are a combination of infrastructure measures offered to interested providers in the tourist industry.

The continuous growth of 24/7 internet connectivity has transformed the tourism industry by making online buying and selling ubiquitous. This shift led to the disintermediation of the traditional travel services model, forcing legacy service providers and Global Distribution Systems (GDS) to migrate to new digital models.

While consumers increasingly prioritize a convenient, comfortable, and burden-free purchasing experience, experts note that the immense scope of technology still comes with distinct limitations. Consequently, the field of travel services research in India remains a vast, open area ripe for mutually beneficial collaborations between industry and academia.

The researchers chose this topic to determine the effect of using Computer Reservation System among Travel Agencies in terms of technical, human and financial aspect. This will help the Internship office to include the identified travel agencies in their linkages where the students will be deployed for internship. The result of this study will also be helpful and can be utilized in the course travel and tour operations with computer reservation system.

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Felicen, S, S., and Ylagan, A, P. (2016). This study aimed to assess Computer Reservation System used in the travel agencies in Batangas City and Lipa City, Philippines. Specifically, it determined the profile of the travel agencies in Batangas, identify the reservation system used by travel agencies, find out the services offered by travel agencies; determine the features of the CRS; and determine the effects of using computer reservation system in terms of human aspect, technical aspect and financial aspect.

Felicen, S, S., and Ylagan, A, P. (2016). A majority of the surveyed travel agencies in Batangas and Lipa City have been operating for 6 to 10 years, employ 3 to 4 people, and utilize the Abacus computer reservation system (CRS) for ticketing, reservations, and tour packaging. This CRS enhances agency efficiency and productivity by securing airline data, managing inventories, and connecting guests to all forms of travel. While the study concludes that technology brings both positive and negative impacts to the workplace, these findings will be integrated into curriculum development and pre-orientation programs to better prepare internship students. However, due to the study's limited geographic scope, future research should expand to include other variables and a wider respondent pool across the CALABARZON region.

Felicen, S, S., and Ylagan, A, P. (2016). Effects of Computer Reservation System in the Operations of Travel Agencies. Asia Pacific Journal of Multidisciplinary Research Vol. 4 No.4, 23-28. E-ISSN 2350-8442.

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2.3 Related Literature

Designing a Web-Based Restaurant Reservation Information System with Requirement Prototyping Method

According to Sutjiadi, R., Rahmawati, T., Kristianto, A., and Kanessa, F, T. (2026). The development of information technology is proliferating, along with the increasing human need for fast, precise, and accurate information. The role of information technology in supporting the business world will make it easier for business people to run their business.

Internet-based digital communication tools are currently increasing. This cannot be separated from several supporting factors, including the increasingly affordable price of digital communication devices, such as smartphones, tablets and laptops. In addition, internet users are currently also experiencing a relatively rapid increase.

The use of information technology in the business world also brings more or less changes in the company's business strategy. Computerised methods make recording data and information more systematised and accessible for further processing. This disruption will change the company's marketing strategy from conventional to digital through information technology and Internet media (Khairunnisa, 2022). With the application of digital marketing, business actors can develop a broader market and add value to increase customer satisfaction,

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which also impacts increasing company turnover (Saputra et al., 2023).

One of the business fields that can utilise information technology is restaurants. In the past, restaurant business actors still used manual records to run their business, such as recording customer orders, reservations, and financial transactions. This manual recording certainly has risks such as data loss, recording errors due to human error, and slow processing of the data. Currently, by utilising information technology infrastructure, data is recorded centrally through a database, where the data can be recalled quickly, system validation to minimise recording errors, and processed more rapidly into reports, which is helpful for further business analysis. Business analysis with computerised data can increase the company's productivity (Ahmad et al., 2022).

The Wisata Kampung Kemiri information system integrates table reservations and food ordering into a single, comprehensive web platform. This digital design ensures the system is easily accessible while providing real-time data regarding available tables and menu orders. Rigorous black-box testing and User Acceptance Testing (UAT) validated the performance of these reservation and ordering features. Additionally, a well-designed user interface helps consumers navigate the platform quickly, which increases user engagement and overall convenience. By employing the Requirement Prototyping software development method, the system was able to significantly accommodate specific user needs. This flexible

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approach allowed users to participate actively in the development process, ensuring the final product fully met their expectations. In the future, this application could be developed into a mobile app that is easier to use on customers' smartphones. Additionally, adding features such as reward points and online payments through a payment gateway would provide added value for customers.

Sutjiadi, R., Rahmawati, T., Kristianto, A., and Kanessa, F, T. (2026). DESIGNING A WEB-BASED RESTAURANT RESERVATION INFORMATION SYSTEM WITH REQUIREMENT PROTOTYPING METHOD. Techno Nusa Mandiri: Journal of Computing and Information Technology. P-ISSN: 1978-2136 | E-ISSN: 2527-676X. Vol. 22. DOI <https://doi.org/10.33480/techno.v22i1.6110>.

Operational Efficiency in Retail: Using Data Analytics to Optimize Inventory and Supply Chain Management

The study of Famoti et al. (2025). Operational Efficiency in Retail: Using Data Analytics to Optimize Inventory and Supply Chain Management. The retail industry faces constant challenges in balancing customer demand with cost-minimization during inventory and supply chain management. To enhance operational efficiency, data-driven approaches leverage historical sales data, seasonal trends, and machine learning to improve demand forecasting accuracy. Retailers can maintain optimal stock levels and mitigate overstocking or stockout risks by applying techniques like ABC analysis, safety stock calculations, and

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Economic Order Quantity (EOQ) optimization. Furthermore, data analytics streamlines supply chain logistics through vendor scorecards that track supplier performance, alongside geographic data used for real-time shipment monitoring and route optimization.

Famoti et al. (2025). Integrating advanced warehouse management systems and robust inventory software further streamlines retail operations. These modern analytics platforms equip retailers with data visualization, reporting tools, and predictive analytics to empower informed, data-driven decision-making. Real-world case studies across both large-scale and small-to-medium enterprises demonstrate the tangible benefits and best practices of successfully adopting these data analytics tools.

Key components of operational efficiency in retail include Inventory Management. Efficiently managing inventory is essential for meeting customer demand while minimizing carrying costs. Retailers need to strike a balance between stocking enough inventory to fulfill orders promptly and avoiding excess inventory that ties up capital and storage space.

Data analytics has become indispensable for modern retail operations by driving operational efficiency, cost savings, and enhanced customer experiences. By analyzing historical data, seasonal trends, and external factors, retailers can achieve accurate demand forecasts that align inventory levels with

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market demand while preventing stockouts or overstocking. Furthermore, these data-driven strategies optimize supplier relationship management through performance tracking while utilizing techniques like ABC analysis, safety stock calculations, and Economic Order Quantity (EOQ) to minimize carrying costs. Famoti et al. (2025).

Logistics optimization through route planning, real-time tracking, and warehouse management system integration leads to faster, more reliable, and cost-effective supply chain operations. Additionally, leveraging data on customer preferences, purchase history, and behavior allows retailers to deliver personalized experiences that enhance loyalty, satisfaction, and sales. Ultimately, data analytics replaces intuition with solid evidence, providing actionable insights that support successful strategic decision-making across pricing, product assortments, and marketing campaigns.

Famoti et al. (2025). In conclusion, the future of retail operations relies heavily on the effective utilization of data analytics. Retailers who embrace data-driven decision-making will enhance their operational efficiency and position themselves for long-term success in a rapidly evolving market. By investing in the right infrastructure, tools, and institutional culture, businesses can unlock the full potential of these analytics to achieve significant competitive advantages. Therefore, the time to act is now for retailers to adopt these data-driven strategies and transform their operations for a prosperous future.

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Famoti, O., Ezechi, O, N., Ewim, C, P, M., Eloho, O., Ajayi, T, P, M., Igwe, A, N., and Ihechere, A, O. (2025). Operational Efficiency in Retail: Using Data Analytics to Optimize Inventory and Supply Chain Management. International Journal of Scientific Research in Computer Science, Engineering and Information Technology. ISSN : 2456- 3307. Volume 11, Issue 1. Page 1483-1494. doi : <https://doi.org/10.32628/CSEIT251112173>

System Development Life Cycle (SDLC)

According to Roger S. Pressman and Maxim (2019) in their book *Software Engineering: A Practitioner's Approach*, structured methodologies like the System Development Life Cycle (SDLC) provide a step-by-step framework for system development. Applying to SDLC promotes quality, reduces risks, and ensures that all user requirements are adequately addressed.

The book stated above is relevant to the proponents' study as it provides the necessary framework for the systematic design and implementation of the study hub's features. By following the SDLC, the proponents can ensure that the real-time scheduling, billing, and reporting modules are thoroughly tested and functional before deployment.

Pressman, R. S., & Maxim, B. R. (2019). Software engineering: A practitioner's approach (9th ed.). McGraw-Hill Education.

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IOT Environmental-monitoring System Development

According to Hur,B (2021) Mosquitoes pose a major threat to humans and animals by transmitting dangerous viruses and parasites like West Nile, dengue, Zika, and malaria. Consequently, research and population control efforts are vital to saving lives from these vector-borne diseases. Researchers frequently analyze climate and environmental data to better understand mosquito breeding habits and manage their populations. However, broad macro weather data often fails to provide accurate information for specific, localized areas. Therefore, monitoring precise microclimate conditions is necessary to achieve the high-resolution data and deep analysis needed for effective control.

For the measurement and monitoring of microclimate environmental data, the development of a custom environmental monitoring system for mosquito research has been in progress by the multidisciplinary research team from three departments: Entomology, Engineering Technology, and Industrial and Systems Engineering at Texas A&M University. An Engineering Technology team at Texas A&M University has been working on the engineering aspect of the research project.

In order to expand and broaden the impact of this research as well as on engineering education, three capstone teams were formed to develop parts of the IOT (Internet of Things) environmental monitoring system for Mosquito research. A total of 12 undergrad engineering students have been working on these

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capstone projects. Each capstone team is composed of four undergraduate students. In addition, two graduate students have been involved as graduate student mentors to assist this effort.

The names of the three capstone teams are Big Dog Engineering (Team 1), AIM-N (Team 2), and Pond Hoppers (Team 3). Big Dog Engineering is developing a mobile weather station that can autonomously navigate to specific locations to collect climate data and return to base. AIM-N is building a customized IOT network and a low-cost cluster server using Raspberry Pi modules to process sensor data and perform machine learning analysis. Pond Hoppers is constructing a specialized quadcopter drone capable of floating on standing water to collect samples and monitor breeding site conditions. Together, these three interconnected projects aim to provide high-resolution, mobile environmental data specifically geared toward advanced mosquito research and population control.

Hur,B (2021) To combat fatal vector-borne diseases, researchers integrated their research and teaching efforts to develop a customized IOT monitoring system for mosquito control. This initiative led to the creation of three distinct capstone projects launched in Fall 2020. Student teams were tasked with designing a mobile weather station, a Raspberry Pi cluster server, and a water-sampling drone. Although the three teams focused on significantly different engineering tasks, they discovered strong, beneficial synergies by working under this common theme. By the conclusion of the projects in Spring 2021, all three teams successfully completed their individual goals. Ultimately, their success proved that integrating these distinct

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systems was possible, and the authors now plan to continue advancing this technology for future mosquito research.

Hur,B (2021). IOT Environmental-monitoring System Development for Mosquito Research Through Capstone Project Integration in Engineering Technology. American Society for Engineering Education. Paper ID #34086.

Reservation System and Customer Management Optimization

According to Siregar, Gading, & Raya, (2026) The entertainment industry, especially karaoke services, is one of the sectors that continues to grow along with the increasing need for recreational activities. At many karaoke venues, the room reservation process is still done manually using book notes or simple forms. This practice often causes various operational problems, such as inaccurate data recording, schedule clashes due to lack of visibility into room availability, cost calculation errors, and difficulties in tracking transaction history. These problems not only hinder the effectiveness of operational staff but also have an impact on the quality of service felt by customers. In addition, manual recording has the risk of data loss, writing errors, and a lack of integration between the information needed in management decision-making.

Seeing these conditions, the use of computer-based information systems is one of the relevant solutions to overcome reservation management problems in karaoke businesses. Desktop-based reservation systems allow for structured data storage,

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easy access to information, and increased efficiency in administrative processes. This digital approach can provide real-time visibility into the status of the room, minimize potential schedule clashes, and speed up the service process. Thus, the implementation of a computerized reservation system can provide added value for business actors, both in terms of operations and customer satisfaction.

This research demonstrates that the application of information technology in karaoke reservation management can significantly enhance operational performance. By replacing manual processes with a computerized system, businesses can achieve greater efficiency, transparency, and reliability in their services. The findings of this study not only provide practical benefits for karaoke operators but also contribute academically by offering a structured model for developing reservation systems in similar service industries. Future research may expand on this work by incorporating online reservation features, payment gateways, or data analytics modules to further improve system functionality and business competitiveness.

When compared to the manual reservation process, the developed system provides significant improvements, especially in terms of speed, accuracy, and data consistency. The time required to complete a reservation transaction is shorter, as staff no longer need to make manual records or check room availability one by one. In addition, data digitization has proven to minimize recording errors, such as customer name errors, data duplication, or schedule inconsistencies. Full

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integration between the customer, room, and transaction modules makes it easier for officers to monitor all operational activities.

Ridho Fadlan Fahira Siregar, Rafli Arya Gading, Fathul Hady Raya (2026). Karaoke Reservation System for Room and Customer Management Optimization. Journal of Artificial Intelligence and Digital Business (RIGGS). P-ISSN: 2963-9298, e-ISSN: 2963- 914X.

Web Development Technologies

According to documentation from MySQL and W3Schools (2023), modern web applications rely on a combination of database management systems (MySQL) and server side scripting (PHP). MySQL is used for storing and retrieving structured data efficiently, while PHP enables dynamic content generation and secure transaction handling.

This information is essential to the proponents as it justifies the choice of the technology stack to be used. Using these tools allows developers to build a system that is scalable, secure, and accessible across devices.

W3Schools. (2023). PHP, HTML, CSS, JavaScript tutorials. <https://www.w3schools.com> MySQL. (2023). MySQL documentation. <https://dev.mysql.com/doc/>

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**Development Student-Teacher Online Booking Appointment System in
Academic Institutions**

According to Bello, R. O., Olugbebi, M., Babatunde, A. O., Bello, B. O., and Bello, S. I. (2016), the study titled "Student-Teacher Online Booking Appointment System in Academic Institutions" was published in the Journal of Computer Science and Control Systems (Volume 9, Issue 2, ISSN 2315-9430). The research explains that traditional manual appointment methods are often time-consuming, disorganized, and prone to conflicts, so they developed a web-based system using PHP, JavaScript, and MySQL to address these issues. The platform allows students to view teachers' available time slots, book, reschedule, or cancel appointments anytime, while teachers can manage their schedules and approve requests easily. It is designed to be user-friendly, accessible on different devices, and capable of providing real-time updates—ultimately making the scheduling process faster, more convenient, and more efficient for both students and educators in academic settings.

Bello et al. (2016) also highlighted that the primary goal of developing such systems is to make appointment processes less time-consuming, easier to navigate, and more user-friendly through simple, click-based operations. The system follows a two-tier client-server architecture that communicates via the Hypertext Transfer Protocol (HTTP), allowing the web server to handle core functions such as user authentication, page generation, and database management. In terms of functionality, students can register, log in, book appointments, and send messages stating the purpose of their visit, while teachers can

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create schedules, approve or decline booking requests, view messages, and manage all appointments efficiently. By centralizing these processes and ensuring data is securely stored and easily retrievable through the database, the system ultimately aims to minimize waiting times and maximize the number of students that can be served effectively – contributing to better service delivery and operational efficiency in educational institutions.

Bello, R. O., Olugbebi, M., Babatunde, A. O., Bello, B. O., & Bello, S. I. (2016). Student-Teacher Online Booking Appointment System in Academic Institutions. Journal of Computer Science and Control Systems, 9(2). Published in 2016; Computer Science and Control Systems; ISSN: 2315-9430

Vega, R., & Generoso II, J. (2025). Ereserve: An Online Facility Reservation System of One Philippine Higher Education Institution. Philippine Journal of Information and Communication Technology. Published in 2025; Philippine Society of Information Technologists; ISSN: 2945-387X

Development of Online Student Course Registration System

According to Singh et al. (2016), manual student course registration processes are labor-intensive, time-consuming, and require students to complete multiple forms and obtain physical signatures from teachers, advisors, deans, and accounting staff. To address these challenges, the authors developed a web-based system using PHP, jQuery, Apache, and MySQL that automates the entire workflow. The system is divided into five main modules –

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Admin, Masters, Transactions, Reports, and Utilities – and supports different user roles including students, faculty, heads of department, registrar, accounts officer, and dean. It allows students to register for courses from any location, enables officials to review and approve requests digitally, maintains data consistency and security through role-based access, and eliminates duplication and errors common in manual record-keeping. The study concludes that the online system is more secure, user-friendly, and efficient, significantly reducing workload for both students and institutional staff while making registration faster and more accessible.

With the advent of Information Technology in the last decade, the major focus has shifted from manual systems to computerized systems. Various systems viz. railway reservation, hospital management etc. involving manual work have been automated efficiently. Student course registration process in colleges involve filling registration forms manually, getting it signed by respective subject teachers, and then getting the documents acknowledged from the concerned Advisors, College Deans and Accounts Officers respectively. Finally the registration forms are submitted in the Administrative Branch. As is evident, this process is very laborious and time consuming. An Online Student Course Registration System has been developed to simplify the current manual procedure. This system has been developed using PHP, jQuery, Apache and MySQL. The front-end is designed using PHP with excerpts of code written using jQuery and back-end is designed and managed through MySQL. This system software is more secured, user-friendly and less time-consuming.

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Singh et al. (2016) further noted that web-based applications provide distinct advantages over traditional methods, including the elimination of physical presence requirements, reduction of paperwork, minimization of data duplication, and real-time accessibility of information. By centralizing records in a structured database, institutions can maintain data consistency, integrity, and security, while generating accurate reports to support decision-making and resource planning. The study highlights that implementing such systems not only reduces the workload of both students and administrative personnel but also enhances transparency, accountability, and the overall quality of academic services. It concludes that online course registration systems are essential tools for modern educational institutions, enabling them to manage academic processes effectively, save valuable time and resources, and adapt to the growing demand for digital solutions in education.

Singh, R., Singh, R., Kaur, H., & Gupta, O. P. (2016). Development of Online Student Course Registration System. Oriental Journal of Computer Science and Technology, 9(2). Oriental Scientific Publishing Co., India; ISSN: 0974-6471; DOI: 10.13005/ojcst/9.02.02

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An Analysis of Restaurant Table Reservations Systems

The study by Mahajan, K., Pati, M., and Patel, J. (2025). Presenting the design and implementation of a restaurant table reservation system with the increasing demand for online services. Efficient table reservation system have become essential for both customer and restaurant owners our system aims to streamline the reservation process, enhance customer experience, and optimize restaurant operations.

The system incorporates various features, including a user-friendly interface for customer to browse available tables, select preferred dining times and make reservation the system utilizes database management to store and retrieve reservation information securely to develop the system, we employed python programming language along with relevant libraries such as web development.

The dining industry is increasingly adopting technology-driven solutions like automated table reservation systems to optimize operational efficiency and enhance the customer experience. These systems stand out as pivotal tools that allow restaurants to streamline booking processes, manage seating arrangements, and improve service quality. This research paper explores the significance of digital reservation systems within the contemporary restaurant landscape through a comprehensive review of existing literature, industry reports, and case studies. Ultimately, the study aims to shed light on the benefits, challenges, and implications of adopting these systems

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for various stakeholders, including restaurant owners, staff, and customers.

Mahajan, K., Pati, M., and Patel, J. (2025). The reservation system influences the overall dining experiences for customer, including case of booking, customization options, and communication with the restaurant considering factors such as increased table utilization it may include features such as managing multiple reservation, handling cancellations. This project may involve developing a user friendly interface for customers to view availability select a desired time and date, provide contact information, and receive confirmation.

The restaurant table reservation project offers significant, long-term benefits to both customers and owners. Through the implementation of this efficient system, customers enjoy greater convenience, reduced waiting times, and an enhanced overall dining experience. Concurrently, restaurant owners are able to optimize table turnover, improve resource allocation, and ultimately increase their profitability. Moving forward, continuous monitoring and regular updates to the system will be necessary to ensure it remains highly effective over time. This ongoing maintenance will allow the platform to keep pace with evolving customer needs and changing industry demands.

Mahajan, K., Pati, M., and Patel, J. (2025). AN ANALYSIS OF RESTAURANT TABLE RESERVATION SYSTEMS. INTERNATIONAL JOURNAL OF PROGRESSIVE RESEARCH IN ENGINEERING MANAGEMENT AND SCIENCE (IJPREAMS). e- ISSN : 2583-1062. pp: 1018-1021.

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**Impact of Business Intelligence and Analytics on Decision
Making in Online Reservation Systems within the Hospitality
Sector**

The Study by Ukandu, O., Falana, T., Adiu, A., Kanu, R., and Ebiesuwa, S. (2025). This study investigates the impact of business intelligence and analytics (BIA) on decision-making in online reservation systems within the hospitality sector. This study presents a comprehensive review of ten research papers in the field of BIA within the hospitality sector. The findings indicate that organizations effectively leveraging BIA experience enhanced decision-making, improved operational efficiency, revenue growth, and customer satisfaction. Integration of BIA leads to improved customer experiences through personalization and recommendation systems. Adopting BIA tools results in significant operational efficiency gains by addressing issues in real-time.

The hospitality sector is a dynamic and expanding industry that encompasses a wide range of businesses, including hotels, restaurants, resorts, and tourism services. This sector greatly impacts the global economy and is driven by a rising demand for travel, leisure activities, and technological advancements. To remain competitive in a crowded marketplace, businesses within this industry must prioritize providing exceptional customer experiences and delivering high-quality services. Successful organizations understand that achieving operational efficiency and profitability relies heavily on adapting to changing trends and embracing innovation. Ultimately, these factors have led the

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hospitality sector to experience notable and transformative changes over the years.

Technological advancements have revolutionized various aspects of the industry, from online reservation systems and mobile applications to personalized marketing and guest engagement. Digital platforms and social media have empowered consumers to share their experiences, shaping the reputation and success of hospitality businesses. As the hospitality sector continues to evolve, organizations are increasingly turning to data-driven decision-making to gain a competitive advantage. Business intelligence and analytics (BIA) have emerged as powerful tools that enable hospitality businesses to extract valuable insights from vast amounts of data.

Ukandu, O., Falana, T., Adiu, A., Kanu, R., and Ebiesuwa, S. (2025). This study highlights the powerful impact of business intelligence and analytics on decisionmaking in online reservation systems. The results demonstrate that organizations that effectively leverage datadriven insights are able to make informed decisions, enhance customer experiences, improve operational efficiency, and drive revenue growth. By adopting appropriate business intelligence tools, integrating data from multiple sources, and employing advanced analytics techniques, organizations can unlock the full potential of their online reservation systems.

Ukandu, O., Falana, T., Adiu, A., Kanu, R., and Ebiesuwa, S. (2025). IMPACT OF BUSINESS INTELLIGENCE AND ANALYTICS ON

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DECISIONMAKING IN ONLINE RESERVATION SYSTEMS WITHIN THE HOSPITALITY SECTOR. Indian Journal of Computer Science and Engineering (IJCSE). e-ISSN : 0976-5166. pp.640 - 651.

Reservation Systems As A Tool of Tourist Services Marketing

The study by Myskiv, G. and Wojtan, S,N. (2022). Modern reservation systems was conducted to determine their role in the tourism industry and to improve understanding of the basics of marketing activities of tourism enterprises. The article aims to analyze the functioning of modern reservation systems in the market of tourist services, to structure the existing reservation systems according to the operation levels and according to the distribution channels of the tourist product.

To remain relevant in a fiercely competitive market, any tourism company must continuously improve its business activities. Meeting the constantly growing requirements of modern customers makes it essential for these companies to have permanent access to up-to-date market information. To achieve this, businesses need to invest in reliable, multifunctional reservation systems that can effectively track customer needs and preferences. Implementing these systems ultimately helps companies attract additional clients and successfully expand their existing distribution channels. Furthermore, this technological investment enables active, high-quality promotion of tourist services across the broader market.

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Reservation systems play an extremely important marketing role in the tourism industry by providing travelers with free access to select, book, and cancel hotels, restaurants, or flights. These integrated digital platforms also significantly increase tourist attraction by offering simplified, secure payment methods. Because reservation technologies are developing rapidly alongside shifting market demands, there is a critical need to continuously study modern global booking networks. This research is highly relevant as it explores how these systems interact, coordinate with one another, and ultimately impact overall company competitiveness. Ultimately, analyzing these modern platforms helps tourism businesses better promote their products and successfully deliver high-quality hospitality services.

The global and domestic tourism industries are actively developing, with computer reservation systems serving as the foundational basis for this growth. Driven by the widespread digitalization of the service sector, these systems simplify how tourists access, select, and book various hospitality services. Ultimately, these multifunctional platforms allow tourism companies to operate with high efficiency, cover vast geographic regions, and successfully attract millions of travelers.

Myskiv, G. and Wojtan, S,N. (2022). This study analyzed the modern global travel reservation market to successfully map out its distinct hierarchical structure. The foundation of this system hierarchy is determined by the geographic scale of coverage and the specific distribution channels utilized for

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tourist services. At the top tiers, Global Distribution Systems (GDS) occupy the first global level, while Online Travel Agencies (OTA) and Restaurant Reservation Systems (RRS) comprise the second regional level. The remaining framework descends from the third level's corporate Central Reservation Systems (CRS) down to local automated websites at the fourth level, and finally to local non-automated call centers at the fifth level.

The extreme ease of booking through various computerized reservation systems successfully attracts a growing number of tourists to both established and newly opened hospitality venues. These digital platforms also allow travelers to effortlessly secure essential transit services, including airline flights, railway transport, and car rentals. Ultimately, reservation systems have evolved into a vital tool for distributing and promoting tourism, which comprehensively drives the marketing development of modern travel companies.

Myskiv, G., & Nycz-Wojtan, S. (2022). Reservation systems as a tool of tourist services marketing. Economics, Entrepreneurship, Management, 9(2), 7-18.

Hotel Reservation Management System

The study by Shun, K.W., and Hassim Y,M,M. (2021). Malaysia is a growing tourist destination, there has been a good rise in the number of hotels and resorts in Malaysia due to the broadening of the tourist sector. The rapid development

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and commercialization of Information and Communication Technologies (ICT) for travel and tourism industry have prompted hotels and other enterprises in this sector to increasingly adopt technologies into their businesses.

Hotel Time Johor Bahru is one of the hotel in Malaysia where located in Johor Bahru, Johor. The hotel currently uses BlogSpot blog for displaying hotel's details only without having a proper reservation system and relied on third-party hotel reservation platform. The problems faced by the hotel currently are there will be fewer profits earned since the hotel needs to pay the commission to the external booking portal each time a guest made a reservation from the respective booking portal. Besides, the hotel looks less impressive without having its own booking engine since some guests prefer the hotel's own booking system for reserving a room instead of using an external booking portal.

The objectives of this project is to design a hotel reservation management system and then to develop a hotel reservation management system for Hotel Time Johor Bahru. In this project, the Waterfall model is applied to assist the project development. The system is a web-based system developed based on Structured approach.

The Hotel Reservation Management System implemented for Hotel Time Johor Bahru automates major operations such as booking, accommodations, accounts, inventory, and reporting,

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while offering online reservation tracking for rooms and halls. To support these operations, the platform integrates five distinct domain modules management, reporting, inventory, reservation, reward, and registration tailored for use across five specific user roles, including administrators, multiple staff tiers, and guests.

The Hotel Reservation Management System allows traveling or local guests to easily compare rates, review services, and securely book rooms online without visiting the property beforehand. Simultaneously, hotel staff can effortlessly manage room inventories, update pricing, process transactions, and access daily financial reports for data analysis. Finally, the platform enables owners to promote their business directly by creating custom marketing packages and promotional deals to attract more customers.

By receiving direct online reservations, hotel owners can bypass costly Online Travel Agency (OTA) commission fees, allowing those savings to go directly to the bottom line as increased profit while enabling closer guest engagement. Furthermore, this user-friendly system reduces the risk of lost bookings, making the reservation process easier for customers and helping hotel staff perform their duties more effectively.

Shun, K,W., and Hassim Y,M,M. (2021). The Hotel Time Johor Bahru Reservation Management System is a web-based platform that boosts operational efficiency for hoteliers while providing a

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convenient room booking process for guests. By utilizing this software, owners avoid paying continuous commissions to third-party portals and only need to cover the basic license cost. However, a major disadvantage of the current system is the lack of a channel manager to connect the platform with external booking networks. This limitation prevents real-time data synchronization during reservations, which can ultimately lead to accidental overbooking issues. To resolve this, future developments should integrate a channel manager for synchronous updates, alongside mobile application support and features like meal or group reservation management.

Shun, K,W., and Hassim Y,M,M. (2021). Hotel Reservation Management System. Faculty of Computer Science and Information Technology, Universiti Tun Hussein Onn Malaysia, 86400, Johor, MALAYSIA. Vol. 2 No. 2 (2021) p. 973-992. DOI: <https://doi.org/10.30880/aitcs.2021.02.02.061>.

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2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local literature and studies indicated that web-based management systems significantly enhance operational efficiency, improve data accuracy, and elevate overall customer satisfaction. The literature consistently stressed the importance of structured software development methodologies, real-time data processing, secure relational databases, and user-centered interface design.

However, a critical gap identified across most existing management frameworks was the lack of specialized integration tailored specifically to the multi-faceted operations of a study hub environment. Most existing systems functioned in isolation—handling basic space booking or simple billing independently—without offering a unified, synchronized mechanism for real-time room allocation, dynamic time duration/expiration tracking, automated membership tier validation, and consolidated sales auditing.

The present study directly addressed these operational gaps through the design, development, and implementation of the Integrated Study Hub Management System for WS Students & Professionals Lounge – La Paz Branch. By consolidating reservation management, automated membership and time tracking, payment billing, and real-time analytical reporting into a single web-based platform, this study bridged the divide found

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in previous literature. Ultimately, this integrated approach successfully replaced manual front-desk friction with a secure, transparent, and frictionless facility environment for both administrative personnel and clients.

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2.5 Conceptual Framework

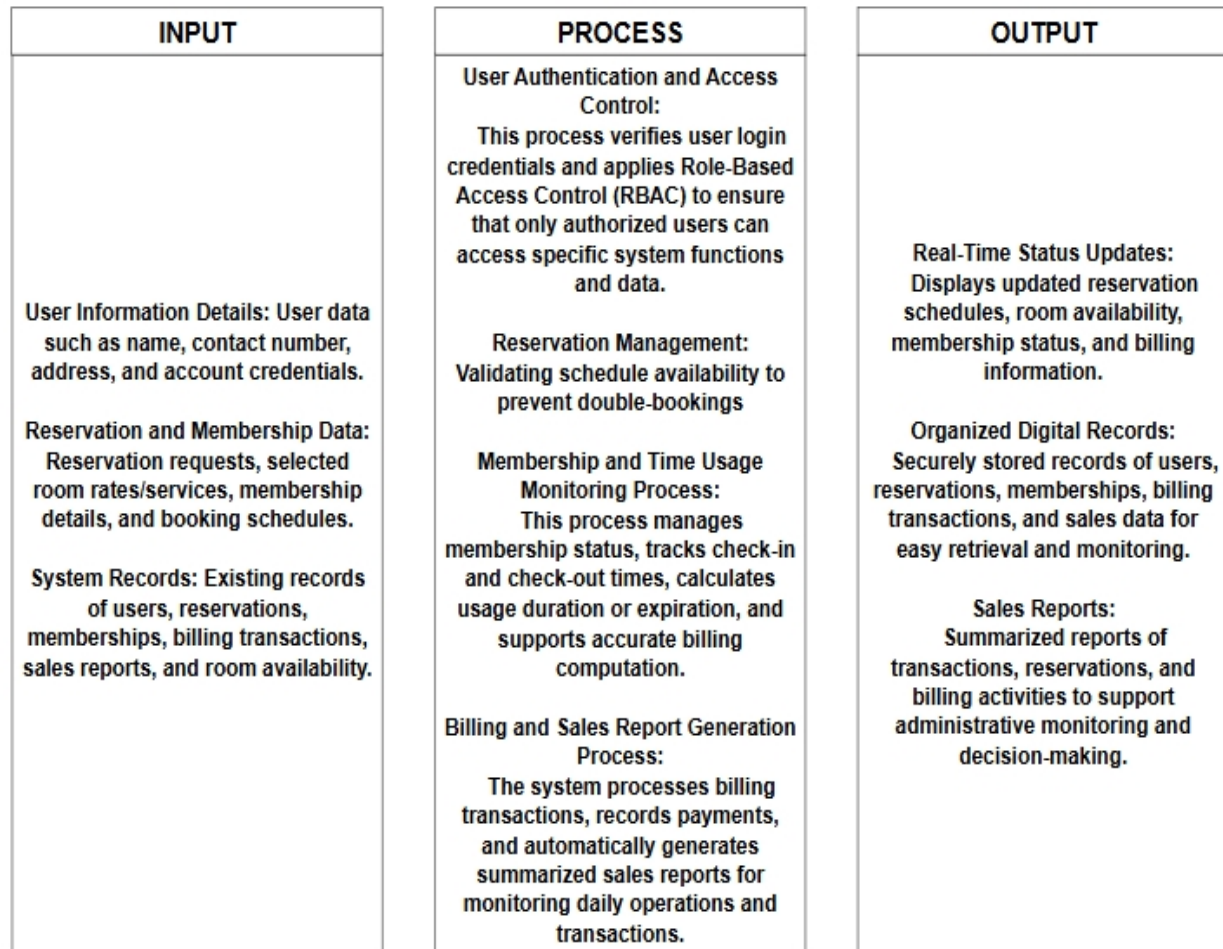


Figure 1. Conceptual Framework of the Integrated Study Hub Management System for WS Students & Professionals Lounge

The conceptual framework of the Integrated Study Hub Management System for WS Students & Professionals Lounge illustrated how manual, fragmented workflows were converted into an automated digital environment to improve operational efficiency, data accuracy, and security. Built upon an Input-

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Process-Output (IPO) model, the framework demonstrated how raw transaction inputs were systematically processed by core software modules to generate centralized management outputs.

The Inputs comprised primary user information, service requests, and existing operational records, including customer profile details, workspace reservation requests, active membership subscriptions, and transaction logs.

The Process phase encompassed the execution of core system functions and database manipulations:

- User Authentication & RBAC - The process initiated when a Student/Professional client or an Admin/Staff member logged into the platform. The system validated user credentials against the database Users table and programmatically enforced Role-Based Access Control (RBAC) to grant appropriate feature access permissions.

- Reservation & Space Allocation - When a user or staff member queried workspace or room availability, the reservation module programmatically checked for timeslot conflicts. Once validated, the system updated the Reservations table in real time, preventing double-bookings and issuing an immediate booking confirmation.

- Check-In & Dynamic Time Tracking - Upon the customer's physical arrival, frontline staff triggered an automated check-in, recording an accurate time_in timestamp in the

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system's Time Logs. Upon session conclusion, a `time_out` timestamp was logged, allowing the billing engine to calculate the exact elapsed duration and compute the itemized charge.

- Data Aggregation & Auditing - At the end of operational cycles, the system harvested all transactional data, automatically compiling high-level metrics—such as `total_check_ins`, `total_logins`, and `total_time_logged`—into the Daily Reports table. Administrators then audited these real-time summaries to monitor daily hub performance, peak space usage patterns, and revenue trends.

The Outputs generated by the system included real-time room availability status updates, conflict-free booking records, accurate time-based invoices, and consolidated, tamper-proof sales reports. Overall, this framework demonstrated how the system successfully transformed unstructured raw inputs into actionable business intelligence, ultimately creating a seamless, efficient, and data-driven environment for both lounge personnel and clients.

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Chapter 3

3.1 Methodology

In this project, the proponents focused on how the management, staff, and customers of WS Students & Professionals Lounge - La Paz Branch handled their daily operations to thoroughly understand their existing operational workflow. By combining primary data gathering methodologies, the proponents collected useful, actionable, and meaningful insights. These findings enabled the proponents to address and resolve key operational issues, such as manual reservation errors, difficulties in tracking memberships, and inaccuracies in billing and sales recording.

The proponents conducted structured surveys, stakeholder interviews, and direct field observations to gain a detailed understanding of the manual tools and processes previously used by the lounge to manage its daily operational data. This comprehensive approach allowed the proponents to quantify how frequently operational issues occurred—such as resource double-bookings, membership expiration oversights, and sales discrepancies—while capturing qualitative feedback regarding staff frustrations with the paper-based setup. Based on these findings, the proponents gathered essential requirements that directly guided the design, engineering, and implementation of the Integrated Study Hub Management System tailored specifically to the needs of the lounge.

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The surveys were utilized to measure the frequency of operational bottlenecks and the specific administrative friction experienced by frontline staff during peak operating hours. This provided a quantitative foundation to evaluate user experience pain points under the legacy setup. The collected survey data served as the basis for assessing administrative challenges and engineering targeted digital solutions for both the lounge administrator and desk employees.

Furthermore, the proponents performed face-to-face interviews and direct operational observations to map out the detailed daily routines of the lounge's frontline personnel. Engaging directly with stakeholders provided firsthand clarity on procedural vulnerabilities across workspace reservations, membership subscription updates, billing calculations, and end-of-day sales report generation. Combining these research methods allowed the proponents to integrate both quantitative metrics and qualitative user insights, ensuring that the developed management system satisfied both technical performance standards and practical user requirements.

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3.2 Methodology Table

METHODS	PARTICIPANTS	PURPOSE
Surveys	WS Students & Professionals Lounge Owner, General Manager, Staff, and Regular Customers	To gather quantitative data regarding the frequency of manual errors and the level of satisfaction with the current system.
Interviews	WS Students & Professionals Lounge Administrator and Employees	To obtain qualitative insights into the specific operational challenges, pain points, and desired features for the new system.
Observations	WS Students & Professionals Lounge Staff and Customers	To observe the actual workflow of reservation, membership management, time room usage, and billing to

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		identify bottlenecks in the manual process.
Data Analysis	The Proponents	To synthesize the collected data, identifying patterns in operational errors, and correlating them with system requirements.
Solution Development	The Proponents	To design and develop the Integrated Study Hub Management System based on the analyzed needs and technical specifications.

Table 1. Methodology Table

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3.3 Research and Development Stage

The Research and Development (R&D) stage was essential in guiding the Integrated Study Hub Management System from initial conceptualization to full system implementation. This stage was dedicated to ensuring that the platform specifically addressed the unique operational challenges of the WS Students & Professionals Lounge - La Paz Branch. The process began with extensive background research to understand the pre-existing manual workflow and the administrative frustrations experienced by staff regarding data integrity and customer service speed.

The proponents analyzed existing workspace management systems to identify industry best practices while maintaining a focus on the specific operational requirements of a study lounge environment. Through primary data-gathering methods—including surveys, interviews, and field observations—the proponents documented the risks associated with the manual setup, particularly concerning user registration, reservation handling, membership tracking, time-based room usage logging (duration and expiration), manual billing, and sales report generation. This collected data served as the formal foundation for defining the software's functional and non-functional requirements.

Following the requirements gathering phase, the proponents transitioned to the Design Stage. In this phase, the proponents engineered the system architecture, focusing on key core modules:

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user registration, interactive reservation bookings, automated membership tracking, time-based room session tracking, dynamic billing computation, and sales reporting. Detailed user interfaces and system database schemas were created to streamline the front-desk workload. Throughout this phase, stakeholder feedback was integrated to ensure the UI/UX design remained intuitive, responsive, and user-friendly.

After completing the design architecture, the proponents proceeded to the Development Phase, where the physical coding and database construction took place. The proponents constructed a fully functional platform that integrated responsive front-end user interfaces with a secure MySQL back-end database engine. Development followed an iterative approach, allowing the proponents to continuously refine system features based on preliminary internal testing to ensure software stability, security, and computational efficiency.

Once the application was fully built, it entered the Testing Phase. The system underwent rigorous evaluation, including unit functionality testing, database integrity checks, and formal User Acceptance Testing (UAT). The proponents engaged the staff and management of WS Students & Professionals Lounge to test the system under simulated operational conditions, ensuring it could reliably handle real-world stress scenarios such as peak-hour simultaneous bookings. Bugs, logical edge cases, and interface inconsistencies identified during testing were rectified prior to final deployment.

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Finally, the R&D process concluded with the Evaluation Stage. The proponents evaluated the overall performance and accuracy of the completed system against the study's initial general and specific objectives. By analyzing UAT feedback and test results, the proponents verified that the Integrated Study Hub Management System successfully resolved the legacy problems of manual transcription errors, scheduling overlaps, and tracking inefficiencies, providing a secure, organized, and reliable digital platform for the lounge.

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3.4 Research Design

This study utilized a developmental research design because the primary goal of the proponents was to design, engineer, and evaluate a functional software system that resolved the operational bottlenecks of the WS Students & Professionals Lounge - La Paz Branch. The proponents focused on creating a practical, reliable, and deployable digital tool tailored for the actual operational environment of the enterprise.

A descriptive research approach was also employed to systematically analyze the pre-existing operational state of the lounge. This involved investigating how frontline personnel previously managed daily transactions, identifying recurring operational bottlenecks, and determining the specific functional requirements necessary for the system. By thoroughly documenting and describing the legacy manual processes, the proponents successfully designed a targeted digital solution that aligned with the operational needs of the lounge administrator, staff, and clients.

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3.5 System Development Methodology

The project followed the System Development Life Cycle (SDLC) methodology, executing the following sequential phases:

- **Planning** - In this phase, the proponents identified the core operational vulnerabilities in the manual workflow of the WS Students & Professionals Lounge - La Paz Branch, such as reservation conflicts and the difficulties of tracking active memberships, calculating time-based billing, and compiling daily sales reports. The proponents defined the system requirements and operational scope to ensure the project directly resolved manual recording errors, data fragmentation, and slow transaction processing speeds.

- **Analysis** - The proponents gathered operational data through structured surveys, stakeholder interviews, and direct field observations involving the administrators, frontline staff, and clients of the lounge. This involved analyzing the systemic risks associated with paper-based logbooks, such as physical data loss, room double-bookings, and the labor-intensive process of manually auditing usage durations and financial transactions.

- **Design** - The proponents engineered an intuitive, user-friendly interface designed for rapid front-desk navigation. This phase

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also involved designing a normalized relational database schema (MySQL) capable of securely storing customer profiles, membership subscription states, reservation records, hourly rate matrices, billing logs, and sales transaction data to enable fast search and retrieval.

- **Implementation** - The proponents developed the software application and configured the backend database infrastructure. Key operational modules were implemented, including user registration, real-time reservation management, automated membership tracking, time-based room usage session monitoring, dynamic billing computation, and sales report generation. System security controls—specifically Role-Based Access Control (RBAC) and user authentication logs—were integrated to protect confidential enterprise data.

- **Testing** - The proponents executed comprehensive software testing across all modules to identify and rectify bugs, logical inconsistencies, or edge cases. Functionality was validated for user registration, space booking, membership verification, billing calculations, and reporting modules. Additionally, frontline personnel and administrators performed hands-on testing to evaluate system usability and confirm alignment with operational workflows.

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- **Deployment** - In this phase, the system was deployed for actual operational use within the WS Students & Professionals Lounge - La Paz Branch. The proponents conducted user training sessions for frontline employees to ensure confident, proficient, and secure system operation. Post-deployment monitoring was conducted to verify operational stability and support personnel during initial business adoption.

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3.6 Agile Development Methodology



Figure 2. Agile Methodology

The Agile Methodology was selected for the development of the Integrated Study Hub Management System because the project required a secured, organized, and highly adaptive framework tailored for the WS Students & Professionals Lounge - La Paz Branch. Agile provided the necessary flexibility to adapt to evolving project requirements, which proved essential as functional rules concerning user registration, interactive reservations, automated membership tracking, time-based usage session logging, billing computations, and sales reporting were continuously refined during development. This iterative approach enabled the proponents to receive continuous feedback from lounge management, ensuring that the final platform directly addressed their operational bottlenecks. Furthermore, it

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fostered close collaboration between the proponents and stakeholders, allowing technical challenges and edge cases to be identified and resolved early in the development lifecycle.

In the Requirements Analysis phase, the proponents gathered detailed operational requirements regarding the lounge's legacy manual setup. This involved identifying specific administrative pain points, such as room double-bookings, manual check-in transcription errors, and difficulties in monitoring active membership validity windows. The proponents formally defined the software requirements and system scope to ensure that the developed platform directly resolved the daily challenges faced by frontline staff and administrators.

During the System Design phase, the proponents engineered the overall software architecture, module workflows, and database structures. This involved designing intuitive graphical user interfaces (GUIs) for the reservation, membership management, dynamic time-tracking, and billing modules. Additionally, a normalized relational database schema (MySQL) was constructed to securely store user credentials, membership statuses, booking calendars, time logs, itemized billing records, and sales summaries. The UI/UX design focused on simplicity and ease of use to minimize the learning curve for staff during the transition from paper-based operations to a digital platform.

In the Development phase, which encompassed active programming and database integration, the proponents constructed

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the system according to the finalized architecture. Core functional modules—including user registration, real-time reservation validation, automated membership tracking, time-duration session logging, dynamic billing computation, and sales report generation—were programmed and integrated. The proponents ensured that the underlying backend architecture was stable, secure, and capable of processing concurrent transactions without data corruption or latency.

During the Testing phase, the proponents conducted rigorous system evaluation to verify functional accuracy and eliminate software defects. Testing validated the precision of automated billing calculations, the reliability of time-based session tracking, and the effectiveness of programmatic validation logic in preventing reservation overlaps. Furthermore, the administrator and frontline personnel of WS Students & Professionals Lounge participated in formal User Acceptance Testing (UAT) to evaluate system usability, verify workflow alignment, and recommend minor interface refinements prior to deployment.

Finally, in the Deployment phase, the proponents delivered the fully functional system to the WS Students & Professionals Lounge. The proponents conducted structured orientation and hands-on training sessions for frontline employees to ensure confident and secure system operation. Initial post-deployment monitoring was maintained to verify system stability, resolve minor operational queries, and confirm that the platform was

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fully prepared to handle the lounge's daily operational transactions.

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3.7 Data Collection Techniques

To design and develop an Integrated Study Hub Management System that addressed the specific operational needs of the management, frontline personnel, and clients of WS Students & Professionals Lounge - La Paz Branch, the proponents gathered primary data and operational insights using a combination of qualitative and quantitative collection methodologies. These techniques ensured that the developed platform was user-friendly, structurally sound, and directly aligned with the operational challenges identified during preliminary investigation.

1. Surveys

The proponents distributed structured survey questionnaires to the target user groups, including the lounge owner, general manager, frontline staff, and registered clients. These surveys evaluated how respondents previously handled space reservations, membership tracking, billing transactions, and sales reporting. The instruments also assessed user perceptions regarding operational friction, task efficiency, and preferred software capabilities, providing quantitative data that guided subsequent system development.

2. Interviews

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The proponents conducted one-on-one structured and semi-structured interviews with the lounge administrator and key staff members involved in managing daily front-desk operations, room scheduling, billing, and record-keeping. This provided detailed qualitative insights into specific operational bottlenecks—such as manual calculation errors and scheduling overlaps—that quantitative surveys alone could not capture. Interviewees shared firsthand experiences, operational concerns, and recommendations for improving workflow efficiency.

3. Focus Group Discussions

The proponents conducted focus group discussions (FGDs) with the lounge staff and management to review proposed system features and workflow designs. These collaborative sessions allowed participants to express opinions, evaluate prototype mockups, and highlight critical operational requirements. The feedback gathered during these discussions enabled the proponents to prioritize essential system features and refine UI/UX elements based on direct user consensus.

4. Observations

To gain an accurate understanding of daily operational routines, the proponents performed direct field observations of the legacy manual processes at the lounge. The proponents monitored how staff recorded member profiles, verified study pod availability, calculated active customer usage session durations, processed manual billing transactions, and compiled end-of-day sales totals. Direct observation enabled the proponents to

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pinpoint precise stages in the manual workflow where delays, mathematical mistakes, and record duplication frequently occurred.

5. Document Analysis

The proponents analyzed existing physical logbooks, manual reservation ledgers, billing receipts, sales summary sheets, and technical literature related to study hub operations and Point of Sale (POS) architecture. Examining these historical records provided the proponents with a clear baseline of the lounge's data tracking structures, facilitating the design of an optimal relational database schema tailored specifically to the operational requirements of the WS Lounge.

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3.8 System Architecture and Design

Overview

The Integrated Study Hub Management System was designed and engineered to modernize, organize, and streamline the daily operational workflows of the WS Students & Professionals Lounge – La Paz Branch. The platform was specifically constructed to eliminate the operational risks associated with legacy manual processes—such as physical record loss, workspace double-bookings, manual check-in calculation errors, and fragmented sales tracking. By transitioning the business from a paper-based setup to a centralized digital framework, the system ensured that all operational data was securely stored, easily retrievable, and programmatically audited, enabling frontline personnel to manage the lounge efficiently while providing clients with a seamless workspace experience.

The platform's architecture supported three distinct user tiers—Administrator, Front-desk Staff, and Customers—with access privileges enforced through Role-Based Access Control (RBAC):

- Administrator – Held primary system configuration privileges, allowing management to oversee user account access, perform revenue and transaction audits, view real-time sales analytics, and monitor overall facility operations.

- Front-desk Staff – Handled daily operational workflows, including processing walk-in and online client check-ins,

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verifying real-time study pod and room availability, monitoring active session durations, and executing billing transactions upon check-out.

- Customers - Utilized the client-facing portal to view workspace availability in real time, manage personal room bookings, track remaining membership validity days, and review itemized billing histories.

This structured operational pipeline ensured seamless task delegation across all user roles, eliminated administrative vulnerabilities, and maintained structural data integrity across the lounge's daily operations.

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3.9 System Architecture

The Integrated Study Hub Management System was developed using a Three-Tier Architecture to ensure a structured, scalable, and organized flow of data across all operations. This architecture consisted of the following three core layers:

- Presentation Layer - Provided the Graphical User Interface (GUI) through which administrators, frontline staff, and customers interacted with the system. It featured a responsive web dashboard for user account registration, real-time room occupancy monitoring, space reservations, and membership management. For administrative personnel and frontline staff, it included dedicated control panels for managing user profiles, auditing reservations, updating subscription tiers, processing check-in/check-out logs, and generating sales reports. This layer enforced Role-Based Access Control (RBAC) to dynamically customize navigation menus and functional views based on authenticated user roles—restricting sensitive configuration functions to authorized personnel while delivering an intuitive, self-service portal for general clients.

- Application / Logic Layer - Served as the core processing engine of the platform, handling all client-server requests and enforcing business logic rules. Key operational modules in this layer included:

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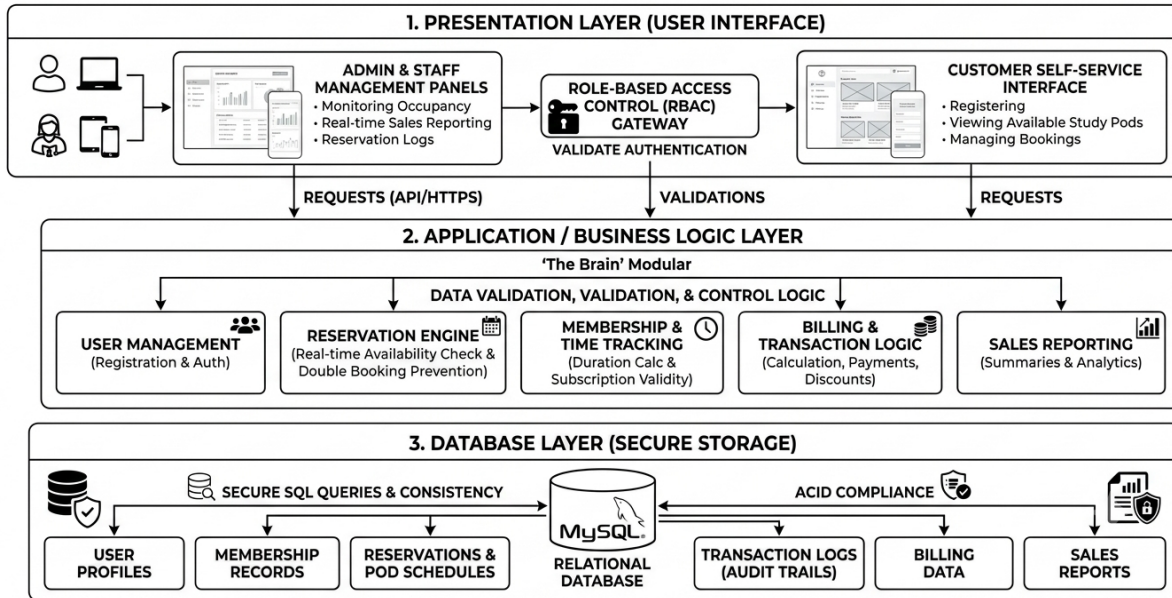
- User Management Module - Handled account registration, credential encryption, and session authentication.
- Reservation Module - Checked real-time workspace availability and programmatically prevented schedule overlaps or double-bookings.
- Membership Module - Managed subscription lifecycles, calculated active session durations, and validated tier discounts.
- Billing Module - Executed automated financial calculations based on time logs, hourly rates, and membership privileges.
- Sales Reporting Module - Aggregated transactional logs to generate itemized financial summaries.

This layer ensured that all user actions were validated prior to querying the database layer and returning processed results to the presentation interface.

- Database Layer - Managed the secure, relational storage of all core system records using MySQL. It housed interconnected tables storing user authentication profiles, membership subscription states, room availability schedules, time-in/time-out logs, billing transaction histories, and sales reports. The database layer maintained strict data integrity, consistency, and security, ensuring that every operational transaction was captured accurately and enabling rapid data retrieval for real-time reporting, auditing, and administrative decision-making.

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3.9.1 System Architecture Diagram



System Architecture Diagram

Figure 3. System Architecture Diagram of the Integrated Study Hub Management System

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3.10 System Components and Modules

The Integrated Study Hub Management System was comprised of interconnected functional modules that operated synchronously to deliver a streamlined, automated operational environment. The platform leveraged automated tracking mechanisms and Role-Based Access Control (RBAC) to ensure operational fluidity, eliminate manual calculation errors, and safeguard sensitive system records against unauthorized access.

User Management Module

This module managed account profiles, credential security, and session privileges for administrators, frontline staff, and clients. Key capabilities included:

- Secure multi-level user authentication and password management.
- Member profile creation and updating.
- Session management to restrict administrative access strictly to authenticated personnel.

Reservation Booking Module

As the core module for physical space allocation, this component managed study pods, open desk areas, and conference rooms. Key capabilities included:

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- Real-time workspace availability validation to programmatically prevent schedule overlaps and double-bookings.
- Walk-in and advance booking forms for both self-service clients and front-desk personnel.
- Interactive visual schedule monitors displaying real-time occupancy and vacancy statuses.
- Automated session duration tracking to assist staff in managing customer turnaround.

Membership Subscription Module

This module managed registration and status monitoring for the lounge's subscription clients. Key capabilities included:

- Digital registration onboarding for new members.
- Automated tracking of membership validity windows and expiration dates.
- Historical visit tracking to identify recurring clients, apply membership tier perks, and automate subscription discounts during checkout.

Sales Reports Module

This module managed transaction logs, revenue compilation, and financial history. Key capabilities included:

- Automated financial transaction logging for every completed checkout.

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- Historical ledger tracking to ensure transaction records were accurate, transparent, and audit-ready.

Admin Dashboard and Reporting Module

This module provided management with a centralized command center to oversee overall business operations and financial health. Key capabilities included:

- On-demand generation of daily, weekly, and monthly consolidated sales reports.
- Monitoring of real-time user session logs and system activities.
- Configuration panels for room rates and subscription packages.
- Analytical reporting on peak usage hours and customer demographics to support administrative scheduling decisions.

Notification and Alert Module

This module facilitated operational communication across staff and clients. Key capabilities included:

- Instant confirmation alerts for successful workspace reservations.
- Time-expiration alerts notifying staff when a client's active usage session approached completion.
- Automated system notifications highlighting operational errors or database validation warnings.

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Security and Access Control Module

This module enforced essential data security and system integrity protocols across the platform:

- Credential Encryption - Employed secure hashing algorithms to encrypt user password credentials stored in the database.
- Role-Based Access Control (RBAC) - Restricted front-desk staff to routine operational tasks (check-ins, reservations, and billing) while reserving high-level financial reporting and system configuration privileges exclusively for the administrator.
- Data Preservation & Recovery - Implemented automated database backup routines to safeguard reservation ledgers and membership records against physical or system data loss.
- Information Privacy - Enforced access restrictions to protect confidential customer profiles against data breaches or unauthorized administrative access.

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3.11 System Architecture and Design

3.11.1 Context Diagram

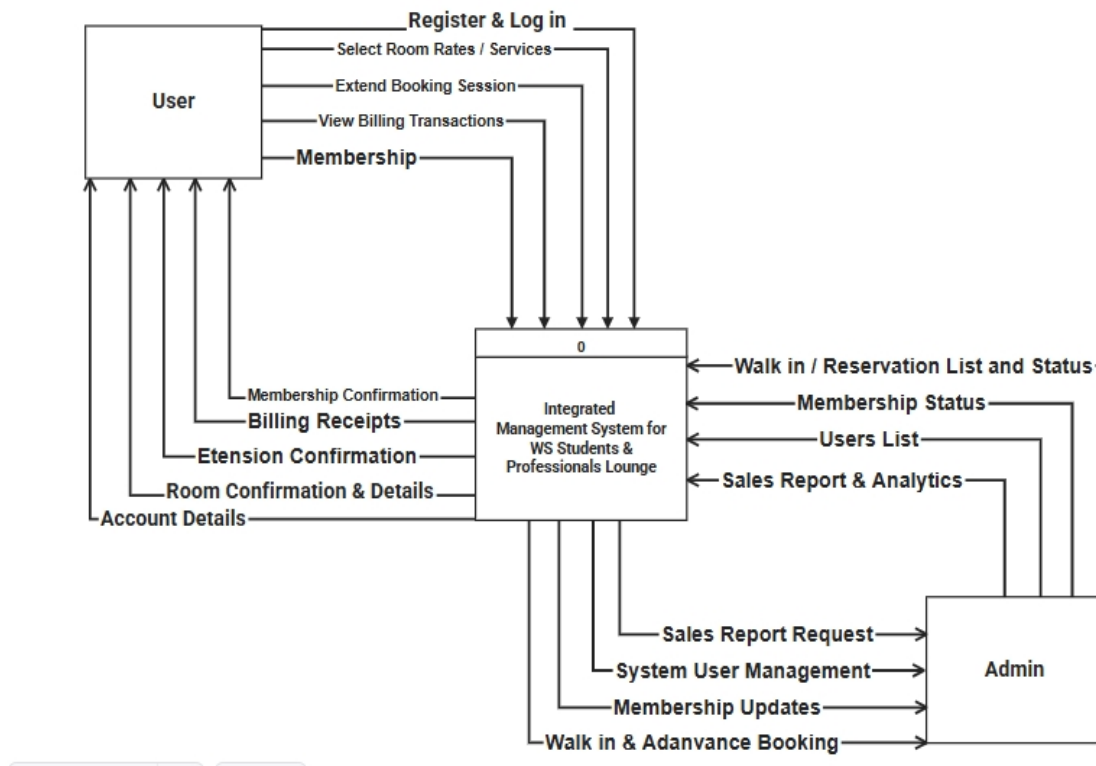


Figure 4. Context Diagram of the Integrated Study Hub Management System for WS Students & Professionals Lounge

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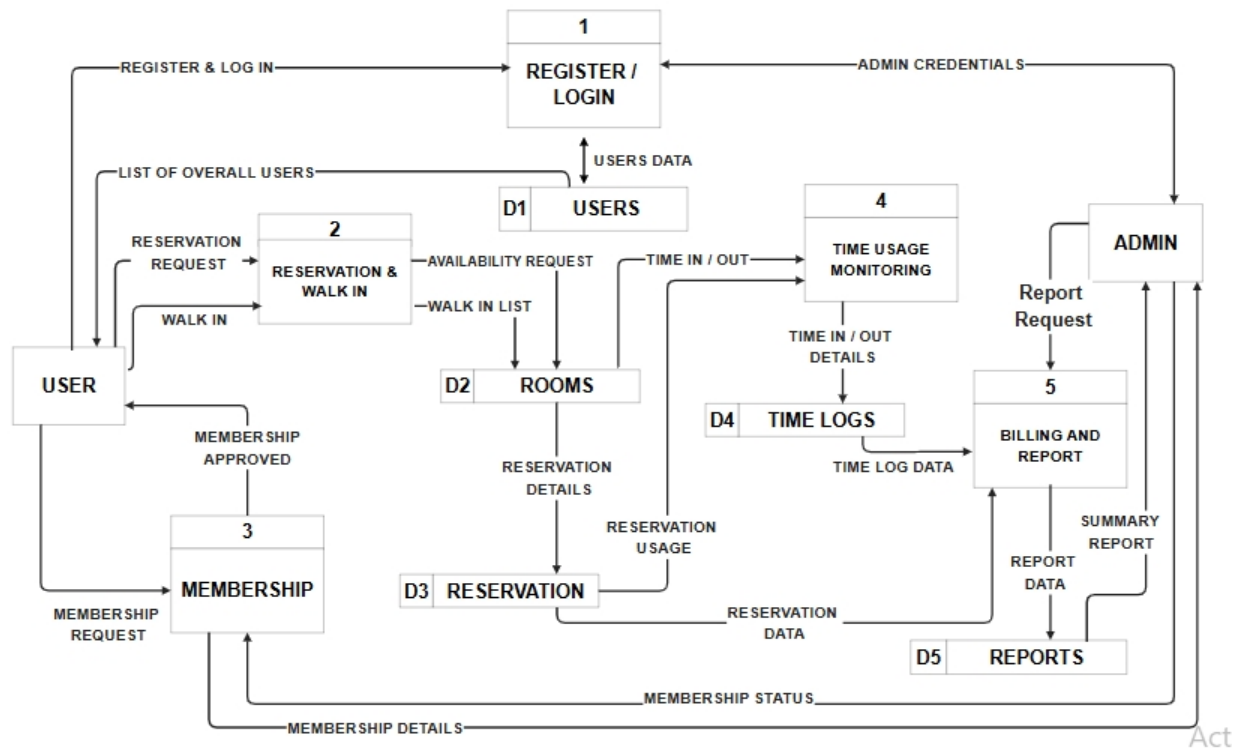


Figure 5. Data Flow Diagram of the Integrated Study Hub Management System for WS Students & Professionals Lounge

Significance of Context and Data Flow Diagram

The hierarchical structure of the Context Diagram and Data Flow Diagrams (DFDs) provided a detailed operational blueprint for the Integrated Study Hub Management System for WS Students & Professionals Lounge - La Paz Branch. This structured approach systematically mapped the flow of data from the high-level conceptual overview down to granular database transactions, ensuring that every operational event—from initial room bookings to end-of-day sales updates—was accurately captured, validated, and logged.

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As illustrated in Figure 4 (Context Diagram), the scope of the system was defined by mapping the external entities that interacted directly with the centralized platform:

- Customers / Users initiated system processes by providing account registration details, authentication credentials, workspace reservation requests, and time-extension queries. In response, the system returned immediate booking confirmations, real-time room status updates, and itemized billing receipts.
- Administrators & Front-Desk Staff interacted with the system by managing room rate matrices, processing customer walk-in check-ins, updating user profile states, and enforcing security policies. In return, the platform generated high-level outputs, including real-time analytic dashboards and consolidated sales reports.

Moving to the Level 1 DFD, these broad interactions were decomposed into discrete functional processes: *Register/Login (1.0)*, *Reservation & Walk-In Management (2.0)*, *Membership Tracking (3.0)*, *Time Usage Monitoring (4.0)*, and *Billing & Sales Reporting (5.0)*. For instance, when a customer initiated a room reservation, the system executed validation logic against the Rooms (D2) data store to prevent scheduling overlaps. Similarly, the *Time Usage Monitoring* process programmatically recorded exact `time_in` and `time_out` timestamps into the Time Logs (D4) data store, guaranteeing transparent usage duration calculations.

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Furthermore, the Level 2 and Level 3 DFDs provided the deepest architectural view of the internal software mechanics, demonstrating the dynamic calculation of total billing charges using raw data retrieved from time logs and active reservation records. This structured DFD modeling validated the software design, ensuring that all functional and operational requirements of the WS Students & Professionals Lounge were met to deliver a reliable, secure, and error-free digital environment.

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3.11 Use-Case Diagram

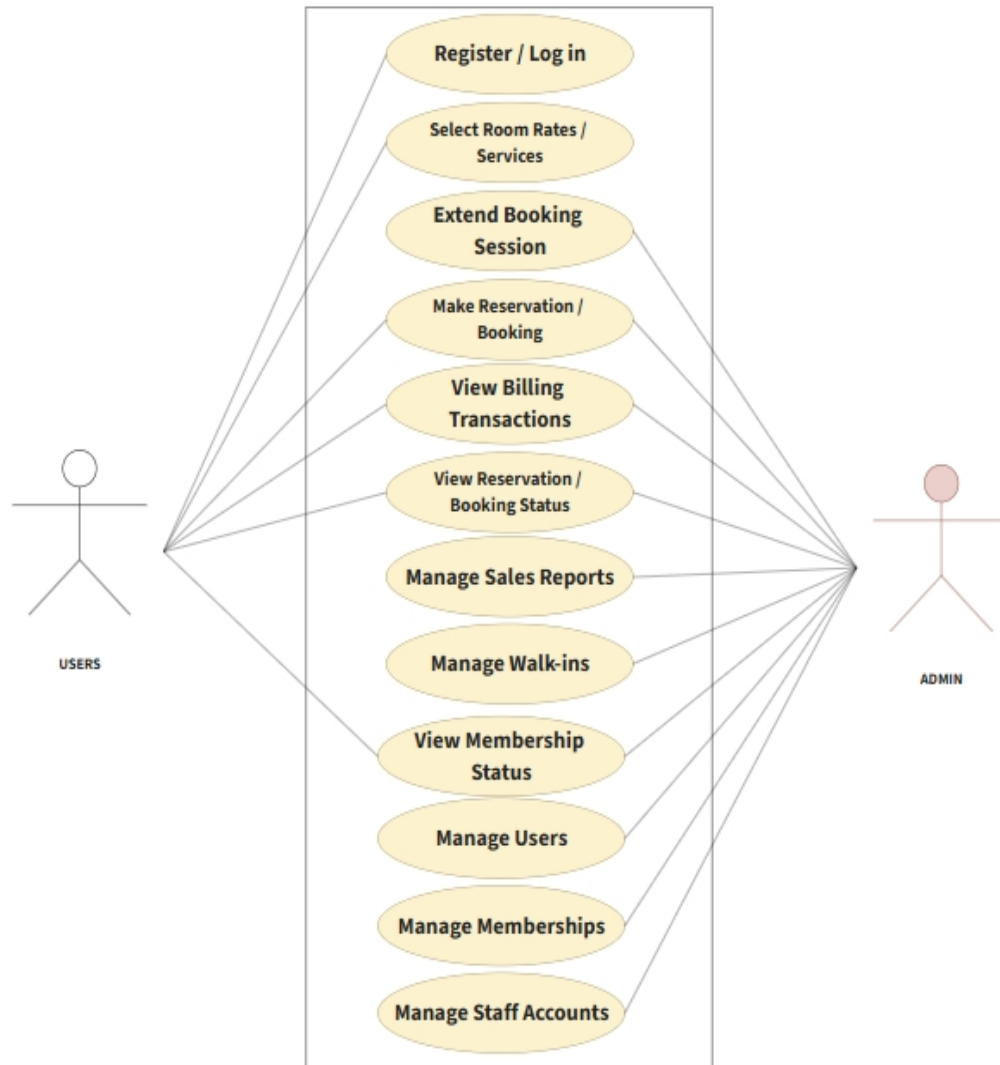


Figure 6. Use-Case Diagram of the Integrated Study Hub Management System for WS Students & Professionals Lounge

Significance of Use-Case Diagram

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The Use-Case Diagram for the Integrated Study Hub Management System for WS Students & Professionals Lounge - La Paz Branch provided a structured behavioral model that outlined the precise interactions between external actors and the internal functional capabilities of the application. By defining a clear system boundary, this diagram served as an essential blueprint for verifying that all functional requirements mapped directly to specific user roles, eliminating structural conflicts prior to software deployment (Sommerville, 2016). The system architecture segregated operational responsibilities into two primary actor categories: Users (representing Students and Working Professionals) and the Admin (representing Front-desk Personnel and Executive Management) of the lounge.

From the Users' operational domain, the interface was engineered to offer an intuitive self-service portal that streamlined client interactions with the facility. Based on the system's behavioral logic, the functions explicitly made available to general clients included:

- Register / Log In - The gateway authentication module for securing user credentials and establishing authenticated sessions.
- Select Room Rates / Services - Allowed users to view real-time workspace availability, amenity options, and pricing matrices.
- Make Reservation / Booking - The primary functional control for securing a future schedule within the hub.
- Extend Booking Session - Enabled dynamic, real-time extensions of active room allocations depending on timeslot availability.

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- View Reservation / Booking Status - Tracked real-time confirmation statuses of pending or active room requests.
- View Membership Status - Displayed active subscription details, remaining validity days, and tier perks.
- View Billing Transactions - Provided a transparent, itemized ledger of personal usage expenses and historical payments.

Conversely, the Admin actor possessed elevated operational permissions designed to govern daily logistical workflows, manage physical facilities, and maintain strict data governance over business assets. As illustrated by the multi-line connection schema, the Administrator could oversee user-level modules while holding exclusive privileges to execute the following administrative use cases:

- Manage Walk-Ins - Facilitated real-time check-ins, time-log initiations, and duration tracking for on-site clients without prior bookings.
- Manage Users - Granted administrative control over customer account profiles, restriction levels, and registration lists.
- Manage Memberships - Governed subscription package distributions, customer tier configurations, and expiration updates.
- Manage Sales Reports - Allowed management to aggregate financial data, track revenue streams, and generate analytical business summaries.
- Manage Staff Accounts - An exclusive backend administrative capability utilized to supervise employee system privileges,

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generate staff credentials, and enforce Role-Based Access Control (RBAC).

In synthesis, the use-case model validated the role-based architecture of the platform. By systematically organizing these lines of interaction, the system guaranteed that while clients enjoyed a frictionless, self-service workspace registration experience, hub administrators were equipped with the backend operational controls required to uphold financial transparency, system security, and high organizational efficiency.

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3.12 Entity Relationship Diagram (ERD)

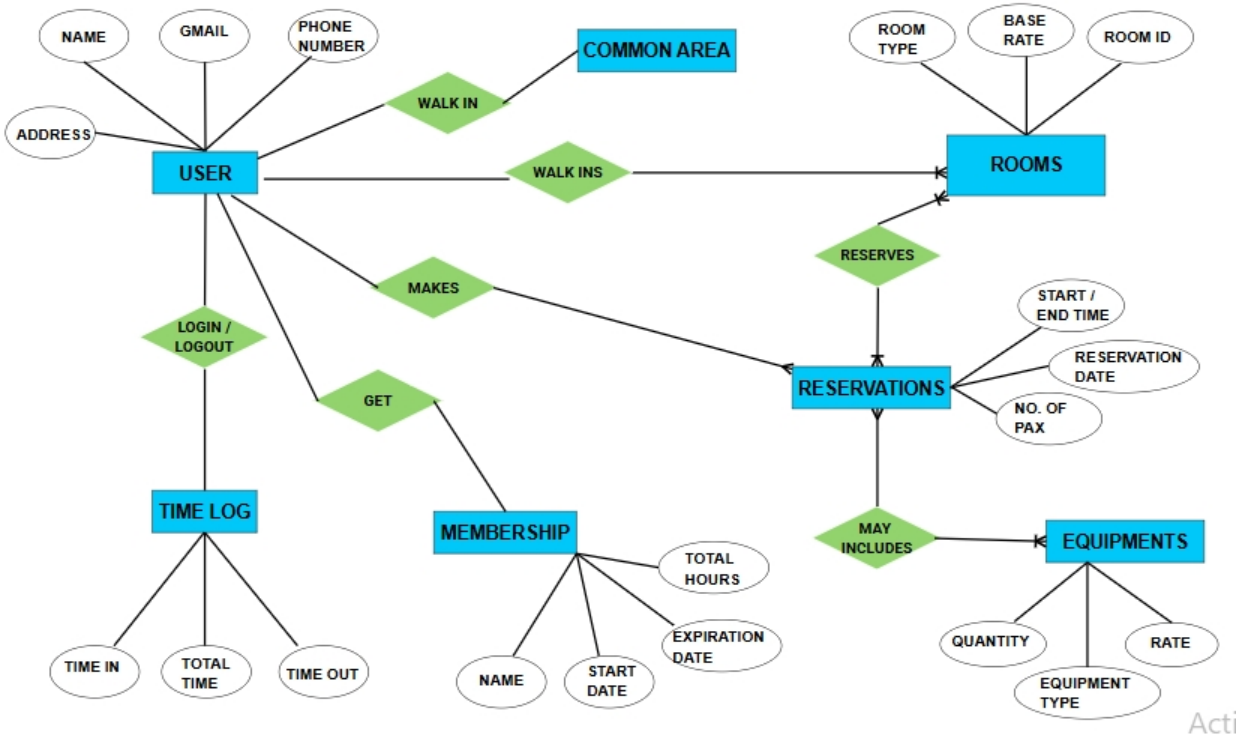


Figure 7. Entity Relationship Diagram of the Integrated Study Hub Management System

Significance of Entity Relationship Diagram

The Entity-Relationship Diagram (ERD) served as the foundational architectural blueprint for the data management layer of the Integrated Study Hub Management System for WS Students & Professionals Lounge - La Paz Branch. By providing a visual and structural representation of the relational database schema, the ERD enabled the proponents to conceptualize and map how complex business operations—such as multi-tier room

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reservations, dynamic time-logging, membership subscription tracking, and itemized payments—were translated into normalized data entities (Pressman & Maxim, 2020).

The primary significance of this diagram lay in its capacity to enforce structural data integrity and prevent relational anomalies across the application. Through the precise application of cardinality rules, primary key (PK) specifications, and foreign key (FK) constraints, the diagram mapped out the direct dependencies between core user activities, physical facility assets, financial logs, and security parameters, ensuring that orphan records were prevented during high-volume operational cycles.

As a formal database design map, each entity defined in the ERD correlated directly to physical relational database tables where attributes, data types, and key constraints were explicitly declared to maintain high data consistency and enable efficient query indexing (Silberschatz et al., 2019). Furthermore, the ERD served as a critical validation tool between the proponents and lounge stakeholders, verifying that the data persistence layer of the application aligned perfectly with the physical business rules, transaction pipelines, and administrative workflows of the lounge.

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3.13 Database Design

The database of the Integrated Study Hub Management System was engineered to securely store, organize, and retrieve critical operational metrics, financial transactions, user accounts, and activity logs. The system utilized a normalized relational database model implemented via MySQL, where tables were interconnected through Primary Key (PK) and Foreign Key (FK) constraints to ensure strict data synchronization, eliminate redundancy, and maintain relational integrity across all modules (Silberschatz et al., 2019).

The following finalized relational tables, structural schemas, and operational roles were derived directly from the system's physical database implementation:

1. users (ws_lounge_lapaz_pro_5_users)

- **Operational Role:** Stores the account details, login credentials, and permission settings for all types of users in the system, including Administrators, Staff, and Customers.
- **Attributes and Schema Specifications:**
 - id: int(11) (PK) – Unique auto-incremented identifier for each user.
 - name: varchar(64) – Full name of the user.
 - email: varchar(120) – Registered email address for notifications and logins.
 - phone: varchar(20) – Active contact number of the user.

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- password: varchar(255) – Secure, hashed authentication password.
- role: varchar(20) – System privilege classification (e.g., Admin, Staff, Customer).
- is_active: tinyint(1) – Binary flag for account status validation.
- created_at: datetime – Timestamp indicating account registration date.
- customer_id: int(11) – External identifier for localized indexing.
- membership_id: varchar(20) – Tracks active subscription numbers.
- expiry_date: datetime – Deadline for account or membership validity.
- last_activity: datetime – Real-time stamp for monitoring active online sessions.
- session_timeout_minutes: int(11) – Security configuration variable for session lifespans.

2. reservations (ws_lounge_lapaz_pro_5_reservations)

- **Operational Role:** Captures the comprehensive life cycle of a workspace booking, managing scheduling details, spatial allocation, and advanced billing computations.
- **Attributes and Schema Specifications:**
 - id: int(11) (PK) – Unique identifier for each booking instance.
 - customer_id: int(11) – Primary customer linking constraint.

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- `user_id: int(11) (FK)` – Relational link to the user table.
- `room_id: int(11) (FK)` – Relational link to the specific room booked.
- `customer_name: varchar(64)` – Explicit name label for rapid display queries.
- `contact_number: varchar(20)` – Temporary or reference phone detail.
- `address: varchar(128)` – Physical address background of the client.
- `pax_count: int(11)` – Total headcount utilizing the reserved space.
- `start_time: datetime` – Slotted timestamp for the commencement of the booking.
- `end_time: datetime` – Slotted timestamp for the expiration of the booking.
- `is_open_time: tinyint(1)` – Variable determining flexible or fixed-hour extensions.
- `status: varchar(20)` – Operational state of the booking (e.g., Pending, Confirmed, Cancelled).
- `total_amount: float` – Computed initial charge for space usage.
- `amount_paid: float` – Actual cash/digital value received from the client.
- `payment_method: varchar(50)` – Transaction mode identifier.
- `receipt_image: varchar(255)` – File path or URL registry for digital payment proof uploads.
- `approved_by_id: int(11) (FK)` – Link to the staff/admin who approved the reservation.

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- `paid: tinyint(1)` – Boolean flag confirming complete payment settlement.
- `added_by: varchar(64)` – Audit trail tracking who encoded the transaction.
- `extra_notes: varchar(255)` – Free-text field for special customer demands.
- `extra_fee: float` – Overtime charges or auxiliary fee additions.
- `discount_rate: float` – Percentages deducted based on dynamic marketing promos or loyalty tiers.
- `created_at: datetime` – System timestamp for record generation.

3.walkin_reservations (ws_lounge_lapaz_pro_5_walkin_reservations)

- **Operational Role:** Specifically tracks immediate, on-the-spot bookings made by customers directly at the physical hub counter.
- **Attributes and Schema Specifications:**
 - `id: int(11) (PK)` – Unique identifier for the walk-in transaction.
 - `reservation_id: int(11) (FK)` – Cross-reference link to the main reservations record.
 - `user_id: int(11) (FK)` – Relational link to the user account handling or using the booking.
 - `room_id: int(11) (FK)` – Relational link to the selected physical room.

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- `customer_name: varchar(64)` – Name of the walk-in customer.
- `contact_number: varchar(20)` – Contact info provided upon arrival.
- `pax_count: int(11)` – Number of individuals in the walk-in group.
- `start_time: datetime` – Actual log-in or entry timestamp.
- `end_time: datetime` – Scheduled or actual departure timestamp.
- `status: varchar(20)` – Current status of the walk-in session.
- `total_amount: float` – Calculated final cost for the walk-in duration.
- `paid: tinyint(1)` – Payment status indicator for the walk-in.
- `extra_fee: float` – Surcharges for extended stay or equipment rentals.
- `added_by: varchar(64)` – Staff account who processed the walk-in guest.
- `extra_notes: varchar(255)` – Remarks or special operational requests.
- `created_at: datetime` – Timestamp when the transaction was saved.

4. `rooms (ws_lounge_lapaz_pro_5_rooms)`

- **Operational Role:** Acts as the primary inventory registry for physical spaces, including individual study pods, general lounges, and corporate meeting rooms.

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- **Attributes and Schema Specifications:**

- `id: int(11) (PK)` – Unique index for each room asset.
- `name: varchar(64)` – Label designating the space name or room number.
- `base_rate: float` – Standard hourly pricing blueprint for the room.
- `category: varchar(50)` – Structural classification (e.g., Private, Shared, Group).
- `status: varchar(20)` – Live availability status indicator (e.g., Available, Occupied, Maintenance).

5. memberships (ws_lounge_lapaz_pro_5_memberships)

- **Operational Role:** Manages long-term customer subscription packages, tracking remaining hours, plan validity, and current checking status.

- **Attributes and Schema Specifications:**

- `id: int(11) (PK)` – Unique identification code for the membership entry.
- `user_id: int(11) (FK)` – Ties the membership directly to a registered user.
- `status: varchar(20)` – Validity indicator of the membership (e.g., Active, Expired, Paused).
- `start_date: datetime` – Activation date of the subscription bundle.
- `expiry_date: datetime` – Programmatic expiration date of the access plan.

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- `total_hours: float` – Total consumable hours granted by the purchased package.
- `hours_left: float` – Remaining balances of usable workspace hours.
- `plan_name: varchar(100)` – Name classification of the membership tier.
- `is_checked_in: tinyint(1)` – Active real-time binary tracker for facility presence.
- `created_at: datetime` – Record creation date.
- `updated_at: datetime` – Log for recent profile modifications.

6. `time_logs (ws_lounge_lapaz_pro_5_time_logs)`

- **Operational Role:** Functions as an immutable data ledger that tracks exact physical arrivals and exits of users, serving as the raw data feed for automated billing engines.
- **Attributes and Schema Specifications:**
 - `id: int(11) (PK)` – Sequential tracking code for every log instance.
 - `user_id: int(11) (FK)` – Links the timestamp directly to a specific user.
 - `plan: varchar(64)` – Tracks what specific pricing archetype or membership is active.
 - `time_in: datetime` – Precise clock-in timestamp.
 - `time_out: datetime` – Precise clock-out timestamp.
 - `total_time: int(11)` – Net computational minutes or hours used.

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7. attendance_logs (ws_lounge_lapaz_pro_5_attendance_logs)

- **Operational Role:** Specifically isolates and monitors the time metrics of subscription members, managing automated deductions against their available package hours.
- **Attributes and Schema Specifications:**
 - id: int(11) (PK) – System key for the membership attendance event.
 - membership_id: int(11) (FK) – Direct connection to the specific user membership.
 - check_in_time: datetime – Verified check-in timestamp at the desk.
 - check_out_time: datetime – Verified check-out timestamp upon leaving.
 - hours_deducted: float – Computed number of hours subtracted from the membership balance.
 - created_at: datetime – Date and time when the log was documented.

8. solo_plans (ws_lounge_lapaz_pro_5_solo_plans)

- **Operational Role:** Manages individualized, non-group workspace subscription plans, tracking user-specific membership duration and active cycles.
- **Attributes and Schema Specifications:**
 - id: int(11) (PK) – Unique primary index for individual plans.
 - customer_id: int(11) – System indexing code for identification.

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- `user_id: int(11) (FK)` – Ties the billing package to an authenticated client account.
- `approved_by_id: int(11) (FK)` – Administrative staff ID who verified and authorized the plan.
- `plan_name: varchar(64)` – Name classification of the subscription package.
- `status: varchar(20)` – Current state of the plan validity (e.g., Active, Expired).
- `payment_method: varchar(50)` – Selected channel for payment collection.
- `receipt_image: varchar(255)` – Digital log file for proof of payment verification.
- `created_at: datetime` – Commencement date of the individual plan subscription.
- `expiry_date: datetime` – Mandatory programmatic cutoff date for the subscription.

9. **daily_reports (ws_lounge_lapaz_pro_5_daily_reports)**

- **Operational Role:** A data warehouse entity that consolidates transaction logs and user entries into high-level business intelligence snapshots for management review.
- **Attributes and Schema Specifications:**
 - `id: int(11) (PK)` – Unique report sequence index.
 - `report_date: date` – Specific operational calendar date being evaluated.
 - `total_check_ins: int(11)` – Aggregated count of successful physical entries for the day.

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- `total_logins: int(11)` – Total authentication attempts tracked within the system.
- `total_timelogged: float` – Cumulative number of workspace hours consumed by clients.
- `generated_at: datetime` – Microsecond timestamp indicating exact system report generation time.

10. `equipment (ws_lounge_lapaz_pro_5_equipment)`

- **Operational Role:** Monitors auxiliary tangible assets that can be rented or utilized within the lounge, such as projectors, extension wires, and adapters.
- **Attributes and Schema Specifications:**
 - `id: int(11) (PK)` – Unique structural ID for every equipment entry.
 - `name: varchar(100)` – Label text for the specific machine or item.
 - `type: varchar(50)` – Equipment classification tag (e.g., Electronics, Peripheral, Furniture).
 - `hourly_rate: decimal(10,2)` – Fixed surcharge cost added when the specific asset is rented out.
 - `quantity_available: int(11)` – Operational inventory stock monitoring variable.

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11. user_activity_logs (ws_lounge_lapaz_pro_5_user_activity_logs)

- **Operational Role:** Serves as a vital security audit trail that archives user actions to detect system exploitation and ensure internal operational transparency.
- **Attributes and Schema Specifications:**
 - id: int(11) (PK) – Automated security record ID.
 - user_id: int(11) (FK) – Connects the system action to an explicit account.
 - activity_type: varchar(50) – System action definition tag (e.g., Login Attempt, Reservation Creation).
 - activity_time: datetime – Definitive date and hour log of the action.
 - ip_address: varchar(45) – Captures the unique network point location of the device used.
 - created_at: datetime – Log baseline generation timestamp.

12. payment_info (ws_lounge_lapaz_pro_5_payment_info)

- **Operational Role:** Manages digital remittance architectures and payment routing channels for customer awareness during checkout phases.
- **Attributes and Schema Specifications:**
 - id: int(11) (PK) – Unique payment channel identifier.
 - method: varchar(32) – Financial gateway type (e.g., GCash, PayMaya, Bank Transfer).

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- `account_name: varchar(128)` – Business register holder name for receipt verification.
- `account_number: varchar(64)` – Explicit account numeric string where cash flows are targeted.
- `qr_image: varchar(255)` – String directory for the graphic display of scanned payment assets.
- `instructions: varchar(255)` – Explicit step-by-step text guiding the user through the transaction process.
- `created_at: datetime` – Blueprint formulation timestamp.
- `updated_at: datetime` – Track log for adjustments in administrative banking structures.

13. `inventory (ws_lounge_lapaz_pro_5_inventory)`

- **Operational Role:** Keeps track of consumable internal resources or items sold within the hub (e.g., snacks, beverages, printing paper).
- **Attributes and Schema Specifications:**
 - `id: int(11) (PK)` – Unique primary catalog index.
 - `item_name: varchar(100)` – Description or name of the physical stock.
 - `category: varchar(50)` – Classification category (e.g., Food, Beverage).
 - `quantity: int(11)` – Numerical data tracking remaining units on hand.

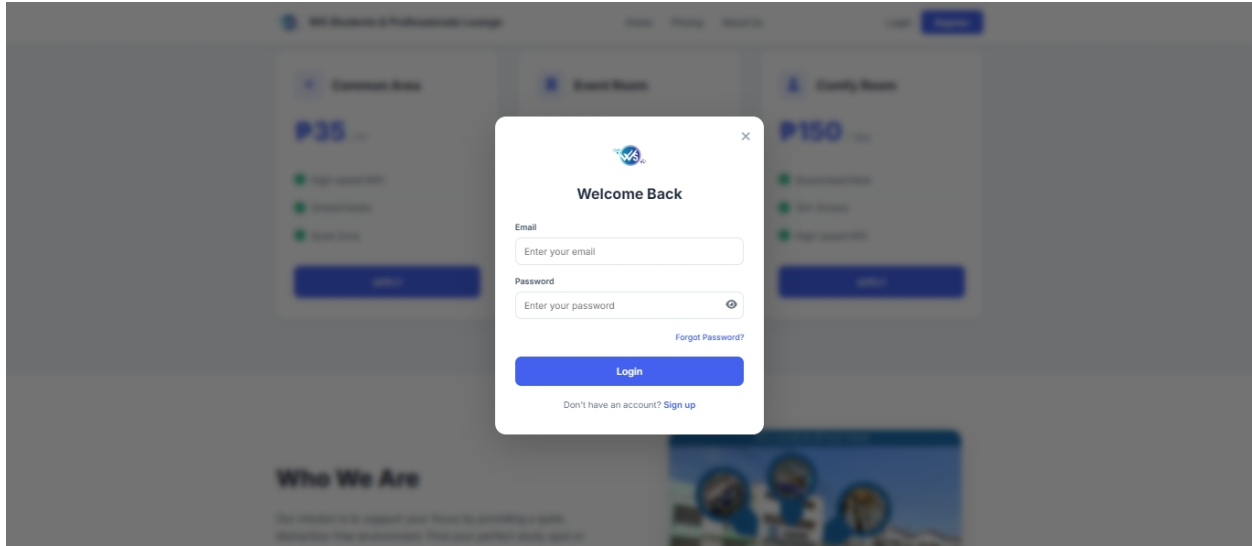
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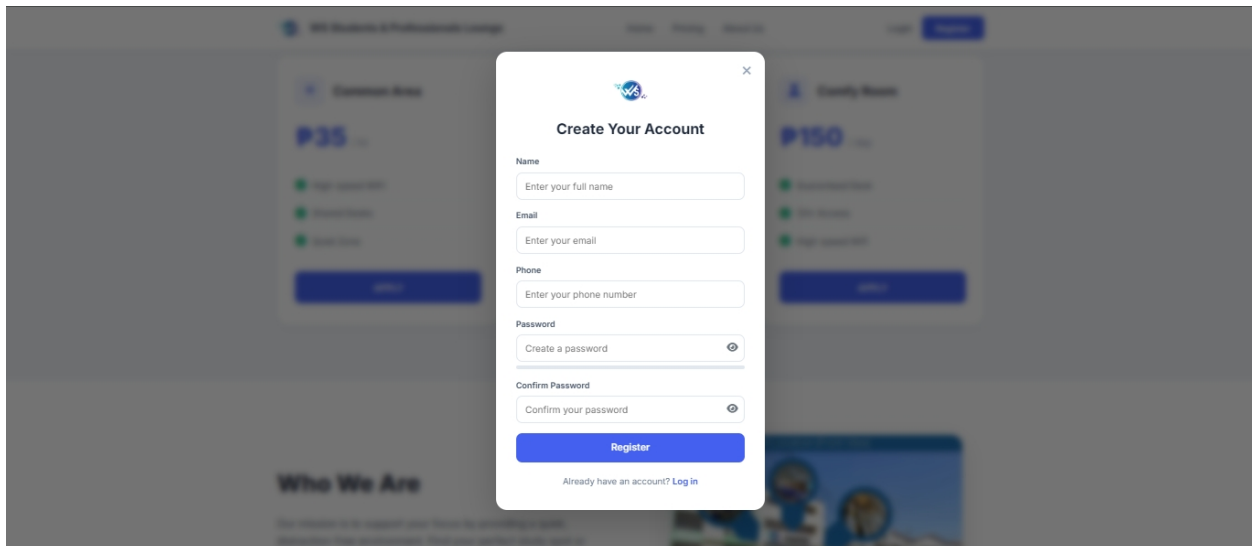
- `price: decimal(10,2)` – Operational retail unit value for point-of-sale calculations.

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3.14 User Interface Design



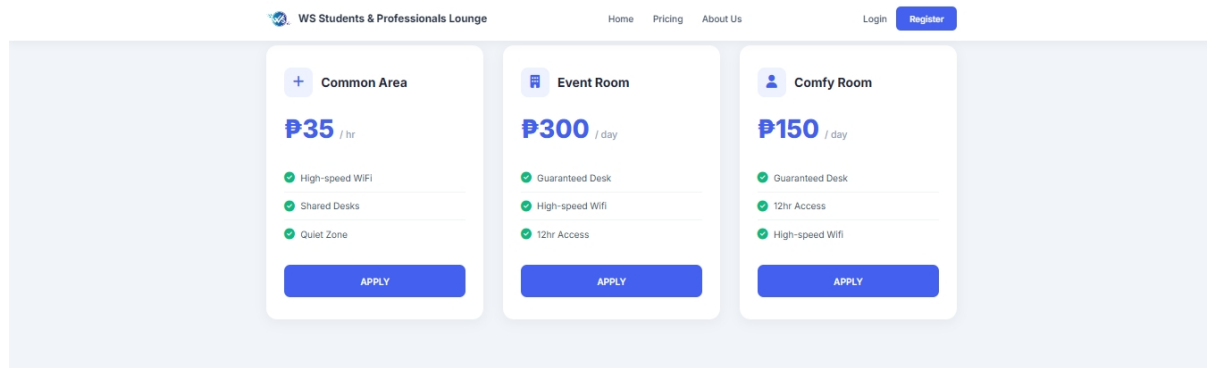
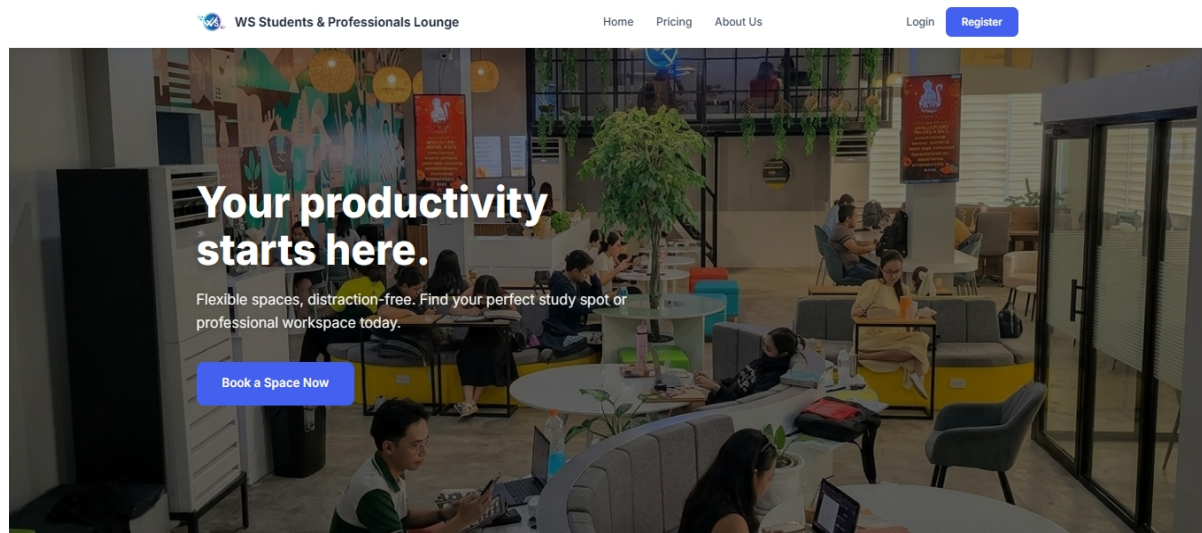
Login Page



Registration Page

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Who We Are

Our mission is to support your focus by providing a quiet, distraction-free environment. Find your perfect study spot or



Landing Page

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Admin Dashboard

Welcome Back, wslounge!

TOTAL MEMBERS
5

ACTIVE PLANS
1

PENDING RESERVATIONS
1

EARNINGS TODAY
₱365.00

Live Room Status

Search customer or room...

WALK-IN

COMMON AREA - OCCUPIED
1

COMMON AREA - AVAILABLE
69

Common Area

OCCUPIED

Occupied
1

Available Slots
69

Common Area

OCCUPIED

Room
Common Area

Start
07:53 PM

End
08:53 PM

Name
Ramil

Customer ID
13

Add Time

End Session

Occupied
0

Available Slots
70

Info
No active Common Area sessions

Small Meeting Room 1

AVAILABLE

Status
Available

Start

End

Name

Small Meeting Room 2

AVAILABLE

Status
Available

Start

End

Name

Lecture Room

AVAILABLE

Status
Available

Start

End

Name

Conference Room

AVAILABLE

Status
Available

Start

End

Name

Comfy Room

AVAILABLE

Status
Available

Start

End

Event Room 1

AVAILABLE

Status
Available

Start

End

Waiting List - Today

Next: Justine Grace Reyes
(Small Meeting Room 2, 08:55 - 09:55 PM)

Common Area Members

1

Ramil Fernandez
Checked in: 07:53 PM
Time Used: 00:10:30

Common Area Members

0

No active common area occupants.

Admin Dashboard

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RESERVATION

Customer Name
e.g. Darwen Salcedo

Contact Number
e.g. 0912-123-4567

Select Area
Common Area (P35.0/hr)

Duration
1 hour

Start Date/Time
dd/mm/yyyy --:-- --

Open Time
dd/mm/yyyy --:-- --

Open Time

STAFF NOTES & EXTRA FEES

Predefined Add-ons
None

Quantity
1

e.g. Extra monitor, HDMI cable...

Empty/invalid values become 1.

Selected Add-on Total: **P0.00**

Additional Fees (Non-Add-on)
0.00

RESERVATION DETAILS

Customer Name: ----

Contact Number: ----

Area: Common Area

Start Date: ----

Start Time: ----

End Time: ----

Duration: ----

Additional Fees: **P0.00**

Add-ons Subtotal: **P0.00**

Discount: 0%

TOTAL PAYABLE
P0.00

CONFIRM & SAVE

CANCEL



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Walk-in Check-in

Use this form for immediate lounge access.

Customer Name
e.g. Darwen Salcedo

Contact Number
e.g. 0912-123-4567

Select Area
Common Area (P35.0/hr)

Duration
1 hour

Start Date/Time
10/08/2026 08:15 pm

Open Time
10/08/2026 09:15 pm

Open Time

STAFF NOTES & EXTRA FEES

Predefined Add-ons
None

Quantity
1

e.g. Extra monitor, HDMI cable...

Empty/invalid values become 1.

Add-on Request (Staff Notes)

Additional Fees (P)

Check-in Details

Customer Name: ----

Contact No: ----

Area: Common Area

Start Time: Aug 10, 2026, 08:15 PM

End Time: Aug 10, 2026, 09:15 PM

Duration: 1.0 Hr/s

Selected Add-on Total: P0.00

Additional Fees: P0.00

Discount: 0%

TOTAL PAYABLE
P35.00

Confirm & Save

Cancel

Reservation Page

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[Logout \(wslounge\)](#)

Reservations List

Manage confirmed and pending reservations. Use the separate confirmation page for pending approvals.

[Confirm Reservation](#)

ID	CUSTOMER	ROOM	DATE & TIME	STATUS	PAYMENT	RECEIPT	AMOUNT	ADDED BY	ACTIONS
#13	Ramil No Address	Common Area	Aug 10, 2026 07:53 PM - 08:53 PM	Walk-in	N/A	None	P35.00	wslounge (Staff)	Add-ons: None Delete

Confirm Reservation

Review pending reservation requests and confirm, hold, or delete them here.

[Back to Reservation List](#)

ID	CUSTOMER	ROOM	DATE & TIME	PAYMENT	RECEIPT	AMOUNT	ADDED BY	ACTIONS
#150	Justine Grace Reyes No Address	Small Meeting Room 2	Aug 10, 2026 08:55 PM - 09:55 PM	N/A	None	P250.00	wslounge (Staff)	Add-ons: None Confirm Hold Delete

Reservation List

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[Logout \(wslounge\)](#)

All Members

Monday, August 10, 2026

Membership Requests Member List Deactivated Accounts

Prioritize by: **Date Requested**

Ramil Fernandez
Email: ramil@gmail.com
Contact Number: 09345678932

Selected Plan
INDIVIDUAL RATE
Date Requested: 08/10/26 - 07:57 PM

Payment Verification
[Check](#)

[Approve](#)
[Reject](#)



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All Members

Monday, August 10, 2026

Membership Requests **Member List** Deactivated Accounts

Official Members

Review and manage current members.

**Active**

Ramil Fernandez
VWS-WLX-107
Email: ramil@gmail.com
Contact: 09345678932

Plan: INDIVIDUAL RATE
Status: **Checked In**

[Checked In](#)
[Check Out Customer](#)
[View Records](#)
[Renew Membership](#)
[Deactivate](#)

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WS LOUNGE

Dashboard

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Logout (wslounge)

All Members

Monday, August 10, 2026

Membership Requests

Member List

Deactivated Accounts

Deactivated Accounts

Soft disabled accounts are preserved for reporting and can be restored.

Search deactivated accounts...

Deactivated since

dd/mm/yyyy

**Inactive**

Ramil Fernandez
WS-WLK-107
Email: ramil@gmail.com
Contact: 09945678932

DEACTIVATED ON

Aug 08, 2026

REASON

Manual soft deactivation

Reactivate

Permanent Delete

Membership Page

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Dashboard

New Reservation

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Logout (wslounge)

127.0.0.1:5000/dashboard

Payment Settings

Manage GCash and Maya account details for customer downpayments.

GCash

Account Name

WS Students & Professionals Lounge

Account Number

0999XXXXXX

Instructions

Please make your payment using the details above and upload your payment receipt.

Upload QR Code

Choose File No file chosen

Current QR



Save GCash Settings

Maya

Account Name

WS Students & Professionals Lounge

Account Number

0999XXXXXX

Instructions

Please make your payment using the details above and upload your payment receipt.

Upload QR Code

Choose File No file chosen

Current QR



Save Maya Settings

Billing Page

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[Logout \(wslounge\)](#)

Daily Reports & Analytics

Start Date
10/08/2026

End Date
10/08/2026

Filter Reports

Completed Sessions

Download Completed Sessions (PDF)

CUSTOMER	CONTACT	EMAIL ADDRESS	ROOM	PAX	STATUS	STAFF	START
Tin	09997672051	wslounge@lounge.com	Small Meeting Room 1	1	Checked-Out	wslounge	Aug 10, 2026 05:47 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Tin	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 05:27 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Justine Reyes	09997672051	ramil@gmail.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 05:25 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Lyza	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 04:00 AM



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[Logout \(wslounge\)](#)

Ramil	03535346342	ramil@gmail.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 03:07 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Justine Reyes	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 02:23 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Flo	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 09, 2026 11:18 PM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Justine Reyes	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 09, 2026 10:59 PM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							

Daily Analytics Summary

REPORT DATE	TOTAL LOGINS	TOTAL CHECK-INS	MEMBER OF THE DAY	GENERATED AT
August 10, 2026	0	10	N/A	05:48 AM

Data automatically stored upon session end.

Report History

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Logout (wslounge)

Manage Staff Accounts

Create new employee accounts and toggle active status without deleting audit history.

Create Staff / Admin Account

Full Name

Email Address

Phone

Password

Role

Create Account

Staff Directory

Name	Email	Phone	Role	Status
Admin Kim	kim@gmail.com	09997672053	Staff	Active
Admin Lorie	lorie@gmail.com	09997672054	Staff	Active
Admin Lyn	lyn@gmail.com	09997672052	Staff	Active
Admin Rose	rose@gmail.com	09997672055	Staff	Active
Admin Smilie	smilie@gmail.com	09997672057	Staff	Inactive
Admin Thyna	thyna@gmail.com	09997672058	Staff	Active
Admin Weyn	weyn@gmail.com	09997672056	Staff	Inactive
admin-t236f8b7	admin-t236f8b7@example.com	N/A	Admin	Active
Maam Smilie	smilesorongon@gmail.com	N/A	Admin	Active

Manage Staff Accounts

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[Dashboard](#)
[My Profile](#)
[Use Lounge \(Time In/Out\)](#)
[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

Member Dashboard

Welcome Back, Ramil !



3
Total Reservations



6.01 hrs
Total Solo Hours



5
Solo Plans

✓ **Official Member**

INDIVIDUAL RATE

STARTED ON	EXPIRES ON	COUNTDOWN	REMAINING HOURS	TOTAL HOURS	STATUS
August 10, 2026 08:23 PM	August 10, 2026 09:23 PM	00h 53m 51s		6.01 hrs	ACTIVE ✓ Checked In

Current Session Duration

00:06:00

Hours Used So Far

0.10 hrs

Recent Attendance

DATE	CHECK-IN	CHECK-OUT	HOURS USED
------	----------	-----------	------------



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WS LOUNGE

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[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

ID	Room	Date & Time	Status
#153	Small Meeting Room 2	08/20 08:25 PM	CONFIRMED
#11	Common Area	08/10 05:25 AM	CHECKED-OUT
#8	Common Area	08/10 03:07 AM	CHECKED-OUT

#152

INDIVIDUAL RATE

Expires: 08/10 2026

#151

INDIVIDUAL RATE

Expires: 08/10 2026

#149

INDIVIDUAL RATE

Expires: 08/10 2026

#147

INDIVIDUAL RATE

Expires: 08/10 2026

#100

INDIVIDUAL RATE (4HRS)

Expires: 08/10 2026

[New Reservation](#)[View Rooms](#)[Solo Plans](#)[Timelogs](#)

User Dashboard

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STUDENT & PROFESSIONALS
WS LOUNGE

[Dashboard](#)
[My Profile](#)
[Use Lounge \(Time In/Out\)](#)
[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

Lounge Usage Logs



5hr 59m
Monthly Total Hours



INDIVIDUAL RATE
Most Used Room



36m
Avg. Session Length



10
Total Completed Sessions

Active Session Countdown

Aug 10, 2026, 8:27:50 PM

00:55:31

Status: ACTIVE

Currently using: **INDIVIDUAL RATE**

[Check Out Now](#)

Usage History

Date	Status	Room	Duration	Total
August 10, 2026	ACTIVE	INDIVIDUAL RATE	08:23 PM - Ongoing	4 mins
August 10, 2026	ENDED	INDIVIDUAL RATE	07:58 PM - 08:15 PM	16 mins
August 10, 2026	ENDED	INDIVIDUAL RATE	05:50 AM - 06:49 AM	59 mins
August 10, 2026	ENDED	INDIVIDUAL RATE	03:04 AM - 05:15 AM	130 mins
August 09, 2026	ENDED	INDIVIDUAL RATE	10:07 PM - 10:07 PM	0 mins
August 09, 2026	ENDED	INDIVIDUAL RATE	06:55 PM - 09:25 PM	149 mins
August 09, 2026	ENDED	INDIVIDUAL RATE	12:55 PM - 12:56 PM	0 mins
August 09, 2026	ENDED	INDIVIDUAL RATE		

Time Logs

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Rooms & Reservations

Book your preferred space with ease

AVAILABLE ROOMS

Common Area
P35.0/hr (1/70 occupied)

AVAILABLE

Small Meeting Room 1
P60.0/hr

AVAILABLE

Small Meeting Room 2
P60.0/hr

AVAILABLE

Lecture Room
P150.0/hr

AVAILABLE

Conference Room
P250.0/hr

AVAILABLE

Comfy Room
P150.0/hr

AVAILABLE

Event Room 1
P300.0/hr

AVAILABLE

Event Room 2
P300.0/hr

AVAILABLE

RESERVATION

CUSTOMER NAME

CONTACT

Ramil

09345128764

SELECT AREA

Small Meeting Room 2 (P50.0/hr)

START DATE/TIME

Duration

20/08/2026 08:25 pm

1 hour

20/08/2026 09:25 pm

Open Time

STAFF NOTES & EXTRA FEES

PREDEFINED ADD-ONS

QUANTITY

Extra Chair (Free)

2

Empty value will be treated as 1.

SPECIAL INSTRUCTIONS

Enter specific requests here...



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VIEW PAYMENT DETAILS

View Payment QR & Details

Grand Total:

P0.00

Downpayment Amount (50%):

P0.00

UPLOAD RECEIPT

Choose File

No file chosen

Verify the amount matches the downpayment (50%) shown above.

RESERVE SPACE

UPCOMING BOOKINGS

dd/mm/yyyy

All Status

EXPORT CSV

ROOMS	START TIME	END TIME	STATUS
Small Meeting Room 2	Aug 20, 2026 08:25 PM	Aug 20, 2026 09:25 PM	CONFIRMED

Reservation Page

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My Profile

Keep your account details current and update your password securely.

Profile Information

FULL NAME
Ramil Fernandez
EMAIL ADDRESS
ramil@gmail.com
CONTACT NUMBER
09345678932

Member account • Updated through your secure profile settings

Update Your Details

Full Name

Email Address

Contact Number

[Save Profile](#)

Change Password

Current Password

New Password

Confirm New Password

Member Profile Page

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Solo Plan Rates

Choose your plan, upload your payment receipt, and wait for admin approval. Your current solo plan status is shown below.

SOLO PLAN STATUS

ACTIVE

Plan: INDIVIDUAL RATE
Expiry: 2026-08-10 09:23 PM
Payment Method: GCash
Status: Approved

MEMBERSHIP INFO

ACTIVE MEMBER PLAN

Plan Name:
Started: 2026-08-10 08:23 PM
Expiration: 2026-08-10 09:23 PM
Remaining days: 0.0
Countdown: 00h 59m 41s

RATE COMPARISON

Compare solo rates at a glance. Click a plan card to submit your payment receipt.

PLAN	PRICE	DETAILS
INDIVIDUAL RATE	P35/HR	✓ Hourly common area usage ✓ WiFi / Charging
INDIVIDUAL RATE (4HRS)	P100	✓ 4hrs common area usage ✓ WiFi & Charging
DAY/NIGHT PASS	P200	✓ Choice of Day (7AM – 10PM) or Night (6PM – 9AM) common area usage ✓ WiFi & Charging



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[Logout \(Ramil Fernandez\)](#)

- WiFi & Charging
- Gold Card

ACTIVE PLAN IN USE

- WiFi & Charging
- Platinum Card

ACTIVE PLAN IN USE

- WiFi, Charging & Exclusive Locker
- Gold Card

ACTIVE PLAN IN USE

MONTHLY PASS (24HRS) P2500

- 24/7 Access to All Branches in Iloilo
- 1 Month (30 days) unlimited common area usage
- WiFi & Charging, Exclusive Locker
- 1hr Sleeping pod & Shower Room use per day (City Proper Branch only)
- Platinum Card

ACTIVE PLAN IN USE

WORKSTATION (24HRS) P3000

- 24/7 Access to dedicated Workstation (Lapaz)
- 24/7 Common area access to all Branches
- 1 Dedicated Workstation
- WiFi & Charging & Exclusive Locker
- 1hr Sleeping pod & Shower Room use per day (City Proper Branch only)
- Platinum Card

ACTIVE PLAN IN USE

PLAN SUMMARY

Choose the rate that fits your needs. After selecting a plan, upload your payment receipt and wait for admin approval.

- Plan approvals are reviewed manually by the admin.
- Your uploaded receipt is stored securely under uploads/receipts.
- Active plans remain valid until expiry.

Membership Plans

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The Integrated Study Hub Management System was comprised of several interconnected modules and Graphical User Interfaces (GUIs) engineered to streamline the daily operations of the WS Students & Professionals Lounge - La Paz Branch. These components operated synchronously to guarantee data integrity, system security, and a seamless operational workflow.

The platform featured the following primary interfaces:

Login Page

Provided a secure authentication gateway that validated user credentials against the system database. It controlled system access by directing authenticated users to their designated role-based dashboards while enforcing session security and data privacy.

Registration Page

Allowed new clients to create an official member account by inputting their personal, contact, and account credential details. It processed data validation rules to prevent duplicate entries and automatically provisioned new member profiles upon successful registration.

Landing Page

Served as the primary digital portal for the platform, designed to direct incoming web traffic based on user intent. It

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provided clear navigation paths for prospective clients seeking workspace availability and a secure login gateway for lounge staff and management.

Admin Dashboard

Served as the primary command center for management and frontline staff, offering real-time visibility into overall hub operations. The interface featured four high-level metrics cards at the top of the view:

- Total Members - Displayed the total count of registered customer accounts in the system database.
- Active Plans - Displayed the total number of clients currently enrolled in active subscription packages.
- Pending Reservations - Highlighted incoming space booking requests awaiting staff verification and approval.
- Earnings Today - Provided an automated, real-time summary of daily revenue generated from space rentals, memberships, and service add-ons.

Directly below these metrics cards, the Live Room Status Grid provided a visual, real-time layout of workspace occupancy. It monitored specific lounge areas—including the Common Area, Small Meeting Room, Lecture Room, Conference Room, and Comfy Room. Each room node displayed live status tags (Available or Occupied), start/end timestamps, active client names, and rapid-action controls (Book Now or Add Session) to enable staff to assign spaces efficiently.

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The right sidebar featured a prominent Walk-In control button for processing on-site arrivals without prior online bookings. Below this button, the Waiting List - Today panel monitored upcoming arrivals, while the Common Area Members List tracked active clients inside the facility, displaying exact check-in timestamps, elapsed session durations, and remaining time to ensure accurate automated billing calculations upon checkout.

Reservation Page (Make Reservation / Bookings)

Facilitated manual booking inputs by staff or advance reservations by clients. The layout was structured into two functional panels:

- Reservation Form - Located on the left panel, this form captured the Customer Name, Contact Number, room category (with associated hourly rates), and No. of Pax. It integrated interactive Start Date/Time and End Date/Time pickers alongside an Open Time toggle for flexible session durations. A dedicated Staff Notes & Extra Fees sub-section allowed staff to record equipment add-ons (such as extra monitors, microphones, chairs, or HDMI cables) and apply custom fee adjustments.
- Reservation Details (Summary) - Located on the right panel, this component acted as a real-time invoice preview. It dynamically computed charges based on selected space rates, session duration, add-on costs, and applicable membership discounts selected from a drop-down menu. Upon validation, the system automatically calculated the Total Payable

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amount, enabling staff to confirm and save the transaction directly to the database.

Reservation Lists (Confirm Reservation)

Served as an administrative management table where staff and administrators reviewed, verified, and processed pending customer booking requests. The interface organized booking data across structured column fields:

- Booking Information Fields - Displayed unique transaction identifiers (e.g., #102), customer contact details, assigned workspace categories, and scheduled reservation timestamps.
- Payment & Audit Columns - Displayed itemized payment breakdowns (such as GCash down-payment percentages, remaining balances, and grand totals), a View Receipt modal link to audit customer-uploaded payment screenshots, total transaction trackers, and an Added By tag identifying the processing staff member.
- Action Controls - Provided one-click administrative options, including a green Confirm button to lock in schedule allocations, a gray Hold button to pause questionable requests, and a red Delete button to cancel invalid bookings. A navigation shortcut at the top allowed staff to return quickly to the main reservation list.

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Membership Page

Served as the centralized administrative module for managing subscription tiers and long-term customer accounts. The module was organized into three sub-tabs:

- Membership Requests - Displayed incoming subscription applications from clients (such as Weekly Pass requests). Staff utilized a Check button under payment verification to review payment screenshots, subsequently clicking Approve to activate the subscription or Reject to dismiss the request.
- Member List (Official Members) - Displayed all active, verified subscription members. Each profile card presented the member's system ID, contact details, active plan type, and live status badge. Operational action controls included a Check In button for duration tracking, a View Records button to audit visit histories, a Renew Membership button for plan extensions, and a Deactivate Account button to suspend access.
- Deactivated Accounts - Functioned as an archive for soft-disabled customer accounts to preserve historical transaction logs for financial audits. Each record displayed deactivation timestamps and recorded justification notes. It featured a Reactivate option to restore account privileges instantly and a Permanent Delete option for permanent database removal when necessary.

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Billing Page (Payment Settings)

Served as the backend financial configuration module where administrators managed digital payment parameters used for online booking down-payments. The interface was split into two primary configuration columns:

- GCash and Maya Configuration Forms - Allowed administrators to update official mobile wallet account names, account numbers, and optional step-by-step payment instructions displayed to clients during checkout.
- QR Code Management - Integrated an image upload component allowing administrators to update official payment QR code graphics. The system displayed a live graphic preview under the Current QR section, while dedicated Save Settings buttons executed real-time database updates.

Report History (Daily Reports & Analytics)

Served as the administrative auditing interface where management reviewed and exported business performance metrics and financial logs. The module was organized into three functional sections:

- Date Filter and Export Controls - Integrated date pickers allowing administrators to specify Start Date and End Date ranges to filter report logs. A Download PDF Report control allowed management to compile and print official hard-copy administrative reports instantly.

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- Session Monitoring Logs - Displayed a complete record of finished study lounge sessions within the filtered date range, enabling management to monitor customer attendance, peak room utilization, and session durations.
- Daily Analytics Summary - A structured data table compiling daily business metrics, automatically recording the Report Date, Total Portal Logins, Total Check-Ins, Member of the Day selections, and execution timestamps (Generated At). The system compiled these metrics automatically upon session completions.

Manage Staff Accounts

Served as the user management portal where administrators generated employee credentials and managed platform access permissions. The layout was divided into two sections:

- Create Staff / Admin Account Form - Located on the left panel, this form captured new employee details, including Full Name, Email Address, Phone Number, and Password. A Role drop-down menu assigned either standard Staff or elevated Admin access rights prior to saving credentials to the database.
- Staff Directory Table - Located on the right panel, this directory listed all registered system operators, displaying their contact profiles and assigned roles. Each row included an interactive status toggle, allowing management to activate or suspend employee platform access without purging their historical transaction logs.

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User Dashboard

Served as the self-service welcome portal for registered clients (Students and Working Professionals), allowing users to monitor their active subscriptions and personal usage histories in real time:

- Personal Summary Cards - Displayed high-level personal metrics at the top of the interface, including Total Reservations, Total Solo Hours, and active plan counts.
- Official Member Tracking Banner - A prominent tracking banner displayed active subscription tiers (such as Individual Rate), highlighting start/expiration timestamps, real-time countdown timers, remaining plan hours, current session durations, and live Checked In status badges.
- Attendance and Plan History - Displayed structured tables detailing recent check-in/check-out timestamps, reservation histories, and active plan expiration dates. Shortcut buttons provided direct navigation to New Reservation, View Rooms, Solo Plans, and Time Logs modules.

Time Logs (Use Lounge - Time In/Out)

Served as the self-service time-tracking interface for active clients within the hub, monitoring stay durations to establish the foundation for automated billing calculations:

- Lounge Session Analytics - Displayed four metrics cards summarizing the client's lifetime usage history, including

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Monthly Total Hours, Most Used Room, Average Session Length, and Total Completed Sessions.

- Live Session Status - Displayed the real-time condition of the client's current stay, showing a live digital clock alongside session status indicators (e.g., "No Active Session - Status: Awaiting Reservation").
- Usage History Card - Displayed ongoing session parameters, logging the check-in date, room category, active status tags, and live duration timestamps.

Reservation Page (Book Space)

Served as the customer-facing online space reservation portal, enabling clients to select, schedule, and pay for workspace allocations independently:

- Available Rooms Panel - Located on the left panel, this vertical list displayed real-time workspace availability alongside hourly rate matrices for the Small Meeting Room, Lecture Room, Conference Room, Comfy Room, Event Rooms 1 & 2, and the Common Area.
- Reservation & Payment Form - Located in the center panel, this form captured client details, guest counts (No. of Pax), scheduled timeslots via date-time pickers, predefined service add-ons, special instructions, and preferred payment channels (GCash or Maya).
- Upcoming Bookings Table - Located at the bottom of the interface, this table enabled clients to track personal reservation schedules and confirmation statuses, with an Export CSV feature for downloading schedule records.

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Member Profile Page

Enabled registered members to manage and personalize their account information. This interface provided self-service features allowing users to update their profile name, contact details, and account passwords while maintaining an updated record of their membership status within the system.

Membership Plans

Served as the self-service subscription catalog where clients reviewed and purchased long-term access plans:

- Available Plans Selection - Displayed active subscription cards (such as the Individual Rate plan at ₱100.00), detailing pricing, validity durations, and included amenities (e.g., high-speed internet, free-flowing coffee, ergonomic seating). Each card featured a Purchase Plan button to initiate onboarding.
- Subscription Checkout Form - Displayed an automated checkout form upon plan selection. The left panel captured the applicant's Full Name, Email Address, Contact Number, and selected Membership Type. The right panel provided a payment summary, step-by-step mobile wallet instructions, and a file attachment control for uploading payment screenshots prior to final submission.

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Role-Based Access Control (RBAC)

Enforced security and data privacy across the entire platform by dynamically restricting menu navigation and feature availability based on authenticated user roles:

- Admin and Staff Access - Configured for lounge operational personnel. Elevated Admin roles held full system permissions, including Manage Sales Reports, Manage Staff Accounts, and Payment Settings. Front-desk Staff roles held operational access to manage walk-ins, approve reservations, and process check-ins, while remaining restricted from altering core system configurations or deleting historical financial records.
- User Access (Students and Professionals) - A restricted client portal where users interacted strictly with personal self-service tools, such as Book Space, Solo Plans, and Use Lounge (Time In/Out). Administrative navigation bars, overall financial reports, daily analytics, and other users' private profiles were completely hidden from client views.

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Technology Stack

For the engineering and deployment of the Integrated Study Hub Management System for WS Students & Professionals Lounge – La Paz Branch, the proponents utilized a structured combination of programming languages, software frameworks, database engines, development libraries, and hardware tools. The selection of these technologies ensured high computational performance, system security, database stability, and an intuitive user experience. The finalized technology stack is detailed below:

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3.15 Programming Languages and Tools Used

Programming Languages	Used For
Python	For the main back end language.
SQL	For database queries and schema definition.
HTML5	For frontend templating and mark up structure for web pages using Jinja2 templates
CSS3	For frontend styling and responsive design across all pages.
JavaScript	For client-side functionality and interactivity.

Table 2. Programming Languages of the Integrated Study Hub Management System for WS Students & Professionals Lounge

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3.15.1 Back-End Frameworks and Libraries

Back-end Frameworks and Library	Purpose or Used for
Flask	The primary web framework is used for routing and request handling.
Flask-SQLAlchemy	Used as an ORM to manage database models and execute queries easily.
Flask-Login	Manages user sessions and authentication for Admin and Staff access.
Flask-WTF / WTForms	Handles secure form processing and provides CSRF protection.
Bcrypt	Used for secure password hashing to protect user credentials.
Flask-SocketIO	Enables real-time, bidirectional communication for live dashboard updates.
MySQL-connector-python	The official driver used to connect the Flask app to the MySQL database.
Flask-Mail	Used for sending automated email notifications to users or administrators.
ReportLab	A library used for generating PDF files and

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	operational reports.
Stripe	Integrated for secure online payment processing for lounge services.
Alembic	Used for database migrations to track changes in the DB schema.
Python-dotenv	Loads environment variables from a .env file for security and configuration.
Werkzeug	Provides WSGI utility functions for the Flask application stack.
Email-validator	Ensures that user-provided email addresses are in the correct format.
Gunicorn	A production-grade WSGI HTTP server used for deploying the application.
Pytest	The framework used for running automated tests to ensure system stability.
Black	A PEP 8 compliant code format used to maintain clean and consistent code.

Table 3. Back-end Frameworks and Libraries of the Integrated Study Hub Management System for WS Students & Professionals Lounge

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3.15.2 Front-End Frameworks and Libraries

Front-end Frameworks and Library	Purpose or Used for
Vanilla HTML / CSS	Used for the core structure and custom styling of the user interface.
Vanilla JavaScript	Handles client-side interactivity and application logic.
Font Awesome (via CDN)	Provides the icon library used throughout the system's UI.
Google Fonts (Inter)	The primary typography used for a clean and professional look.
Chart.js (via CDN)	Used for creating visual charts and data visualizations for reports.
Socket.IO (Client)	Handles real-time notifications and live updates on the dashboard.
Icons8	Source of external decorative icons for the system's statistics and menus.
Moqups	Used for creating wire frames and initial UI designs of the system.
Canva	Used for designing graphic

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	assets, presentations, and branding elements.
Favicon	Used to provide a small, iconic image for the browser tab of the system.
Jinja2	Server-side template engine for dynamic HTML rendering.

Table 4. Front-End Frameworks and Libraries of the Integrated Study Hub Management System for WS Students & Professionals Lounge

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3.15.3 Software Tools

- Visual Studio Code - The primary source code editor used for writing and managing the system's code.
- GitHub - For version control and collaborative management of the project's source code.
- Figma / Canva / Moqups - Used for creating diagrams, flowcharts, and the final presentation.
- XAMPP / MySQL Workbench - For managing the local database and testing queries.

3.15.4 Software Specifications

- Database: MySQL, SQLite3, SQLAlchemy
- Python Version: 3.12
- Web Framework: Flask 3.0.3

3.15.5 Hardware Specifications

- Processor: Intel(R) Celeron(R) N4020 CPU @ 1.10GHz (1.10 GHz)
- RAM: 4.00 GB
- Storage: 256 GB

3.15.6 Minimum Hardware Requirements for User (Client)

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- Device: Any smartphone, tablet, laptop, or PC with an active internet connection.
- Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge.

3.15.7 Software Requirements

- Operating System: 64-bit operating system, x64-based processor
- Python: 3.8+ version.
- Database Drivers: MySQL-Connector-Python for MySQL and SQLite3 (Native Python

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3.16 Survey Questions for System Assessment

This survey instrument was designed and deployed to systematically assess the user requirements, operational friction, and functional expectations regarding the Integrated Study Hub Management System. The primary data gathered through this evaluation enabled the proponents to analyze and address the core operational vulnerabilities inherent in the pre-existing manual setup of the WS Students & Professionals Lounge – La Paz Branch. The instruments were administered to the lounge administrator, frontline staff, and registered clients, with all collected responses treated with strict data confidentiality in compliance with research ethics standards.

Demographic Information

What is your role in WS Students & Professionals Lounge?

- ☐ Staff / Owner
- ☐ Regular Customer (Student/Professional)
- ☐ Occasional Visitor

For Students/Professionals/Visitors Survey

1. How long have you been using study hubs or similar lounge facilities?

- ☐ Less than 1 year
- ☐ 1-2 years
- ☐ 3-4 years
- ☐ 5-6 years
- ☐ More than 6 years

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2. Which area in the lounge do you prefer to use?

- ☐ Private Study Pods
- ☐ Group Meeting Rooms
- ☐ Common Area
- ☐ Other: _____

3. How many times do you visit the lounge per month?

- 1. Less than 5
- 2. 5-10
- 3. 11-20
- 4. 21-30
- 5. More than 30

4. What type of transaction method have you been using?

- ☐ Face-to-face (Manual)
- ☐ Online Booking
- ☐ Phone-based Reservation
- ☐ Walk-in

5. What is your average duration of stay in the lounge?

- ☐ Less than an hour
- ☐ 2-5 hours
- ☐ 6-8 hours
- ☐ Whole day

6. Do you prefer getting a membership?

- ☐ Yes
- ☐ No

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7. What would convince you to use a new online booking system instead of making a manual booking? (Select all that apply)

- ☐ Fast and instant booking confirmation (Email / SMS)
- ☐ Real-time view of available rooms, photos, and exact prices
- ☐ Ability to book anytime (24/7) without waiting for a reply
- ☐ Convenient online payment options (e.g., E-Wallets, Credit Card)
- ☐ Ability to easily modify or cancel my reservation online

8. If this establishment launches a fully automated online reservation platform, how likely are you to use it for your next booking?

- ☐ Extremely Likely
- ☐ Likely
- ☐ Neutral
- ☐ Not Likely
- ☐ Extremely UnLikely

9. How satisfied are you with the current reservation/booking method you are using?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Not Satisfied

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10. How would you rate the support and service provided for your lounge concerns?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

For Staff/Owner

1. How long have you been working in WS Students & Professionals Lounge?

- ☐ Less than a year
- ☐ 1-2 years
- ☐ 3-4 years
- ☐ 5-6 years
- ☐ More than 6 years
- ☐

2. Have you ever experienced missing or unrecorded transaction logs?

- ☐ Yes, frequently during peak hours
- ☐ Yes, occasionally
- ☐ Rarely
- ☐ No, records are always complete

3. What issues have you been experienced with current booking or check-in methods? (Select all that apply)

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- ☐ Double Bookings
- ☐ Uncertainty if seats/spaces are actually available
- ☐ Delayed responses when inquiring about availability
- ☐ Slow Billing Process
- ☐ None / Satisfied with current method
- ☐ Other: _____

4. How are active customer session times currently monitored?

- ☐ Staff manually check a spreadsheet
- ☐ Relying on customers to keep track of their own time
- ☐ Staff manually check paper logbooks periodically
- ☐ Verbal reminders from staff members

5. How do you currently reserve a room reservation/booking to a customer?

- ☐ Social Media Platform
- ☐ Email/Contact Form
- ☐ Phone Call/Message
- ☐ Walk-in

6. How do staff currently generate and submit daily/weekly/monthly transaction reports to management?

- ☐ Handwritten summary sheets / Physical logbooks
- ☐ Manually filled-out digital spreadsheets
- ☐ Sending photos of physical receipt logs or logbook pages via chat apps
- ☐ Verbal updates at the end of the shift or day
- ☐ Other: _____

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7. How open are you to replacing your current manual booking process with an automated online reservation system?

- ☐ Very Open
- ☐ Somewhat Open
- ☐ Neutral
- ☐ Not Open

8. What is your primary goal for adopting an automated reservation system?

(Select up to 2)

- ☐ Reduce manual errors / double-bookings
- ☐ Save time on administrative tasks
- ☐ Provide 24/7 availability for customers
- ☐ Accept real-time online payments
- ☐ Improve customer tracking and reporting

9. How easy was it to navigate and use the new automated system compared to your previous manual process?

- ☐ Significantly easier
- ☐ Slightly easier
- ☐ About the same
- ☐ More difficult
- ☐ Significantly more difficult

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10. How would you rate the Integrated Study Hub Management System for WS Students & Professionals Lounge?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

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3.17 Sampling Methods for Respondent

To ensure a balanced, objective, and comprehensive evaluation of the system, the proponents implemented a stratified random sampling technique. This approach guaranteed that the diverse user base of the WS Students & Professionals Lounge - La Paz Branch—ranging from regular student clients to frontline administrative personnel—was accurately represented during the data collection process. The execution of this sampling strategy followed three primary phases:

1. Definition of Strata: Respondents were categorized into distinct subgroups based on their operational roles and frequency of interaction with the lounge facility. These strata comprised Students, Working Professionals/Freelancers, Walk-In Visitors, Frontline Staff, and Business Management. Segmenting the sample population enabled the proponents to isolate and analyze specific user requirements, software usability preferences, and operational pain points from multiple perspectives.

2. Sampling Within Strata: A random selection method was applied within each user stratum to eliminate selection bias and maintain objectivity. For client-side categories (Students and Working Professionals), the proponents conducted survey distribution across both peak and off-peak operational hours to capture a broad cross-section of client experiences. For the organizational strata, the study involved the actual active staff members and managers responsible for processing daily lounge transactions.

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3. Sample Size & Distribution: The proponents successfully secured a total sample size within the target range of 30 to 50 respondents, which provided sufficient statistical reliability to identify recurring operational themes, validate functional requirements, and evaluate system usability. Data gathering was conducted using a hybrid distribution approach:

- Digital Administration: An online version of the survey instrument was deployed via Google Forms, accessible across Android, iOS, and Windows platforms.

- On-Site Administration: Physical questionnaires and tablet-assisted digital forms were facilitated directly at the lounge front desk for walk-in clients and staff members during active business hours.

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3.18 Data Analysis Techniques

To evaluate the efficiency, functionality, and operational impact of the Integrated Study Hub Management System, the proponents applied a dual-framework approach utilizing both quantitative and qualitative data analysis. This structured analytical framework enabled the conversion of raw primary data—collected through structured surveys, semi-structured stakeholder interviews, field observations, and system audit logs—into actionable empirical findings regarding system performance and usability.

Quantitative Analysis

Quantitative data gathered from the structured Likert-scale questionnaires were processed and summarized using Descriptive Statistics:

- **Frequency and Percentage Distribution:** Utilized to profile respondent demographics and quantify user experiences, such as the proportion of clients satisfied with pre-existing manual transactions versus those who encountered operational bottlenecks (e.g., booking errors, check-in delays).
- **Measures of Central Tendency (Weighted Mean, Median, and Mode):** Calculated to determine average baseline ratings for pre-existing manual issues as well as post-implementation usability and functionality scores across the ISO 25010 software quality characteristics.
- **Cross-Tabulation Analysis:** Applied to evaluate variations in user requirements and satisfaction levels across

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different respondent strata (e.g., comparing the functional requirements and usability feedback of Student clients against those of Working Professionals).

To enhance data readability and interpretation, quantitative findings were organized into structured summary tables and presented through visual charts—specifically Bar Graphs and Pie Charts—facilitating clear comparative analysis between the lounge's legacy manual setup and the newly implemented digital platform.

Qualitative Analysis

Qualitative data derived from open-ended survey comments, one-on-one administrator interviews, and focus group discussions were processed using Thematic Analysis. Qualitative feedback was transcribed, coded, and categorized into recurring operational themes to contextualize the quantitative findings, refine UI/UX components, and validate that all functional requirements of the WS Lounge were met.

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3.19 Methods for Processing Raw Data

To uphold the accuracy, validity, and reliability of the research findings, a systematic multi-stage protocol was strictly executed to process all raw primary data prior to formal statistical computation and thematic analysis. Raw data gathered through structured surveys, stakeholder interviews, field observations, and system audit logs underwent rigorous data preparation procedures to ensure structural integrity and readiness for interpretation:

1. Data Scrubbing and Auditing: The initial stage involved a meticulous audit of all completed survey submissions to detect anomalies, incomplete questionnaires, or duplicate entries. Submissions with unengaged responses or significant missing fields were isolated and excluded from the data pool to preserve the overall quality and validity of the dataset.

2. Systematic Data Coding: Following data scrubbing, a standardized coding schema was applied to both quantitative Likert-scale responses and qualitative open-ended interview transcripts. Numerical values were assigned to ordinal scale selections (e.g., \$1 = \text{Strongly Disagree}\$ to \$5 = \text{Strongly Agree}\$), while thematic codes were assigned to qualitative feedback to facilitate structured categorization and statistical aggregation.

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3. Data Tabulation and Centralized Management: Coded datasets were digitized and migrated into centralized analytical spreadsheets (Microsoft Excel / IBM SPSS). Data tables were organized logically by user strata (Students, Working Professionals, Staff, and Management) to enable rapid statistical computation, cross-tabulation, and database cross-referencing.
4. Data Verification and Validation: A double-entry validation audit was conducted to confirm that the digitized values accurately reflected the original inputs recorded on physical survey forms and digital Google Forms submissions. This step eliminated clerical transcription errors and ensured complete fidelity to the original participant responses.
5. Final Synthesis and Aggregation: In the final stage, processed data were summarized and converted into structured statistical tables, weighted mean matrices, bar graphs, and pie charts. This synthesis prepared the empirical findings for a comprehensive evaluation against the system's operational objectives, functional specifications, and ISO 25010 software quality standards.

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3.20 Frameworks for Evaluating System Effectiveness

To verify that the Integrated Study Hub Management System met all technical specifications and user requirements for the WS Students & Professionals Lounge - La Paz Branch, the proponents adopted two standardized evaluation models. These frameworks provided structured criteria for assessing software quality, system performance, operational security, and overall user adoption readiness:

1. ISO/IEC 25010 Software Quality Model:

The ISO/IEC 25010 standard served as the primary framework for technical evaluation. This model enabled the proponents and evaluators to measure software quality across key characteristics, including Functional Suitability, Performance Efficiency, Compatibility, Usability, Security, Maintainability, and Portability. By applying this model, the proponents verified that the system effectively executed its core tasks, operated reliably under varying user loads, and maintained compatibility across the various hardware and software environments utilized in the hub.

2. Technology Acceptance Model (TAM):

The Technology Acceptance Model was implemented to gauge the adoption readiness of both frontline staff and lounge

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clients. Through the analysis of user feedback, the proponents evaluated two core metrics:

- Perceived Usefulness (PU) - Assessed the degree to which users believed the digital system simplified booking workflows, eliminated manual errors, and enhanced overall operational efficiency.
- Perceived Ease of Use (PEOU) - Evaluated user perceptions regarding interface intuitiveness, learning effort, and task navigation simplicity during daily lounge operations.

The proponents maintained complete transparency, research integrity, and strict ethical standards throughout the development, testing, and data gathering lifecycle. To safeguard participant rights and protect institutional assets, the following ethical protocols were strictly observed:

- Voluntary Participation & Informed Consent: All survey respondents, interview subjects, and testing participants took part in the study voluntarily. The proponents strictly avoided any form of coercion or exploitation. Participants were fully informed of the research nature and objectives prior to evaluation, retaining the right to withdraw from the study at any time without penalty or prejudice.
- Academic and Professional Approval: Institutional approval and ethical clearance were secured from the academic review board and executive management of the WS Students & Professionals Lounge prior to the commencement of field observations, system evaluation, and data gathering.

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- Confidentiality and Data Privacy: All contributions and responses from participants were handled with strict confidentiality and respect. Personal data were anonymized, and all compiled findings were reported with full research integrity, ensuring no manipulation or misrepresentation of results occurred.

- Accountability in Development: The finalized software was engineered to promote a secure, accountable digital environment, ensuring that the lounge's operational data were handled responsibly and that the platform fostered long-term trust among its users

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3.21 Ethical Considerations

The proponents maintained an unyielding commitment to upholding the highest standards of research ethics, data privacy, and software integrity throughout the development, testing, and implementation of the Integrated Study Hub Management System. The execution of the project was guided by the following core ethical principles:

- **Commitment to Data Privacy:** Secure multi-level authentication and encryption measures were implemented to ensure that sensitive operational records, customer credentials, and financial metrics were accessible strictly to authorized personnel. User profiles and transactional records were protected against unauthorized access, potential data breaches, and accidental information leaks
- **System Integrity and Security:** Robust data protection protocols were integrated to maintain the absolute confidentiality, accuracy, and immutability of all financial and operational records. The proponents conducted continuous software audits and vulnerability evaluations to identify and resolve security risks, maintaining a safe digital environment for all lounge transactions.
- **Informed Consent and Transparency:** Clear information regarding platform functions, data collection scopes, and storage policies was provided to participants prior to

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system interaction and survey administration. All respondents and system users were fully informed of how their activity logs, personal details, and transactional data were managed, processed, and stored.

- **Prevention of System Misuse:** To discourage fraudulent usage and unauthorized administrative overrides, the system integrated strict accountability features, including automated audit logs, timestamped transaction trails, and Role-Based Access Control (RBAC). These safeguards ensured that management tools were utilized strictly for professional, verifiable operational activities within the lounge.
- **Inclusive and Universal Access:** The graphical user interface was engineered following universal usability guidelines, ensuring accessibility for all students and professionals regardless of technical proficiency. Automated pricing, subscription handling, and session duration calculations were designed to execute impartially and without algorithmic bias.
- **Compliance with Regulatory Standards:** All data handling, storage mechanisms, and administrative reporting procedures strictly adhered to the Data Privacy Act of 2012 (Republic Act No. 10173) of the Philippines. The proponents ensured that personal information was collected for specified, legitimate operational purposes

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and processed with complete legal compliance and confidentiality.

- **Professional and Ethical Responsibility:** The proponents demonstrated objective, responsible, and transparent decision-making across all phases of the Software Development Life Cycle (SDLC). Priority was given to protecting stakeholder welfare, maintaining data accuracy, and delivering a secure, reliable, and trustworthy study hub management platform

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Chapter IV

4.1 Result and Interpretation

This chapter presents the data gathered, analyzed, and interpreted during the development, implementation, and evaluation phases of the Integrated Study Hub Management System. It highlights the key findings and evaluates the system's effectiveness in achieving its core objectives, specifically in automating transaction flows, implementing Role-Based Access Control (RBAC), and resolving the operational bottlenecks at the WS Students & Professionals Lounge.

The presented outcomes were derived from comprehensive system performance evaluations, User Acceptance Testing (UAT), and operational workflow assessments conducted during the system's deployment. Ultimately, this chapter demonstrates how the newly developed platform addressed the logistical vulnerabilities and administrative challenges identified in the earlier parts of the study, thereby validating the proponents' goal of establishing a streamlined, accurate, and efficient management solution for the study hub.

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4.2 Task Manager:

4.2.1 Gantt Chart / Project Timetable

Week	Task	Programmer	UI/UX Designer	Documenter
Week 1	Project Planning & Requirements Gathering	Defined the technical stack and system requirements.	Conducted user interviews and established the customer journey map.	Conducted user interviews; Created initial drafts of Chapters 1 and 3.
Week 2-3	Research & System Architecture Planning	Finalized the backend framework and determined the database logic data flow.	Researched best practices for lounge booking interfaces and planned wireframes.	Completed the literature review for Chapter 2.
Week 2-3	Agile Framework Development	Assisted in mapping out sprint cycles.	Assisted in defining user stories for the hub.	Created and structured the Agile Scrum Framework documentation.

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Week 4-5	Use Case Diagram	Assisted in technical scoping.	Contributed to user role definitions.	Developed the system's Use Case Diagram.
Week 4-5	Data Flow Diagram	Mapped the data input/output routes.	Assisted in workflow visualization.	Formulated the system's Data Flow Diagram (DFD).
Week 4-5	Entity Relationship Diagram	Designed the Entity Relationship Diagram (ERD) for data optimization.	Formulated user interface schema dependencies.	Documented the ERD and detailed the specific data attributes.
Week 6-7	Database Design & Schema Creation	Developed the final database schema, including tables for Users, Reservations, Memberships, and Time Logs.	Reviewed data input field structures for the UI.	Documented the Database Design and created the necessary table specifications.
Week	UI	Reviewed	Developed	Assisted in

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8-10	Wireframes and Mockups	technical feasibility of the layouts.	and finalized interactive mockups for the Landing Page, Dashboards, and Booking Page.	organizing visual design documentation.
Week 10-12	Frontend Development	Began developing core system logic and API connection points.	Implemented approved UI mockups using HTML, CSS (Bootstrap), and JavaScript.	Documented the frontend architecture and user interface manual.
Week 12-16	Backend Development	Coded backend system architecture, including RBAC engines, reservation logics, and automated billing.	Refined components to match backend data responses.	Tracked development milestones and compiled technical notes.
Week	Integratio	Integrated	Coordinated	Documented

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16-18	n of Frontend and Backend	backend APIs with the interactive frontend modules.	with the programmer to adjust responsive layouts.	integration test cases and data flows.
Week 18-19	System Security Testing & Debugging	Debugged backend logic and validated transaction processing data.	Patched responsive UI/UX flaws and navigation bugs.	Logged active test cases, system errors, and resolution results.
Week 20	System Evaluation	Optimized database response times and query speeds.	Implemented aesthetic refinements based on initial user feedback.	Summarized evaluation outcomes and final performance metrics.
Week 21-22	Documentat ion Finalizati on	Reviewed technical descriptions in the paper.	Finalized system figures, diagrams, and captions.	Compiled and completed the Final Research Documentatio n.
Week 23-24	Presentati on and Submission	Set up the backend environment for live demonstrati	Polished the client-side presentation interface.	Formulated the presentation structure and

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		on.		finalized all project deliverables .
--	--	-----	--	---

**Table 5. Gantt Chart of the Integrated Study Hub Management
System of WS Students & Professionals Lounge**

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4.3 Presentation of Results

The development of the Integrated Study Hub Management System was driven by the necessity to transition WS Students & Professionals Lounge into a fully digitalized environment, optimizing how user interactions and workspace assets were administered. To validate its efficacy, the software underwent a series of technical evaluations that tested its resilience against common operational friction. The engineering of the platform emphasized data governance through a structured user hierarchy, ensuring that administrative privileges were securely partitioned.

During the operational trials, the application was subjected to stress testing to observe its stability in managing synchronous reservation inputs, logging actual client check-ins without database lag, and ensuring that time-dependent fee calculations remained accurate. Furthermore, a user-centric assessment was executed involving active front-desk operators, business managers, and daily clients. These participants navigated through the client portal, validated the automated billing adjustments for varying membership statuses, and audited the generation of financial history sheets. The empirical feedback gathered from these assessments demonstrated that the application completely eradicated the operational risks of overlapping schedules and manual calculation errors, successfully achieving the project's benchmark for institutional modernization.

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4.4 Impact of Integrated Study Hub Management System of WS Students & Professionals Lounge on the Information Technology Field

The architecture of this project offered an architectural case study in the field of localized corporate automation and specialized database management. It confronted a prevalent challenge in small-scale hospitality and co-working sectors: the deployment of a centralized transaction framework that balanced client data integrity with dynamic, time-sensitive space allocation.

By engineering a solution that displaced fragmented administrative methods—such as paper-based logs, decentralized messaging systems, and manual timekeeping—this research provided a concrete methodology for small-to-medium enterprise (SME) software integration. It introduced a practical paradigm for multi-tier system layouts, illustrating how consumer-facing dashboards, real-time employee interfaces, and administrative reporting engines could co-exist within a single relational database. Consequently, this system elevated the regional benchmarks for operational technology, establishing that cloud-ready workflows could dramatically suppress human oversight errors while remaining aligned with the compliance directives of the Data Privacy Act of 2012 (RA 10173).

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4.5 Contribution to Technological Advancement

The principal innovation of this system centered on its capability to align abstract digital scheduling with concrete physical facility tracking into an automated operational lifecycle. It introduced an automated bridge where a consumer's booking intent directly triggered computational time logs and algorithmic invoice creation.

By employing a customized role-based architecture, the software introduced a strict security boundary, ensuring that high-level administrative intelligence and overall business audits remained strictly confidential to executive accounts, while maintaining a transparent self-service interface for the general user base. Moreover, the algorithmic execution of automated pricing tiers within the subscription module demonstrated how business operations could be optimized through programmatic workflow controls. This system served as a functional model of how modern software methodologies could re-engineer contemporary commercial workspaces, establishing a foundation for more agile, accountable, and secure transaction tracking systems.

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4.6 Code Impression Management

1. App setup and registration

run.py

"""

run.py

App factory, CLI commands, and entry point.

"""

import os

from datetime import datetime, timedelta

from database_fixed import (

 Config,

 DailyReport,

 PaymentInfo,

 Reservation,

 Room,

 SoloPlan,

 TimeLog,

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```
User,

db,

login_manager,

mail,

socketio,

)

from flask import Flask

from sqlalchemy import inspect, text

from time_utils import format_checkin_time, format_checkout_time,
format_date, decimal_hours_to_readable

def create_app(config_class=Config):

    root_dir = os.path.dirname(os.path.abspath(__file__))

    app = Flask(

        __name__,

        template_folder=os.path.join(root_dir, "app",
"templates"),

        static_folder=os.path.join(root_dir, "static"),

        static_url_path="/static",

    )
```

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```
app.config.from_object(config_class)

app.config['WTF_CSRF_ENABLED'] = False


db.init_app(app)

login_manager.init_app(app)

login_manager.login_view = "auth.login"

login_manager.login_message = "Login required"

socketio.init_app(app)

mail.init_app(app)


# Register Jinja2 filters for time formatting

app.jinja_env.filters['format_checkin'] =
format_checkin_time

app.jinja_env.filters['format_checkout'] =
format_checkout_time

app.jinja_env.filters['format_date_time'] = format_date

app.jinja_env.filters['format_hours'] =
decimal_hours_to_readable


with app.app_context():
```


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```
db.create_all()

def ensure_default_rooms():

    with app.app_context():

        default_rooms = [

            {"name": "Common Area", "base_rate": 35.0,
"category": "solo"},

            {"name": "Small Meeting Room 1", "base_rate":
50.0, "category": "meeting"},

            {"name": "Small Meeting Room 2", "base_rate":
50.0, "category": "meeting"},

            {"name": "Lecture Room", "base_rate": 150.0,
"category": "lecture"},

            {"name": "Conference Room", "base_rate": 250.0,
"category": "conference"},

            {"name": "Comfy Room", "base_rate": 150.0,
"category": "comfy"},

            {"name": "Event Room 1", "base_rate": 300.0,
"category": "event"},

            {"name": "Event Room 2", "base_rate": 300.0,
"category": "event"},

        ]
```

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```
for room_data in default_rooms:

    if                                                                 not
Room.query.filter_by(name=room_data["name"]).first():

        db.session.add(

            Room(

                name=room_data["name"],

                base_rate=room_data["base_rate"],

                category=room_data["category"],

            )

        )

    db.session.commit()


def cleanup_test_rooms():

    with app.app_context():

        test_rooms = Room.query.filter(Room.name.ilike('test
room%')).all()

        for room in test_rooms:

            has_reservation =
Reservation.query.filter_by(room_id=room.id).first()

            if not has_reservation:
```

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```
db.session.delete(room)

db.session.commit()

def ensure_payment_columns():

    with app.app_context():

        inspector = inspect(db.engine)

        if not inspector.has_table("reservations"):

            return

        columns = {column["name"] for column in
inspector.get_columns("reservations")}

        if "payment_method" not in columns:

            db.session.execute(text("ALTER          TABLE
reservations ADD COLUMN payment_method VARCHAR(50)"))

        if "payment_type" not in columns:

            db.session.execute(text("ALTER          TABLE
reservations  ADD  COLUMN  payment_type  VARCHAR(20)  DEFAULT
'Downpayment'"))

        if "amount_paid" not in columns:

            db.session.execute(text("ALTER          TABLE
reservations ADD COLUMN amount_paid FLOAT DEFAULT 0.0"))

        if "receipt_image" not in columns:
```

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```
db.session.execute(text("ALTER                                TABLE
reservations ADD COLUMN receipt_image VARCHAR(255)"))

if "approved_by_id" not in columns:

    db.session.execute(text("ALTER                                TABLE
reservations ADD COLUMN approved_by_id INTEGER"))

if "addon_subtotal" not in columns:

    db.session.execute(text("ALTER                                TABLE
reservations ADD COLUMN addon_subtotal FLOAT DEFAULT 0.0"))

if inspector.has_table("walkin_reservations"):

    walkin_columns = {column["name"] for column in
inspector.get_columns("walkin_reservations")}

    if "addon_subtotal" not in walkin_columns:

        db.session.execute(text("ALTER                                TABLE
walkin_reservations ADD COLUMN addon_subtotal FLOAT DEFAULT
0.0"))

if inspector.has_table("solo_plans"):

    solo_columns = {column["name"] for column in
inspector.get_columns("solo_plans")}

    if "approved_by_id" not in solo_columns:
```

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```
        db.session.execute(text("ALTER          TABLE
solo_plans ADD COLUMN approved_by_id INTEGER"))

        if "payment_method" not in solo_columns:

            db.session.execute(text("ALTER          TABLE
solo_plans ADD COLUMN payment_method VARCHAR(50)"))

            if "receipt_image" not in solo_columns:

                db.session.execute(text("ALTER          TABLE
solo_plans ADD COLUMN receipt_image VARCHAR(255)"))

        if inspector.has_table("users"):

            user_columns = {column["name"] for column in
inspector.get_columns("users")}

            if "is_active" not in user_columns:

                db.session.execute(text("ALTER  TABLE  users
ADD COLUMN is_active BOOLEAN DEFAULT 1"))

            db.session.commit()

def ensure_membership_tables():

    """Ensure memberships and attendance_logs tables have
all required columns"""

    with app.app_context():

        inspector = inspect(db.engine)
```

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```
# Ensure memberships table columns

if inspector.has_table("memberships"):

    membership_columns = {column["name"] for column
in inspector.get_columns("memberships")}

    if "is_checked_in" not in membership_columns:

        db.session.execute(text("ALTER          TABLE
memberships ADD COLUMN is_checked_in BOOLEAN DEFAULT 0"))

    if "updated_at" not in membership_columns:

        db.session.execute(text("ALTER          TABLE
memberships  ADD      COLUMN      updated_at      DATETIME      DEFAULT
CURRENT_TIMESTAMP"))

db.session.commit()

def ensure_default_admin():

    with app.app_context():

        if not
User.query.filter_by(email="wslounge@lounge.com").first():

            admin = User(

                name="wslounge",
```

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```
        email="wslounge@lounge.com",

        role="admin",

        phone="09171111111",

    )

    admin.set_password("ws12345")

    db.session.add(admin)

    db.session.commit()


ensure_payment_columns()

ensure_membership_tables()

ensure_default_rooms()

cleanup_test_rooms()

ensure_default_admin()


@login_manager.user_loader

def load_user(user_id):

    return User.query.get(int(user_id))


from admin import admin_bp
```

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```
from client_fixed import api_bp, auth_bp, main_bp

from membership_routes import membership_bp


app.register_blueprint(auth_bp, url_prefix="/auth")

app.register_blueprint(main_bp)

app.register_blueprint(admin_bp)

app.register_blueprint(api_bp, url_prefix="/api")

app.register_blueprint(membership_bp)


return app


app = create_app()


@app.cli.command()

def init_db():

    """Create database tables."""

    db.create_all()
```


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```
print("Tables created, including solo_plans")

@app.cli.command()

def seed_db():

    """Seed database with sample data."""

    # Rooms

    if not Room.query.first():

        rooms = [

            Room(name="Common Area", base_rate=35.0,
category="solo"),

            Room(name="Small Meeting Room 1", base_rate=50.0,
category="meeting"),

            Room(name="Small Meeting Room 2", base_rate=50.0,
category="meeting"),

            Room(name="Lecture Room", base_rate=150.0,
category="lecture"),

            Room(name="Conference Room", base_rate=250.0,
category="conference"),

            Room(name="Comfy Room", base_rate=150.0,
category="comfy"),
```

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```
        Room(name="Event Room 1", base_rate=300.0,
category="event"),

        Room(name="Event Room 2", base_rate=300.0,
category="event"),

    ]

    for room in rooms:

        db.session.add(room)

    db.session.commit()


# Admin

if not
User.query.filter_by(email="wslounge@lounge.com").first():

    admin = User(

        name="wslounge",

        email="wslounge@lounge.com",

        role="admin",

        phone="09171111111",

    )

    admin.set_password("ws12345")

    db.session.add(admin)
```

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```
db.session.commit()

# Members

members = User.query.filter_by(role="member").all()

if len(members) == 0:

    cedrick = User(

        name="cedrick",

        email="cedrick@lounge.com",

        role="member",

        phone="09171222222",

    )

    cedrick.set_password("member123")

    db.session.add(cedrick)

    db.session.commit()

    member1 = cedrick

else:

    member1 = members[0]


# Sample payment settings
```

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```
if not PaymentInfo.query.first():

    payment_settings = [

        PaymentInfo(

            method="GCash",

            account_name="High End WS Lounge",

            account_number="09171234567",

            instructions="Send 50% downpayment and upload
the receipt.",

        ),

        PaymentInfo(

            method="Maya",

            account_name="High End WS Lounge",

            account_number="09179876543",

            instructions="Please upload the receipt after
payment.",

        ),

    ]

    for setting in payment_settings:

        db.session.add(setting)

    db.session.commit()
```

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```
# Sample reservations

if Reservation.query.count() == 0:

    res1 = Reservation(

        user_id=member1.id,

        room_id=1,

        customer_name="cedrick",

        contact_number=member1.phone,

        pax_count=6,

        start_time=datetime.now() + timedelta(hours=2),

        end_time=datetime.now() + timedelta(hours=4),

        status="Pending",

        total_amount=100.0,

        added_by="seed",

    )

    db.session.add(res1)

    db.session.commit()
```

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```
@app.cli.command()

def migrate_payment_columns():

    """Add missing payment-related columns (safe, idempotent).

    Use this CLI command when you prefer an explicit migration
    step

    instead of relying on the runtime ensure function.

    """

    with app.app_context():

        inspector = inspect(db.engine)

        if not inspector.has_table("reservations"):

            print("No reservations table found; nothing to
migrate.")

            return

        columns = {col["name"] for col in
inspector.get_columns("reservations")}

        added = False

        if "payment_type" not in columns:

            try:
```

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```
db.session.execute(text("ALTER          TABLE
reservations ADD COLUMN payment_type VARCHAR(20) DEFAULT
'Downpayment'"))

db.session.commit()

print("Added column: payment_type")

added = True

except Exception as e:

    db.session.rollback()

    print("Failed to add payment_type:", e)

else:

    print("Column payment_type already exists")

if not added:

    print("Migration finished; no changes applied or
already up-to-date.")

else:

    print("Migration finished; database updated.")

# Backfill expiry for approved plans

plan_durations = {
```

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```
"INDIVIDUAL RATE": timedelta(hours=1),

"INDIVIDUAL RATE (4HRS)": timedelta(hours=4),

"DAY/NIGHT PASS": timedelta(days=1),

"WEEKLY PASS (DAY/NIGHT)": timedelta(days=7),

"WEEKLY PASS (24HRS)": timedelta(days=7),

"MONTHLY PASS (DAY/NIGHT)": timedelta(days=30),

"MONTHLY PASS (24HRS)": timedelta(days=30),

"WORKSTATION (24HRS)": timedelta(days=30),

"Active Plan": timedelta(days=30),

}

approved_plans =
SoloPlan.query.filter_by(status="approved").all()

for plan in approved_plans:

    if plan.expiry_date is None:

        duration = plan_durations.get(plan.plan_name,
timedelta(days=30))

        plan.expiry_date = plan.created_at + duration

        db.session.add(plan)

db.session.commit()
```


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```
print(f"Backfilled expiry for {len(approved_plans)} approved
plans")

# Solo plans seed if none

if SoloPlan.query.count() == 0:

    solo1 = SoloPlan(

        user_id=member1.id,    plan_name="Active    Plan",
status="approved"

    )

    solo1.expiry_date = solo1.created_at + timedelta(days=30)

    db.session.add(solo1)

    db.session.commit()


print(

    "Enhanced database seeded: Conference Room, cedrick,
Active Plan with expiry, reservations!"

)


if __name__ == "__main__":
```

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```
socketio.run(app, debug=True)
```

2. Database models for the system

```
database_fixed.py
```

```
"""
```

```
Fixed database.py - clean imports for membership features
```

```
"""
```

```
import os
```

```
import sqlite3
```

```
from datetime import datetime, timedelta
```

```
from flask import Flask
```

```
from flask_mail import Mail
```

```
from dotenv import load_dotenv
```

```
from flask_wtf import FlaskForm
```

```
from flask_socketio import SocketIO
```

```
from flask_sqlalchemy import SQLAlchemy
```

```
from flask_login import LoginManager, UserMixin
```

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```
from sqlalchemy import and_, create_engine, func, inspect,
or_, text

from werkzeug.security import check_password_hash,
generate_password_hash

from wtforms import (

    DateField,

    TimeField,

    HiddenField,

    StringField,

    PasswordField,

    SubmitField,

    SelectField,

    IntegerField,

    BooleanField,

    TextAreaField,

)

from wtforms.fields import DateTimeField, DecimalField

from wtforms.validators import (

    DataRequired,

    ValidationError,
```

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Email,

EqualTo,

Length,

Regex ,

Optional,

NumberRange,

DataRequired,

)

`load_dotenv()`

Config

```
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
```

```
def resolve_database_uri():
```

```
env_uri = os.environ.get("SQLALCHEMY_DATABASE_URI")
```

```
preferred_local_uri = f"sqlite:/// {os.path.join(BASE_DIR,
'app.db')} "
```

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```
if not env_uri:

    return preferred_local_uri

if env_uri.startswith("sqlite"):

    return env_uri

if env_uri.startswith("mysql") or
env_uri.startswith("postgres"):

    try:

        engine = create_engine(env_uri)

        with engine.connect() as conn:

            conn.execute(text("SELECT 1"))

        return env_uri

    except Exception as exc:

        print(

            "Warning: configured SQLALCHEMY_DATABASE_URI

is not reachable; "

            "falling back to local SQLite."

        )
```

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```
        print(f"Database error: {exc}")

        return preferred_local_uri

    return env_uri


class Config:

    SECRET_KEY = os.environ.get("SECRET_KEY") or "dev-
secret-change-in-prod"

    SQLALCHEMY_DATABASE_URI = resolve_database_uri()

    SQLALCHEMY_TRACK_MODIFICATIONS = False

    MAIL_SERVER = os.environ.get("MAIL_SERVER")

    MAIL_PORT = int(os.environ.get("MAIL_PORT") or 587)

    MAIL_USE_TLS = os.environ.get("MAIL_USE_TLS",
"true").lower() == "true"

    MAIL_USERNAME = os.environ.get("MAIL_USERNAME")

    MAIL_PASSWORD = os.environ.get("MAIL_PASSWORD")

    STRIPE_SECRET_KEY = os.environ.get("STRIPE_SECRET_KEY")
```

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```
STRIPE_PUBLIC_KEY = os.environ.get("STRIPE_PUBLIC_KEY")

POSTS_PER_PAGE = 25


# Extensions

db = SQLAlchemy()

login_manager = LoginManager()

socketio = SocketIO(cors_allowed_origins="*")

mail = Mail()


# ID Generator Helper Functions

def generate_membership_id():

    """Generate WS-WLK-XXXX sequential ID for members."""

    max_seq =
db.session.query(func.max(func.cast(func.replace(User.membership
_id, 'WS-WLK-', ''), db.Integer)))\
```

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```
        .filter(User.membership_id.like('WS-WLK-%')).scalar()
or 0

seq = max_seq + 1

return f"WS-WLK-{str(seq).zfill(4)}"


def generate_customer_id(room_type="common"):

    """

    Generate a customer-facing ID based on the type.

    - Common Area: 1-50

    - Other rooms and monthly passes: 100-999

    """

    if room_type.lower() == "common area":

        existing_ids = set([

            u.customer_id          for          u          in
User.query.filter(User.customer_id.between(1,      50)).all() if
u.customer_id

            ]          +          [r.customer_id          for          r          in
Reservation.query.filter(Reservation.customer_id.between(1,
50)).all() if r.customer_id] + [
```


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```
        s.customer_id          for          s          in
SoloPlan.query.filter(SoloPlan.customer_id.between(1, 50)).all()
if s.customer_id

    ])

    for i in range(1, 51):

        if i not in existing_ids:

            return i

        raise ValueError("Common Area capacity reached")

    else:

        existing_ids = set([

            u.customer_id          for          u          in
User.query.filter(User.customer_id.between(100, 999)).all() if
u.customer_id

        ] + [r.customer_id          for          r          in
Reservation.query.filter(Reservation.customer_id.between(100,
999)).all() if r.customer_id] + [

            s.customer_id          for          s          in
SoloPlan.query.filter(SoloPlan.customer_id.between(100,
999)).all() if s.customer_id

        ])

    for i in range(100, 1000):

        if i not in existing_ids:
```

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```
        return i

        raise ValueError("ID range exhausted")


def get_common_area_count():

    return Reservation.query.filter(

        Reservation.customer_id.between(1, 50),

        Reservation.status.in_(["Confirmed",          "Pending",
"Walk-in"]))

    ).count()


def get_user_by_email(email):

    """Resolve a user by email from the active database or
the local SQLite fallback."""

    normalized_email = (email or "").strip().lower()

    if not normalized_email:

        return None
```

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```
        user = User.query.filter(func.lower(User.email) ==
normalized_email).first()

    if user:

        return user

    sqlite_db_path = os.path.join(BASE_DIR, "app.db")

    if not os.path.exists(sqlite_db_path):

        return None

    try:

        active_db_name = getattr(db.engine.url, "database",
None)

    except Exception:

        active_db_name = None

    if active_db_name and os.path.abspath(sqlite_db_path) ==
os.path.abspath(active_db_name):

        return None

    try:
```

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```
conn = sqlite3.connect(sqlite_db_path)

cur = conn.cursor()

cur.execute(

    "SELECT  name,  email,  phone,  password,  role,
is_active FROM users WHERE lower(email)=?",

    (normalized_email,),

)

row = cur.fetchone()

conn.close()

except Exception:

    return None


if not row:

    return None


fallback_user = User.query.filter(func.lower(User.email)
== normalized_email).first()

if fallback_user:

    return fallback_user
```

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```
imported_user = User(

    name=row[0] or "Imported User",

    email=row[1],

    phone=row[2],

    password=row[3],

    role=row[4] or "member",

    is_active=(row[5] if row[5] is not None else True),

)

db.session.add(imported_user)

db.session.commit()

return imported_user
```

Models

```
class User(UserMixin, db.Model):

    __tablename__ = "users"

    id = db.Column(db.Integer, primary_key=True)
```

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```
customer_id = db.Column(db.Integer, unique=True,
nullable=True)

membership_id = db.Column(db.String(20), unique=True,
nullable=True)

name = db.Column(db.String(64), nullable=False)

email = db.Column(db.String(120), unique=True,
nullable=False)

phone = db.Column(db.String(20))

password = db.Column(db.String(255))

role = db.Column(db.String(20), default="member")

is_active = db.Column(db.Boolean, default=True)

expiry_date = db.Column(db.DateTime, nullable=True)

created_at = db.Column(db.DateTime,
default=datetime.utcnow)

reservations = db.relationship("Reservation",
backref="user", lazy="dynamic",
foreign_keys='Reservation.user_id')

approved_reservations = db.relationship("Reservation",
foreign_keys='Reservation.approved_by_id', backref='approved_by')

solo_plans = db.relationship(

    "SoloPlan",
```

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```
        backref="user",

        lazy="dynamic",

        foreign_keys='SoloPlan.user_id',

    )

    time_logs = db.relationship("TimeLog", backref="user",
                                lazy="dynamic")

    membership = db.relationship("Membership",
                                back_populates="user", uselist=False, cascade="all, delete-orphan")

def set_password(self, password):

    self.password = generate_password_hash(password)

def check_password(self, password):

    if not self.password:

        return False

    if not isinstance(self.password, str):

        return False
```

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```
if password is None:

    return False


    if self.password.startswith(("scrypt:", "pbkdf2:",
"bcrypt:", "argon2")):

        return check_password_hash(self.password,
password)


if self.password == password:

    return True


if self.password == password.strip():

    return True


return False


@property

def total_mins(self):

    total =
db.session.query(func.coalesce(func.sum(TimeLog.total_time),
0)).filter(TimeLog.user_id == self.id).scalar()
```


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```
return total or 0
```

```
@property
```

```
def total_hours(self):
```

```
    return self.total_mins // 60
```

```
@property
```

```
def total_minutes(self):
```

```
    return self.total_mins % 60
```

```
@property
```

```
def membership_display_id(self):
```

```
    return self.membership_id or f"WS-WLK-  
{str(self.id).zfill(4)}"
```

```
@property
```

```
def is_membership_active(self):
```

```
    if not self.expiry_date:
```

```
        return self.is_active
```

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```
        return self.is_active and datetime.now() <
self.expiry_date
```

```
@property
```

```
def total_timelogs(self):
```

```
    total =
db.session.query(func.sum(TimeLog.total_time)).filter(TimeLog.us
er_id == self.id).scalar()
```

```
    return total or 0
```

```
def __repr__(self):
```

```
    return f"<User {self.name}>"
```

```
class Room(db.Model):
```

```
    __tablename__ = "rooms"
```

```
    id = db.Column(db.Integer, primary_key=True)
```

```
    name = db.Column(db.String(64), nullable=False)
```

```
    base_rate = db.Column(db.Float, nullable=False)
```

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```
category = db.Column(db.String(50), default="standard")

status = db.Column(db.String(20), default="available") #
Manual override status (e.g. Under Maintenance)


reservations = db.relationship(

    "Reservation",

    backref="room",

    lazy="dynamic",

    cascade="all, delete-orphan"

)


def is_currently_occupied(self):

    # Adjusted to Philippine Standard Time (+8 Hours)

    now = datetime.utcnow() + timedelta(hours=8)

    active_res = Reservation.query.filter(

        Reservation.room_id == self.id,

        # GINKAKAS: "Pending" gin-dula ta para indi ma-
lock ang room sang tapos na nga pending requests
```

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```
Reservation.status.in_(["Confirmed", "Walk-in",
"Checked-in"]),

Reservation.start_time <= now,

or_(

    Reservation.end_time >= now,

    Reservation.is_open_time == True

)

).first()

return active_res is not None

@property

def is_available_now(self):

    # Dynamic check: Kon manual "Maintenance" /
    "Unavailable" ang DB status OR occupied subong, False.

    if self.status and self.status.lower() in
["maintenance", "out_of_service", "unavailable"]:

        return False

    return not self.is_currently_occupied()
```

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```
end time = db.Column(db.DateTime, nullable=True)
```

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```
is_open_time = db.Column(db.Boolean, default=False)

status = db.Column(db.String(20), default="Pending") #
Options: Pending, Confirmed, Cancelled, Completed, Expired

total_amount = db.Column(db.Float, default=0.0)

amount_paid = db.Column(db.Float, default=0.0)

payment_method = db.Column(db.String(50))

payment_type = db.Column(db.String(20),
default="Downpayment")

receipt_image = db.Column(db.String(255))

approved_by_id = db.Column(db.Integer,
db.ForeignKey("users.id"), nullable=True)

paid = db.Column(db.Boolean, default=False)

added_by = db.Column(db.String(64))

extra_notes = db.Column(db.String(255))

extra_fee = db.Column(db.Float, default=0.0)

addon_subtotal = db.Column(db.Float, default=0.0)

discount_rate = db.Column(db.Float, default=0.0)
```

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```
created_at = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))
```

```
reservation_addons = db.relationship(

    "ReservationAddOn",

    backref="reservation",

    lazy="dynamic",

    cascade="all, delete-orphan",

    passive_deletes=True,

)
```

```
walkin = db.relationship(

    "WalkinReservation",

    backref="reservation",

    uselist=False,

    cascade="all, delete-orphan",

    passive_deletes=True,

)
```

```
@staticmethod
```

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```
def check_conflict(room_id, start_dt, end_dt,
exclude_id=None):

    # Current PH Time (+8 Hours)

    now = datetime.utcnow() + timedelta(hours=8)

    # GIN-UPDATE: Dulaon ang mga lumang Pending
    reservations nga naglabay na ang start_time

    query = Reservation.query.filter(

        Reservation.room_id == room_id,

        Reservation.status.in_(["Confirmed", "Pending",
"Walk-in", "Checked-in"]),

    )

    if exclude_id:

        query = query.filter(Reservation.id !=
exclude_id)

    if end_dt is None:

        # Open time check

        query = query.filter(

            or_(
```


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```
        Reservation.end_time > start_dt,

        Reservation.is_open_time == True

    )

)

else:

    # Fixed duration check: Siguraduhon nga ang
    active reservation may overlap

    query = query.filter(

        or_(

            and_(

                Reservation.is_open_time == False,

                Reservation.start_time < end_dt,

                Reservation.end_time > start_dt

            ),

            and_(

                Reservation.is_open_time == True,

                Reservation.start_time < end_dt

            )

        )

    )

)
```

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```
        return query.first()

def __repr__(self):

    return f"<Reservation {self.id}>"


class PaymentInfo(db.Model):

    __tablename__ = "payment_info"

    id = db.Column(db.Integer, primary_key=True)

    method = db.Column(db.String(32), unique=True,
nullable=False)

    account_name = db.Column(db.String(128))

    account_number = db.Column(db.String(64))

    qr_image = db.Column(db.String(255))

    instructions = db.Column(db.String(255))

    # FIX: PH Timezone (+8 hours)
```

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```
        created_at = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))

        updated_at = db.Column(

            db.DateTime,

            default=lambda: datetime.utcnow() +
timedelta(hours=8),

            onupdate=lambda: datetime.utcnow() +
timedelta(hours=8)

        )

    def __repr__(self):

        return f"<PaymentInfo {self.method}>"

class WalkinReservation(db.Model):

    __tablename__ = "walkin_reservations"

    id = db.Column(db.Integer, primary_key=True)

    reservation_id = db.Column(db.Integer,
db.ForeignKey("reservations.id",
ondelete="CASCADE"),
nullable=False)
```

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```
user_id = db.Column(db.Integer,
db.ForeignKey("users.id"), nullable=False)

room_id = db.Column(db.Integer,
db.ForeignKey("rooms.id"), nullable=False)

customer_name = db.Column(db.String(64))

contact_number = db.Column(db.String(20))

pax_count = db.Column(db.Integer, default=1)

start_time = db.Column(db.DateTime, nullable=False)

end_time = db.Column(db.DateTime, nullable=True)

status = db.Column(db.String(20), default="Walk-in")

total_amount = db.Column(db.Float, default=0.0)

paid = db.Column(db.Boolean, default=False)

extra_fee = db.Column(db.Float, default=0.0)

addon_subtotal = db.Column(db.Float, default=0.0)

added_by = db.Column(db.String(64))

extra_notes = db.Column(db.String(255))


# FIX: Philippine Standard Time (+8 hours)

created_at = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))
```

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```
walkin_addons = db.relationship(

    "WalkinAddOn",

    backref="walkin_reservation",

    lazy="dynamic",

    cascade="all, delete-orphan",

    passive_deletes=True,

)


def __repr__(self):

    return f"<WalkinReservation {self.id} -- Res

{self.reservation_id}>"


class SoloPlan(db.Model):

    __tablename__ = "solo_plans"


    id = db.Column(db.Integer, primary_key=True)

    customer_id = db.Column(db.Integer, nullable=True)
```

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```
        user_id = db.Column(db.Integer,
db.ForeignKey("users.id"), nullable=False)

        approved_by_id = db.Column(db.Integer,
db.ForeignKey("users.id"), nullable=True)

        plan_name = db.Column(db.String(64), nullable=False)

        status = db.Column(db.String(20), default="Pending")

        payment_method = db.Column(db.String(50))

        receipt_image = db.Column(db.String(255))

        created_at = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))

        expiry_date = db.Column(db.DateTime)

        approved_by_user = db.relationship(

            "User",

            foreign_keys=[approved_by_id],

            backref="approved_solo_plans",

        )

    @property

    def is_active(self):
```

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```
        if self.status and self.status.lower() in
["checked_out", "checked-out", "completed", "expired"]:

            return False

        now_ph = datetime.utcnow() + timedelta(hours=8)

        if self.expiry_date:

            return now_ph < self.expiry_date

        return self.status.lower() in ["approved", "active"]

def set_expiry_date(self, start_time=None):

    durations = {

        "INDIVIDUAL RATE": timedelta(hours=1),

        "INDIVIDUAL RATE (4HRS)": timedelta(hours=4),

        "DAY/NIGHT PASS": timedelta(days=1),

        "WEEKLY PASS (DAY/NIGHT)": timedelta(days=7),

        "WEEKLY PASS (24HRS)": timedelta(days=7),

        "MONTHLY PASS (DAY/NIGHT)": timedelta(days=30),

        "MONTHLY PASS (24HRS)": timedelta(days=30),

        "WORKSTATION (24HRS)": timedelta(days=30),

        "ACTIVE PLAN": timedelta(days=30),

    }
```

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```
now_ph = datetime.utcnow() + timedelta(hours=8)

start = start_time if start_time is not None else
now_ph

# Case-insensitive lookup para indi mag-fallback sa
30 days kon iba ang capitalizations

clean_plan_name = (self.plan_name or
"").strip().upper()

duration = durations.get(clean_plan_name,
timedelta(days=30))

self.expiry_date = start + duration

def __repr__(self):

    return f"<SoloPlan {self.plan_name}>"

def to_dict(self):

    return {

        'id': self.id,

        'customer_id': self.customer_id,
```


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```
        'user_id': self.user_id,

        'user_name': self.user.name if self.user else
None,

        'plan_name': self.plan_name,

        'status': self.status,

        'created_at': self.created_at.isoformat() if
self.created_at else None,

        'expiry_date': self.expiry_date.isoformat() if
self.expiry_date else None,

        'is_active': self.is_active
    }

    def check_out(self):

        self.status = "checked_out"

class TimeLog(db.Model):

    __tablename__ = "time_logs"

    id = db.Column(db.Integer, primary_key=True)

    user_id = db.Column(db.Integer, db.ForeignKey("users.id",
ondelete="CASCADE"), nullable=False)
```

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```
plan = db.Column(db.String(64))

# FIX: Dynamic Philippine Standard Time (+8 Hours)

time_in = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))

time_out = db.Column(db.DateTime, nullable=True)

total_time = db.Column(db.Integer, default=0) # Total
in minutes

def calculate_total_time(self):

    """Helper method para mag-compute sang total minutes
upon check-out."""

    if self.time_in and self.time_out:

        delta = self.time_out - self.time_in

        self.total_time = int(delta.total_seconds()) //

60)

        return self.total_time

    return 0

def __repr__(self):

    return f"<TimeLog {self.id} - User {self.user_id}>"
```

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```
class DailyReport(db.Model):

    __tablename__ = "daily_reports"

    id = db.Column(db.Integer, primary_key=True)

    report_date = db.Column(db.Date, unique=True)

    total_check_ins = db.Column(db.Integer, default=0)

    total_logins = db.Column(db.Integer, default=0)

    total_timelogged = db.Column(db.Float, default=0)

    generated_at = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))


class UserActivityLog(db.Model):

    __tablename__ = "user_activity_logs"

    id = db.Column(db.Integer, primary_key=True)

    # Foreign Key cascade o nullable handler

    user_id = db.Column(db.Integer, db.ForeignKey("users.id",
ondelete="SET NULL"), nullable=True)
```

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```
activity_type = db.Column(db.String(50))

activity_time = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))

ip_address = db.Column(db.String(45))

class Membership(db.Model):

    __tablename__ = "memberships"

    id = db.Column(db.Integer, primary_key=True)

    # ondelete="CASCADE" para indi mag-crash sa MySQL kon
mag-deactivate/delete sang User

    user_id = db.Column(

        db.Integer,

        db.ForeignKey("users.id", ondelete="CASCADE"),

        nullable=False,

        unique=True

    )
```

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```
status = db.Column(db.String(20), default="pending") #
pending, active, expired

# Dynamic Philippine Standard Time (+8 Hours)

start_date = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))

expiry_date = db.Column(db.DateTime, nullable=False)

total_hours = db.Column(db.Float, default=0.0) # Total
hours credited

hours_left = db.Column(db.Float, default=0.0) #
Remaining hours

plan_name = db.Column(db.String(100)) # Solo plan name

is_checked_in = db.Column(db.Boolean, default=False) #
Current session status

is_checked_out = db.Column(db.Boolean, default=False)

created_at = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))

updated_at = db.Column(

    db.DateTime,
```

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```
        default=lambda: datetime.utcnow() +
timedelta(hours=8),

        onupdate=lambda: datetime.utcnow() +
timedelta(hours=8)

    )

    user = db.relationship("User",
back_populates="membership", uselist=False)

    attendance_logs = db.relationship(

        "AttendanceLog",

        backref="membership",

        lazy="dynamic",

        cascade="all, delete-orphan"

    )

    @property

    def is_active(self):

        """Check if membership is active based on
expiry_date (PH Timezone Alignment)"""

        now_ph = datetime.utcnow() + timedelta(hours=8)
```

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```
        return self.status == "active" and now_ph <
self.expiry_date
```

```
@property
```

```
    def accumulated_hours(self):
```

```
        """Calculate total hours used from all completed
attendance logs"""
```

```
        logs =
self.attendance_logs.filter(AttendanceLog.check_out_time.isnot(N
one)).all()
```

```
        total = sum(log.hours_deducted for log in logs if
log.hours_deducted)
```

```
        return round(total, 2)
```

```
    def __repr__(self):
```

```
        return f"<Membership {self.user_id} -
{self.status}>"
```

```
class AttendanceLog(db.Model):
```

```
    __tablename__ = "attendance_logs"
```

```
    id = db.Column(db.Integer, primary_key=True)
```

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```
# ondelete="CASCADE" para sa safe deletions/cleanup

membership_id = db.Column(

    db.Integer,

    db.ForeignKey("memberships.id", ondelete="CASCADE"),

    nullable=False

)


check_in_time = db.Column(db.DateTime, nullable=False)

check_out_time = db.Column(db.DateTime)

hours_deducted = db.Column(db.Float, default=0.0)      #
Calculated on check-out


created_at = db.Column(db.DateTime, default=lambda:
datetime.utcnow() + timedelta(hours=8))


@property

def session_duration_hours(self):

    """Calculate duration in hours between check-in and
check-out"""
```


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```
if self.check_out_time and self.check_in_time:

    delta = self.check_out_time - self.check_in_time

    return delta.total_seconds() / 3600    # Convert
seconds to hours

    return 0.0


def __repr__(self):

    return f"<AttendanceLog {self.membership_id} -
{self.check_in_time}>"


class AddOn(db.Model):

    """

    Master list of available add-ons for reservations.

    Defines the name, unit price, and availability.

    """

    __tablename__ = "addons"


    id = db.Column(db.Integer, primary_key=True)
```

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```
name      =      db.Column(db.String(128),      nullable=False,
unique=True)

description = db.Column(db.String(255))

unit_price = db.Column(db.Float, nullable=False)

requires_quantity = db.Column(db.Boolean, default=True)
# Whether quantity input is needed

min_quantity = db.Column(db.Integer, default=1)

max_quantity = db.Column(db.Integer, default=100)

is_active = db.Column(db.Boolean, default=True)


created_at = db.Column(

    db.DateTime,

    default=lambda:      datetime.utcnow()      +
timedelta(hours=8)

)

updated_at = db.Column(

    db.DateTime,

    default=lambda:      datetime.utcnow()      +
timedelta(hours=8),

    onupdate=lambda:      datetime.utcnow()      +
timedelta(hours=8)
```

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)

```
reservation_addons = db.relationship(
```

```
    "ReservationAddOn",
```

```
    backref="addon",
```

```
    lazy="dynamic",
```

)

```
walkin_addons = db.relationship(
```

```
    "WalkinAddOn",
```

```
    backref="addon",
```

```
    lazy="dynamic",
```

)

```
def __repr__(self):
```

```
    return f"<AddOn {self.name} - ₱{self.unit_price}>"
```

```
def to_dict(self):
```

```
    return {
```

```
        'id': self.id,
```

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```
'name': self.name,

'description': self.description,

'unit_price': float(self.unit_price),

'requires_quantity': self.requires_quantity,

'min_quantity': self.min_quantity,

'max_quantity': self.max_quantity,

}
```

```
class ReservationAddOn(db.Model):

    """

    Tracks selected add-ons for a specific reservation.

    Stores quantity, unit price, and subtotal for each add-
on.

    """

    __tablename__ = "reservation_addons"

    id = db.Column(db.Integer, primary_key=True)

    reservation_id = db.Column(
```

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```
        db.Integer,

        db.ForeignKey("reservations.id", ondelete="CASCADE"),

        nullable=False

    )

    # RESTRICT para indi ma-erase ang financial history o
    resibo

    addon_id = db.Column(

        db.Integer,

        db.ForeignKey("addons.id", ondelete="RESTRICT"),

        nullable=False

    )

    quantity = db.Column(db.Integer, default=1,
nullable=False)

    unit_price = db.Column(db.Float, nullable=False) #
    Price at time of booking

    subtotal = db.Column(db.Float, default=0.0) # quantity
    * unit_price

    created_at = db.Column(
```

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```
db.DateTimeField,

    default=lambda: datetime.utcnow() +
timedelta(hours=8)

)

def __repr__(self):

    return f"<ReservationAddOn {self.addon_id}
qty={self.quantity}>"

def calculate_subtotal(self):

    """Calculate subtotal: quantity * unit_price"""

    self.subtotal = self.quantity * self.unit_price

    return self.subtotal

def to_dict(self):

    return {

        'id': self.id,

        'addon_id': self.addon_id,

        'addon_name': self.addon.name if self.addon else
None,
```

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```
        'quantity': self.quantity,  
  
        'unit_price': float(self.unit_price),  
  
        'subtotal': float(self.subtotal),  
  
    }
```

```
class WalkinAddOn(db.Model):  
  
    """  
  
    Tracks selected add-ons for a specific walk-in  
reservation.  
  
    Stores quantity, unit price, and subtotal for each add-  
on.  
  
    """  
  
    __tablename__ = "walkin_addons"  
  
    id = db.Column(db.Integer, primary_key=True)  
  
    walkin_reservation_id = db.Column(  
  
        db.Integer,  
  
        db.ForeignKey("walkin_reservations.id",  
ondelete="CASCADE"),
```

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```
        nullable=False

    )

    addon_id = db.Column(

        db.Integer,

        db.ForeignKey("addons.id", ondelete="RESTRICT"),

        nullable=False

    )

    quantity = db.Column(db.Integer, default=1,
nullable=False)

    unit_price = db.Column(db.Float, nullable=False)

    subtotal = db.Column(db.Float, default=0.0)

    created_at = db.Column(

        db.DateTime,

        default=lambda: datetime.utcnow() +
timedelta(hours=8)

    )
```


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```
def __repr__(self):

    return f"<WalkinAddOn {self.addon_id}
qty={self.quantity}>"

def calculate_subtotal(self):

    """Calculate subtotal: quantity * unit_price"""

    self.subtotal = self.quantity * self.unit_price

    return self.subtotal

def to_dict(self):

    return {

        'id': self.id,

        'addon_id': self.addon_id,

        'addon_name': self.addon.name if self.addon else
None,

        'quantity': self.quantity,

        'unit_price': float(self.unit_price),

        'subtotal': float(self.subtotal),

    }
```

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```
# Forms

class LoginForm(FlaskForm):

    email = StringField("Email", validators=[DataRequired(),
Email()])

    password = PasswordField("Password",
validators=[DataRequired()])

    remember_me = BooleanField("Remember Me")

    submit = SubmitField("Sign In")


class RegistrationForm(FlaskForm):

    name = StringField("Name", validators=[DataRequired(),
Length(min=2, max=64)])

    email = StringField("Email", validators=[DataRequired(),
Email()])

    phone = StringField("Phone", validators=[Length(max=20)])

    # Length (min=8) & Regexp for Alphanumeric Password

    password = PasswordField(
```

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```
"Password",

validators=[

    DataRequired(),

    Length(min=8, message="Password must be at least
8 characters long."),

    Regexp(

        r'^(?=.*[A-Za-z])(?=.*\d).+$',

        message="Password must contain at least one
letter and one number."

    )

]

)

password2 = PasswordField("Repeat Password",
validators=[DataRequired(), EqualTo("password",
message="Passwords must match")])

submit = SubmitField("Register")


class ProfileForm(FlaskForm):

    name = StringField("Full Name",
validators=[DataRequired(), Length(max=64)])
```

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```
email = StringField("Email Address",  
validators=[DataRequired(), Email()])
```

```
phone = StringField("Contact Number",  
validators=[Length(max=20)])
```

```
profile_submit = SubmitField("Save Profile")
```

```
class ChangePasswordForm(FlaskForm):  
  
    current_password = PasswordField("Current Password",  
validators=[DataRequired()])
```

```
    # Regexp for Alphanumeric Password  
  
    new_password = PasswordField(  
  
        "New Password",  
  
        validators=[  
  
            DataRequired(),  
  
            Length(min=8, message="Password must be at least  
8 characters long."),  
  
            Regexp(  
  
                r'^(?=.*[A-Za-z])(?=.*\d).+$',
```

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```
        message="Password must contain at least one
letter and one number."

    )

]

)

confirm_password = PasswordField(

    "Confirm New Password",

    validators=[DataRequired(),    EqualTo("new_password",
message="Passwords must match")],

)

password_submit = SubmitField("Change Password")


class ReservationForm(FlaskForm):

    room_id      =      SelectField("Room",      coerce=int,
validators=[DataRequired()])

    customer_name      =      StringField("Customer      Name",
validators=[DataRequired(), Length(min=2)])

    contact_number      =      StringField("Contact",
validators=[DataRequired()])

    pax_count      =      IntegerField("Pax      Count",      default=1,
validators=[NumberRange(min=1)])
```

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```
start_time = DateTimeField("Start Time", format="%Y-%m-%dT%H:%M", validators=[DataRequired()])

end_time = DateTimeField("End Time", format="%Y-%m-%dT%H:%M", validators=[DataRequired()])

# Support open-time toggle on client reservation form
(template expects `is_open_time`)

is_open_time = BooleanField("Open Time")

extra_notes = TextAreaField("Notes")

# Payment fields expected by client-side booking flow

payment_method = SelectField("Payment Method",
choices=[("GCash", "GCash"), ("Maya", "Maya")],
validators=[Optional()])

payment_type = SelectField("Payment Type",
choices=[("Downpayment", "Downpayment"), ("Full Payment", "Full Payment")], default="Downpayment", validators=[Optional()])

submit = SubmitField("Reserve")

class AdminReservationForm(FlaskForm):

    room_id = SelectField("Room", coerce=int,
validators=[DataRequired()])

    customer_name = StringField("Customer Name",
validators=[DataRequired(), Length(min=2)])
```

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```
contact_number = StringField("Contact",
validators=[Optional()])

pax_count = IntegerField("Pax Count", default=1,
validators=[NumberRange(min=1)])

start_time = DateTimeField("Start Date/Time",
format="%Y-%m-%dT%H:%M", validators=[DataRequired()])

end_time = DateTimeField("End Date/Time", format="%Y-%m-%dT%H:%M",
validators=[Optional()])

extra_fee = DecimalField("Extra Fee (P)", default=0.00,
validators=[Optional()])

addon_subtotal = HiddenField("Add-on Subtotal",
default="0.00")

addons_json = HiddenField("Add-ons JSON") # JSON string
with selected add-ons

total_price = HiddenField("Total Price")

open_time = BooleanField("Open Time")

discount = SelectField("Discount", choices=[(str(i /
100), f"{i}%") for i in range(0, 101, 5)], default="0.0")

payment_method = SelectField("Payment Method",
choices=[("Cash", "Cash"), ("GCash", "GCash"), ("Bank Transfer",
"Bank Transfer")], default="Cash")

extra_notes = TextAreaField("Notes")

submit = SubmitField("Reserve")
```

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```
class WalkinForm(FlaskForm):

    customer_name = StringField("Customer Name",
validators=[DataRequired()])

    contact_number = StringField("Contact Number",
validators=[Optional()])

    room_id = SelectField("Select Area", coerce=int,
validators=[DataRequired()])

    pax_count = IntegerField("No. of Pax", default=1)

    start_time = DateTimeField("Start Time", format='%Y-%m-%dT%H:%M',
validators=[Optional()])

    end_time = DateTimeField("End Time", format='%Y-%m-%dT%H:%M',
validators=[Optional()])

    extra_notes = TextAreaField("Staff Notes")

    extra_fee = DecimalField("Additional Fees", default=0.0)

    addon_subtotal = HiddenField("Add-on Subtotal",
default="0.00")

    addons_json = HiddenField("Add-ons JSON") # JSON string
with selected add-ons

    open_time = BooleanField("Open Time")
```


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```
discount = SelectField("Discount", choices=[(str(i /
100), f"{i}%") for i in range(0, 101, 5)], default="0.0")

payment_method = SelectField("Payment Method",
choices=[("Cash", "Cash"), ("GCash", "GCash"), ("Bank Transfer",
"Bank Transfer")], default="Cash")

total_price = HiddenField("Total Price")

submit = SubmitField("Save Walk-in")
```

3. Admin-side management logic

admin.py

```
"""
```

admin.py

Contains the admin blueprint with all admin routes and daily report helpers.

```
"""
```

```
import os
```

```
from io import BytesIO
```

```
from datetime import datetime, timezone, timedelta
```

```
import pytz
```

```
from zoneinfo import ZoneInfo
```

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```
from reportlab.lib import colors

from reportlab.lib.enums import TA_LEFT

from reportlab.lib.pagesizes import letter

from reportlab.lib.styles import ParagraphStyle,
getSampleStyleSheet

from reportlab.platypus import Paragraph, SimpleDocTemplate,
Spacer, Table, TableStyle


from database_fixed import AdminReservationForm,
AttendanceLog, DailyReport, Membership, PaymentInfo, Reservation,
\

Room, SoloPlan, TimeLog, User, UserActivityLog,
WalkinForm, WalkinReservation, db, generate_customer_id, \

get_common_area_count, mail

from flask_mail import Message

from werkzeug.utils import secure_filename

from flask_login import current_user, login_required

from sqlalchemy import and_, func, inspect, or_, text

from time_utils import format_checkin_time,
format_checkout_time, format_date, decimal_hours_to_readable

from flask import Blueprint, current_app, flash, jsonify,
redirect, render_template, request, send_file, url_for
```

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```
admin_bp = Blueprint("admin", __name__, url_prefix="/admin")


def calculate_admin_total_amount(

    room_rate=0.0,

    duration_hours=1.0,

    extra_fee=0.0,

    addon_subtotal=0.0,

    discount_rate=0.0,

    is_open_time=False,

):

    """Calculate an admin reservation/walk-in total using
the shared billing formula."""

    rate = float(room_rate or 0.0)

    duration = 1.0 if is_open_time else float(duration_hours
or 1.0)

    discount = float(discount_rate or 0.0)

    room_cost = rate * duration * (1 - discount)
```

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```
total = room_cost + float(extra_fee or 0.0) +  
float(addon_subtotal or 0.0)
```

```
return round(total, 2)
```

```
def require_super_admin():
```

```
    if current_user.role != "admin":
```

```
        flash("Access Restricted: Super Admin Account  
Required", "danger")
```

```
        return redirect(url_for("main.dashboard"))
```

```
def require_admin_or_staff():
```

```
    if current_user.role not in ("admin", "staff"):
```

```
        flash("Access Restricted: Super Admin Account  
Required", "danger")
```

```
        return redirect(url_for("main.dashboard"))
```

```
def require_super_admin_json():
```

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```
if current_user.role != "admin":  
  
    return jsonify({"status": "error", "message": "Admin  
access required"}), 403
```

```
def require_admin_or_staff_json():  
  
    if current_user.role not in ("admin", "staff"):  
  
        return jsonify({"status": "error", "message":  
"Access restricted"}), 403
```

```
def _solo_plan_credit_hours(plan_name):  
  
    plan_key = (plan_name or "").strip().upper()  
  
    hour_mapping = {  
  
        "INDIVIDUAL RATE": 1.0,  
  
        "INDIVIDUAL RATE (4HRS)": 4.0,  
  
        "DAY/NIGHT PASS": 24.0,  
  
        "WEEKLY PASS (DAY/NIGHT)": 168.0,  
  
        "WEEKLY PASS (24HRS)": 168.0,  
  
        "MONTHLY PASS (DAY/NIGHT)": 720.0,
```

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```
"MONTHLY PASS (24HRS)": 720.0,

"WORKSTATION (24HRS)": 720.0,

"ACTIVE PLAN": 720.0,

}

return hour_mapping.get(plan_key, 24.0)


def _ensure_approved_solo_plan_membership(member):

    ph_tz = pytz.timezone("Asia/Manila")

    now_naive = datetime.now(ph_tz).replace(tzinfo=None)

    # LATEST nga approved SoloPlan sang user nga active pa

    latest_approved_plan = (

        SoloPlan.query.filter(

            SoloPlan.user_id == member.id,

            SoloPlan.status.ilike("approved"),

            SoloPlan.expiry_date > now_naive      # Dapat WALA

PA NAG-EXPIRE

        )

        .order_by(SoloPlan.created_at.desc())
```

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```
        .first()

    )

    if not latest_approved_plan:

        return False

    membership = Membership(
Membership.query.filter_by(user_id=member.id).first()

    # if no membership record, i-create sang bag-o

    if not membership:

        membership = Membership(

            user_id=member.id,

            plan_name=latest_approved_plan.plan_name,

            status="active",

            start_date=latest_approved_plan.created_at or
now_naive,

            expiry_date=latest_approved_plan.expiry_date,

        )

        db.session.add(membership)
```

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```
        return True

    else:

        #Kon may daan na nga membership (bisan expired pa),
        I-UPDATE sa BAG-O NGA PLAN DATES!

        membership.plan_name =
        latest_approved_plan.plan_name

        membership.status = "active"    # Maga-aktibo liwat
        ang Check-In button!

        membership.start_date =
        latest_approved_plan.created_at or now_naive

        membership.expiry_date =
        latest_approved_plan.expiry_date

        return True


def _expire_membership_if_needed(membership):

    if not membership or membership.status != "active" or
    not membership.expiry_date:

        return


# Philippine Standard Time (+8 Hours)

ph_tz = pytz.timezone("Asia/Manila")
```


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```
now_ph = datetime.now(ph_tz).replace(tzinfo=None)

if now_ph >= membership.expiry_date:

    membership.status = "expired"

    membership.hours_left = 0.0

    membership.is_checked_in = False

    # Auto check-out kon nag-expire ang plan samtang
naka-check in

    active_log =
membership.attendance_logs.filter(AttendanceLog.check_out_time.is_
s_(None)).first()

    if active_log:

        active_log.check_out_time =
membership.expiry_date

        active_log.hours_deducted = round(

            (active_log.check_out_time -
active_log.check_in_time).total_seconds() / 3600,

            2,

        )
```

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```
db.session.commit()
```

```
# Daily Report Helpers
```

```
@admin_bp.route("/api/rooms")
```

```
@login_required
```

```
def admin_rooms_api():
```

```
    redirect_response = require_admin_or_staff()
```

```
    if redirect_response:
```

```
        return (
```

```
            jsonify(
```

```
                {
```

```
                    "status": "error",
```

```
                    "message": "Access Restricted: Executive  
Admin Account Required",
```

```
                }
```

```
            ),
```

```
            403,
```

```
        )
```

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```
rooms = Room.query.all()

room_data = [

    {

        "id": room.id,

        "name": room.name,

        "base_rate": room.base_rate,

        "status": room.status,

        "is_common_area": room.name.strip().lower() ==

"common area",

    }

    for room in rooms

]

room_data = sorted(

    room_data, key=lambda r: 0 if r["is_common_area"]

else 1

)

return jsonify(room_data)
```

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```
def get_or_create_daily_report(report_date):
```

report =

```
DailyReport.query.filter_by(report_date=report_date).first()
```

```
if not report:
```

```
report = DailyReport(
```

```
report_date=report_date,
```

```
total_check_ins=0,
```

```
total_logins=0,
```

```
total_timelogs=0,
```

)

```
db.session.add(report)
```

```
db.session.commit()
```

```
return report
```

```
def ensure_address_column():
```

```
try:
```

```
inspector = inspect(db.engine)
```

```
if inspector.has_table("reservations"):
```

```
columns = [
```

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```
        column["name"]

        for                column                in

    inspector.get_columns("reservations")

    ]

    if "address" not in columns:

        db.session.execute(

            text(

                "ALTER TABLE reservations ADD COLUMN

address VARCHAR(128) "

            )

        )

        db.session.commit()

    except Exception:

        pass


def refresh_daily_report(report_date):

    report = get_or_create_daily_report(report_date)

    start                =                datetime.combine(report_date,

datetime.min.time())
```

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```
end = datetime.combine(report_date, datetime.max.time())

report.total_check_ins = Reservation.query.filter(

    Reservation.end_time >= start,

    Reservation.end_time <= end,

    Reservation.status == "Checked-Out",

    Reservation.paid == True,

).count()

report.total_logins = TimeLog.query.filter(

    TimeLog.time_in >= start, TimeLog.time_in <= end

).count()

total_revenue = (

db.session.query(func.coalesce(func.sum(Reservation.total_amount)

, 0))

    .filter(

        Reservation.end_time >= start,

        Reservation.end_time <= end,
```

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```
        Reservation.status == "Checked-Out",

        Reservation.paid == True,

    )

    .scalar()

    or 0

)

report.total_timelogs = total_revenue

ph_tz = ZoneInfo("Asia/Manila")

report.generated_at = datetime.now(ph_tz)


db.session.add(report)

try:

    db.session.commit()

except Exception:

    db.session.rollback()

return report
```

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```
def get_active_room_reservations(rooms):

    room_ids = [room.id for room in rooms]

    now = datetime.now()

    reservations = (

        Reservation.query.filter(

            Reservation.room_id.in_(room_ids),

            Reservation.status.in_(["Confirmed", "Walk-in"]),

        )

        .order_by(Reservation.start_time.asc())

        .all()

    )


def is_current(reservation):

    if reservation.start_time and reservation.end_time:

        return reservation.start_time <= now <=
reservation.end_time

    if reservation.start_time and
reservation.is_open_time:

        return reservation.start_time <= now

    return False
```


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```
room_reservations = {}

for res in reservations:

    existing = room_reservations.get(res.room_id)

    if existing is None:

        room_reservations[res.room_id] = res

        continue

    existing_current = is_current(existing)

    res_current = is_current(res)

    if res_current and not existing_current:

        room_reservations[res.room_id] = res

        continue

    if existing_current and not res_current:

        continue

    if res.start_time and existing.start_time:

        if res.start_time < existing.start_time:
```

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```
        room_reservations[res.room_id] = res

    return room_reservations


@admin_bp.route("/check_availability")
@login_required
def check_availability():

    redirect_response = require_admin_or_staff()

    if redirect_response:

        return jsonify({"status": "error", "message":
"Access Restricted: Executive Admin Account Required"}), 403


    room_id = request.args.get("room_id", type=int)

    date_str = request.args.get("date", "")

    start_str = request.args.get("start", "")

    end_str = request.args.get("end", "")

    open_time = request.args.get("open_time",
"false").lower() in ["1", "true", "yes"]


    try:
```

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```
if start_str and "T" in start_str:

    start_dt = datetime.strptime(start_str, "%Y-%m-%dT%H:%M")

else:

    start_date = datetime.strptime(date_str, "%Y-%m-%d").date() if date_str else datetime.now().date()

    if start_str:

        start_time = datetime.strptime(start_str, "%H:%M").time()

        start_dt = datetime.combine(start_date, start_time)

    else:

        start_dt = datetime.now()

if open_time:

    end_dt = start_dt + timedelta(hours=8)

elif end_str:

    if "T" in end_str:

        end_dt = datetime.strptime(end_str, "%Y-%m-%dT%H:%M")

    else:
```

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```
        end_time      =      datetime.strptime(end_str,
"%H:%M").time()

        end_dt  =  datetime.combine(start_dt.date(),
end_time)

        if end_dt <= start_dt:

            end_dt += timedelta(days=1)

        else:

            end_dt = start_dt + timedelta(hours=1)


        conflict      =      Reservation.check_conflict(room_id,
start_dt, end_dt)

        if conflict:

            return jsonify(

                {

                    "status": "conflict",

                    "message":  "Selected    room    is    not
available for the requested time.",

                }

            )

    except Exception as error:

        return jsonify(
```

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```
        {"status": "error", "message": f"Invalid  
availability check data: {error}"}  
  
    ), 400
```

```
    return jsonify({"status": "ok"})
```

```
# Admin Routes
```

```
@admin_bp.route("/dashboard")
```

```
@login_required
```

```
def dashboard():
```

```
    redirect_response = require_admin_or_staff()
```

```
    if redirect_response:
```

```
        return redirect_response
```

```
    ph_tz = pytz.timezone("Asia/Manila")
```

```
    now = datetime.now(ph_tz).replace(tzinfo=None)
```

```
    today_start = now.replace(hour=0, minute=0, second=0,  
microsecond=0)
```

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```
today_end = today_start + timedelta(days=1)

form = WalkinForm()

rooms      =      Room.query.filter(~Room.name.ilike('Test
Room%')).all()

unique_rooms = []

seen_room_names = set()

for room in rooms:

    room_name = room.name.strip().lower()

    if room_name in seen_room_names:

        continue

    seen_room_names.add(room_name)

    unique_rooms.append(room)


form.room_id.choices = [

    (room.id, f"{room.name} (P{room.base_rate}/hr)") for
room in rooms

]
```

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```
room_reservations                                     =
get_active_room_reservations(unique_rooms)

recent_members = (

    User.query.filter_by(role="member")

    .order_by(User.created_at.desc())

    .limit(8)

    .all()

)

incomplete_count = (

    User.query.filter(

        User.role == "member",

        or_(

            User.phone.is_(None),

            User.phone == "",

            User.email.is_(None),

            User.email == "",

        ),

    )

    .count()

)
```

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```
total_members =
User.query.filter_by(role="member").count()

active_plans = (
    SoloPlan.query.filter_by(status="approved")
    .filter(SoloPlan.expiry_date >= now)
    .count()
)

revenue_today = (
    db.session.query(func.coalesce(func.sum(Reservation.total_amount)
, 0))
    .filter(
        Reservation.end_time >= today_start,
        Reservation.end_time < today_end,
        Reservation.paid == True,
        Reservation.status == "Checked-Out",
    )
    .scalar()
```


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```
        or 0

    )

    updated = False

    expired_sessions = Reservation.query.filter(

        Reservation.end_time <= now,

        Reservation.status.in_(["Confirmed", "Walk-in"]),

        Reservation.is_open_time == False,

    ).all()

    for res in expired_sessions:

        res.status = "Ended"

        db.session.add(res)

        updated = True

    if updated:

        db.session.commit()

    reservations_today = (
```

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```
Reservation.query.filter(

    Reservation.status.in_(["Pending", "Confirmed"]),

    Reservation.start_time >= today_start,

    Reservation.start_time < today_end,

    Reservation.end_time >= now,

)

.count()

)

all_res = Reservation.query.filter(

    Reservation.start_time <= now,

    Reservation.end_time >= today_start,

    Reservation.status.in_(["Confirmed", "Walk-in",
"Ended"]),

).all()

active_reservations = sorted(all_res, key=lambda x:
(x.status != "Ended", x.end_time))

pending_reservations = (
```

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```
Reservation.query.filter(

    Reservation.status.in_(["Pending", "Confirmed"]),

    Reservation.start_time >= today_start,

    Reservation.start_time < today_end,

    Reservation.end_time >= now,

)

.order_by(Reservation.start_time.asc())

.limit(7)

.all()

)

active_walkins = [

    r

    for r in active_reservations

    if r.status in ("Walk-in", "Ended") and r.room is

not None

]

common_area_reservations = [

    r
```

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```
        for r in active_reservations

            if r.room and r.room.name.strip().lower() == "common
area"

                ]

# Fetch active AttendanceLog entries for checked-in
members to bind real-time check-in stamp

for res in common_area_reservations:

    if hasattr(res, 'user_id') and res.user_id:

        active_log =
AttendanceLog.query.join(Membership).filter(

            Membership.user_id == res.user_id,

            AttendanceLog.check_out_time.is_(None)

        ).first()

        if active_log and active_log.check_in_time:

            # Direct overwrite sang display start_time
gamit ang actual AttendanceLog check-in time

            res.actual_check_in =
active_log.check_in_time
```

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```
common_area_occupancy = get_common_area_count()

common_area_available_slots = max(0, 70 -
common_area_occupancy)

return render_template(

    "admin/admin_dashboard.html",

    form=form,

    total_members=total_members,

    active_plans=active_plans,

    reservations_today=reservations_today,

    revenue_today=revenue_today,

    rooms=unique_rooms,

    room_reservations=room_reservations,

    recent_members=recent_members,

    incomplete_count=incomplete_count,

    active_reservations=active_reservations,

    pending_reservations=pending_reservations,

    active_walkins=active_walkins,

    common_area_reservations=common_area_reservations,

    common_area_occupancy=common_area_occupancy,
```

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```
common_area_available_slots=common_area_available_slots,

    now=now,

)

@admin_bp.route('/payment_settings', methods=['GET', 'POST'])
@login_required
def payment_settings():

    redirect_response = require_super_admin()

    if redirect_response:

        return redirect_response

    payment_infos = {info.method: info for info in
PaymentInfo.query.all()}

    if request.method == 'POST':

        method = request.form.get('method')

        account_name = request.form.get('account_name',
'').strip()
```

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```
account_number = request.form.get('account_number',
''.strip())

instructions = request.form.get('instructions',
''.strip())

qr_file = request.files.get('qr_image')

if method not in ['GCash', 'Maya']:

    flash('Invalid payment method.', 'danger')

    return
redirect(url_for('admin.payment_settings'))

payment_info =
PaymentInfo.query.filter_by(method=method).first()

if not payment_info:

    payment_info = PaymentInfo(method=method)

payment_info.account_name = account_name

payment_info.account_number = account_number

payment_info.instructions = instructions

if qr_file and qr_file.filename:
```

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```
        filename = secure_filename(f'{method.lower()}_{int(datetime.now().timestamp())}_{qr_file.filename}')

        upload_folder = os.path.join(current_app.root_path, 'static/uploads/payment')

        os.makedirs(upload_folder, exist_ok=True)

        qr_path = os.path.join(upload_folder, filename)

        qr_file.save(qr_path)

        payment_info.qr_image = filename

    db.session.add(payment_info)

    db.session.commit()

    flash(f'{method} payment settings updated.', 'success')

    return redirect(url_for('admin.payment_settings'))

payment_infos = {info.method: info for info in PaymentInfo.query.all()}

return render_template('admin/payment_settings.html', payment_infos=payment_infos)
```


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```
@admin_bp.route("/manage_staff", methods=["GET", "POST"])

@login_required

def manage_staff():

    redirect_response = require_super_admin()

    if redirect_response:

        return redirect_response


message = None

if request.method == "POST":

    name = request.form.get("name", "").strip()

    email = request.form.get("email", "").strip().lower()

    phone = request.form.get("phone", "").strip()

    password = request.form.get("password", "")

    role = request.form.get("role", "staff")


    # Strict Alphanumeric Regex (At least 1 letter, 1
number, and ONLY letters & numbers)

    alphanumeric_regex = r'^(?=.*[A-Za-z])(?=.*\d)[A-Za-
z0-9]+$'
```

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```
        if not name or not email or not password or role not
in ["admin", "staff"]:

            message = "Please fill in all fields and select
a valid role."

            elif User.query.filter(func.lower(User.email) ==
email).first():

                message = "A staff account with that email
already exists."

            else:

                new_user = User(name=name, email=email,
phone=phone, role=role, is_active=True)

                new_user.set_password(password)

                db.session.add(new_user)

                try:

                    db.session.commit()

                    flash(f"{role.title()} account created for
{name}.", "success")

                    return
redirect(url_for("admin.manage_staff"))

                except Exception as err:

                    db.session.rollback()
```

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```
        message = "Failed to create staff account.  
Check server logs for details."
```

```
        flash(message, "danger")
```

```
        staff_users = User.query.filter(User.role.in_(["admin",  
"staff"]))
```

```
        staff_users = staff_users.order_by(User.role.desc(),  
User.name).all()
```

```
        return render_template(  
            "admin/manage_staff.html",  
  
            staff_users=staff_users,  
  
            message=message,  
  
        )
```

```
@admin_bp.route("/toggle_staff_status/<int:user_id>",  
methods=["POST"])
```

```
@login_required
```

```
def toggle_staff_status(user_id):
```

```
    redirect_response = require_super_admin()
```

```
    if redirect_response:
```

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```
        return redirect_response

    user = User.query.get_or_404(user_id)

    # 1. Validation checks gamit ang flash messages kag
redirect

    if user.role not in ["admin", "staff"]:

        flash("Unsupported user type.", "danger")

        return redirect(url_for("admin.manage_staff"))

    if user.id == current_user.id:

        flash("You cannot change your own status.",
"warning")

        return redirect(url_for("admin.manage_staff"))

    # 2. Toggle Status & Activity Log

    user.is_active = not user.is_active

    action = "reactivated" if user.is_active else
"deactivated"

    log_message = f"{action} by {current_user.name}"
```

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```
log = UserActivityLog(

    user_id=user.id,

    activity_type=log_message,

    ip_address=request.remote_addr or "unknown",

)

db.session.add(log)

db.session.commit()


flash(f"Staff account {action} successfully.", "success")

return redirect(url_for("admin.manage_staff"))


@admin_bp.route("/walkin_checkin", methods=["POST"])

@login_required

def walkin_checkin_modal():

    form = WalkinForm()

    rooms      =      Room.query.filter(~Room.name.ilike('Test

Room%')).all()

    form.room_id.choices = [(room.id, room.name) for room in

rooms]
```

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```
current_app.logger.debug("walkin_checkin request.form at  
entry: %s", request.form.to_dict())
```

```
current_app.logger.debug("csrf_token present: %s",  
'csrf_token' in request.form)
```

```
if form.validate_on_submit():
```

```
    room = Room.query.get(form.room_id.data)
```

```
    if not room:
```

```
        flash("Please select a valid room.")
```

```
        return redirect(url_for("admin.dashboard"))
```

```
now = datetime.now()
```

```
is_open_time = form.open_time.data
```

```
if is_open_time:
```

```
    start_time = now
```

```
else:
```

```
    if form.start_time.data:
```

```
        if isinstance(form.start_time.data, str):
```

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```
        try:

            start_time =
datetime.strptime(form.start_time.data, "%Y-%m-%dT%H:%M")

        except ValueError:

            start_time = now

    else:

        start_time = form.start_time.data

    else:

        start_time = now

if is_open_time:

    end_time = start_time + timedelta(hours=8)

else:

    if form.end_time.data:

        if isinstance(form.end_time.data, str):

            try:

                end_time =
datetime.strptime(form.end_time.data, "%Y-%m-%dT%H:%M")

            except ValueError:
```

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```

                                end_time      =      start_time      +
timedelta(hours=1)

    else:

        end_time = form.end_time.data

        if end_time <= start_time:

            end_time += timedelta(days=1)

    else:

        end_time = start_time + timedelta(hours=1)


if room.name.strip().lower() != "common area":

    conflict = Reservation.query.filter(

        Reservation.room_id == room.id,

        Reservation.status.in_( ["Confirmed", "Walk-
in", "Pending"] ),

        Reservation.start_time < end_time,

        Reservation.end_time > start_time,

    ).first()


if conflict:

    flash(
```


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```
        f"Conflict! {room.name} is already
occupied or reserved by {conflict.customer_name} from
{conflict.start_time.strftime('%I:%M %p')} to
{conflict.end_time.strftime('%I:%M %p')}."

    )

    return redirect(url_for("admin.dashboard"))

else:

    if get_common_area_count() >= 50:

        flash("Common Area has reached maximum
capacity of 50 people.")

        return redirect(url_for("admin.dashboard"))

# Generate customer_id based on room type

room_type = "common area" if
room.name.strip().lower() == "common area" else "other"

try:

    customer_id = generate_customer_id(room_type)

except ValueError as e:

    flash(str(e))

    return redirect(url_for("admin.dashboard"))
```

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```
total_from_js = request.form.get("total_price")

extra_fee      =      float(form.extra_fee.data)      if
form.extra_fee.data else 0.0

addon_subtotal = float(form.addon_subtotal.data) if
form.addon_subtotal.data else 0.0

room_rate = room.base_rate or 0

total_amount = calculate_admin_total_amount(

    room_rate=room_rate,

    duration_hours=1.0,

    extra_fee=extra_fee,

    addon_subtotal=addon_subtotal,

    discount_rate=float(form.discount.data)      if
getattr(form, "discount", None) else 0.0,

    is_open_time=is_open_time,

)

new_walkin = Reservation(

    user_id=current_user.id,

    customer_id=customer_id,

    room_id=form.room_id.data,
```

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```
customer_name=form.customer_name.data,

contact_number=form.contact_number.data or "N/A",

pax_count=form.pax_count.data,

extra_notes=form.extra_notes.data,

extra_fee=extra_fee,

addon_subtotal=addon_subtotal,

start_time=start_time,

end_time=end_time,

is_open_time=is_open_time,

status="Walk-in",

added_by=current_user.name,

total_amount=round(total_amount, 2),

paid=False,

)

if room and room.name.strip().lower() != "common
area" and start_time <= now:

    room.status = "unavailable"

db.session.add(new_walkin)
```

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```
db.session.commit()
```

```
walkin_entry = WalkinReservation(  
    reservation_id=new_walkin.id,  
    user_id=new_walkin.user_id,  
    room_id=new_walkin.room_id,  
    customer_name=new_walkin.customer_name,  
    contact_number=new_walkin.contact_number,  
    pax_count=new_walkin.pax_count,  
    start_time=new_walkin.start_time,  
    end_time=new_walkin.end_time,  
    status="Walk-in",  
    total_amount=new_walkin.total_amount,  
    extra_fee=new_walkin.extra_fee,  
    addon_subtotal=new_walkin.addon_subtotal,  
    paid=False,  
    added_by=new_walkin.added_by,  
)  
db.session.add(walkin_entry)
```

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```
db.session.commit()

    current_app.logger.debug("Walk-in      created:      %s",
new_walkin.customer_name)

    flash(f"Walk-in      {new_walkin.customer_name}      added
successfully!")

    return redirect(url_for("admin.dashboard"))


    current_app.logger.debug("WalkinForm      validation
failed: %s", form.errors)

    current_app.logger.debug("walkin_checkin request.form at
failure: %s", request.form.to_dict())

    return redirect(url_for("admin.dashboard"))


@admin_bp.route("/walkin_checkout/<int:res_id>",
methods=["GET", "POST"])

@login_required

def walkin_checkout(res_id):

    res = Reservation.query.get_or_404(res_id)

    now = datetime.now()
```

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```
        final_bill_arg    =    request.args.get("final_bill")    or
request.form.get("final_bill")

        final_bill = None

        if final_bill_arg:

            try:

                final_bill = float(final_bill_arg)

            except (TypeError, ValueError):

                final_bill = None


        if final_bill is not None and final_bill > 0:

            res.total_amount = round(final_bill, 2)

            res.end_time = now

        elif res.is_open_time:

            duration_seconds          =          (now          -
res.start_time).total_seconds()

            duration_minutes = max(1, int(duration_seconds / 60))

            res.total_amount = calculate_admin_total_amount(

                room_rate=res.room.base_rate or 0,

                duration_hours=duration_minutes / 60,
```

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```
        extra_fee=res.extra_fee or 0,

        addon_subtotal=res.addon_subtotal or 0,

        discount_rate=getattr(res, "discount_rate", 0)
or 0,

        is_open_time=True,

    )

    res.end_time = now

else:

    if not res.total_amount or res.total_amount == 0:

        try:

            if res.end_time and res.start_time:

                diff_hours = (res.end_time -
res.start_time).total_seconds() / 3600

                diff_hours = max(diff_hours, 0)

                res.total_amount =
calculate_admin_total_amount(

                    room_rate=res.room.base_rate or 0,

                    duration_hours=diff_hours,

                    extra_fee=res.extra_fee or 0,

                    addon_subtotal=res.addon_subtotal or
0,
```

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```
        discount_rate=getattr(res,
"discount_rate", 0) or 0,

        is_open_time=False,

    )

    except Exception:

        pass

    res.end_time = now

    report_date = res.end_time.date()

    res.status = "Checked-Out"

    res.paid = True

    if res.room and res.room.name.strip().lower() != "common
area":

        res.room.status = "available"

    walkin_entry =
WalkinReservation.query.filter_by(reservation_id=res.id).first()

    if walkin_entry:
```


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```
walkin_entry.status = "Checked-Out"

walkin_entry.paid = True

walkin_entry.total_amount = res.total_amount

walkin_entry.end_time = res.end_time


db.session.commit()

refresh_daily_report(report_date)


if (

    request.headers.get("X-Requested-With") ==
    "XMLHttpRequest"

    or request.is_json

):

    return jsonify(

        {

            "status": "success",

            "message": f"Payment of
P{res.total_amount:.2f} received from {res.customer_name}.",

        }

    )
```

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```
flash(f"Payment of ₱{res.total_amount:.2f} received from  
{res.customer_name}.")
```

```
return redirect(url_for("admin.dashboard"))
```

```
@admin_bp.route("/process_payment/<int:reservation_id>",  
methods=["GET", "POST"])
```

```
@login_required
```

```
def process_payment(reservation_id):
```

```
    res = Reservation.query.get_or_404(reservation_id)
```

```
    now = datetime.now()
```

```
    amount_passed = request.form.get("total_bill") or  
request.args.get("total_bill")
```

```
if amount_passed and float(amount_passed) > 0:
```

```
    res.total_amount = round(float(amount_passed), 2)
```

```
    if res.is_open_time or not res.end_time:
```

```
        res.end_time = now
```

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```
elif res.is_open_time:

    duration_minutes = max(

        1, int((now - res.start_time).total_seconds() /

60)

    )

    res.total_amount = calculate_admin_total_amount(

        room_rate=res.room.base_rate or 0,

        duration_hours=duration_minutes / 60,

        extra_fee=res.extra_fee or 0,

        addon_subtotal=res.addon_subtotal or 0,

        discount_rate=getattr(res, "discount_rate", 0)

or 0,

        is_open_time=True,

    )

    res.end_time = now


if not res.is_open_time and (not res.total_amount or
res.total_amount == 0):

    try:

        if res.end_time and res.start_time:
```

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```
        diff_hours = (res.end_time -
res.start_time).total_seconds() / 3600

        diff_hours = max(diff_hours, 0)

        res.total_amount =
calculate_admin_total_amount(

            room_rate=res.room.base_rate or 0,

            duration_hours=diff_hours,

            extra_fee=res.extra_fee or 0,

            addon_subtotal=res.addon_subtotal or 0,

            discount_rate=getattr(res,
"discount_rate", 0) or 0,

            is_open_time=False,

        )

    except Exception:

        pass

report_date = res.end_time.date()

res.status = "Checked-Out"

res.paid = True
```

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```
        if res.room and res.room.name.strip().lower() != "common
area":

            res.room.status = "available"


        walkin_entry =
WalkinReservation.query.filter_by(reservation_id=res.id).first()

        if walkin_entry:

            walkin_entry.status = "Checked-Out"

            walkin_entry.paid = True

            walkin_entry.total_amount = res.total_amount

            walkin_entry.end_time = res.end_time


        db.session.commit()

        refresh_daily_report(report_date)


        flash(

            f"Payment of ₱{res.total_amount:.2f} received from
{res.customer_name}. Dashboard updated!"

        )
```

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```
return redirect(url_for("admin.dashboard"))

@admin_bp.route('/extend_time/<int:res_id>',
methods=['POST'])

@login_required

def extend_time(res_id):

    redirect_response = require_admin_or_staff_json()

    if redirect_response:

        return redirect_response

    res = Reservation.query.get_or_404(res_id)

    data = request.get_json(silent=True) or {}

    try:

        added_hours = int(data.get('added_hours', 0))

    except (TypeError, ValueError):

        added_hours = 0

    if added_hours <= 0:
```

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```
        return jsonify({"status": "error", "message":  
"Please add at least 1 hour."}), 400
```

```
    if not res.end_time:  
  
        return jsonify({"status": "error", "message":  
"Reservation end time is missing."}), 400
```

```
    if res.status == 'Ended':  
  
        res.status = 'Walk-in'
```

```
    room_rate = res.room.base_rate or 0
```

```
    added_cost = round(room_rate * added_hours, 2)
```

```
    if res.is_open_time:  
  
        res.extra_fee = (res.extra_fee or 0.0) + added_cost  
  
        res.total_amount = round((res.total_amount or 0.0) +  
added_cost, 2)
```

```
    else:
```

```
        previous_end = res.end_time
```

```
        current_total = res.total_amount or 0.0
```

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```
        if current_total == 0 and res.start_time and
previous_end:

            original_duration = max(0, (previous_end -
res.start_time).total_seconds() / 3600)

            current_total = round(original_duration *
room_rate + (res.extra_fee or 0), 2)

            new_end = previous_end + timedelta(hours=added_hours)

            res.end_time = new_end

            res.total_amount = round(current_total + added_cost,
2)

        if res.room and res.room.name.strip().lower() != 'common
area':

            res.room.status = 'unavailable'

        walkin_entry =
WalkinReservation.query.filter_by(reservation_id=res.id).first()

        if walkin_entry:

            walkin_entry.end_time = res.end_time

            walkin_entry.total_amount = res.total_amount

            walkin_entry.status = 'Walk-in'
```


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```
db.session.commit()

return jsonify(

    {

        "status": "success",

        "message": f"Extended {added_hours} hour(s) for
{res.customer_name}. New total is ₱{res.total_amount:.2f}.",

        "new_total": res.total_amount,

        "new_end_time": res.end_time.isoformat(),

        "new_extra_fee": res.extra_fee or 0.0,

    }

)


@admin_bp.route("/reservations", methods=["GET", "POST"])

@login_required

def admin_reservations():

    redirect_response = require_admin_or_staff()

    if redirect_response:
```

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```
        return redirect_response

    rooms = Room.query.filter(~Room.name.ilike('Test
Room%')).all()

    form = AdminReservationForm()

    form.room_id.choices = [

        (r.id, f"{r.name} (P{r.base_rate}/hr)") for r in
rooms

    ]

    ensure_address_column()

    if request.method == "POST":

        selected_room_id = request.form.get("room_id")

        if selected_room_id:

            try:

                selected_room_id = int(selected_room_id)

            except (TypeError, ValueError):

                selected_room_id = None
```

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```
        if selected_room_id and selected_room_id not in
[choice[0] for choice in form.room_id.choices]:

            selected_room = Room.query.get(selected_room_id)

            if selected_room:

                form.room_id.choices.append(

                    (selected_room.id, f"{selected_room.name}
(₱{selected_room.base_rate}/hr)")

                )

total_from_js = request.form.get("total_price")

is_open_time = form.open_time.data

if form.validate_on_submit():

    room = Room.query.get(form.room_id.data)

    if not room:

        flash("Please select a valid room.")

        return render_template(

            "admin/admin_reservations.html",
form=form, rooms=rooms

        )
```

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```
try:

    if is_open_time:

        full_start = datetime.now()

        full_end      =      full_start      +
timedelta(hours=12)

    else:

        if form.start_time.data:

            if isinstance(form.start_time.data,
str):

                full_start      =
datetime.strptime(form.start_time.data, "%Y-%m-%dT%H:%M")

            else:

                full_start      =
form.start_time.data

        else:

            full_start = datetime.now()

        if form.end_time.data:

            if isinstance(form.end_time.data,
str):
```

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```

        full_end =
datetime.strptime(form.end_time.data, "%Y-%m-%dT%H:%M")

    else:

        full_end = form.end_time.data

        if full_end <= full_start:

            full_end += timedelta(days=1)

    else:

        full_end = full_start +
timedelta(hours=1)

    if room.name.strip().lower() != "common
area":

        conflict = Reservation.query.filter(

            Reservation.room_id == room.id,

            Reservation.status.in_(["Confirmed",
"Walk-in", "Pending"]),

            Reservation.start_time < full_end,

            Reservation.end_time > full_start,

        ).first()

    if conflict:
```

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```
        flash(

            f"Cannot book: {room.name} is
already reserved by {conflict.customer_name} from
{conflict.start_time.strftime('%I:%M %p')} to
{conflict.end_time.strftime('%I:%M %p')}."

        )

        return render_template(

            "admin/admin_reservations.html",
form=form, rooms=rooms

        )

    except Exception as e:

        flash(f"Time Format Error: {str(e)}")

        return render_template(

            "admin/admin_reservations.html",
form=form, rooms=rooms

        )

    # Generate customer_id based on room type

    room_type = "common area" if
room.name.strip().lower() == "common area" else "other"

    try:
```

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```
        customer_id = generate_customer_id(room_type)

    except ValueError as e:

        flash(str(e))

    return render_template(

        "admin/admin_reservations.html",
form=form, rooms=rooms

    )


reservation = Reservation(

    user_id=current_user.id,

    customer_id=customer_id,

    room_id=form.room_id.data,

    customer_name=form.customer_name.data,

contact_number=request.form.get("contact_number", "N/A"),

    pax_count=form.pax_count.data,

    start_time=full_start,

    end_time=full_end,

    is_open_time=is_open_time,

    status="Pending",
```

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```
        added_by=current_user.name,

        extra_notes=form.extra_notes.data,

        extra_fee=float(form.extra_fee.data)            if
form.extra_fee.data else 0.0,

addon_subtotal=float(form.addon_subtotal.data)            if
form.addon_subtotal.data else 0.0,

        total_amount=round(float(total_from_js or 0),
2),

        discount_rate=(

            float(form.discount.data)

            if getattr(form, "discount", None)

            else 0.0

        ),

        paid=False,

    )

now = datetime.now()

if (

    room
```


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```
        and room.name.strip().lower() != "common
area"

        and full_start <= now

    ):

        room.status = "unavailable"

db.session.add(reservation)

if not is_open_time and (

        not reservation.total_amount or
reservation.total_amount == 0

):

    try:

        base_rate = room.base_rate or 0

        diff_hours = (

            reservation.end_time -
reservation.start_time

        ).total_seconds() / 3600

        diff_hours = max(diff_hours, 0)

        discount = reservation.discount_rate or
0.0

        room_cost = base_rate * diff_hours
```

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```
room_cost = room_cost * (1 - (discount
or 0))

reservation.total_amount = round(

    room_cost + (reservation.extra_fee
or 0) + (reservation.addon_subtotal or 0), 2

)

except Exception:

    reservation.total_amount =
round(float(total_from_js or 0), 2)

db.session.commit()

flash("Reservation added! It will appear on
dashboard once it starts.")

return redirect(url_for("admin.dashboard"))

return render_template("admin/admin_reservations.html",
form=form, rooms=rooms)

@admin_bp.route("/reservations_list")
```

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```
@login_required

def admin_reservations_list():

    redirect_response = require_admin_or_staff()

    if redirect_response:

        return redirect_response

    search = request.args.get("search", "")

    # Only show reservations that are clearly legitimate in
the main list:

    # - Walk-ins (staff-registered in-person)

    # - On Hold (clients who paid a downpayment and were
placed on hold)

    # - Confirmed & paid reservations created by staff (not
online full-pay clients)


    base_query = Reservation.query

    # Build filter: include Walk-in and On Hold always;
include Confirmed+paid only if created by staff or no user

    query = base_query.filter(

        or_(

            Reservation.status == "On Hold",

            Reservation.status == "Walk-in",
```

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```
        and_(
            Reservation.status == "Confirmed",
            Reservation.paid == True,
            or_(
                Reservation.user_id.is_(None),

Reservation.user.has(User.role.in_(["admin", "staff"])),

            ),
        ),
    )
)

if search:
    query = query.filter(
        or_(
Reservation.customer_name.ilike(f"%{search}%"),

            Reservation.added_by.ilike(f"%{search}%"),

            Reservation.status.ilike(f"%{search}%"),

        )
    )
```

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)

```
        reservations =  
query.order_by(Reservation.start_time.asc()).all()  
  
    return render_template(  
        "admin/admin_reservations_list.html",  
        reservations=reservations,  
        search=search,  
    )
```

```
@admin_bp.route("/confirm_reservations")
```

```
@login_required
```

```
def admin_confirm_reservations():
```

```
    redirect_response = require_admin_or_staff()
```

```
    if redirect_response:
```

```
        return redirect_response
```

```
pending_reservations = (
```

```
    Reservation.query.filter_by(status="Pending")
```

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```
        .order_by(Reservation.created_at.desc() ,
Reservation.start_time.desc())

        .all()

    )

    return render_template(

        "admin/confirm_reservations.html",

        reservations=pending_reservations,

    )
```

```
@admin_bp.route("/solo_applications")

@login_required

def solo_applications():

    redirect_response = require_super_admin()

    if redirect_response:

        return redirect_response

    pending_plans = (

        SoloPlan.query.filter_by(status="pending")

        .order_by(SoloPlan.created_at.desc())
```

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```
        .all()

    )

    return render_template(

        "admin/solo_applications.html",
pending_plans=pending_plans

    )


@admin_bp.route("/approve_membership/<int:req_id>",
methods=["POST"])

@login_required

def approve_membership(req_id):

    redirect_response = require_super_admin()

    if redirect_response:

        return redirect_response


    plan = SoloPlan.query.get_or_404(req_id)

    plan.approved_by_id = current_user.id

    plan.status = "approved"
```

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```
ph_tz = pytz.timezone('Asia/Manila')

now_ph = datetime.now(ph_tz)

plan.set_expiry_date(now_ph)


if not plan.customer_id:

    plan.customer_id = generate_customer_id("other")


# Ensure approved solo plan users receive a matching
active membership record.

_ensure_approved_solo_plan_membership(plan.user)


db.session.commit()

flash(f"Membership approved for {plan.user.name}")

return redirect(url_for("admin.members", tab="list"))


@admin_bp.route("/reject_membership/<int:req_id>",
methods=["POST"])

@login_required

def reject_membership(req_id):
```


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```
redirect_response = require_super_admin()

if redirect_response:

    return redirect_response


plan = SoloPlan.query.get_or_404(req_id)

plan.status = "rejected"

db.session.commit()

flash(f"Membership rejected for {plan.user.name}")

return redirect(url_for("admin.members", tab="requests"))


@admin_bp.route("/renew_member", methods=["POST"])

@admin_bp.route("/admin/renew_member", methods=["POST"])

@login_required

def renew_member():

    try:

        data = request.get_json(silent=True) or request.form
or {}

        user_id = data.get("user_id")

        if not user_id:
```

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```
        return jsonify({"status": "error", "message":  
"Missing user ID."}), 400
```

```
    user = User.query.get(int(user_id))
```

```
    if not user:
```

```
        return jsonify({"status": "error", "message":  
"Member not found."}), 404
```

```
    # Reject renewal kon kasamtangan nga Checked In
```

```
    membership =  
Membership.query.filter_by(user_id=user.id).first()
```

```
    # ga check sa status sang membership o sa latest  
solo plan
```

```
    if membership and membership.status and  
membership.status.lower() in ["checked in", "checked_in"]:
```

```
        return jsonify(  
  
            "status": "error",  
  
            "message": "Cannot renew membership while  
customer is currently Checked In. Please check out the customer  
first."
```

```
        ), 400
```

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```
# 1. Kuhaon ang SoloPlan entry sang user

latest_plan = (

    SoloPlan.query.filter_by(user_id=user.id)

    .order_by(SoloPlan.id.desc())

    .first()

)

now_ph = datetime.utcnow() + timedelta(hours=8)

# 2. Kon wala sing plan record, himuan sang default
nga bag-o

if not latest_plan:

    latest_plan = SoloPlan(

        user_id=user.id,

        plan_name="INDIVIDUAL RATE",

        status="approved",

        created_at=now_ph

    )

    db.session.add(latest_plan)
```

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```
        if latest_plan.expiry_date and
latest_plan.expiry_date > now_ph:

            base_time = latest_plan.expiry_date

        else:

            base_time = now_ph

# Dynamic duration base sa SoloPlan Model method

latest_plan.set_expiry_date(start_time=base_time)

latest_plan.status = "approved"

# Sync status kag expiry sa Membership Table

if membership:

    membership.is_checked_out = False

    membership.status = "active"

    membership.expiry_date = latest_plan.expiry_date

db.session.commit()

return jsonify({
```

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```
        "status": "success",

        "message": "Please Check In the customer again
to start the membership renewal."

    }), 200

except Exception as e:

    db.session.rollback()

    return jsonify({"status": "error", "message":
str(e)}), 500


@admin_bp.route("/deactivate_member", methods=["POST"])
@login_required
def deactivate_member():

    redirect_response = require_super_admin_json()

    if redirect_response:

        return redirect_response


data = request.get_json(silent=True) or request.form
```

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```
        user_id = data.get("user_id") or
request.form.get("user_id")

        user = User.query.get(int(user_id))

        if not user:

            return jsonify({"status": "error", "message":
"Member not found."}), 404

        # Soft-deactivate the user

        user.is_active = False

        # Gamiton ang local server time para sakto ang adlaw

        user.last_deactivated_at = datetime.now()

        # I-update ang membership status kon may ara

        membership =
Membership.query.filter_by(user_id=user.id).order_by(Membership.
id.desc()).first()

        if membership:

            membership.status = 'deactivated'

        log = UserActivityLog(
```

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```
        user_id=user.id,

        activity_type=f"deactivated|Account deactivated by
{current_user.name}",

        ip_address=request.remote_addr or "unknown",

    )

    db.session.add(log)

    db.session.commit()


    return jsonify({"status": "success", "message": "Member
account deactivated successfully."})


@admin_bp.route("/reactivate_member", methods=["POST"])

@login_required

def reactivate_member():

    redirect_response = require_super_admin_json()

    if redirect_response:

        return redirect_response


    data = request.get_json(silent=True) or request.form
```

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```
        user_id = data.get("user_id") or
request.form.get("user_id")

        if not user_id:

            return jsonify({"status": "error", "message":
"Missing user ID."}), 400

        user = User.query.get(int(user_id))

        if not user:

            return jsonify({"status": "error", "message":
"Member not found."}), 404

        # 1. Clear soft-deactivation flags sa User

        user.is_active = True

        user.last_deactivated_at = None

        # 2. Re-activate associated membership (status lang ang
bag-ohon)

        membership =
Membership.query.filter_by(user_id=user.id).order_by(Membership.
id.desc()).first()

        if membership:
```


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```
membership.status = 'active'

# 3. Log activity

log = UserActivityLog(

    user_id=user.id,

    activity_type=f"reactivated|Account reactivated by
{current_user.name}",

    ip_address=request.remote_addr or "unknown",

)

db.session.add(log)

db.session.commit()

return jsonify({"status": "success", "message": "Member
account reactivated successfully."})


@admin_bp.route("/approve_solo_plan/<int:plan_id>",
methods=["POST"])

@login_required

def approve_solo_plan(plan_id):
```

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```
redirect_response = require_super_admin()

if redirect_response:

    return redirect_response


plan = SoloPlan.query.get_or_404(plan_id)

plan.status = "approved"

plan.approved_by_id = current_user.id

plan.set_expiry_date() # Set expiry date on SoloPlan


# Siguraduhon nga nakakabit ang membership record

_ensure_approved_solo_plan_membership(plan.user)


# Reset checkout & checkin flags

membership =
Membership.query.filter_by(user_id=plan.user.id).first()

if membership:

    membership.is_checked_in = False

    membership.is_checked_out = False

    membership.status = "active"
```

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```
db.session.commit()

flash(f"Plan approved for {plan.user.name}", "success")

return redirect(url_for("admin.solo_applications"))


@admin_bp.route("/reject_solo_plan/<int:plan_id>",
methods=["POST"])

@login_required

def reject_solo_plan(plan_id):

    redirect_response = require_super_admin()

    if redirect_response:

        return redirect_response


    plan = SoloPlan.query.get_or_404(plan_id)

    plan.status = "rejected"

    db.session.commit()

    flash(f"Plan rejected for {plan.user.name}", "info")

    return redirect(url_for("admin.solo_applications"))
```

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```
@admin_bp.route("/checkout_user/<int:user_id>",
methods=["POST"])

@login_required

def admin_checkout_user(user_id):

    if current_user.role != "admin":

        flash("Unauthorized access.", "danger")

        return redirect(url_for("main.dashboard"))


ph_tz = pytz.timezone("Asia/Manila")

now_naive      =      datetime.now(ph_tz).replace(second=0,
microsecond=0, tzinfo=None)


# 1. Papatyon ang active SoloPlan sang user

active_plan = SoloPlan.query.filter(

    SoloPlan.user_id == user_id,

    SoloPlan.status.ilike("approved"),

    SoloPlan.expiry_date > now_naive

).first()


if active_plan:
```

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```
active_plan.expiry_date = now_naive

active_plan.status = "completed"


# 2. Papatyon man ang Membership status sang user

membership =
Membership.query.filter_by(user_id=user_id).first()

if membership:

    membership.expiry_date = now_naive

    membership.status = "expired"


db.session.commit()

flash("Customer has been successfully checked out by
Admin.", "success")

return redirect(url_for("admin.manage_users"))


@admin_bp.route("/confirm_reservation/<int:res_id>",
methods=["POST"])

@login_required

def confirm_reservation(res_id):
```

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```
    redirect_response = require_admin_or_staff()

    if redirect_response:

        return redirect_response

    res = Reservation.query.get_or_404(res_id)

    res.status = "Confirmed"

    res.approved_by_id = current_user.id

    if res.room and res.room.name.strip().lower() != "common
area":

        now = datetime.now()

        if res.start_time <= now <= res.end_time:

            res.room.status = "unavailable"

    db.session.commit()

    flash(f"Reservation confirmed for {res.customer_name}")

    return redirect(url_for("admin.admin_reservations_list"))


@admin_bp.route("/hold_reservation/<int:res_id>",
methods=["POST"])

@login_required
```

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```
def hold_reservation(res_id):

    redirect_response = require_admin_or_staff()

    if redirect_response:

        return redirect_response


    res = Reservation.query.get_or_404(res_id)

    if res.status not in ["Pending", "Confirmed"]:

        flash("Reservation cannot be held in its current
state.")

        return
    redirect(url_for("admin.admin_reservations_list"))


    res.status = "On Hold"

    if res.room and res.room.name.strip().lower() != "common
area":

        res.room.status = "available"

    db.session.commit()

    flash(f"Reservation for {res.customer_name} has been
placed on hold.")

    return redirect(url_for("admin.admin_reservations_list"))
```

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```
@admin_bp.route("/cancel_reservation/<int:res_id>",
methods=["POST"])

@login_required

def cancel_reservation(res_id):

    redirect_response = require_admin_or_staff()

    if redirect_response:

        return redirect_response


    res = Reservation.query.get_or_404(res_id)

    if res.status == "Confirmed":

        res.room.status = "available"

    res.status = "Cancelled"

    db.session.commit()

    flash(f"Reservation cancelled for {res.customer_name}")

    return redirect(url_for("admin.admin_reservations_list"))


@admin_bp.route("/delete_reservation/<int:res_id>",
methods=["POST"])
```


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```
@login_required

def delete_reservation(res_id):

    redirect_response = require_admin_or_staff()

    if redirect_response:

        return redirect_response


    res = Reservation.query.get_or_404(res_id)

    # If reservation has associated walkin entries, delete
    them first to avoid FK constraint errors

    try:

        walkin_entries =
WalkinReservation.query.filter_by(reservation_id=res.id).all()

        for w in walkin_entries:

            db.session.delete(w)

    except Exception:

        # ignore if table or relation unavailable; will
        surface on commit if real problem

        pass


    if res.status in ["Pending", "Confirmed", "Walk-in",
"Ended"]:
```

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```
        if res.room and res.room.name.strip().lower() !=
"common area":

        res.room.status = "available"


        next_url = request.form.get("next") or
request.args.get("next") or request.referrer

        db.session.delete(res)

        db.session.commit()

        flash(f"Reservation for {res.customer_name} has been
permanently deleted.")

        return redirect(next_url or url_for("admin.dashboard"))


@admin_bp.route("/reports", methods=["GET", "POST"])
@login_required
def reports():

    redirect_response = require_super_admin()

    if redirect_response:

        return redirect_response
```

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```
start_date = datetime.utcnow().date()

end_date = datetime.utcnow().date()


if request.method == "POST":

    start_date_str = request.form.get("start_date")

    end_date_str = request.form.get("end_date")

    try:

        if start_date_str:

            start_date = \
datetime.strptime(start_date_str, "%Y-%m-%d").date()

            if end_date_str:

                end_date = datetime.strptime(end_date_str,
"%Y-%m-%d").date()

        except ValueError:

            flash("Invalid date format.")


    query = DailyReport.query.filter(

        DailyReport.report_date >= start_date,
        DailyReport.report_date <= end_date

    )
```

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```
daily_reports =  
query.order_by(DailyReport.report_date.desc()).all()  
  
reports_with_member = []  
  
for report in daily_reports:  
    member_log = (  
        TimeLog.query.join(User)  
        .filter(  
            TimeLog.time_in >= datetime.combine(  
                report.report_date, datetime.min.time()  
            ),  
            TimeLog.time_in <= datetime.combine(  
                report.report_date, datetime.max.time()  
            ),  
            User.role == "member",  
        )  
        .first()  
    )  
  
    reports_with_member.append(  
        {
```

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```
        "report_date": report.report_date,

        "total_logins": report.total_logins,

        "total_check_ins": report.total_check_ins,

        "total_revenue": report.total_timelogs,

        "generated_at": report.generated_at,

        "member_name":      member_log.user.name      if
member_log else "N/A",

    }

)

customer_sessions = (

    Reservation.query.filter(

        func.date(Reservation.end_time) >= start_date,

        func.date(Reservation.end_time) <= end_date,

        Reservation.status.in_(["Checked-Out",
"Cancelled"]),

    )

    .order_by(Reservation.end_time.desc())

    .all()

)
```

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```
return render_template(  
    "admin/reports.html",  
    daily_reports=reports_with_member,  
    customer_sessions=customer_sessions,  
    start_date=start_date,  
    end_date=end_date,  
)
```

```
@admin_bp.route("/generate_pdf")  
  
@login_required  
  
def generate_pdf():  
    redirect_response = require_super_admin()  
  
    if redirect_response:  
        return redirect_response  
  
    flash("PDF Generation feature is being initialized.")  
  
    return redirect(url_for("admin.reports"))
```

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```
@admin_bp.route("/generate_completed_sessions_pdf")

@login_required

def generate_completed_sessions_pdf():

    redirect_response = require_super_admin()

    if redirect_response:

        return redirect_response


    start_date = request.args.get("start_date", "")

    end_date = request.args.get("end_date", "")


    parsed_start_date = None

    parsed_end_date = None


    if start_date:

        try:

            parsed_start_date = datetime.strptime(start_date,
"%Y-%m-%d").date()

        except ValueError:
```

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```
        parsed_start_date = None

    if end_date:

        try:

            parsed_end_date = datetime.strptime(end_date,
"%Y-%m-%d").date()

        except ValueError:

            parsed_end_date = None

    query = Reservation.query.filter(

        Reservation.status.in_(["Checked-Out", "Cancelled"])

    )

    if parsed_start_date:

        query = query.filter(func.date(Reservation.end_time)
        parsed_start_date)

    if parsed_end_date:

        query = query.filter(func.date(Reservation.end_time)
        <= parsed_end_date)
```


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```
sessions =
query.order_by(Reservation.end_time.desc()).all()

buffer = BytesIO()

doc = SimpleDocTemplate(
    buffer,
    pagesize=letter,
    rightMargin=36,
    leftMargin=36,
    topMargin=36,
    bottomMargin=36,
)

styles = getSampleStyleSheet()

title_style = ParagraphStyle(
    "ReportTitle",
    parent=styles["Heading1"],
    fontSize=16,
    leading=20,
    spaceAfter=8,
    textColor=colors.HexColor("#1f4e79"),
```

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```
)

subtitle_style = ParagraphStyle(

    "ReportSubtitle",

    parent=styles["BodyText"],

    fontSize=9,

    leading=11,

    textColor=colors.grey,

    spaceAfter=10,

)

body_style = styles["BodyText"]


ph_tz = timezone(timedelta(hours=8))

now_ph = datetime.now(ph_tz)


elements = [

    Paragraph("Completed Sessions Report", title_style),

    Paragraph(

        f"Generated:                {now_ph.strftime('%Y-%m-%d %I:%M %p')}",

        subtitle_style,
```

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```
    ),  
  
    Paragraph(  
  
        f"Date range: {parsed_start_date or 'Beginning'}  
to {parsed_end_date or 'Today'}",  
  
        subtitle_style,  
  
    ),  
  
    Spacer(1, 8),  
  
]  
  
if not sessions:  
  
    elements.append(Paragraph("No    completed    sessions  
found for the selected date range.", body_style))  
  
else:  
  
    table_data = [[  
  
        "Customer",  
  
        "Contact",  
  
        "Room",  
  
        "Status",  
  
        "Staff",  
  
        "Start",
```

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```
        "End",

        "Total Bill",

    ]]

    for session in sessions:

        table_data.append([

            session.customer_name or "N/A",

            session.contact_number or "N/A",

            session.room.name if session.room else

            "Unknown",

            session.status or "N/A",

            (session.approved_by.name if

            session.approved_by else (session.user.name if session.user else

            session.added_by or "Unknown")),

            session.start_time.strftime("%b %d, %Y %I:%M %p") if

            session.start_time else "N/A",

            session.end_time.strftime("%b %d, %Y %I:%M %p") if

            session.end_time else "N/A",

            f"PHP{session.total_amount:,.2f}" if

            session.total_amount is not None else "PHP0.00",
```

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```
  ])
```

```
table = Table(table_data, repeatRows=1)

table.setStyle(

    TableStyle([

        ("BACKGROUND",      (0,      0),      (-1,      0),
colors.HexColor("#1f4e79")),

        ("TEXTCOLOR", (0, 0), (-1, 0), colors.white),

        ("FONTNAME", (0, 0), (-1, 0), "Helvetica-
Bold"),

        ("FONTSIZE", (0, 0), (-1, 0), 9),

        ("BOTTOMPADDING", (0, 0), (-1, 0), 8),

        ("TOPPADDING", (0, 0), (-1, 0), 8),

        ("GRID", (0, 0), (-1, -1), 0.5, colors.grey),

        ("ALIGN", (0, 1), (-1, -1), "LEFT"),

        ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),

        ("FONTNAME", (0, 1), (-1, -1), "Helvetica"),

        ("FONTSIZE", (0, 1), (-1, -1), 8),

        ("ROWBACKGROUNDS", (0, 1), (-1, -1),
[colors.whitesmoke, colors.white]),
```


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```
def members():

    redirect_response = require_admin_or_staff()

    if redirect_response:

        return redirect_response


ph_tz = pytz.timezone("Asia/Manila")

now_naive = datetime.now(ph_tz).replace(tzinfo=None)


active_tab = request.args.get("tab", "requests")

search = request.args.get("search", "")


approved_solo_users =
db.session.query(SoloPlan.user_id).filter(

    SoloPlan.status.ilike('approved'),

    or_(SoloPlan.expiry_date.is_(None),
SoloPlan.expiry_date > now_naive)

)


query = User.query.filter(
```

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```
        or_(User.role == "member",
User.id.in_(approved_solo_users))

    )

    if search:

        search_pattern = f"%{search}%"

        filters = [

            User.name.ilike(search_pattern),

            User.email.ilike(search_pattern),

            User.phone.ilike(search_pattern),

        ]

        if search.isdigit():

            filters.append(User.id == int(search))

        query = query.filter(or_(*filters))

    approved_solo_user_ids = [r[0] for r in
approved_solo_users.distinct().all()]

    approved_solo_user_ids_set = set(approved_solo_user_ids)
```


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```
all_members =
query.order_by(User.created_at.desc()).all()

# Siguraduhon nga ma-sync ang membership status sa bag-o
nga SoloPlan

created_membership = False

for member in all_members:

    if member.id in approved_solo_user_ids_set:

        created_membership =
        _ensure_approved_solo_plan_membership(member) or
        created_membership

    if created_membership:

        db.session.commit()

        all_members =
        query.order_by(User.created_at.desc()).all()

membership_requests = (

    SoloPlan.query.filter_by(status="pending")

    .order_by(SoloPlan.created_at.desc())

    .all())
```

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```
)

return render_template(

    "admin/members.html",

    members=all_members,

    search=search,

    membership_requests=membership_requests,

    active_tab=active_tab,

    approved_solo_user_ids=approved_solo_user_ids,

)


# PERMANENT DELETE ROUTE

@admin_bp.route("/delete_member/<int:user_id>",
methods=["POST"])

@login_required

def delete_member(user_id):

    redirect_response = require_super_admin_json()

    if redirect_response:

        return redirect_response
```

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```
user = User.query.get_or_404(user_id)

if user.role == "admin":

    return jsonify({"status": "error", "message":
"Cannot delete admin account."}), 400

try:

    # 1. Clear attendance_logs nga nakakabit sa
memberships sang user

    db.session.execute(

        text("""

            DELETE FROM attendance_logs

            WHERE membership_id IN (

                SELECT id FROM memberships WHERE user_id

= :uid

            )

        """),

        {"uid": user.id}

    )

    # 2. Clear attendance_logs kon may direct user_id
column man ini
```

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```
try:

    db.session.execute(text("DELETE          FROM
attendance_logs WHERE user_id = :uid"), {"uid": user.id})

except Exception:

    pass # Sagap lang kon wala sang direct user_id
column

# 3. Clear child tables sang user

    db.session.execute(text("DELETE      FROM      solo_plans
WHERE user_id = :uid"), {"uid": user.id})

    db.session.execute(text("DELETE          FROM
user_activity_logs WHERE user_id = :uid"), {"uid": user.id})

    db.session.execute(text("DELETE      FROM      memberships
WHERE user_id = :uid"), {"uid": user.id})

# 4. Permanent Delete sa User

    db.session.delete(user)

    db.session.commit()

    return jsonify({"status": "success", "message":
"Member account permanently deleted."})
```

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```
except Exception as e:

    db.session.rollback()

    print(f"Permanent Delete Error: {str(e)}")

    return jsonify({"status": "error", "message":
f"Failed to delete: {str(e)}"}), 500


# Membership Management Routes

@admin_bp.route("/official_members")

@login_required

def official_members():

    """Display all members with active/expired memberships
in card layout."""

    redirect_response = require_admin_or_staff()

    if redirect_response:

        return redirect_response


    search = request.args.get("search", "")


# Get all memberships with their user info
```

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```
query =
Membership.query.join(User).order_by(Membership.status.desc(),
User.name.asc())

if search:

    search_pattern = f"%{search}%"

    query = query.filter(or_(

        User.name.ilike(search_pattern),

        User.email.ilike(search_pattern),

        User.phone.ilike(search_pattern)

    ))

memberships = query.all()

# Separate active, pending, and expired

active_memberships = [m for m in memberships if
m.is_active]

pending_memberships = [m for m in memberships if
m.status == "pending"]

expired_memberships = [m for m in memberships if
m.status == "expired"]
```

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```
return render_template(

    "admin/official_members.html",

    active_memberships=active_memberships,

    pending_memberships=pending_memberships,

    expired_memberships=expired_memberships,

    search=search

)


@admin_bp.route("/api/member/<int:user_or_membership_id>/check-in", methods=["POST"])

@login_required

def membership_check_in(user_or_membership_id):

    """Check in a member or solo user dynamically."""

    redirect_response = require_admin_or_staff_json()

    if redirect_response:

        return redirect_response


# A. Pangitaon ang Membership O User
```

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```
membership =
Membership.query.filter_by(id=user_or_membership_id).first()

if membership:

    user = membership.user

else:

    user = User.query.get_or_404(user_or_membership_id)

    membership =
Membership.query.filter_by(user_id=user.id).first()

now_ph = datetime.utcnow() + timedelta(hours=8)

plan_hours_map = {

    "INDIVIDUAL RATE": 1.0,

    "INDIVIDUAL RATE (4HRS)": 4.0,

    "DAY/NIGHT PASS": 24.0,

    "WEEKLY PASS (DAY/NIGHT)": 168.0,

    "WEEKLY PASS (24HRS)": 168.0,

    "MONTHLY PASS (DAY/NIGHT)": 720.0,

    "MONTHLY PASS (24HRS)": 720.0,

    "WORKSTATION (24HRS)": 720.0,
```


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```
"ACTIVE PLAN": 720.0

}

latest_plan = SoloPlan.query.filter(

    SoloPlan.user_id == user.id

).order_by(SoloPlan.id.desc()).first()

clean_plan_name      =      (latest_plan.plan_name      or
"".strip().upper() if latest_plan else "INDIVIDUAL RATE"

allocated_hours      =      plan_hours_map.get(clean_plan_name,
1.0)

# Siguraduhon nga may Membership row baseline

if not membership:

    membership = Membership(

        user_id=user.id,

        status="active",

        start_date=now_ph,

        expiry_date=latest_plan.expiry_date      if
(latest_plan and latest_plan.expiry_date) else (now_ph +
timedelta(days=30)),
```

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```
        total_hours=allocated_hours,

        hours_left=allocated_hours,

        plan_name=latest_plan.plan_name    if    latest_plan
else "INDIVIDUAL RATE",

        is_checked_in=False

    )

    db.session.add(membership)

    db.session.commit()


if membership.hours_left <= 0 and latest_plan:

    membership.plan_name = latest_plan.plan_name

    membership.total_hours = float(allocated_hours)

    membership.hours_left = float(allocated_hours)

    membership.status = "active"

    if latest_plan.expiry_date:

        membership.expiry_date = latest_plan.expiry_date

    db.session.commit()


# Update status to Active

if latest_plan:
```

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```
latest_plan.status = "approved"

latest_plan.updated_at = now_ph


# Create Attendance Log

log = AttendanceLog(

    membership_id=membership.id,

    check_in_time=now_ph

)

membership.is_checked_in = True

membership.updated_at = now_ph

membership.is_checked_out = False


db.session.add(log)

db.session.commit()


return jsonify({

    "status": "success",

    "message": f"{user.name} checked in successfully!",

    "check_in_time": now_ph.isoformat(),
```

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```
        "hours_left": membership.hours_left

    })

@admin_bp.route("/api/member/<int:user_or_membership_id>/check-out", methods=["POST"])

@login_required

def membership_check_out(user_or_membership_id):

    """Check out a member/solo user, end active session, and deduct hours dynamically."""

    redirect_response = require_admin_or_staff_json()

    if redirect_response:

        return redirect_response

    # Pangitaon ang Membership O ang User ID

    membership = Membership.query.filter_by(id=user_or_membership_id).first()

    if not membership:

        membership = Membership.query.filter_by(user_id=user_or_membership_id).first()
```

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```
target_user_id = membership.user_id if membership else  
user_or_membership_id
```

```
# Pangitaon ang tanan nga active / approved SoloPlan  
sang user
```

```
active_solos = SoloPlan.query.filter(  
  
    SoloPlan.user_id == target_user_id,  
  
    SoloPlan.status.ilike("approved")  
  
) .all()
```

```
now_ph = datetime.utcnow() + timedelta(hours=8)
```

```
# Close Attendance Log & Deduct Hours
```

```
m_id = membership.id if membership else None
```

```
if m_id:
```

```
    current_log = AttendanceLog.query.filter_by(  
  
        membership_id=m_id,  
  
        check_out_time=None  
  
    ) .first()
```

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```
if current_log:

    current_log.check_out_time = now_ph

    duration = (now_ph -
current_log.check_in_time).total_seconds() / 3600

    hours_deducted = round(duration, 2)

if membership and membership.hours_left is not
None:

    if hours_deducted > membership.hours_left:

        hours_deducted = membership.hours_left

        membership.hours_left = round(max(0.0,
membership.hours_left - hours_deducted), 2)

    if membership.hours_left <= 0:

        membership.status = "expired"

current_log.hours_deducted = hours_deducted

# I-set ang status flags sa CHECKED OUT

if membership:

    membership.is_checked_in = False
```

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```
membership.is_checked_out = True

    if membership.status and membership.status.lower()
in ["checked in", "checked_in"]:

        membership.status = "active"

membership.updated_at = now_ph

for solo in active_solos:

    solo.status = "checked_out"

    solo.updated_at = now_ph

db.session.commit()

    user_name = membership.user.name if (membership and
membership.user) else "User"

return jsonify({

    "status": "success",

    "message": f"{user_name} checked out successfully!",
```

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```
        "hours_left": membership.hours_left if membership
else 0.0

    })
```

```
@admin_bp.route("/api/member/<int:membership_id>/attendance",
methods=["GET"])
```

```
@login_required
```

```
def member_attendance_history(membership_id):
```

```
    """Get attendance history using direct raw database
formatting."""
```

```
    redirect_response = require_admin_or_staff_json()
```

```
    if redirect_response:
```

```
        return redirect_response
```

```
    membership = Membership.query.get_or_404(membership_id)
```

```
    logs =
membership.attendance_logs.order_by(AttendanceLog.check_in_time.
desc()).all()
```


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```
attendance_data = []

for log in logs:

    c_in = log.check_in_time

    c_out = log.check_out_time

    # Direct String Formatting (%b %d, %Y kag %I:%M %p)

    date_str = c_in.strftime("%b %d, %Y") if c_in else
    "-"

    check_in_str = c_in.strftime("%I:%M %p") if c_in
    else "-"

    check_out_str = c_out.strftime("%I:%M %p") if c_out
    else "-"

    attendance_data.append({

        "id": log.id,

        "date": date_str,

        "check_in": check_in_str,

        "check_out": check_out_str,
```

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```
        "hours":
decimal_hours_to_readable(log.hours_deducted)                                if
(log.hours_deducted and log.hours_deducted > 0) else "-"

    })

    return jsonify({

        "status": "success",

        "member_name": membership.user.name,

        "total_hours": membership.total_hours,

        "hours_left": membership.hours_left,

        "attendance": attendance_data

    })


@admin_bp.route("/api/dashboard/common-area-occupants",
methods=["GET"])

@login_required

def common_area_occupants():

    """Get list of members currently checked in (for Common
Area display on dashboard)."""

    redirect_response = require_admin_or_staff_json()
```

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```
if redirect_response:

    return redirect_response


checked_in = db.session.query(Membership).filter(

    Membership.is_checked_in == True

).all()


occupants = []


for membership in checked_in:

    _expire_membership_if_needed(membership)

    if membership.status != "active" or not
membership.is_checked_in:

        continue


    active_log = membership.attendance_logs.filter(

        AttendanceLog.check_out_time.is_(None)

    ).first()


    if active_log and active_log.check_in_time:
```

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```
        formatted_check_in                                =  
active_log.check_in_time.strftime("%I:%M %p")
```

```
        epoch_time_ms                                    =  
int(active_log.check_in_time.timestamp() * 1000)
```

```
        occupants.append(  
            "id": membership.id,  
            "name": membership.user.name,  
            "check_in_time":  
active_log.check_in_time.isoformat(),  
            "check_in_ms": epoch_time_ms,  
            "formatted_check_in": formatted_check_in,  
            "hours_left": membership.hours_left  
        )
```

```
    return jsonify(  
        "status": "success",  
        "count": len(occupants),  
        "occupants": occupants
```

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```
})
```

4. Member/client booking and usage flow

```
client_fixed.py
```

```
"""
```

```
client.py
```

```
Contains all client-facing blueprints: auth, main, and api.
```

```
"""
```

```
import os
```

```
from datetime import datetime, timedelta
```

```
import pytz
```

```
from database_fixed import (
```

```
    db,
```

```
    Room,
```

```
    User,
```

```
    TimeLog,
```

```
    SoloPlan,
```

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```
LoginForm,  
  
Membership,  
  
PaymentInfo,  
  
ProfileForm,  
  
Reservation,  
  
AttendanceLog,  
  
ReservationForm,  
  
RegistrationForm,  
  
ChangePasswordForm,  
  
generate_customer_id,  
  
get_user_by_email,  
  
)  
  
from flask import (  
  
    flash,  
  
    jsonify,  
  
    request,  
  
    session,  
  
    url_for,  
  
    redirect,
```

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```
Blueprint,

current_app,

render_template,

)

from sqlalchemy import func, or_

from werkzeug.utils import secure_filename

from flask_login import current_user, login_required,
login_user, logout_user


# Auth Blueprint

auth_bp = Blueprint("auth", __name__)

main_bp = Blueprint('main', __name__,
template_folder='templates')


@auth_bp.route("/login", methods=["GET", "POST"])

def login():

    if current_user.is_authenticated:

        return redirect(url_for("main.dashboard"))
```

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```
if request.method == "GET":

    return redirect(url_for("main.index",
show_login="true"))

form = LoginForm(meta={"csrf": False})

if form.validate():

    email = form.email.data.strip().lower()

    user = get_user_by_email(email)

    if user is None or not
user.check_password(form.password.data):

        flash("Invalid email or password", "danger")

        return redirect(url_for("main.index",
show_login="true"))

    if not user.is_active:

        flash("Access Restricted: Account is inactive.
Contact your administrator.", "danger")

        return redirect(url_for("main.index",
show_login="true"))

    login_user(user, remember=form.remember_me.data)

    session["user_id"] = user.id
```


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```
session["user_name"] = user.name

session["user_role"] = user.role

next_page = request.args.get("next")

return redirect(next_page) or
url_for("main.dashboard"))

flash("Please enter valid credentials and try again.",
"danger")

return redirect(url_for("main.index", show_login="true"))


@auth_bp.route("/register", methods=["GET", "POST"])
def register():

    if current_user.is_authenticated:

        return redirect(url_for("main.dashboard"))

    if request.method == "GET":

        return redirect(url_for("main.index",
show_register="true"))
```

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```
form = RegistrationForm()

if form.validate_on_submit():

    email = form.email.data.strip().lower()

    # Check if email already exists

    existing_user =
User.query.filter_by(email=email).first()

    if existing_user:

        flash("Email already registered. Please login or
use a different email.", "danger")

        return redirect(url_for("main.index",
show_register="true"))

    try:

        user = User(name=form.name.data, email=email,
phone=form.phone.data)

        user.set_password(form.password.data)

        db.session.add(user)

        db.session.commit()

    except Exception as e:

        db.session.rollback()
```

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```
flash("Registration failed. Please try again  
with a different email.", "danger")
```

```
return redirect(url_for("main.index",  
show_register="true"))
```

```
flash("Account created successfully! Please log in  
with your credentials.", "success")
```

```
return redirect(url_for("main.index",  
show_login="true"))
```

```
flash("Please complete all required fields correctly.",  
"danger")
```

```
return redirect(url_for("main.index",  
show_register="true"))
```

```
@auth_bp.route("/logout")
```

```
@login_required
```

```
def logout():
```

```
    logout_user()
```

```
    session.pop("user_id", None)
```

```
    session.pop("user_name", None)
```

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```
session.pop("user_role", None)

return redirect(url_for("main.index"))


# Main Blueprint

main_bp = Blueprint("main", __name__)


def get_admin_stats():

    ph_tz = pytz.timezone("Asia/Manila")

    now = datetime.now(ph_tz)

    total_members =
User.query.filter_by(role="member").count()

    active_timelogs = (

db.session.query(TimeLog).filter(TimeLog.time_out.is_(None)).count()

    )
```

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```
# Calculate today's boundary based on PHT

today_start      =      ph_tz.localize(datetime(now.year,
now.month, now.day))

today_end = today_start + timedelta(days=1)


res_today = (

    Reservation.query.filter(

        Reservation.start_time >= today_start,

        Reservation.start_time < today_end,

        Reservation.status == "Confirmed",

    ).count()

)


revenue_today = (

    db.session.query(func.sum(Reservation.total_amount))

    .filter(

        Reservation.start_time >= today_start,

        Reservation.start_time < today_end,

        Reservation.status == "Confirmed",

    )

)
```

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```
        .scalar()

        or 0

    )

    return    total_members,    active_timelogs,    res_today,
revenue_today


def _expire_membership_if_needed(membership):

    if not membership or membership.status != 'active' or
not membership.expiry_date:

        return

    ph_tz = pytz.timezone("Asia/Manila")

    now = datetime.now(ph_tz)

    expiry_dt = membership.expiry_date

    if expiry_dt.tzinfo is None:

        expiry_dt = ph_tz.localize(expiry_dt)
```

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```
if now >= expiry_dt:

    membership.status = 'expired'

    membership.hours_left = 0.0

    membership.is_checked_in = False


    active_log = membership.attendance_logs.filter(AttendanceLog.check_out_time.is_(None)).first()

    if active_log:

        # Check-out time set to expiry date

        active_log.check_out_time = membership.expiry_date


        # Synchronize active_log dates for subtraction
if needed

    check_in = active_log.check_in_time

    check_out = active_log.check_out_time


    if check_in.tzinfo is None:

        check_in = ph_tz.localize(check_in)

    if check_out.tzinfo is None:
```

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```
        check_out = ph_tz.localize(check_out)

        # Calculate exact hours deducted

        time_diff = check_out - check_in

        active_log.hours_deducted =
round(time_diff.total_seconds() / 3600, 2)

        db.session.commit()

@main_bp.route("/")

def index():

    if current_user.is_authenticated:

        return redirect(url_for("main.dashboard"))

    login_form = LoginForm()

    register_form = RegistrationForm()

    show_register = request.args.get("show_register",
"false").lower() == "true"
```


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```
show_login = request.args.get("show_login",  
"false").lower() == "true"
```

```
return render_template(  
    "landing.html",  
    login_form=login_form,  
    register_form=register_form,  
    show_register=show_register,  
    show_login=show_login,  
)
```

```
@main_bp.route("/dashboard")  
  
@login_required  
  
def dashboard():  
    ph_tz = pytz.timezone("Asia/Manila")  
    now_ph = datetime.now(ph_tz)  
    now_naive = now_ph.replace(second=0, microsecond=0,  
tzinfo=None)  
  
    reservations = (
```

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```
Reservation.query.filter_by(user_id=current_user.id)

.order_by(Reservation.created_at.desc())

.limit(10)

.all()

)

latest_log = (

    TimeLog.query.filter_by(user_id=current_user.id)

    .order_by(TimeLog.time_in.desc())

    .first()

)

if current_user.role in ["admin", "staff"]:

    return redirect(url_for("admin.dashboard"))

# 1. Fetch Membership Data

membership =
Membership.query.filter_by(user_id=current_user.id).first()

attendance_logs = []

remaining_days = None
```

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```
if membership:

    _expire_membership_if_needed(membership)

    attendance_logs = (

AttendanceLog.query.filter_by(membership_id=membership.id)

        .order_by(AttendanceLog.check_in_time.desc())

        .limit(20)

        .all()

    )

    if membership.expiry_date:

        expiry_dt = membership.expiry_date

        if expiry_dt.tzinfo is None:

            expiry_dt = ph_tz.localize(expiry_dt)

        diff = expiry_dt - now_ph

        if diff.total_seconds() > 0:

            remaining_days =
max(round(diff.total_seconds() / 86400, 1), 0)

        else:

            remaining_days = 0
```

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```
# 2. Fetch Solo Plans
```

```
solo_plans = (  
  
    SoloPlan.query.filter_by(user_id=current_user.id)  
  
    .order_by(SoloPlan.created_at.desc())  
  
    .limit(5)  
  
    .all()  
  
)
```

```
# CALCULATE TOTAL SOLO HOURS (Safe Join & Status Check)
```

```
try:
```

```
    all_user_solo_plans =   
SoloPlan.query.join(Membership).filter(  
  
    Membership.user_id == current_user.id  
  
).all()
```

```
except Exception:
```

```
    all_user_solo_plans = SoloPlan.query.filter(  
  
    SoloPlan.user_id == current_user.id  
  
).all()
```

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```
total_solo_hours = 0.0

for plan in all_user_solo_plans:

    p_status      =      str(getattr(plan,      'status',
    '')).lower().strip()

    if p_status in ['approved', 'active', 'completed',
    'ended', 'checked_out', 'checked-out', 'used', 'expired']:

        p_hours = (

            getattr(plan, 'hours', None) or

            getattr(plan, 'duration_hours', None) or

            getattr(plan, 'duration', None) or

            getattr(plan, 'hours_left', 0) or 0

        )

        try:

            total_solo_hours += float(p_hours)

        except (ValueError, TypeError):

            pass

total_solo_hours = round(total_solo_hours, 2)
```

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```
# Strict Attendance Log & Is_Checked_In Validation para  
sa Active Session
```

```
open_log = None
```

```
if membership:
```

```
    open_log = AttendanceLog.query.filter_by(
```

```
        membership_id=membership.id,
```

```
        check_out_time=None
```

```
    ).first()
```

```
    is_user_checked_in = (membership and  
membership.is_checked_in) or (open_log is not None)
```

```
active_solo = None
```

```
if is_user_checked_in:
```

```
    active_solo = SoloPlan.query.filter(
```

```
        SoloPlan.user_id == current_user.id,
```

```
        SoloPlan.status.ilike("approved"),
```

```
        SoloPlan.expiry_date > now_naive,
```

```
        SoloPlan.status.notin_(["checked_out", "checked-  
out", "completed"]))
```

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```
        ).order_by(SoloPlan.created_at.desc()).first()

    active_session = None

    if is_user_checked_in:

        if active_solo:

            active_session = active_solo

            elif membership and membership.expiry_date and
membership.expiry_date > now_naive and membership.status not in
["checked_out", "completed"]:

                active_session = membership

        else:

            active_session = latest_log

    return render_template(

        "dashboard/member_dashboard.html",

        reservations=reservations,

        active_plan=active_session,

        active_session=active_session,

        remaining_days=remaining_days,

        solo_plans=solo_plans,
```

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```
total_solo_hours=total_solo_hours,

membership=membership,

attendance_logs=attendance_logs,

now_ph=now_ph,

)


@main_bp.route("/profile", methods=["GET", "POST"])

@login_required

def profile():

    profile_form = ProfileForm(obj=current_user)

    password_form = ChangePasswordForm()


    if request.method == "POST" and
request.form.get("profile_submit") is not None:

        if profile_form.validate_on_submit():

            email = profile_form.email.data.strip().lower()

            existing_user =
User.query.filter(func.lower(User.email) == email).first()
```


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```
        if existing_user and existing_user.id !=
current_user.id:

            profile_form.email.errors.append("Email
already in use by another account.")

        else:

            current_user.name =
profile_form.name.data.strip()

            current_user.email = email

            current_user.phone =
profile_form.phone.data.strip() or None

            db.session.commit()

            flash("Profile updated successfully.",
"success")

            return redirect(url_for("main.profile"))

    if request.method == "POST" and
request.form.get("password_submit") is not None:

        if password_form.validate_on_submit():

            if not
current_user.check_password(password_form.current_password.data):

password_form.current_password.errors.append("Current password
is incorrect.")
```

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```

        elif password_form.new_password.data !=
password_form.confirm_password.data:

password_form.confirm_password.errors.append("Passwords do not
match.")

        else:

current_user.set_password(password_form.new_password.data)

        db.session.commit()

        flash("Password changed successfully.",
"success")

        return redirect(url_for("main.profile"))

    return render_template("profile.html",
profile_form=profile_form, password_form=password_form)

@main_bp.route("/rooms", methods=["GET", "POST"])

@login_required

def rooms():

    ph_tz = pytz.timezone("Asia/Manila")
```

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```
form = ReservationForm()

rooms = Room.query.filter(~Room.name.ilike('Test
Room%')).order_by(Room.id).all()

now_ph = datetime.now(ph_tz).replace(tzinfo=None)

active_occupied_reservations = Reservation.query.filter(

    Reservation.room_id.isnot(None),

    Reservation.status.in_(["Confirmed", "Occupied",
"Walk-in", "Pending", "Checked-in"]),

    Reservation.start_time <= now_ph,

    or_(

        Reservation.end_time >= now_ph,

        Reservation.is_open_time == True

    )

).all()

room_pax_count = {}

occupied_room_ids = set()
```

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```
for res in active_occupied_reservations:

    if res.room_id:

        occupied_room_ids.add(res.room_id)

        room_pax_count[res.room_id] =
room_pax_count.get(res.room_id, 0) + (res.pax_count or 1)


available_rooms = []

for r in rooms:

    # Check kon DB status is available

    if r.status and r.status.lower() == "available":

        # Kon Common Area / Shared Space (Max Capacity
e.g. 70)

        if "common area" in r.name.lower():

            max_capacity = 70 # Ukon r.capacity kon may
column kamo sa DB

            current_pax = room_pax_count.get(r.id, 0)

            if current_pax < max_capacity:

                available_rooms.append(r)

        else:

            if r.id not in occupied_room_ids:
```

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```
        available_rooms.append(r)

    selected_room_id = None

    try:

        selected_room_id = int(form.room_id.data)    if
form.room_id.data is not None else None

    except (TypeError, ValueError):

        selected_room_id = None

    if selected_room_id is not None:

        selected_room = Room.query.get(selected_room_id)

        if selected_room and selected_room.id not in
{room.id for room in available_rooms}:

            available_rooms.append(selected_room)

    available_rooms = sorted(available_rooms, key=lambda
room: room.id)

    form.room_id.choices = [

        (room.id, f"{room.name} - ₱{room.base_rate}/hr") for
room in available_rooms

    ]
```

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```
payment_info = {

    info.method: {

        "account_number": info.account_number,

        "account_name": info.account_name,

        "qr_image":                                url_for("static",
filename=f"uploads/payment/{info.qr_image}")    if    info.qr_image
else None,

        "instructions": info.instructions,

    }

    for info in PaymentInfo.query.all()

}


bookings = (

    db.session.query(

        Reservation.start_time,        Reservation.end_time,
Reservation.status, Room.name

    )

    .join(Room)

    .filter(
```

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```
        Reservation.status == "Confirmed",

        ~Room.name.ilike('Test Room%'),

        Reservation.end_time >= now_ph

    )

    .order_by(Reservation.start_time)

    .all()

)

if form.validate_on_submit() and current_user.role ==
"member":

    payment_method = request.form.get("payment_method",
"")

    receipt_file = request.files.get("receipt_image")

    room = Room.query.get(form.room_id.data)

    start_time = form.start_time.data

    end_time = form.end_time.data

    is_open_time = form.is_open_time.data

    if payment_method not in ["GCash", "Maya"]:
```

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```
flash("Please select GCash or Maya as payment
method.", "danger")

return render_template("rooms.html", rooms=rooms,
bookings=bookings,      form=form,      payment_info=payment_info,
occupied_room_ids=occupied_room_ids)

if not receipt_file or receipt_file.filename == "":

    flash("Please upload your payment receipt for
verification.", "danger")

    return render_template("rooms.html", rooms=rooms,
bookings=bookings,      form=form,      payment_info=payment_info,
occupied_room_ids=occupied_room_ids)

ALLOWED_EXTENSIONS = {'.png', '.jpg', '.jpeg',
'.jpe', '.gif', '.webp', '.pdf'}

original_name = receipt_file.filename or ''

ext = os.path.splitext(original_name)[1].lower()

if ext not in ALLOWED_EXTENSIONS:

    flash("Unsupported receipt file type. Allowed:
PNG, JPG, JPEG, JPE, GIF, WEBP, PDF.", "danger")
```


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```
        return render_template("rooms.html", rooms=rooms,
bookings=bookings,        form=form,        payment_info=payment_info,
occupied_room_ids=occupied_room_ids)

    receipt_file.seek(0, os.SEEK_END)

    size = receipt_file.tell()

    receipt_file.seek(0)

    max_bytes = 5 * 1024 * 1024

    if size > max_bytes:

        flash("Receipt file too large (max 5MB).",
"danger")

        return render_template("rooms.html", rooms=rooms,
bookings=bookings,        form=form,        payment_info=payment_info,
occupied_room_ids=occupied_room_ids)

    if not room:

        flash("Please select a valid room.", "danger")

        return render_template("rooms.html", rooms=rooms,
bookings=bookings,        form=form,        payment_info=payment_info,
occupied_room_ids=occupied_room_ids)
```

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```
if not end_time and not is_open_time:

    flash("Please choose a valid reservation end
time.", "danger")

    return render_template("rooms.html", rooms=rooms,
bookings=bookings,      form=form,      payment_info=payment_info,
occupied_room_ids=occupied_room_ids)

if not end_time:

    end_time = start_time + timedelta(hours=8)

conflict = Reservation.query.filter(

    Reservation.room_id == form.room_id.data,

    Reservation.status.in_(["Pending",    "Confirmed",
"Occupied"])),

    or_(

        (Reservation.start_time  <=  start_time)    &
(Reservation.end_time >= start_time),

        (Reservation.start_time  <=  end_time)      &
(Reservation.end_time >= end_time),

        (start_time  <=  Reservation.start_time)    &
(end_time >= Reservation.end_time),

    ),
```

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```
) .first()

if conflict:

    flash("Room unavailable for selected time.",
"danger")

    return render_template("rooms.html", rooms=rooms,
bookings=bookings, form=form, payment_info=payment_info,
occupied_room_ids=occupied_room_ids)

hours = (end_time - start_time).total_seconds() /
3600

total = max(room.base_rate * hours, room.base_rate)

extra_fee_total = 0.0

try:

    extra_fee_total =
float(request.form.get('extra_fee_total', 0) or 0)

except ValueError:

    extra_fee_total = 0.0

total = round(total + extra_fee_total, 2)
```

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```
        payment_type = form.payment_type.data if
hasattr(form, 'payment_type') else "Downpayment"

        if payment_type == "Full Payment":

            amount_paid = round(total, 2)

        else:

            amount_paid = round(total * 0.5, 2)

    try:

        customer_id = generate_customer_id(

            "common area" if room.name.strip().lower()
== "common area" else "other"

        )

    except ValueError as e:

        flash(str(e), "danger")

        return render_template("rooms.html", rooms=rooms,
bookings=bookings, form=form, payment_info=payment_info,
occupied_room_ids=occupied_room_ids)

    now_timestamp = int(datetime.now(ph_tz).timestamp())

    filename = secure_filename(
```

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```
f"receipt_{current_user.id}_{now_timestamp}_{receipt_file.filename}"}

    )

    upload_folder = os.path.join(current_app.root_path,
    "static/uploads/receipts")

    os.makedirs(upload_folder, exist_ok=True)

    file_path = os.path.join(upload_folder, filename)

    receipt_file.save(file_path)


reservation = Reservation(

    user_id=current_user.id,

    customer_id=customer_id,

    room_id=form.room_id.data,

    customer_name=form.customer_name.data,

    contact_number=form.contact_number.data,

    pax_count=form.pax_count.data,

    start_time=start_time,

    end_time=end_time,

    status="Pending",
```

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```
        added_by=current_user.name,

        extra_notes=form.extra_notes.data,

        extra_fee=extra_fee_total,

        total_amount=total,

        amount_paid=amount_paid,

        payment_method=payment_method,

        payment_type=form.payment_type.data,

        receipt_image=filename,

        paid=False,

    )

    db.session.add(reservation)

    db.session.commit()

    flash(

        "Reservation created and payment receipt
uploaded. Awaiting admin approval.",

        "success",

    )

    return redirect(url_for("main.rooms"))
```

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```
return render_template(

    "rooms.html",

    rooms=rooms,

    bookings=bookings,

    form=form,

    payment_info=payment_info,

    occupied_room_ids=occupied_room_ids,

    room_pax_count=room_pax_count

)


@main_bp.route("/timelog")

@login_required

def timelog():

    ph_tz = pytz.timezone("Asia/Manila")

    now_ph = datetime.now(ph_tz)

    now_naive = now_ph.replace(second=0, microsecond=0,
tzinfo=None)

    month_start = ph_tz.localize(datetime(now_ph.year,
now_ph.month, 1))
```

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```
# 1. FETCH MEMBERSHIP & ATTENDANCE LOGS
```

```
membership =  
Membership.query.filter_by(user_id=current_user.id).first()
```

```
# 2. DYNAMIC ACTIVE SESSION CHECK (Checked-In Validation)
```

```
open_log = None
```

```
if membership:
```

```
    open_log = AttendanceLog.query.filter_by(  
        membership_id=membership.id,  
        check_out_time=None  
    ).first()
```

```
# Dapa checked in gid man sa Membership flag 0 may  
nabilin nga open AttendanceLog
```

```
is_checked_in = (membership and membership.is_checked_in)  
or (open_log is not None)
```

```
active_session = None
```

```
if is_checked_in:
```


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```
active_solo = SoloPlan.query.filter(

    SoloPlan.user_id == current_user.id,

    SoloPlan.status.ilike("approved"),

    SoloPlan.expiry_date > now_naive,

    SoloPlan.status.notin_(["checked_out", "checked-
out", "completed"])

).order_by(SoloPlan.id.desc()).first()


if active_solo:

    active_session = active_solo

    elif membership and membership.expiry_date and
membership.expiry_date > now_naive and membership.status not in
["checked_out", "completed"]:

        active_session = membership


logs = []

completed_sessions_count = 0

total_time_all = 0

total_time_month = 0

today_logs = 0
```

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```
plan_counts = {}

# 3. COMPUTE ATTENDANCE LOGS (If exists)

if membership:

    raw_logs = (

AttendanceLog.query.filter_by(membership_id=membership.id)

        .order_by(AttendanceLog.check_in_time.desc())

        .limit(10)

        .all()

    )

    for l in raw_logs:

        check_in = l.check_in_time

        if check_in and check_in.tzinfo is None:

            check_in = ph_tz.localize(check_in)

        if l.check_out_time:

            check_out = l.check_out_time

            if check_out.tzinfo is None:
```

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```
        check_out = ph_tz.localize(check_out)

        duration_hours = l.session_duration_hours or
((check_out - check_in).total_seconds() / 3600)

    else:

        duration_hours = max((now_ph -
check_in).total_seconds() / 3600, 0)

    p_name = membership.plan_name or 'INDIVIDUAL
RATE'

    plan_counts[p_name] = plan_counts.get(p_name, 0)
+ 1

    logs.append({

        'time_in': check_in,

        'time_out': l.check_out_time,

        'status': 'Ended' if l.check_out_time else
'Active',

        'plan': p_name,

        'total_time': int(duration_hours * 60),

    })
```

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```
completed_sessions_count =
AttendanceLog.query.filter(

    AttendanceLog.membership_id == membership.id,

    AttendanceLog.check_out_time != None

).count()


all_attendance_logs =
AttendanceLog.query.filter_by(membership_id=membership.id).all()


for log in all_attendance_logs:

    c_in = log.check_in_time

    if c_in and c_in.tzinfo is None:

        c_in = ph_tz.localize(c_in)


    if log.check_out_time:

        c_out = log.check_out_time

        if c_out.tzinfo is None:

            c_out = ph_tz.localize(c_out)

        dur = (c_out - c_in).total_seconds() / 3600

    else:
```

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```
dur = max((now_ph - c_in).total_seconds() /  
3600, 0)
```

```
dur_minutes = int(dur * 60)
```

```
total_time_all += dur_minutes
```

```
if c_in >= month_start:
```

```
    total_time_month += dur_minutes
```

```
if c_in.date() == now_ph.date():
```

```
    today_logs += dur_minutes
```

```
# 4. COMPUTE SOLO PLANS (If Attendance Logs are empty or  
to complement history)
```

```
solo_history =  
SoloPlan.query.filter_by(user_id=current_user.id).order_by(SoloP  
lan.created_at.desc()).all()
```

```
for sp in solo_history:
```

```
    p_name = sp.plan_name or 'INDIVIDUAL RATE'
```

```
    plan_counts[p_name] = plan_counts.get(p_name, 0) + 1
```

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```
        if not logs:      # Populated lang kun wala pa ma-
populate sang AttendanceLog

            is_ended = sp.status in ['completed', 'ended',
'checked_out'] or (sp.expiry_date and sp.expiry_date <=
now_naive)

            logs.append({

                'time_in': getattr(sp, 'created_at', None)
or getattr(sp, 'start_time', None),

                'time_out': sp.expiry_date if is_ended else
None,

                'status': 'Ended' if is_ended else 'Active',

                'plan': p_name,

                'total_time': 0,

            })

    if not membership and solo_history:

        completed_sessions_count = SoloPlan.query.filter(

            SoloPlan.user_id == current_user.id,

            SoloPlan.status.in_(['completed',          'ended',
'checked_out'])

        ).count()
```

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```
most_frequent_plan = db.session.query(

    SoloPlan.plan_name,

    func.count(SoloPlan.id).label('plan_count')

).filter(

    SoloPlan.user_id == current_user.id

).group_by(

    SoloPlan.plan_name

).order_by(

    db.desc('plan_count')

).first()

if most_frequent_plan:

    most_used_room = most_frequent_plan.plan_name

else:

    # Fallback sa plan_counts dict kon walang query
match

    most_used_room = max(plan_counts,
key=plan_counts.get) if plan_counts else "None"
```

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```
# CALCULATE AVG SESSION LENGTH (Safe Join sa Membership)

all_user_logs = AttendanceLog.query.join(Membership).filter(

    Membership.user_id == current_user.id

).all()

completed_session_minutes = []

for log in all_user_logs:

    c_in = getattr(log, 'check_in_time', None)

    c_out = getattr(log, 'check_out_time', None)

    if c_in and c_out:

        diff_seconds = (c_out - c_in).total_seconds()

        if diff_seconds > 0:

            completed_session_minutes.append(diff_seconds / 60)

        elif getattr(log, 'duration_minutes', 0) and \
            getattr(log, 'duration_minutes', 0) > 0:

            completed_session_minutes.append(float(log.duration_minutes))
```


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```
elif getattr(log, 'duration', 0) and getattr(log, 'duration', 0) > 0:
```

```
completed_session_minutes.append(float(log.duration))
```

```
if len(completed_session_minutes) > 0:
```

```
    avg_session_length =  
round(sum(completed_session_minutes) /  
len(completed_session_minutes))
```

```
else:
```

```
    avg_session_length = 0
```

```
totals = {
```

```
    "all_time_minutes": total_time_all,
```

```
    "month_time_minutes": total_time_month,
```

```
    "today_time_minutes": today_logs,
```

```
    "plan_totals": plan_counts,
```

```
    "most_used_room": most_used_room,
```

```
    "avg_session_length": avg_session_length
```

```
}
```

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```
return render_template(

    "timelog.html",

    logs=logs[:10],

    current_plan=membership,

    totals=totals,

    active_session=active_session,

    completed_sessions=completed_sessions_count

)


@main_bp.route("/timein", methods=["POST"])

@login_required

def time_in():

    ph_tz = pytz.timezone("Asia/Manila")

    now_ph = datetime.now(ph_tz)


    active = (

        TimeLog.query.filter_by(user_id=current_user.id)

        .filter(TimeLog.time_out.is_(None))
```

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```
        .first()

    )

    if active:

        flash("You are already timed in. Time out first.",
"warning")

        return redirect(url_for("main.timelog"))


approved_plan = (

    SoloPlan.query.filter(

        SoloPlan.user_id == current_user.id,

        SoloPlan.status == "approved",

        SoloPlan.expiry_date >
now_ph.replace(tzinfo=None) ,

    )

    .order_by(SoloPlan.created_at.desc())

    .first()

)

if not approved_plan:

    flash(
```

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```
        "No active approved solo plan found. Please  
select, get approval, and check expiry.",
```

```
        "danger",
```

```
    )
```

```
    return redirect(url_for("main.solo_rates"))
```

```
    timelog = TimeLog(
```

```
        user_id=current_user.id,
```

```
        plan=approved_plan.plan_name,
```

```
        time_in=now_ph,
```

```
    )
```

```
    db.session.add(timelog)
```

```
    db.session.commit()
```

```
    flash(f"Timed in with {approved_plan.plan_name}!",  
"success")
```

```
    return redirect(url_for("main.timelog"))
```

```
@main_bp.route("/timeout", methods=["POST"])
```

```
@login_required
```

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```
def time_out():

    ph_tz = pytz.timezone("Asia/Manila")

    now_ph = datetime.now(ph_tz)

    active = (

        TimeLog.query.filter_by(user_id=current_user.id)

        .filter(TimeLog.time_out.is_(None))

        .first()

    )

    if not active:

        flash("No active time session found. Time in first.",
"warning")

        return redirect(url_for("main.timelog"))

    # Synchronize active.time_in timezone if naive

    time_in_dt = active.time_in

    if time_in_dt and time_in_dt.tzinfo is None:

        time_in_dt = ph_tz.localize(time_in_dt)

    # Calculate exact duration in minutes based on PHT
```

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```
duration_seconds = (now_ph - time_in_dt).total_seconds()

duration = max(int(duration_seconds / 60), 0)

active.time_out = now_ph.replace(tzinfo=None)

active.total_time = duration

db.session.commit()


flash(f"Timed out. Session duration: {duration} minutes",
"info")

return redirect(url_for("main.timelog"))


@main_bp.route("/reservations")

@login_required

def reservations():

    if current_user.role != "member":

        flash("Members only")

        return redirect(url_for("main.dashboard"))


ph_tz = pytz.timezone("Asia/Manila")

now_ph = datetime.now(ph_tz)
```

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```
reservations = (  
  
    db.session.query(Reservation)  
  
    .join(Room)  
  
    .filter(Reservation.user_id == current_user.id)  
  
    .order_by(Reservation.start_time.desc())  
  
    .all()  
  
)  
  
for res in reservations:  
  
    end_time_dt = res.end_time  
  
    if end_time_dt and end_time_dt.tzinfo is None:  
  
        end_time_dt = ph_tz.localize(end_time_dt)  
  
  
    # Attach dynamic property for template rendering if  
needed  
  
    res.is_past = end_time_dt < now_ph if end_time_dt  
else False
```

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```
        return render_template("reservations.html",
reservations=reservations)
```

```
@main_bp.route("/get_time_inside")
```

```
@login_required
```

```
def get_time_inside():
```

```
    ph_tz = pytz.timezone("Asia/Manila")
```

```
    now_ph = datetime.now(ph_tz)
```

```
    latest = (
```

```
        TimeLog.query.filter_by(user_id=current_user.id)
```

```
        .order_by(TimeLog.time_in.desc())
```

```
        .first()
```

```
)
```

```
if latest and latest.time_out is None:
```

```
    # Localize time_in if timezone-naive
```

```
    time_in_dt = latest.time_in
```

```
    if time_in_dt.tzinfo is None:
```


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```
time_in_dt = ph_tz.localize(time_in_dt)

# Compute exact elapsed seconds based on PHT

diff = int((now_ph - time_in_dt).total_seconds())

diff = max(diff, 0) # Prevent negative values

return jsonify({"status": "inside", "seconds": diff})


return jsonify({"status": "outside", "seconds": 0})


@main_bp.route("/solo_rates", methods=["GET", "POST"])
@login_required
def solo_rates():

    if current_user.role != "member":

        flash("Unauthorized. Members only.", "warning")

        return redirect(url_for("main.dashboard"))


ph_tz = pytz.timezone("Asia/Manila")
```

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```
now_ph = datetime.now(ph_tz)

now_naive = now_ph.replace(tzinfo=None)


plans = Plan.query.all() if 'Plan' in globals() else []


# Check attendance / check-in flag sang membership

membership_record =
Membership.query.filter_by(user_id=current_user.id).first()

open_log = None

if membership_record:

    open_log = AttendanceLog.query.filter_by(

        membership_id=membership_record.id,

        check_out_time=None

    ).first()


# User is only considered checked-in if flag is true or
open log exists

is_user_checked_in = (membership_record and
membership_record.is_checked_in) or (open_log is not None)
```

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```
active_membership = None

if is_user_checked_in:

    active_membership = (

        Membership.query.filter(

            Membership.user_id == current_user.id,

            Membership.status.ilike("active"),

            Membership.status.notin_(["checked_out",
"checked-out", "completed"]),

            Membership.expiry_date > now_naive

        )

        .order_by(Membership.start_date.desc())

        .first()

    )

active_plan = active_membership

remaining_days = expiration = None

if active_plan and active_plan.expiry_date:

    expiration = active_plan.expiry_date
```

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```
if expiration.tzinfo is None:

    expiration = ph_tz.localize(expiration)

diff = expiration - now_ph

remaining_days = round(diff.total_seconds() / 86400,
1) if diff.total_seconds() > 0 else 0

active_solo_plan = None

if is_user_checked_in:

    active_solo_plan = (

        SoloPlan.query.filter(

            SoloPlan.user_id == current_user.id,

            SoloPlan.status.ilike("approved"),

            SoloPlan.expiry_date > now_naive,

            SoloPlan.status.notin_([ "checked_out",
"checked-out", "completed", "CHECKED_OUT" ])

        )

        .order_by(SoloPlan.created_at.desc())

        .first()

    )
```

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```
plans = [  
    {  
        "title": "INDIVIDUAL RATE",  
        "price": "P35/HR",  
        "details": ["Hourly common area usage", "WiFi /  
Charging"],  
    },  
    {  
        "title": "INDIVIDUAL RATE (4HRS)",  
        "price": "P100",  
        "details": ["4hrs common area usage", "WiFi &  
Charging"],  
    },  
    {  
        "title": "DAY/NIGHT PASS",  
        "price": "P200",  
        "details": [  
            "Choice of Day (7AM - 10PM) or Night (6PM -  
9AM) common area usage",
```

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```
        "WiFi & Charging",
    ],
},
{
    "title": "WEEKLY PASS (DAY/NIGHT)",
    "price": "P800",
    "details": [
        "1 Week (7DAYS) common area usage",
        "Choice of Day (7AM - 10PM) or Night (6PM -
9AM) ",
        "WiFi & Charging",
        "Gold Card",
    ],
},
{
    "title": "WEEKLY PASS (24HRS)",
    "price": "P1000",
    "details": [
        "1 Week (7DAYS) common area usage",
        "24/7 Usage",
    ],
}
```

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```
        "WiFi & Charging",
        "Platinum Card",
    ],
},
{
    "title": "MONTHLY PASS (DAY/NIGHT)",
    "price": "P1999",
    "details": [
        "1 Month (30 DAYS) unlimited common area
usage",
        "Choice of Day (7AM - 10PM) or Night (6PM -
9AM) ",
        "WiFi, Charging & Exclusive Locker",
        "Gold Card",
    ],
},
{
    "title": "MONTHLY PASS (24HRS)",
    "price": "P2500",
    "details": [
```

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```
"24/7 Access to All Branches in Iloilo",  
  
"1 Month (30 days) unlimited common area  
usage",  
  
"WiFi & Charging, Exclusive Locker",  
  
"1hr Sleeping pod & Shower Room use per day  
(City Proper Branch only)",  
  
"Platinum Card",  
  
],  
  
},  
  
{  
  
  "title": "WORKSTATION (24HRS)",  
  
  "price": "P3000",  
  
  "details": [  
  
    "24/7 Access to dedicated Workstation  
(Lapaz)",  
  
    "24/7 Common area access to all Branches",  
  
    "1 Dedicated Workstation",  
  
    "WiFi & Charging & Exclusive Locker",  
  
    "1hr Sleeping pod & Shower Room use per day  
(City Proper Branch only)",  
  
    "Platinum Card",
```


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```
        ],  
    },  
]  
  
has_pending = (  
    SoloPlan.query.filter_by(user_id=current_user.id,  
status="pending").first()  
    is not None  
)  
  
message = None  
  
if request.method == "POST":  
    selected_plan = request.form.get("plan_name")  
  
    if selected_plan:  
        if active_plan or active_solo_plan:  
            flash(  
                "You currently have an active plan.  
Please wait for it to expire before purchasing a new one.",  
                "warning",  
            )
```

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```
elif has_pending:

    flash(

        "You already have a pending plan
application. Please wait for admin approval.",

        "warning",

    )

else:

    try:

        customer_id =
generate_customer_id("monthly")

    except ValueError as e:

        flash(str(e), "danger")

    return
redirect(url_for("main.solo_rates"))


solo_plan = SoloPlan(

    user_id=current_user.id,

    customer_id=customer_id,

    plan_name=selected_plan,

    status="pending",
```

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```
        created_at=now_naive,

    )

    db.session.add(solo_plan)

    db.session.commit()


    message = (

        f"Plan      '{selected_plan}'      selected!
Waiting for admin approval. Your plan ID is: {customer_id}"

    )

    flash(message, "success")


return render_template(

    "solo_rates.html",

    plans=plans,

    active_plan=active_plan,

    remaining_days=remaining_days,

    expiration=expiration,

    message=message,

    active_solo_plan=active_solo_plan,

    has_pending=has_pending,
```

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```
)

# API Blueprint

api_bp = Blueprint("api", __name__)

@api_bp.route('/payment-info')

def payment_info():

    payment_data = {

        info.method: {

            'account_name': info.account_name,

            'account_number': info.account_number,

            'qr_image': url_for('static',
filename=f'uploads/payment/{info.qr_image}') if info.qr_image
else None,

            'instructions': info.instructions,

        }

        for info in PaymentInfo.query.all()

    }

    return jsonify(payment_data)
```

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```
@api_bp.route("/stats")

def stats():

    ph_tz = pytz.timezone("Asia/Manila")

    now_ph = datetime.now(ph_tz)

    total_members =
User.query.filter_by(role="member").count()

    active_plans = (

db.session.query(TimeLog).filter(TimeLog.time_out.is_(None)).cou
nt()

    )

    today_start = ph_tz.localize(datetime(now_ph.year,
now_ph.month, now_ph.day))

    today_end = today_start + timedelta(days=1)

    res_today = Reservation.query.filter(

        Reservation.start_time >= today_start,
```

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```
Reservation.start_time < today_end,  
  
Reservation.status == "Confirmed",  
  
) .count()  
  
revenue_today = (  
  
    db.session.query(func.sum(Reservation.total_amount))  
  
    .filter(  
  
        Reservation.start_time >= today_start,  
  
        Reservation.start_time < today_end,  
  
        Reservation.status == "Confirmed",  
  
    )  
  
    .scalar()  
  
    or 0  
  
)  
  
return jsonify(  
  
    {  
  
        "total_members": total_members,  
  
        "active_timelogs": active_plans,
```

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```
        "reservations_today": res_today,

        "revenue_today": float(revenue_today),

    }

)

@api_bp.route("/submit-solo-payment", methods=["POST"])

@login_required

def submit_solo_payment():

    """Handle GCash/Maya Receipt Uploads for Solo Plans"""

    ph_tz = pytz.timezone("Asia/Manila")

    now_ph = datetime.now(ph_tz)

    plan_name = request.form.get('plan_name')

    payment_method = request.form.get('payment_method')

    receipt_file = request.files.get('receipt_image')

    if not receipt_file or not plan_name:

        return jsonify({'success': False, 'message':
'Missing data or receipt.'}), 400
```

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```
        allowed_ext = ['.png', '.jpg', '.jpeg', '.jpe', '.webp',  
' .gif', '.pdf']  
  
        original_name = receipt_file.filename or ''  
  
        ext = os.path.splitext(original_name)[1].lower() or  
' .png'  
  
        if ext not in allowed_ext:  
  
            return jsonify({'success': False, 'message':  
'Unsupported file type.'}), 400  
  
        receipt_file.seek(0, os.SEEK_END)  
  
        size = receipt_file.tell()  
  
        receipt_file.seek(0)  
  
        if size > 5 * 1024 * 1024:  
  
            return jsonify({'success': False, 'message': 'File  
too large (max 5MB).'}), 400  
  
        mimetype = receipt_file.mimetype or ''  
  
        if ext == '.pdf':  
  
            if mimetype != 'application/pdf':
```


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```
        return jsonify({'success': False, 'message':
'Uploaded file is not a valid PDF.'}), 400

    else:

        if not mimetype.startswith('image/'):

            return jsonify({'success': False, 'message':
'Uploaded file is not a valid image.'}), 400

    try:

        timestamp = int(now_ph.timestamp())

        filename =
secure_filename(f"receipt_{current_user.id}_{timestamp}{ext}")

        upload_folder = os.path.join(current_app.root_path,
'static/uploads/receipts')

        if not os.path.exists(upload_folder):

            os.makedirs(upload_folder)

        file_path = os.path.join(upload_folder, filename)

        receipt_file.save(file_path)

    # Generate Customer ID for the Solo Plan
```

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```
customer_id = generate_customer_id("monthly")

new_plan = SoloPlan(

    user_id=current_user.id,

    customer_id=customer_id,

    plan_name=plan_name,

    status="pending",

    receipt_image=filename,

    payment_method=payment_method,

    created_at=now_ph

)

db.session.add(new_plan)

db.session.commit()

return jsonify({'success': True, 'message': 'Payment
submitted for verification.'})

except Exception as e:

    db.session.rollback()
```

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```
        return jsonify({'success': False, 'message':  
str(e)}), 500
```

```
@api_bp.route("/create-checkout-session", methods=["POST"])
```

```
@login_required
```

```
def create_checkout_session():
```

```
    import stripe
```

```
    from database_fixed import Config
```

```
    stripe.api_key = Config.STRIPE_SECRET_KEY
```

```
    ph_tz = pytz.timezone("Asia/Manila")
```

```
    now_ph = datetime.now(ph_tz)
```

```
    data = request.json or {}
```

```
    amount = int(data.get('amount', 0) * 100)
```

```
    plan_name = data.get('plan_name', 'Reservation')
```

```
    customer_email = current_user.email
```

```
    try:
```

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```
session = stripe.checkout.Session.create(

    payment_method_types=['card'],

    line_items=[{

        'price_data': {

            'currency': 'php',

            'product_data': {

                'name': f"{plan_name} Payment"

            },

            'unit_amount': amount,

        },

        'quantity': 1,

    ]],

    mode='payment',

    success_url=url_for('main.solo_rates',
_external=True) + '?session_id={CHECKOUT_SESSION_ID}',

    cancel_url=url_for('main.solo_rates',
_external=True),

    customer_email=customer_email,

    metadata={

        'user_id': current_user.id,
```

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```
        'plan': plan_name,

        'created_at_pht':      now_ph.strftime('%Y-%m-
%d %H:%M:%S'),

        'timestamp_pht': int(now_ph.timestamp())

    }

)

return jsonify({'id': session.id})

except Exception as e:

    return jsonify({'error': str(e)}), 400


@main_bp.route("/checkout_solo_plan", methods=["POST"])

@login_required

def checkout_solo_plan():

    ph_tz = pytz.timezone("Asia/Manila")

    now_ph = datetime.now(ph_tz)

    now_naive    =    now_ph.replace(second=0,    microsecond=0,
tzinfo=None)


# 1. Update Membership Check-in status
```

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```
membership = Membership.query.filter_by(user_id=current_user.id).first()

if membership:

    membership.is_checked_in = False

    membership.is_checked_out = True

    membership.is_status = "active"

# Close open attendance log

open_log = AttendanceLog.query.filter(

    AttendanceLog.membership_id == membership.id,

    AttendanceLog.check_out_time.is_(None)

).order_by(AttendanceLog.check_in_time.desc()).first

()

if open_log:

    open_log.check_out_time = now_naive

    c_in = open_log.check_in_time

    if c_in and c_in.tzinfo is None:

        c_in = ph_tz.localize(c_in)
```

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```
        dur_hours = max((now_ph - c_in).total_seconds()  
/ 3600, 0) if c_in else 0.0
```

```
        open_log.hours_deducted = round(dur_hours, 2)
```

```
        user_plans = SoloPlan.query.filter(  
            SoloPlan.user_id == current_user.id,  
            SoloPlan.status.in_(["Approved", "approved",  
"APPROVED", "Pending", "pending"])  
        ).all()
```

```
        for plan in user_plans:
```

```
            plan.status = "checked_out"
```

```
        db.session.commit()
```

```
        flash("Successfully checked out!", "success")
```

```
        return redirect(url_for("main.solo_rates"))
```

```
@api_bp.route("/membership/status", methods=["GET"])
```

```
@login_required
```

```
def membership_status():
```

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```
"""Get current user's membership status"""

ph_tz = pytz.timezone("Asia/Manila")

now_ph = datetime.now(ph_tz)


membership =
Membership.query.filter_by(user_id=current_user.id).first()


if not membership:

    return jsonify({"status": "error", "message": "No
membership found"})


_expire_membership_if_needed(membership)


expiry_iso = None

if membership.expiry_date:

    expiry_dt = membership.expiry_date

    if expiry_dt.tzinfo is None:

        expiry_dt = ph_tz.localize(expiry_dt)

    expiry_iso = expiry_dt.isoformat()
```


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```
return jsonify({

    "status": "success",

    "hours_left": membership.hours_left,

    "is_checked_in": membership.is_checked_in,

    "is_active": membership.is_active,

    "plan_name": membership.plan_name,

    "expiry_date": expiry_iso,

    "accumulated_hours": membership.accumulated_hours

})


@api_bp.route("/membership/current-session", methods=["GET"])

@login_required

def membership_current_session():

    """Get current user's active session (check-in time)"""

    ph_tz = pytz.timezone("Asia/Manila")

    now_ph = datetime.now(ph_tz)

    membership = Membership.query.filter_by(user_id=current_user.id).first()
```

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```
if not membership:

    return jsonify({"status": "error", "message": "No
membership found"})

_expire_membership_if_needed(membership)

if not membership.is_checked_in:

    return jsonify({"status": "error", "message": "No
active session"})

# Get the most recent attendance log that doesn't have a
check_out_time

current_session = (

    AttendanceLog.query.filter(

        AttendanceLog.membership_id == membership.id,

        AttendanceLog.check_out_time.is_(None)

    )

    .order_by(AttendanceLog.check_in_time.desc())

    .first())
```

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```
)

if not current_session:

    return jsonify({"status": "error", "message": "No
active session found"})

# Localize check_in_time and compute exact live duration
in seconds

check_in_dt = current_session.check_in_time

if check_in_dt and check_in_dt.tzinfo is None:

    check_in_dt = ph_tz.localize(check_in_dt)

elapsed_seconds = int((now_ph -
check_in_dt).total_seconds())

elapsed_seconds = max(elapsed_seconds, 0)

return jsonify({

    "status": "success",

    "check_in_time": check_in_dt.isoformat(),

    "elapsed_seconds": elapsed_seconds,
```

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```
"membership_id": membership.id,  
  
"hours_left": membership.hours_left  
  
})
```

5. Main dashboard and UI templates

admin_dashboard.html

```
{% extends "base.html" %}  
  
  
  
{% block extra_css %}  
  
    <link    rel="stylesheet"    href="{{    url_for('static',  
filename='css/admin/admin_dashboard.css')    }}?v=3">  
  
    <link    rel="stylesheet"    href="{{    url_for('static',  
filename='css/modals/admin_walkin_modal.css')    }}?v=3">  
  
    <link    rel="stylesheet"    href="{{    url_for('static',  
filename='css/rooms.css')    }}">  
  
{% endblock %}  
  
  
  
{% block content %}  
  
<div class="dashboard-wrapper">  
  
    <div class="dashboard-header">
```

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```
<div>

    <h1>Admin Dashboard</h1>

    <h2      id="adminWelcome"      class="welcome-
text">Welcome                      Back,          <span
id="adminUserName">{{ current_user.name }}</span>!</h2>

</div>

<div      id="dashboardClock"      class="dashboard-
clock">00:00:00</div>

</div>

<div class="stats-grid">

    <div class="stat-card">

        <div class="stat-icon-wrapper">

        </div>

        <div class="stat-info">

            <span                      class="stat-label">Total
Members</span>

            <h2                      class="stat-
value">{{ total_members }}</h2>
```

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```
</div>

</div>

<div class="stat-card">

    <div class="stat-icon-wrapper">

    </div>

    <div class="stat-info">

        <span class="stat-label">Active Plans</span>

        <h2                                class="stat-
value">{{ active_plans }}</h2>

    </div>

</div>

<div class="stat-card">

    <div class="stat-icon-wrapper">

    </div>

    <div class="stat-info">
```

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```

        <span                                class="stat-label">Pending
Reservations</span>

        <h2                                class="stat-
value">{{ reservations_today }}</h2>

    </div>

</div>

<div class="stat-card">

    <div class="stat-icon-wrapper">

    </div>

    <div class="stat-info">

        <span                                class="stat-label">Earnings
Today</span>

        <h2                                class="stat-
value">P{{ "{:,.2f}".format(revenue_today) }}</h2>

    </div>

</div>

</div>

<div class="main-dashboard-layout">
```

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```
<div class="left-section">

    <div class="section-header">

        <h2 class="section-title">Live Room
        Status</h2>

        <div class="search-container">

            <input type="text" id="dashboardSearch"
placeholder="Search customer or room..." class="search-input">

        </div>

    </div>

    <!-- Slot Occupancy Summary -->

    <div class="occupancy-summary-section">

        <div class="summary-card">

            <div class="summary-item">

                <span class="summary-label">Common
                Area - Occupied</span>

                <span class="summary-
value">{{ common_area_occupancy }}</span>

            </div>

            <div class="summary-divider"></div>

            <div class="summary-item">
```


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```

                                <span      class="summary-label">Common
Area - Available</span>

                                <span                                class="summary-
value">{{ common_area_available_slots }}</span>

                                </div>

                                </div>

                                </div>

                                <div class="room-grid common-area-grid">

                                    <div class="room-card dashboard-card common-
area-summary" data-room="common area">

                                        <div class="room-card-header">

                                            <span      class="room-name">Common
Area</span>

                                            <span                                class="status-pill
{{ 'occupied'   if   common_area_available_slots   ==   0   or
common_area_occupancy > 0 else 'available' }}">

                                                {{                                'Full'                                if
common_area_available_slots   ==   0   else   ('Occupied'   if
common_area_occupancy > 0 else 'Available') }}

                                            </span>

                                        </div>

```

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```
<div class="room-card-body">

    <div class="info-row">

        <span                                class="info-
label">Occupied</span>

        <span                                class="info-
value">{{ common_area_occupancy }}</span>

    </div>

    <div class="info-row">

        <span                                class="info-
label">Available Slots</span>

        <span                                class="info-
value">{{ common_area_available_slots }}</span>

    </div>

    {% if common_area_occupancy == 0 %}

    <div class="info-row name-row">

        <span                                class="info-
label">Info:</span>

        <span                                class="info-value">No
active Common Area sessions</span>

    </div>

    {% endif %}

</div>
```

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</div>

```
{% if common_area_reservations %}

    {% for res in common_area_reservations |
sort(attribute='end_time') %}

        {% set is_expired = false %}

        {% if res.end_time and not
res.is_open_time and res.end_time < now %}

            {% set is_expired = true %}

        {% endif %}

        <div class="room-card dashboard-card
{{ 'session-expired' if is_expired }}"

            data-
customer="{{ res.customer_name|lower }}"

            data-
room="{{ res.room.name|lower }}"

            data-isopentime="{{ 'true' if
res.is_open_time else 'false' }}"

            data-
endtime="{{ res.end_time.isoformat() if res.end_time else
'' }}">
```

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```
        <div class="end-time-indicator">TIME  
ENDED - PLEASE CHECKOUT</div>
```

```
        <div class="room-card-header">  
            <span class="room-  
name">{{ res.room.name }}</span>  
            <span class="status-pill  
occupied">Occupied</span>  
        </div>
```

```
        <div class="room-card-body">  
            <div class="info-row">  
                <span class="info-  
label">Room</span>  
                <span class="info-  
value">{{ res.room.name }}</span>  
            </div>
```

```
            <div class="info-row">  
                <span class="info-  
label">Start</span>  
                <span class="info-  
value">{{ res.start_time.strftime('%I:%M %p') }}</span>
```

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```
</div>

{% if not res.is_open_time %}

<div class="info-row">

    <span                class="info-
label">End</span>

    <span                class="info-
value">{{ res.end_time.strftime('%I:%M %p') if res.end_time else
'---' }}</span>

</div>

{% endif %}

{% if res.is_open_time %}

<div class="info-row">

    <span                class="info-
label">Live Timer</span>

    <span class="timer-text"

        data-
starttime="{{ res.start_time.isoformat() }}"

        data-isopentime="true">

        (00:00:00)
```

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```

        </span>

    </div>

    {% endif %}

    <div class="info-row name-row">

        <span                class="info-
label">Name:</span>

        <span                class="info-
value">{{ res.customer_name }}</span>

    </div>

    <div class="info-row">

        <span                class="info-
label">Customer ID:</span>

        <span                class="info-
value">{{ res.customer_id or 'N/A' }}</span>

    </div>

</div>

<div class="room-card-footer card-
action-row">

    {% if not res.is_open_time %}
```

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```

                                <button      class="btn-end-session
add-time-btn" type="button"

                                data-reservation-

id="{{ res.id }}"

                                data-customer-

name="{{ res.customer_name }}"

                                data-room-

name="{{ res.room.name }}"

                                data-start-

time="{{ res.start_time.isoformat() }}"

                                data-end-

time="{{ res.end_time.isoformat() if res.end_time else '' }}"

                                data-room-

rate="{{ res.room.base_rate or 0 }}"

                                data-total-

amount="{{ res.total_amount or 0 }}"

                                data-is-open-

time="{{ 'true' if res.is_open_time else 'false' }}">Add
Time</button>

                                {% endif %}

                                <button      class="btn-end-session

end-session-btn"

                                data-reservation-

id="{{ res.id }}"
```

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```

data-customer-
name="{{ res.customer_name }}"

data-room-
name="{{ res.room.name }}"

data-start-
time="{{ res.start_time.isoformat() }}"

data-room-
rate="{{ res.room.base_rate or 0 }}"

data-extra-
fee="{{ res.extra_fee or 0 }}"

data-total-
amount="{{ res.total_amount or 0 }}"

data-is-open-
time="{{ 'true' if res.is_open_time else 'false' }}">End
Session</button>

<form method="post"
action="{{ url_for('admin.delete_reservation',
res_id=res.id) }}" class="confirm-delete-form">

<button type="submit"
class="btn-end-session btn-error">X</button>

</form>

</div>
```


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```
</div>

{% endfor %}

{% endif %}

</div>


<div class="room-grid" id="roomGridContainer">

    {%      for      room      in      rooms      if
room.name.strip().lower() != 'common area' %}

        {%      set      active_res      =
room_reservations[room.id] if room.id in room_reservations else
none %}

        {% set is_expired = false %}

        {% if active_res and active_res.end_time
and not active_res.is_open_time %}

            {% if active_res.end_time < now %}

                {% set is_expired = true %}

            {% endif %}

        {% endif %}

    {% endif %}


    <div      class="room-card      dashboard-card
{{ 'session-expired' if is_expired }}"
```

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```
data-
customer="{{ active_res.customer_name|lower if active_res else
'' }}"
```

```
data-
room="{{ room.name.strip().lower() }}"
```

```
data-
endtime="{{ active_res.end_time.isoformat() if (active_res and
active_res.end_time) else '' }}"
```

```
data-isopentime="{{ 'true' if
(active_res and active_res.is_open_time) else 'false' }}"
```

```
<div class="end-time-indicator">TIME
ENDED - {{ room.name }}</div>
```

```
<div class="room-card-header">

    <span class="room-
name">{{ room.name }}</span>

    <span class="status-pill
{{ 'occupied' if active_res else 'available' }}">

        {{ 'Occupied' if active_res
else 'Available' }}

    </span>

</div>
```

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```
<div class="room-card-body">

    <div class="info-row">

        <span                class="info-
label">Status</span>

        <span                class="status-text
{{ 'occupied' if active_res else 'available' }}">

            {{                'Occupied'        if
active_res else 'Available' }}

        </span>

    </div>

    <div class="info-row">

        <span                class="info-
label">Start</span>

        <span                class="info-
value">{{                active_res.start_time.strftime('%I:%M
%p')                if
active_res else '---' }}</span>

    </div>

    {% if active_res %}

        {%                if                not
active_res.is_open_time %}
```

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```
<div class="info-row">

    <span          class="info-
label">End</span>

    <span          class="info-
value">{{          active_res.end_time.strftime('%I:%M          %p')          if
active_res.end_time else '---' }}</span>

</div>

{% else %}

<div class="info-row">

    <span          class="info-
label">Live Timer</span>

    <span class="timer-text"

        data-
starttime="{{ active_res.start_time.isoformat() }}"

        data-
isopentime="true">

            (00:00:00)

    </span>

</div>

{% endif %}

{% else %}

<div class="info-row">
```

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```

                                <span                class="info-
label">End</span>

                                <span  class="info-value">---
</span>

                                </div>

                                {% endif %}

                                <div class="info-row name-row">

                                <span                class="info-
label">Name:</span>

                                <span                class="info-
value">{{  active_res.customer_name  if  active_res  else  '---
'  }}</span>

                                </div>

                                </div>

                                <div class="room-card-footer">

                                {% if active_res %}

                                <div                class="card-action-
row">

                                <button  class="btn-end-
session add-time-btn" type="button"

```

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```

data-
reservation-id="{{ active_res.id }}"

data-customer-
name="{{ active_res.customer_name }}"

data-room-
name="{{ room.name }}"

data-start-
time="{{ active_res.start_time.isoformat() }}"

data-end-
time="{{ active_res.end_time.isoformat() if active_res.end_time
else '' }}"

data-room-
rate="{{ active_res.room.base_rate or 0 }}"

data-total-
amount="{{ active_res.total_amount or 0 }}"

data-extra-
fee="{{ active_res.extra_fee or 0 }}"

data-is-open-
time="{{ 'true' if active_res.is_open_time else 'false' }}">

Add Time

</button>

<button class="btn-end-
session end-session-btn"
```

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```

data-
reservation-id="{{ active_res.id }}"

data-customer-
name="{{ active_res.customer_name }}"

data-room-
name="{{ room.name }}"

data-start-
time="{{ active_res.start_time.isoformat() }}"

data-room-
rate="{{ active_res.room.base_rate or 0 }}"

data-extra-
fee="{{ active_res.extra_fee or 0 }}"

data-total-
amount="{{ active_res.total_amount or 0 }}"

data-is-open-
time="{{ 'true' if active_res.is_open_time else 'false' }}">End
Session</button>

<form          method="post"
action="{{          url_for('admin.delete_reservation',
res_id=active_res.id) }}" class="confirm-delete-form">

<button
type="submit" class="btn-end-session btn-error">X</button>

</form>
```

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</div>

{% endif %}

</div>

</div>

{% endfor %}

</div>

<div id="noSearchResults" class="no-results-message">No matching customer or room found.</div>

</div>

<div class="right-section">

<div class="walkin-header">

<button type="button" id="openWalkinBtn" class="btn-walkin">Walk-in</button>

</div>

<div class="waiting-list-card">

<div class="waiting-header">

<h3>Waiting List - Today</h3>

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```

    </div>

    <div class="waiting-items">

        {% for res in pending_reservations %}

            <div class="waiting-item">

                <p                class="waiting-name">Next:
{{ res.customer_name }}</p>

                <p                class="waiting-
details">({{                res.room.name                }},
{{                res.start_time.strftime('%I:%M')                }}                -
{{                res.end_time.strftime('%I:%M    %p')                }}                if                res.end_time                else
'Open' }})</p>

            </div>

            {% else %}

                <p                class="no-data">No                upcoming
reservations.</p>

            {% endfor %}

        </div>

    </div>
</div>

```

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```
<div class="common-area-occupants-card">

    <div class="occupants-header">

        <h3>👥 Common Area Members</h3>

        <span                                class="occupants-count"
id="occupantsCount">0</span>

    </div>

    <div                                class="occupants-list"
id="occupantsList">

        <p class="loading-text">Loading...</p>

    </div>

</div>

</div>

</div>

</div>

<div id="walkinModal" class="modal-overlay">

    <div class="modal-container">

        <div class="modal-form-side">
```

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`<h3 class="section-title">Walk-in Check-in</h3>`

`<p class="text-muted modal-helper-text">Use this
form for immediate lounge access.</p>`

`<form id="walkinForm" method="POST"
action="{{ url_for('admin.walkin_checkin_modal') }}">`

`{{ form.hidden_tag() }}`

`<input type="hidden" name="total_price"
id="total_price" value="0">`

`<div class="input-grid-2">`

`<div class="input-group">`

`<label class="form-label">Customer
Name</label>`

`{{ form.customer_name(class_="modern-input", id="f-name",
placeholder="e.g Darwen Salcedo") }}`

`</div>`

`<div class="input-group">`

`<label class="form-label">Contact
Number</label>`

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```
{{    form.contact_number(class_="modern-input",    id="f-contact",  
placeholder="e.g 0912-123-4567")  }}
```

```
</div>
```

```
</div>
```

```
<div class="input-grid-2">
```

```
<div class="input-group">
```

```
<label        class="form-label">Select  
Area</label>
```

```
{{        form.room_id(class_="modern-  
select", id="f-area")  }}
```

```
</div>
```

```
</div>
```

```
<div class="input-grid-3">
```

```
<div class="input-group">
```

```
<label        class="form-label">Start  
Date/Time</label>
```

```
<input  type="datetime-local"  id="f-  
start-time" class="modern-input" name="start_time">
```

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```
</div>

<div class="input-group">

    <label class="form-label" for="f-
duration-select">Duration</label>

    <select id="f-duration-select"
class="modern-select">

        {% for hours in range(1, 25) %}

            <option
value="{{ hours }}">{{ hours }} hour{{ 's' if hours > 1 else
' ' }}</option>

        {% endfor %}

    </select>

    <input type="datetime-local" id="f-
end-display" class="modern-input mt-10" readonly
placeholder="Auto-calculated end time">

    <input type="hidden" id="f-end-time"
name="end_time">

</div>

<div class="input-group align-items-
center">

    <label class="form-label">Open
Time</label>

    <label class="switch">
```

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```

                                <input    type="checkbox"    id="f-
open-time" name="open_time">

                                <span                                class="slider
round"></span>

                                </label>

                                </div>

                                </div>

                                </div>

                                <div class="input-group">

                                    <label    class="form-label    form-label-
uppercase">Staff Notes & Extra Fees</label>

                                    <label        class="form-label">Predefined
Add-ons</label>

                                    <select            id="f-predefined-addon"
name="predefined_addon_name" class="modern-select">

                                        <option value="None">None</option>

                                        <option value="Projector" data-unit-
price="100">Projector (+P100/3hrs)</option>

                                        <option value="Extra Chairs" data-
unit-price="0">Extra Chairs (Free)</option>

                                        <option value="Extension Cord" data-
unit-price="0">Extension Cord (Free)</option>

```

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```
                <option    value="Microphone"    data-
unit-price="0">Microphone (Event Room only)</option>
```

```
                <option    value="Speaker"    data-unit-
price="0">Speaker (Event Room only)</option>
```

```
            </select>
```

```
        </div>
```

```
        <div class="input-group">
```

```
                <label                                class="form-
label">Quantity</label>
```

```
                <input    type="number"    id="f-predefined-
addon-qty"    name="predefined_addon_qty"    class="modern-input"
min="1" value="1" step="1" placeholder="1">
```

```
                <small    class="text-muted">Empty/invalid
values become 1.</small>
```

```
        </div>
```

```
                <input    type="hidden"    name="addons_json"
id="addons_json" value="[]">
```

```
                <input    type="hidden"    name="addon_subtotal"
id="addon_subtotal" value="0.00">
```

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```
<div class="input-group">

    <label class="form-label">Add-on Request
    (Staff Notes)</label>

    {{      form.extra_notes(class_="modern-
input", id="f-notes", rows="2", placeholder="e.g. Extra monitor,
HDMI cable...") }}

</div>
```

```
<div class="input-group half-width">

    <label      class="form-label">Additional
Fees (₱)</label>

    {{  form.extra_fee(class_="modern-input",
id="f-fees", placeholder="0.00") }}

</div>
```

```
</form>

</div>
```

```
<div class="modal-preview-side">

    <div class="preview-border-box">

        <h3      class="section-title">Check-in
Details</h3>
```


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```
<div class="preview-list">

    <div class="preview-item"><span
class="preview-label">Customer Name:</span> <span id="p-
customer">----</span></div>

    <div class="preview-item"><span
class="preview-label">Contact No:</span> <span id="p-contact">--
--</span></div>

    <div class="preview-item"><span
class="preview-label">Area:</span> <span id="p-area">----
</span></div>

    <div class="preview-item"><span
class="preview-label">Start Time:</span> <span id="p-start">----
</span></div>

    <div class="preview-item"><span
class="preview-label">End Time:</span> <span id="p-end">----
</span></div>

    <div class="preview-item"><span
class="preview-label">Duration:</span> <span id="p-duration">---
-</span></div>

    <div class="preview-item"><span
class="preview-label">Selected Add-on Total:</span> <span id="p-
addon-total-display">P0.00</span></div>
```

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```

                <div
                    class="preview-item"><span
class="preview-label">Additional Fees:</span> <span id="p-fees-
display">P0.00</span></div>

```

```

        </div>

```

```

        <div class="discount-row">

```

```

            <label>Discount:</label>

```

```

                {{
                    form.discount(class_="discount-
select", id="f-discount") }}

```

```

        </div>

```

```

        <div class="total-box-updated">

```

```

            <p class="total-label">TOTAL PAYABLE</p>

```

```

            <h2 class="total-amount-updated" id="p-
total">P0.00</h2>

```

```

        <div class="modal-btn-group-row">

```

```

                <button
                    type="submit"
form="walkinForm"
                    class="btn-confirm-blue">Confirm
                    &
Save</button>

```

```

                <button
                    type="button"
id="closeModalBtn"
                    class="btn-cancel-red">Cancel</button>

```

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```

    </div>

  </div>

</div>

</div>

</div>

</div>

<div id="addTimeModal" class="modal-overlay">

  <div class="modal-container">

    <div class="modal-form-side">

      <h3 class="section-title" id="extendModalTitle"
style="font-size: 1.25rem; font-weight: 700; color: #1e293b;
margin-bottom: 0.25rem;">Extend Session</h3>

      <p class="text-muted modal-helper-text"
id="extendModalNote" style="margin-bottom: 1.5rem; font-size:
0.85rem; color: #64748b;">Add extra time or billing adjustments
for active sessions.</p>

      <!-- Customer -->

      <div class="input-group" style="display:
flex !important; justify-content: space-between !important;
```

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```
align-items: center !important; width: 100% !important; margin-  
bottom: 0.85rem !important;">"
```

```
        <label class="form-label" style="position:  
relative !important; float: none !important; margin:  
0 !important; color: #64748b !important; font-size:  
0.8rem !important; font-weight: 600 !important; text-transform:  
uppercase;">">Customer</label>
```

```
        <p id="extendCustomer" class="info-value"  
style="position: relative !important; float: none !important;  
margin: 0 !important; font-weight: 700 !important; color:  
#0f172a !important;">">---</p>
```

```
</div>
```

```
<!-- Room -->
```

```
        <div class="input-group" style="display:  
flex !important; justify-content: space-between !important;  
align-items: center !important; width: 100% !important; margin-  
bottom: 0.85rem !important;">"
```

```
        <label class="form-label" style="position:  
relative !important; float: none !important; margin:  
0 !important; color: #64748b !important; font-size:  
0.8rem !important; font-weight: 600 !important; text-transform:  
uppercase;">">Room</label>
```

```
        <p id="extendRoom" class="info-value"  
style="position: relative !important; float: none !important;
```

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```
margin: 0 !important; font-weight: 700 !important; color:
#0f172a !important;">---</p>
```

```
</div>
```

```
<!-- Current End Time -->
```

```
<div      class="input-group"      style="display:
flex !important; justify-content: space-between !important;
align-items: center !important; width: 100% !important; margin-
bottom: 1rem !important;">
```

```
<label  class="form-label"  style="position:
relative !important; float: none !important; margin:
0 !important; color: #64748b !important; font-size:
0.8rem !important; font-weight: 600 !important; text-transform:
uppercase;">Current End Time</label>
```

```
<p id="extendCurrentEnd" class="info-value"
style="position: relative !important; float: none !important;
margin: 0 !important; font-weight: 700 !important; color:
#0f172a !important;">---</p>
```

```
</div>
```

```
<!-- Add Hours Input Box -->
```

```
<div      class="input-group"      style="display:
flex !important; flex-direction: column !important; align-items:
flex-start !important; width: 100% !important; margin-bottom:
1rem !important;">
```

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```
        <label class="form-label" style="position:
relative !important; float: none !important; margin-bottom:
0.35rem !important; color: #64748b !important; font-size:
0.8rem !important; font-weight: 600 !important; text-transform:
uppercase;">Add Hours</label>
```

```
        <input type="number" id="extendHoursInput"
class="modern-input" min="1" step="1" value="1" style="width:
100% !important; padding: 0.5rem 0.75rem !important; border: 1px
solid #cbd5e1 !important; border-radius: 6px !important;">
```

```
    </div>
```

```
    <!-- Rate per Hour -->
```

```
        <div class="input-group" style="display:
flex !important; justify-content: space-between !important;
align-items: center !important; width: 100% !important; margin-
bottom: 0.85rem !important;">
```

```
            <label class="form-label" style="position:
relative !important; float: none !important; margin:
0 !important; color: #64748b !important; font-size:
0.8rem !important; font-weight: 600 !important; text-transform:
uppercase;">Rate per Hour</label>
```

```
            <p id="extendRate" class="info-value"
style="position: relative !important; float: none !important;
margin: 0 !important; font-weight: 700 !important; color:
#0f172a !important;">₱0.00</p>
```

```
    </div>
```

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```
<!-- Original Total -->

<div      class="input-group"      style="display:
flex !important; justify-content: space-between !important;
align-items: center !important; width: 100% !important; margin-
bottom: 0.85rem !important;">

      <label class="form-label" style="position:
relative !important; float: none !important; margin:
0 !important; color: #64748b !important; font-size:
0.8rem !important; font-weight: 600 !important; text-transform:
uppercase;">Original Total</label>

      <p id="extendOriginalTotal" class="info-
value" style="position: relative !important; float:
none !important; margin: 0 !important; font-weight:
700 !important; color: #0f172a !important;">₱0.00</p>

</div>

<!-- New Total -->

<div      class="input-group"      style="display:
flex !important; justify-content: space-between !important;
align-items: center !important; width: 100% !important; margin-
bottom: 1.5rem !important;">

      <label class="form-label" style="position:
relative !important; float: none !important; margin:
0 !important; color: #64748b !important; font-size:
```

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```
0.8rem !important; font-weight: 600 !important; text-transform:
uppercase;">New Total</label>
```

```
        <p      id="extendUpdatedTotal"      class="info-
value"      style="position:      relative      !important;      float:
none      !important;      margin:      0      !important;      font-weight:
700 !important; color: #0f172a !important;">₱0.00</p>
```

```
</div>
```

```
<div class="modal-btn-group-row">
```

```
        <button type="button" id="confirmExtendBtn"
class="btn-confirm-blue">Confirm Extend</button>
```

```
        <button                                type="button"
id="closeExtendModalBtn" class="btn-cancel-red">Cancel</button>
```

```
</div>
```

```
</div>
```

```
<div class="modal-preview-side">
```

```
    <div class="preview-border-box">
```

```
        <h3                                class="section-title">Extension
Summary</h3>
```

```
    <div class="preview-list">
```

```
        <div class="preview-item d-flex justify-
content-between align-items-center border-bottom py-2">
```


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```

                <span    class="preview-label    text-
muted          small    position-relative"        style="position:
static !important; float: none !important;">Customer</span>

```

```

                <span          id="extendSummaryCustomer"
class="fw-bold">---</span>

```

```

</div>

```

```

<div class="preview-item d-flex justify-
content-between align-items-center border-bottom py-2">

```

```

                <span    class="preview-label    text-
muted          small    position-relative"        style="position:
static !important; float: none !important;">Room</span>

```

```

                <span          id="extendSummaryRoom"
class="fw-bold">---</span>

```

```

</div>

```

```

<div class="preview-item d-flex justify-
content-between align-items-center border-bottom py-2">

```

```

                <span    class="preview-label    text-
muted          small    position-relative"        style="position:
static !important; float: none !important;">Added Hours</span>

```

```

                <span          id="extendSummaryHours"
class="fw-bold">1</span>

```

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</div>

```
<div class="preview-item d-flex justify-
content-between align-items-center border-bottom py-2">
```

```
    <span class="preview-label text-
muted small position-relative" style="position:
static !important; float: none !important;">New End Time</span>
```

```
    <span id="extendSummaryEnd"
class="fw-bold">---</span>
```

</div>

```
<div class="preview-item d-flex justify-
content-between align-items-center py-2">
```

```
    <span class="preview-label text-
muted small position-relative" style="position:
static !important; float: none !important;">New Total
Bill</span>
```

```
    <span id="extendSummaryTotal"
class="fw-bold text-primary">P0.00</span>
```

</div>

</div>

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```
<p class="text-muted modal-helper-text mt-3
small">This updates the reservation end time and total bill
immediately for the fixed session.</p>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
{% endblock %}
```

```
{% block extra_js %}
```

```
<script src="{% url_for('static',
filename='js/admin/admin_dashboard.js') %}"></script>
```

```
<script src="{% url_for('static',
filename='js/modals/admin_walkin_modal.js') %}"></script>
```

```
{% endblock %}
```

```
admin_reservations.html
```

```
{% extends "base.html" %}
```

```
{% block extra_css %}
```

```
<link rel="preconnect" href="https://fonts.googleapis.com">
```

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```
<link      rel="preconnect"      href="https://fonts.gstatic.com"
crossorigin>
```

```
<link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800;900&family=Roboto:wght@400;500;700&display=swap"
rel="stylesheet">
```

```
<link      rel="stylesheet"      href="{{      url_for('static',
filename='css/admin/admin_reservations.css')  }}">
```

```
<link      rel="stylesheet"      href="{{      url_for('static',
filename='css/admin/addons.css')  }}">
```

```
<link      rel="stylesheet"      href="{{      url_for('static',
filename='css/rooms.css')  }}">
```

```
{% endblock %}
```

```
{% block content %}
```

```
<div class="reservation-page-admin-container1">
```

```
<div      class="reservation-page-admin-thq-reservation-page-
admin-elm">
```

```
<div class="reservation-page-admin-thq-group42-elm">
```

```
<div class="reservation-page-admin-thq-group41-elm">
```

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```
<div class="reservation-page-admin-thq-reservation-
form-elm">

    <span class="reservation-page-admin-thq-text-
elm13">RESERVATION</span>

    <form id="resForm" method="POST"
action="{{ url_for('admin.admin_reservations') }}">

        {{ form.hidden_tag() }}

        <div class="form-grid">

            <div class="input-group-modern">

                <label class="form-label-
modern">Customer Name</label>

                {{ form.customer_name(class_="modern-
input", placeholder="e.g. Darwen Salcedo", id="customer_name") }}

            </div>

            <div class="input-group-modern">

                <label class="form-label-modern">Contact
Number</label>

                {{ form.contact_number(class_="modern-
input", placeholder="e.g. 0912-123-4567", id="contact_number") }}
```

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</div>

</div>

<div class="form-grid">

<div class="input-group-modern">

<label class="form-label-modern">Select
Area</label>

{{ form.room_id(class_="modern-select",
id="room_id") }}

</div>

</div>

<div class="input-grid-3 date-time-grid">

<div class="input-group">

<label class="form-label">Start
Date/Time</label>

{{ form.start_time(class_="modern-input",
type="datetime-local", id="start_time") }}

</div>

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```
<div class="input-group">

    <label class="form-label">Duration</label>

    <select id="duration-select" class="modern-
select">

        {% for hours in range(1, 25) %}

            <option value="{{ hours }}">{{ hours }}
hour{{ 's' if hours > 1 else '' }}</option>

        {% endfor %}

    </select>

    <input type="datetime-local" id="end-
display" class="modern-input mt-10" readonly placeholder="Auto-
calculated end time">

    {{ form.end_time(type="hidden",
id="end_time") }}

</div>

<div class="input-group align-items-center">

    <label class="form-label">Open Time</label>

    <label class="switch">

        {{ form.open_time(id="openTimeToggle") }}

        <span class="slider round"></span>
```

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```
        </label>

    </div>

</div>


    <span      class="reservation-page-admin-thq-text-elm14 section-title-break">STAFF NOTES & EXTRA FEES</span>


    <!-- Predefined Add-ons -->

    <div class="form-grid mt-20">

        <div class="input-group-modern">

            <label              class="form-label-modern">Predefined Add-ons</label>

            <select              id="f-predefined-addon"
name="predefined_addon_name" class="modern-select">

                <option value="None">None</option>

                <option value="Projector" data-unit-price="100">Projector (+P100/3hrs)</option>

                <option value="Extra Chair" data-unit-price="0">Extra Chair (Free)</option>

                <option value="Extension Cord" data-unit-price="0">Extension Cord (Free)</option>

            </select>

        </div>

    </div>
```


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```

                                <option    value="Microphone"    data-
unit-price="0">Microphone (Event Room only)</option>

                                <option    value="Speaker"      data-unit-
price="0">Speaker (Event Room only)</option>

                                </select>

                                </div>

                                <div class="input-group-modern">

                                <label class="form-label-modern" for="f-
predefined-addon-qty">Quantity</label>

                                <input type="number" id="f-predefined-
addon-qty" name="predefined_addon_qty" class="modern-input"
min="1" value="1" step="1" placeholder="1">

                                <small class="text-muted">Empty/invalid
values become 1.</small>

                                </div>

                                </div>

                                <div class="reservation-page-admin-thq-group33-
elm">

                                {{ form.extra_notes(class_="modern-input",
id="f-notes", rows="2", placeholder="e.g. Extra monitor, HDMI
cable...") }}

```

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```

        <span          class="reservation-page-admin-thq-
text-elm46">Selected Add-on Total:</span>

```

```

        <span          class="reservation-page-admin-thq-
text-elm47                                value-text"
id="preview_addon_total_display">₱0.00</span>

```

```

    </div>

```

```

    <!-- Hidden fields for add-ons data (existing
persistence) -->

```

```

    {{ form.addons_json(id="addons_json") }}

```

```

    {{ form.addon_subtotal(id="addon_subtotal") }}

```

```

    <div          class="reservation-page-admin-thq-pax-
input-elm2 mt-20 partial-width">

```

```

        <span      class="reservation-page-admin-thq-text-
elm29">Additional Fees (Non-Add-on)</span>

```

```

        {{          form.extra_fee(class_="reservation-page-
admin-thq-rectangle15-elm5                                full-width-input",
id="extra_fee_input",          placeholder="0.00",          type="number",
step="0.01") }}

```

```

    </div>

```

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```
                <input          type="hidden"          name="total_price"
id="hidden_total_price" value="0">
```

```
            </form>
```

```
    </div>
```

```
        <div    class="reservation-page-admin-thq-reservation-
details-elm">
```

```
            <span          class="reservation-page-admin-thq-text-
elm31">RESERVATION DETAILS</span>
```

```
        <div class="summary-details-list">
```

```
            <div          class="reservation-page-admin-thq-
group26-elm">
```

```
                <span          class="reservation-page-admin-thq-
text-elm32">Customer Name:</span>
```

```
                    <span          class="reservation-page-admin-thq-
text-elm33 value-text" id="preview_name">----</span>
```

```
            </div>
```

```
            <div          class="reservation-page-admin-thq-
group27-elm">
```

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```
        <span          class="reservation-page-admin-thq-  
text-elm34">Contact Number:</span>
```

```
        <span          class="reservation-page-admin-thq-  
text-elm35 value-text" id="preview_contact">----</span>
```

```
    </div>
```

```
    <div              class="reservation-page-admin-thq-  
group28-elm">
```

```
        <span          class="reservation-page-admin-thq-  
text-elm36">Area:</span>
```

```
        <span          class="reservation-page-admin-thq-  
text-elm37 value-text" id="preview_room">----</span>
```

```
    </div>
```

```
    <div              class="reservation-page-admin-thq-  
group29-elm">
```

```
        <span          class="reservation-page-admin-thq-  
text-elm38">Start Date:</span>
```

```
        <span          class="reservation-page-admin-thq-  
text-elm39 value-text" id="preview_date">----</span>
```

```
    </div>
```

```
    <div              class="reservation-page-admin-thq-  
group29-elm">
```

```
        <span          class="reservation-page-admin-thq-  
text-elm38">Start Time:</span>
```

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```
        <span          class="reservation-page-admin-thq-  
text-elm39 value-text" id="preview_start">----</span>
```

```
    </div>
```

```
    <div          class="reservation-page-admin-thq-  
group30-elm">
```

```
        <span          class="reservation-page-admin-thq-  
text-elm40">End Time:</span>
```

```
        <span          class="reservation-page-admin-thq-  
text-elm41 value-text" id="preview_end">----</span>
```

```
    </div>
```

```
    <div          class="reservation-page-admin-thq-  
group32-elm">
```

```
        <span          class="reservation-page-admin-thq-  
text-elm44">Duration:</span>
```

```
        <span          class="reservation-page-admin-thq-  
text-elm45 value-text" id="preview_duration">----</span>
```

```
    </div>
```

```
    <div          class="reservation-page-admin-thq-  
group33-elm">
```

```
        <span          class="reservation-page-admin-thq-  
text-elm46">Additional Fees:</span>
```

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```
        <span          class="reservation-page-admin-thq-  
text-elm47 value-text" id="preview_extra">----</span>
```

```
    </div>
```

```
    <div          class="reservation-page-admin-thq-  
group33-elm">
```

```
        <span          class="reservation-page-admin-thq-  
text-elm46">Add-ons Subtotal:</span>
```

```
        <span          class="reservation-page-admin-thq-  
text-elm47 value-text" id="preview_addon_subtotal">----</span>
```

```
    </div>
```

```
    <div          class="reservation-page-admin-thq-  
group33-elm discount-row">
```

```
        <span          class="reservation-page-admin-thq-  
text-elm46 discount-label">Discount:</span>
```

```
        {{      form.discount(class_="discount-select",  
id="f-discount") }}
```

```
    </div>
```

```
</div>
```

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```
<div      class="reservation-page-admin-thq-group36-elm">

    <span      class="reservation-page-admin-thq-text-elm51">TOTAL PAYABLE</span>

    <span      class="reservation-page-admin-thq-text-elm50" id="preview_total">P0.00</span>


    <div class="action-buttons-wrapper">

        <button  type="button"  class="reservation-page-admin-thq-button-elm1" id="confirmReservation">

            <span      class="reservation-page-admin-thq-text-elm52">CONFIRM & SAVE</span>

        </button>

        <button  type="button"  class="reservation-page-admin-thq-button-elm2" id="cancelReservation">

            <span      class="reservation-page-admin-thq-text-elm53">CANCEL</span>

        </button>

    </div>

</div>

</div>
```

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```
</div>

</div>

</div>

</div>

{% endblock %}


{% block extra_js %}

<script src="{ { url_for('static',
filename='js/admin/admin_reservations.js') }}"></script>

<script>

    // Predefined Add-ons -> write to existing
    addons_json/addon_subtotal hidden fields

    document.addEventListener('DOMContentLoaded', function()
    {

        const addonSelect = document.getElementById('f-
predefined-addon');

        const addonQtyInput = document.getElementById('f-
predefined-addon-qty');

        const addonsJsonField =
document.getElementById('addons_json');

        const addonSubtotalField =
document.getElementById('addon_subtotal');
```


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```
const previewAddonTotalDisplay =
document.getElementById('preview_addon_total_display');

let addonNameToIdCache = null;

async function ensureAddonIdCache() {
    if (addonNameToIdCache) return
addonNameToIdCache;

    try {
        const res = await fetch('/api/addons',
{    method:    'GET',    headers:    {    'Content-Type':
'application/json' } }));

        if (!res.ok) throw new Error('Failed to load
addons');

        const data = await res.json();

        addonNameToIdCache = new Map((data.addons ||
[]).map(a => [a.name, a.id]));

        return addonNameToIdCache;
    } catch (e) {

        addonNameToIdCache = new Map();

        return addonNameToIdCache;
    }
}
```

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```
}

function normalizeQty(raw) {

    if (raw === null || raw === undefined) return 1;

    if (typeof raw === 'string' && raw.trim() === '')
return 1;

    const n = parseInt(raw, 10);

    if (!Number.isFinite(n) || n < 1) return 1;

    return n;

}

function getSelectedAddonUnitPrice() {

    if (!addonSelect) return 0;

    const opt =
addonSelect.options[addonSelect.selectedIndex];

    return parseFloat(opt?.dataset?.unitPrice) || 0;

}

async function syncPredefinedAddonFields() {

    const selected = (addonSelect?.value || 'None');
```

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```
const qty = normalizeQty(addonQtyInput?.value);

if (addonQtyInput) addonQtyInput.value = qty;


if (!selected || selected === 'None') {

    if (previewAddonTotalDisplay)
previewAddonTotalDisplay.textContent = '₱0.00';

    if (addonSubtotalField)
addonSubtotalField.value = '0.00';

    if (addonsJsonField) addonsJsonField.value =
JSON.stringify([]);

    if
(document.getElementById('preview_addon_subtotal')) {

document.getElementById('preview_addon_subtotal').textContent =
'₱0.00';

    }

    if (typeof updateTotalPrice === 'function')
updateTotalPrice();

    return;

}


const cache = await ensureAddonIdCache();
```

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```
const addonId = cache.get(selected);

const unitPrice = getSelectedAddonUnitPrice();

const subtotal = qty * unitPrice;

const subtotalPayload = [

    {  addon_name:  selected,  quantity:  qty,
unit_price: unitPrice, subtotal: subtotal }

];

if (!addonId) {

    if (previewAddonTotalDisplay)
previewAddonTotalDisplay.textContent = '₱' + subtotal.toFixed(2);

    if (addonSubtotalField)
addonSubtotalField.value = subtotal.toFixed(2);

    if (addonsJsonField) addonsJsonField.value =
JSON.stringify(subtotalPayload);

    if
(document.getElementById('preview_addon_subtotal')) {

document.getElementById('preview_addon_subtotal').textContent  =
'₱' + subtotal.toFixed(2);
```

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```
        }

        if (typeof updateTotalPrice === 'function')
updateTotalPrice();

        return;

    }

    if (previewAddonTotalDisplay)
previewAddonTotalDisplay.textContent = '₱' + subtotal.toFixed(2);

    if (addonSubtotalField) addonSubtotalField.value
= subtotal.toFixed(2);

    if (addonsJsonField) {

        addonsJsonField.value = JSON.stringify([

            {   addon_id:   addonId,   addon_name:
selected,   quantity:   qty,   unit_price:   unitPrice,   subtotal:
subtotal }

        ]);

    }

    if
(document.getElementById('preview_addon_subtotal')) {
```

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```
document.getElementById('preview_addon_subtotal').textContent    =
'₱' + subtotal.toFixed(2);

    }

    if (typeof updateTotalPrice === 'function')
updateTotalPrice();

    }

    if (addonSelect)
addonSelect.addEventListener('change',
syncPredefinedAddonFields);

    if (addonQtyInput)
addonQtyInput.addEventListener('input',
syncPredefinedAddonFields);

    syncPredefinedAddonFields();

});

</script>

{% endblock %}
```

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```
rooms.html

{% extends "base.html" %}


{% block extra_css %}

    <link      rel="stylesheet"      href="{{      url_for('static',
filename='css/rooms_modern.css')  }}">

{% endblock %}


{% block content %}

<div class="rooms-container">

    <h2 class="page-title">Rooms & Reservations</h2>

    <p class="page-subtitle">Book your preferred space with
ease</p>


    <!-- Form Errors Block -->

    {% if form and form.errors %}

        <div class="modern-alert alert-danger">

            <button      type="button"      class="alert-dismiss"
onclick="this.parentElement.remove();">&times;</button>

            {% for field, errors in form.errors.items() %}
```

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```

                <strong>{{          field.replace('_', '
').title() }}:</strong> {{ errors[0] }}<br>

                {% endfor %}

        </div>

    {% endif %}


<div class="rooms-flex">

    <div class="rooms-sidebar">

        <div class="room-list-card">

            <h5    class="sidebar-section-title">Available
Rooms</h5>

            <ul class="room-list">

                {% set unique_rooms = [] %}

                {% for room in rooms %}

                    {%      if      room.name      not      in
unique_rooms %}

                                {% set is_common_area = 'common
area' in room.name.lower() %}

                                {%      set      current_pax      =
room_pax_count[room.id]    if    (room_pax_count    is    defined    and
room.id in room_pax_count) else 0 %}

```


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```
{% set max_pax = 70 %}

{# Dynamic Status Validation #}

{% if is_common_area %}

    {% set is_available =
(current_pax < max_pax) and (room.status and room.status.lower()
== 'available') %}

    {% else %}

        {% set is_occupied =
occupied_room_ids is defined and room.id in occupied_room_ids %}

        {% set is_available = (not
is_occupied) and (room.status and room.status.lower() ==
'available') %}

    {% endif %}

<li class="room-item">

    <div>

        <div class="room-
name">{{ room.name }}</div>

        <div class="room-rate">

P{{ room.base_rate }}/hr
```

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```

                                {%                               if
is_common_area %}

                                <small
class="text-muted">({{      current_pax      }}/{{      max_pax      }}
occupied)</small>

                                {% endif %}

                                </div>

                                </div>

                                <span      class="room-status
status-{{ 'available' if is_available else 'unavailable' }}">

                                {{      'AVAILABLE'      if
is_available else 'UNAVAILABLE' }}

                                </span>

                                </li>

                                {%      set      _      =
unique_rooms.append(room.name) %}

                                {% endif %}

                                {% endfor %}

                                </ul>

                                </div>

                                </div>

```

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```
<div class="reservation-form-card">

    <h5 class="form-section-title">Reservation</h5>

    <form          method="POST"          id="booking-form"
enctype="multipart/form-data">

        {{ form.hidden_tag() }}

        {{          form.is_open_time(class="d-none",
id="form-open-time") }}

        <div class="form-grid">

            <div class="input-group-modern">

                {{ form.customer_name.label(class="form-label-modern") }}

                {{ form.customer_name(class="modern-
input", placeholder="e.g Darwen Salcedo", required=True) }}

            </div>

            <div class="input-group-modern">

                {{ form.contact_number.label(class="form-label-modern") }}

                {{ form.contact_number(class="modern-input", placeholder="e.g
09121234567", required=True) }}

            </div>

        </div>

    </form>

</div>
```

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</div>

<div class="form-grid">

<div class="input-group-modern">

{{ form.room_id.label(class="form-label-modern", text="Select Area") }}

<select
name="{{ form.room_id.name }}" class="modern-select" required
id="room-selector">

<option value="" disabled
selected>Choose a room...</option>

{% set unique_select = [] %}

{% for room in rooms %}

{% if room.name not in
unique_select %}

{# Tsek: Diri ginacheck
kun ang room HORAS-HORAS nga occupied o kon nakaset gid nga
maintenance sa DB #}

{% if room.id not in
occupied_room_ids and room.status and room.status.lower() ==
'available' %}

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```

                                <option
value="{{ room.id }}" data-rate="{{ room.base_rate }}">
                                {{ room.name }}
(P{{ room.base_rate }}/hr)

                                </option>

                                {% else %}

                                <option
value="{{ room.id }}" data-rate="{{ room.base_rate }}" disabled
style="color: #94a3b8; background-color: #f8fafc;">
                                {{ room.name }}
(Unavailable)

                                </option>

                                {% endif %}

                                {%
set
_
=
unique_select.append(room.name) %}

                                {% endif %}

                                {% endfor %}

                                </select>

                                </div>

                                </div>

                                <div class="input-grid-3 date-time-grid">
```

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```
<div class="input-group">

{{ form.start_time.label(class="form-label-modern", text="Start
Date/Time") }}

        {{ form.start_time(class="modern-
input", type="datetime-local", id="start-time") }}

</div>

<div class="input-group">

        <label                                class="form-
label">Duration</label>

        <select                                id="duration-select"
class="modern-select">

                {% for hours in range(1, 25) %}

                        <option
value="{{ hours }}">{{ hours }} hour{{ 's' if hours > 1 else
'' }}</option>

                {% endfor %}

        </select>

        <input                                type="datetime-local"
id="end-display"      class="modern-input      mt-10"      readonly
placeholder="Auto-calculated end time">

        {{ form.end_time(type="hidden",
id="end-time") }}


```

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```

    </div>

    <div    class="input-group    align-items-
center">

        <label        class="form-label">Open
Time</label>

        <label class="switch">

            <input type="checkbox" id="open-
time-toggle">

                <span                class="slider
round"></span>

            </label>

        </div>

    </div>

</div>

<div class="notes-section">

    <h6    class="notes-title">STAFF    NOTES    &
EXTRA FEES</h6>

    <div class="form-grid">

        <div class="input-group-modern">

            <label        class="form-label-
modern">Predefined Add-ons</label>

```

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```

                                <select      id="extra-fees-select"
name="predefined_addon_name" class="modern-select">

                                <option
value="None">None</option>

                                <option      value="Projector"
data-unit-price="100">Projector (+P100/3hrs)</option>

                                <option value="Extra Chair"
data-unit-price="0">Extra Chair (Free)</option>

                                <option      value="Extension
Cord" data-unit-price="0">Extension Cord (Free)</option>

                                <option      value="Microphone"
data-unit-price="0">Microphone (Event Room only)</option>

                                <option      value="Speaker"
data-unit-price="0">Speaker (Event Room only)</option>

                                </select>

                                </div>

                                <div class="input-group-modern">

                                <label class="form-label-modern"
for="addon-quantity">Quantity</label>

                                <input type="number" id="addon-
quantity"      name="predefined_addon_qty"      class="modern-input"
min="1" value="1" step="1">

```


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```

                                <small    class="text-muted">Empty
value will be treated as 1.</small>

                                </div>

                                <input
                                type="hidden"
name="extra_fee_total" id="extra-fee-total" value="0">

                                <div class="input-group-modern">

{{                                form.extra_notes.label(class="form-label-modern",
text="Special Instructions") }}

{{                                form.extra_notes(class="modern-textarea    modern-input",
placeholder="Enter specific requests here...") }}

                                </div>

                                </div>

                                </div>

                                </div>

                                <div class="payment-section">

                                <h6                                class="notes-title">Payment
Method</h6>

                                <p class="small text-muted">
```

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Select your payment method and payment type. For downpayment, a 50% advance is required.

</p>

<div class="payment-method-grid">

<label class="payment-method-option"
for="pm-gcash">

<input type="radio"
name="payment_method" id="pm-gcash" value="GCash">

GCash

</label>

<label class="payment-method-option"
for="pm-maya">

<input type="radio"
name="payment_method" id="pm-maya" value="Maya">

Maya

</label>

</div>

<div id="payment-type-section"
class="payment-type-section d-none">

<h6 class="notes-title payment-type-
title">Payment Type</h6>

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```
<div class="payment-type-grid">

    <label          class="payment-type-
option" for="pt-downpayment">

        <input          type="radio"
name="payment_type"    id="pt-downpayment"    value="Downpayment"
checked>

            <span>Downpayment
(50%)</span>

        </label>

    <label          class="payment-type-
option" for="pt-fullpayment">

        <input          type="radio"
name="payment_type" id="pt-fullpayment" value="Full Payment">

            <span>Full Payment</span>

        </label>

    </div>

</div>

<div          id="payment-info-panel"
class="payment-info-panel d-none">

    <div class="payment-info-row">
```

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```

                                <strong          id="payment-method-
label"></strong>

                                </div>

                                <div class="payment-info-row">

                                    <small  id="payment-instructions"
class="text-muted"></small>

                                </div>

                                <div class="payment-info-row">

                                    <button  type="button"  id="show-
payment-modal"  class="modal-open-btn">View    Payment    QR    &
Details</button>

                                </div>

                                </div>

                                <div          id="payment-amount-display"
class="payment-amount-display d-none">

                                    <div class="amount-row">

                                        <strong>Grand      Total:</strong>

                                        <span id="display-total-amount">₱0.00</span>

                                    </div>

                                    <div      class="amount-row      d-none"
id="downpayment-row">
```

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```

                                <strong>Downpayment          Amount
(50%) :</strong>              <span          id="display-downpayment-
amount">₱0.00</span>

                                </div>

                                <div          class="amount-row          d-none"
id="fullpayment-row">

                                <strong>Full                    Payment
Amount:</strong>              <span          id="display-fullpayment-
amount">₱0.00</span>

                                </div>

                                </div>

                                <div          class="input-group-modern          receipt-
upload-group d-none" id="receipt-upload-section">

                                <label          class="form-label-modern"
for="receipt-upload">Upload Receipt</label>

                                <input          type="file"
name="receipt_image"          id="receipt-upload"          class="modern-input"
accept="image/*,application/pdf">

                                <small          class="text-muted">Verify the
amount          matches          the          <strong          id="receipt-amount-
label">downpayment</strong> shown above.</small>

                                </div>
```

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```
<div id="payment-modal" class="payment-
modal-overlay d-none">

    <div class="payment-modal-content">

        <button                type="button"
class="modal-close" id="payment-modal-close">&times;</button>

        <h3            id="modal-payment-method"
class="modal-title">Payment Details</h3>

        <p><strong>Account:</strong>
<span id="modal-account-info"></span></p>

        <div class="modal-qr-row">

            <img            id="modal-qr-image"
src="" alt="QR Code" class="modal-qr-image d-none">

        </div>

        <p            id="modal-instructions"
class="modal-text"></p>

    </div>

</div>

</div>

</div>

<!-- Live Booking Summary Section -->
```

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```
<div                                id="booking-summary-container"
class="booking-summary-card d-none">

    <h6                                class="summary-title">Booking
Summary</h6>

    <div class="summary-details">

        <p>Selected Room: <span id="summary-
room">-</span></p>

        <p>Duration:      <span      id="summary-
hours">0</span> hour(s)</p>

        <p>Add-ons:      <span      id="summary-
addons">None</span></p>

        <p>Total Cost: <span class="summary-
price">P<span id="summary-total">0.00</span></span></p>

        <p      class="deposit-note      d-
none">Downpayment      Due:      <strong      id="summary-
deposit">P0.00</strong></p>

    </div>

</div>

{%      if      current_user.is_authenticated      and
current_user.role != 'member' %}
```

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```
                <button type="button" class="form-submit-  
btn"          id="submit-booking"          disabled="disabled">Reserve  
Space</button>
```

```
{% else %}
```

```
                <button      type="button"      class="form-  
submit-btn" id="submit-booking">Reserve Space</button>
```

```
{% endif %}
```

```
</form>
```

```
</div>
```

```
</div>
```

```
<div class="bookings-section">
```

```
    <div class="section-header-flex">
```

```
        <h3                class="upcoming-title">Upcoming  
Bookings</h3>
```

```
    <div class="table-filters">
```

```
        <input      type="date"      id="filter-date"  
class="modern-input filter-input">
```

```
        <select  id="filter-status"  class="modern-  
select filter-input">
```

```
            <option value="">All Status</option>
```


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```
        <option
value="Confirmed">Confirmed</option>

        <option value="Pending">Pending</option>

    </select>

    <button    type="button"    class="export-btn"
id="export-csv">Export CSV</button>

</div>

</div>

<div class="table-container">

    <table    class="booking-table"    id="bookings-
table">

        <thead>

            <tr>

                <th>Rooms</th>

                <th>Start Time</th>

                <th>End Time</th>

                <th>Status</th>

            </tr>

        </thead>
```

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```
<tbody>

    {% if bookings %}

        {% for booking in bookings %}

            <tr      class="booking-row"      data-
status="{{ booking[2] }}" data-date="{{ booking[0] }}">

                <!-- Room Name -->

                <td>{{ booking[3] }}</td>


                <!-- Start Time in 12-Hour AM/PM
Format -->

                <td>

                    {% if booking[0] is string %}

                        {{ booking[0] }}

                    {% else %}

                        {{ booking[0].strftime('%b %d, %Y %I:%M %p') }}

                    {% endif %}

                </td>


                <!-- End Time in 12-Hour AM/PM
Format -->
```

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```
<td>

    {% if booking[1] is string %}

        {{ booking[1] }}

    {% else %}

{{ booking[1].strftime('%b %d, %Y %I:%M %p') }}

    {% endif %}

</td>

<!-- Status -->

<td>

    <span    class="booking-status
status-{{ booking[2].lower() }}">

        {{ booking[2] }}

    </span>

</td>

</tr>

{% endfor %}

{% else %}

    <tr class="empty-row">
```

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```

                                <td      colspan="4">No      upcoming
bookings found.</td>

                                </tr>

                                {% endif %}

                                </tbody>

                                </table>

                                </div>

                                </div>

                                </div>

                                {% endblock %}


                                <script                                type="application/json"
id="paymentInfoData">{{ payment_info | tojson | safe }}</script>

                                <div                                id="currentUserRole"                                data-
role="{{ current_user.role }}" hidden></div>

                                {% block extra_js %}

                                <script                                src="{{                                url_for('static',
filename='js/rooms.js') }}"></script>

                                {% endblock %}

```

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```
solo_rates.html

{% extends "base.html" %}


{% block title %}Solo Plans & Rates{% endblock %}


{% block extra_css %}

    <link      rel="stylesheet"      href="{{      url_for('static',
filename='css/solo_plans_modern.css')  }}">

{% endblock %}


{% block content %}

<div class="solo-page container">

    <div class="solo-header">

        <div>

            <h2>Solo Plan Rates</h2>

            <p class="solo-intro">Choose your plan, upload
your payment receipt, and wait for admin approval. Your current
solo plan status is shown below.</p>

        </div>

    </div>

</div>
```

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```
<!-- Solo Status and Membership Info Cards -->

<div class="solo-status-grid">

    <div class="status-card">

        <div class="status-card-header">

            <h5>Solo Plan Status</h5>

            {% if active_solo_plan and
active_solo_plan.is_active %}

                <span class="badge badge-
success">Active</span>

            {% elif has_pending %}

                <span class="badge badge-
warning">Pending</span>

            {% else %}

                <span class="badge badge-
secondary">None</span>

            {% endif %}

        </div>

        <div class="status-card-body">

            {% if active_solo_plan %}
```

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```
<p><strong>Plan:</strong>
{{ active_solo_plan.plan_name }}</p>

<p><strong>Expiry:</strong>
{{ active_solo_plan.expiry_date.strftime('%Y-%m-%d %I:%M %p') if
active_solo_plan.expiry_date else 'Not set' }}</p>

<p><strong>Payment          Method:</strong>
{{ active_solo_plan.payment_method or 'N/A' }}</p>

<p><strong>Status:</strong>
{{ (active_solo_plan.status or 'N/A')|title }}</p>

{% elif has_pending %}

    <p>Your plan request is waiting for
admin approval.</p>

{% else %}

    <p>No solo plan is currently active.</p>

{% endif %}

</div>

</div>

<div class="status-card">

    <div class="status-card-header">

        <h5>Membership Info</h5>

        {% if active_plan %}
```

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```

                                <span    class="badge    badge-info">Active
Member Plan</span>

                                {% else %}

                                <span    class="badge    badge-secondary">No
Active Membership</span>

                                {% endif %}

</div>

<div class="status-card-body">

                                {% if active_plan %}

                                <p><strong>Plan                                Name:</strong>
{{ active_plan.plan }}</p>

                                <p>

                                <strong>Started:</strong>

                                {%          if          active_plan          and
active_plan.start_date %}

                                {{ active_plan.start_date.strftime('%Y-%m-%d %I:%M %p') }}

                                {% else %}

                                N/A

                                {% endif %}

                                </p>
```


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```
<p>

<strong>Expiration:</strong>

{% if expiration %}

        {{      expiration.strftime('%Y-%m-
%d %I:%M %p') }}

        {%      elif      active_plan      and
active_plan.expiry_date %}

        {{ active_plan.expiry_date.strftime('%Y-%m-%d %I:%M %p') }}

        {% else %}

        N/A

        {% endif %}

</p>

<p><strong>Remaining      days:</strong>

{{ remaining_days }}</p>

<!-- PAUSE-AWARE COUNTDOWN -->

<p>

<strong>Countdown:</strong>

<span id="membership-countdown"
```

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```
data-
expiry="{ { expiration.isoformat() if expiration else ' ' } }"

data-is-paused="{ { 'true' if
active_reservation and active_reservation.is_paused else
'false' } }"

data-remaining-
seconds="{ { active_reservation.remaining_seconds if
active_reservation and active_reservation.remaining_seconds else
0 } }"

style="font-weight: bold;">

</span>

</p>

{% else %}

    <p>You do not have a currently active
membership plan.</p>

{% endif %}

</div>

</div>

</div>

<!-- Rate Comparison Table -->

<div class="plan-comparison-card">
```

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```
<div class="plan-comparison-header">

    <h4>Rate Comparison</h4>

    <p>Compare solo rates at a glance. Click a plan
card to submit your payment receipt.</p>

</div>

<div class="table-responsive">

    <table class="table plan-table">

        <thead>

            <tr>

                <th>Plan</th>

                <th>Price</th>

                <th>Details</th>

            </tr>

        </thead>

        <tbody>

            {% for plan in plans %}

            <tr>

                <td>{{ plan.title }}</td>

                <td>{{ plan.price }}</td>

                <td>
```

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```

                                <ul class="plan-details-list">
                                    {%      for      detail      in
plan.details %}

                                <li>{{ detail }}</li>

                                    {% endfor %}

                                </ul>

                                </td>

                                </tr>

                                {% endfor %}

                                </tbody>

                                </table>

                                </div>

</div>

<!-- Plan Cards -->

<div class="plan-cards-grid">

    {% for plan in plans %}

        <div class="plan-card">

            <div class="plan-card-top">

```

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```
<div class="plan-card-
title">{{ plan.title }}</div>

<div class="plan-card-
price">{{ plan.price }}</div>

</div>

<div class="plan-card-details">

  <ul>

    {% for detail in plan.details %}

      <li>{{ detail }}</li>

    {% endfor %}

  </ul>

</div>

<div class="plan-card-footer">

  {% if active_solo_plan %}

    <button class="btn btn-secondary btn-
block" disabled>Active plan in use</button>

  {% elif has_pending %}

    <button class="btn btn-secondary btn-
block" disabled>Pending approval</button>

  {% else %}
```

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```

                                <button class="btn btn-primary btn-block
purchase-plan-btn"           type="button"           data-plan-
name="{{ plan.title          }}"           data-plan-
price="{{ plan.price }}">Payment Options</button>

                                {% endif %}

                                </div>

                                </div>

                                {% endfor %}

                                </div>

<!-- Plan Summary Info -->

<div class="plan-summary-card">

    <h4>Plan Summary</h4>

    {% if message %}

        <div class="alert alert-
success">{{ message }}</div>

    {% endif %}

    <p>Choose the rate that fits your needs. After
selecting a plan, upload your payment receipt and wait for admin
approval.</p>

    <ul>
```

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```
<li>Plan approvals are reviewed manually by the  
admin.</li>
```

```
<li>Your uploaded receipt is stored securely  
under uploads/receipts.</li>
```

```
<li>Active plans remain valid until expiry.</li>
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
<!-- Plan Purchase Modal -->
```

```
<div class="modal-backdrop d-none" id="plan-selection-modal"  
aria-hidden="true">
```

```
<div class="modal-card">
```

```
<button type="button" class="modal-close" id="plan-  
modal-close">&times;</button>
```

```
<div class="modal-content-inner">
```

```
<form id="solo-plan-form" method="POST"  
enctype="multipart/form-data">
```

```
{% if csrf_token is defined and csrf_token %}
```

```
<input type="hidden" name="csrf_token"  
value="{{ csrf_token() }}">
```

```
{% endif %}
```

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```
        <input    type="hidden"    id="modal-plan-name-  
input" name="plan_name" value="">
```

```
    <h3 id="modal-plan-title">Purchase Plan</h3>
```

```
    <p      class="modal-price"      id="modal-plan-  
price"></p>
```

```
    <div class="modal-row">
```

```
        <label class="modal-label" style="text-  
transform: uppercase; font-weight: bold; font-size: 0.85rem;  
color: #2c3e50;">Payment Method</label>
```

```
        <div class="modal-radio-group">
```

```
            <label><input          type="radio"  
name="modal_payment_method"      value="GCash"      required>  
GCash</label>
```

```
            <label><input          type="radio"  
name="modal_payment_method" value="Maya"> Maya</label>
```

```
        </div>
```

```
    </div>
```

```
    <!-- Payment Details -->
```


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```
        <div    class="modal-payment-details    d-none"
id="modal-payment-details">
```

```
                <div                                class="payment-info-
line"><strong>Account    Name:</strong>    <span    id="modal-account-
name">WS Students & Professionals Lounge</span></div>
```

```
                <div                                class="payment-info-
line"><strong>Account    Number:</strong>    <span    id="modal-account-
number">0999XXXXXXX</span></div>
```

```
                <div                                class="payment-info-
line"><strong>Instructions:</strong>                <span                id="modal-
instructions">Please    make    your    payment    using    the    details    above
and    upload    your    payment    receipt.</span></div>
```

```
                <div    class="modal-qr-wrap    d-none"
id="modal-qr-wrap"    style="text-align:    center;    margin-top:
10px;">
```

```
                        <img                                id="modal-qr-image"
alt="Payment QR" class="modal-qr" src="" style="max-width: 200px;
border-radius: 8px; border: 1px solid #ccc;">
```

```
                </div>
```

```
        </div>
```

```
                <div    class="modal-row"    style="margin-top:
15px;">
```

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```
        <label    class="modal-label"    for="modal-  
receipt"    style="text-transform: uppercase; font-weight: bold;  
font-size: 0.85rem; color: #2c3e50;">Upload Receipt</label>
```

```
        <input    type="file"    id="modal-receipt"  
name="receipt_image"  
accept="image/*, .jpg, .jpeg, .jpe, .png, .gif, .webp,  
application/pdf" required>
```

```
    </div>
```

```
        <div        class="modal-row        modal-terms-row"  
style="margin-top: 10px;">
```

```
        <input    type="checkbox"    id="modal-terms"  
required>
```

```
        <label    for="modal-terms"    style="font-  
size: 0.85rem; color: #333;">I agree to upload a valid receipt  
and follow payment instructions.</label>
```

```
    </div>
```

```
        <div        class="modal-error"        id="modal-error"  
style="color: #e74c3c; font-size: 0.85rem; font-weight: 600;  
margin-top: 8px;"></div>
```

```
        <div class="modal-actions" style="margin-top:  
20px; display: flex; gap: 10px; justify-content: flex-end;">
```

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```
        <button type="button" class="btn btn-  
secondary" id="modal-cancel">CANCEL</button>
```

```
        <button type="submit" class="btn btn-  
primary" id="modal-submit" style="background-color: #1a4a75;  
color: #fff; font-weight: bold;">PROCEED TO PAYMENT</button>
```

```
    </div>
```

```
</form>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
{% endblock %}
```

```
{% block extra_js %}
```

```
    <script src="{{ url_for('static',  
filename='js/solo_rates.js') }}"></script>
```

```
{% endblock %}
```

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4.7 Project Structure

```
e:\final4_xample.new.1/
├─ Root Configuration & Scripts
│   ├── run.py                                (Main entry point)
│   └── requirements.txt                      (Python
dependencies)
│   └── .env                                (Environment
variables)
│   ├── .git/                               (Git repository)
│   └── .venv/                             (Python virtual
environment)
│   ├── .vscode/                           (VS Code settings)
│   └── .pytest_cache/                     (Pytest cache)
│
├─ Documentation
│   ├── DESIGN_IMPLEMENTATION_SUMMARY.md
│   ├── IMPLEMENTATION_SUMMARY.md
│   ├── MEMBERSHIP_SYSTEM_GUIDE.md
│   ├── PAYMENT_FEATURE_SUMMARY.md
│   ├── TODO.md
│   ├── TODO_dashboard_fix.md
│   └── TODO_DATABASE_RECREATE.md
│
├─ Root-Level Utility Scripts
│   ├── apply_fk.py
│   ├── fix_db.py
│   ├── inspect_users.py
│   ├── print_fk.py
│   └── test_receipt.txt
```

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```
|   └─ test_receipt.png
|
└─ high_end_ws_lounge/
    └─ Backend Core Files
        └─ run.py                                (Flask app
initialization)
            └─ database.py                      (Database models)
            └─ admin.py                        (Admin routes)
            └─ membership_routes.py            (Membership
features)
                └─ addons_routes.py            (Add-ons features)
                └─ addons_helper.py            (Add-ons helper
functions)
                    └─ time_utils.py            (Time utilities)
                    └─ time_utils_clean.py      (Clean time
utilities)
                        └─ .env                  (Environment config)
                        |
                        └─ Database Files
                            └─ app.db            (SQLite database)
                            └─ test_sqlite.db    (Test database)
                            └─ ws_lounge_lapaz.db (Alternative
database)
                                └─ migrations_addons.sql (Database
migrations)
                                    |
                                    └─ Docker Configuration
                                        └─ Dockerfile
                                        └─ docker-compose.yml
                                        |
```

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```
|— Frontend - Templates (app/templates/)
|   |— base.html                      (Base template)
|   |— landing.html                  (Landing page)
|   |— profile.html                  (User profile)
|   |— rooms.html                    (Rooms listing)
|   |— solo_rates.html                (Solo rates page)
|   |— timelog.html                  (Time logging)
|   |— admin/
|       |— admin_dashboard.html
|       |— admin_reservations.html
|       |— admin_reservations_list.html
|       |— confirm_reservations.html
|       |— manage_staff.html
|       |— members.html
|       |— member_dashboard.html
|       |— official_members.html
|       |— payment_settings.html
|       |— reports.html
|       |— solo_applications.html
|   |— dashboard/
|       |— member_dashboard.html
|
|— Static Assets (static/)
|   |— css/
|       |— base.css
|       |— landing.css
|       |— rooms.css
|       |— rooms_modern.css
|       |— solo_plans_modern.css
|       |— solo_rates.css
```

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```

|   |   └─ timelog.css
|   |   └─ usage_history.css
|   |   └─ admin/                (Admin CSS)
|   |   └─ dashboard/           (Dashboard CSS)
|   |   └─ modals/              (Modal CSS)
|   └─ js/
|   |   └─ app.js
|   |   └─ base.js
|   |   └─ landing.js
|   |   └─ rooms.js
|   |   └─ solo_rates.js
|   |   └─ timelog.js
|   |   └─ admin/                (Admin JS)
|   |   └─ auth/                 (Auth JS)
|   |   └─ dashboard/           (Dashboard JS)
|   |   └─ modals/              (Modal JS)
|   └─ img/                     (Images)
|   └─ uploads/                 (User uploads)
|
└─ Tests (tests/)
|   └─ admin_flow_test.py
|   └─ integration_test.py
|   └─ test_admin_billing.py
|   └─ test_admin_pdf_export.py
|   └─ test_app.py
|   └─ test_app_corrupted.py
|   └─ test_app_new.py
|   └─ test_app_new2.py
|   └─ test_db_registration_flow.py
|   └─ test_profile_feature.py

```

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```
|   └─ hello_tool_test.txt
|
|─ Instance Folder (instance/)
|   └─ (Runtime instance data)
|
|─ Temporary/Debug Scripts
|   └─ admin_fixed.py
|   └─ client_fixed.py
|   └─ database_fixed.py
|   └─ check_admin_walkin.py
|   └─ check_file_size.py
|   └─ check_imports.py
|   └─ debug_api_rooms.py
|   └─ debug_repro.py
|   └─ debug_reservation.py
|   └─ final_fix.py
|   └─ fix_database.py
|   └─ fix_imports.py
|   └─ inspect_db.py
|   └─ migrate_customer_ids.py
|   └─ migrate_receipt_image.py
|   └─ test_payment_feature.py
|   └─ verify_total_flow.py
|   └─ tmp_*.py (various debug scripts)
|
|─ Build/Cache Directories
|   └─ .pytest_cache/
|   └─ venv/
|   └─ __pycache__/
|   └─ Duration folders (0 mins, 30 mins, etc.)
```


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```
|
|└─ Notes & Documentation
|   └─ admin_membership_requests_notes.txt
|   └─ admin_membership_routes_temp.txt
|   └─ TODO.md
|   └─ out_shell.txt
|   └─ pytest_output_debug.txt
|   └─ shell_test.txt
|
└─ Configuration Files
    └─ requirements.txt
    └─ .env.example
    └─ migrations_addons.sql
```

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CHAPTER V

5.1 Summary, Conclusion and Recommendation

5.1.1 Summary

The Integrated Study Hub Management System was conceptualized and developed to resolve the operational inefficiencies at the WS Students & Professionals Lounge - La Paz Branch. Prior to the study, the lounge heavily suffered from front-desk bottlenecks, data fragmentation, and service delays due to its reliance on non-integrated tools like physical handwritten logbooks, standalone spreadsheets, and Facebook Messenger.

To bridge these operational gaps, the proponents aimed to build a unified, web-based platform that streamlined administrative workflows through interactive room reservations, automated billing, and Role-Based Access Control (RBAC). The development process utilized a systematic engineering approach, supplemented by extensive system testing and user evaluations to measure the platform's core functionalities.

The collected data demonstrated that the system effectively minimized administrative workloads, eliminated resource double-bookings, and provided a highly secure environment for financial and customer records. Ultimately, the results validated that the developed system was robust, practical, and highly suitable for optimization in commercial study hub operations.

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5.1.2 Conclusion

The successful implementation of the Integrated Study Hub Management System effectively resolved the primary administrative vulnerabilities encountered by the WS Students & Professionals Lounge - La Paz Branch. By replacing decentralized legacy methods with a database-driven architecture, the system systematically addressed critical issues such as manual tracking errors, delayed response times, and disorganized data retrieval.

The inclusion of features like real-time workspace tracking, instant billing computation, and role-based access restrictions substantially elevated the lounge's operational speed, transactional accuracy, and data security. The favorable feedback gathered from the system's end-users proved that the platform satisfied the precise demands of the study hub's staff and customers.

In conclusion, the project underscored that integrating a centralized digital system was crucial for modernizing service-oriented businesses and achieving long-term operational excellence.

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5.1.3 Recommendation

1. **Conclude Continuous Staff Training** - Provide ongoing orientation and quick-reference user guides for current and future frontline personnel to ensure smooth system navigation, especially during high-traffic peak hours.

2. **Execute Preventive System Maintenance** - Conduct periodic database optimization, security patch updates, and automated data backups to sustain the system's stability and safeguard transaction records.

3. **Incorporate Next-Generation Features** - For future iterations, developers may integrate digital payment gateways (like GCash and Maya) for cashless reservation processing and deploy automated SMS or email alerts for real-time customer booking updates.

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5.2 Final Thoughts

This innovation highlighted how integrating modern technology could completely reshape the operations of contemporary workspaces. By replacing manual, error-prone workflows with a database-driven infrastructure, the project proved that service efficiency and data security could go hand in hand. For the WS Students & Professionals Lounge, this system marked a major step forward, transforming how they served their clients and managed their daily business affairs. The proponents were confident that if the management continued to embrace digital upgrades based on customer experiences, this platform would remain a vital tool in keeping the business highly competitive, organized, and ready for future expansion.

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(Use-Case Diagram)

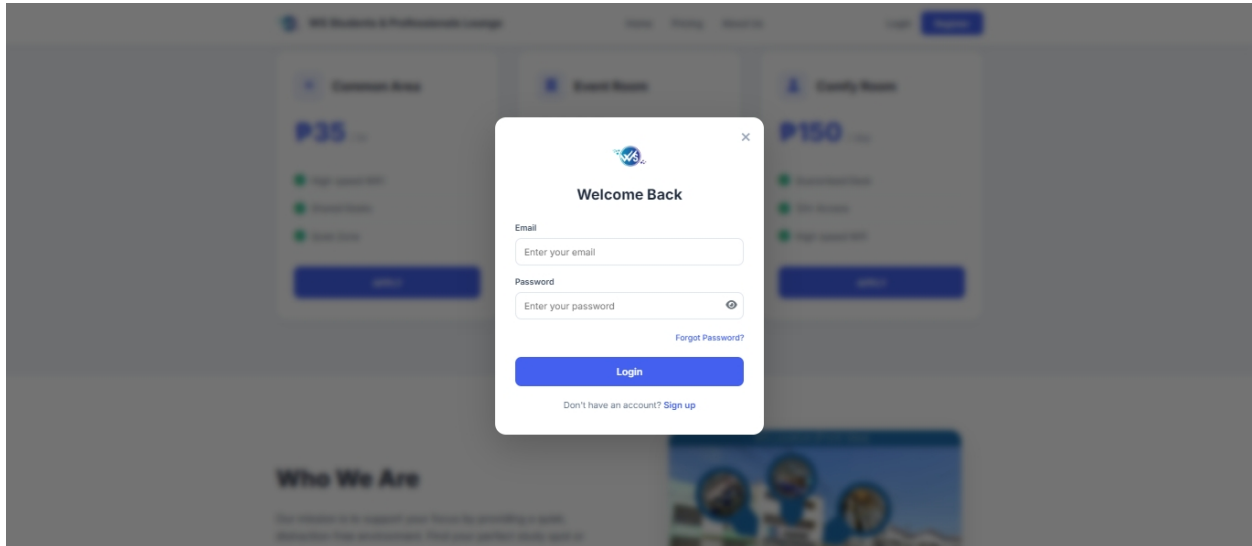
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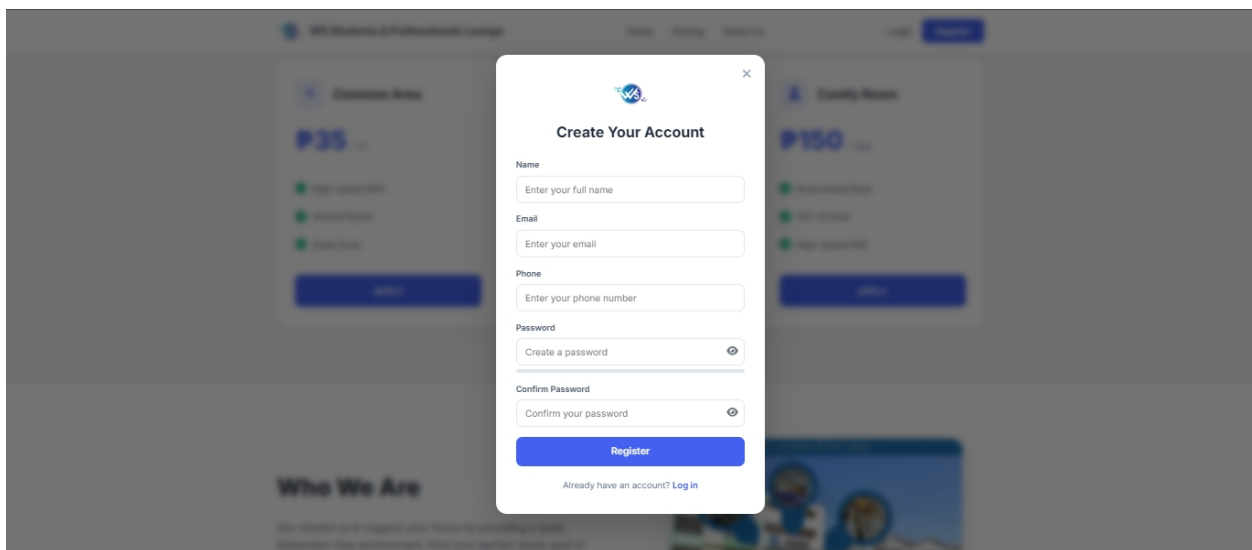
5.4 Appendices

5.4.1 Appendix A: System Interface Screenshot

- Login Page



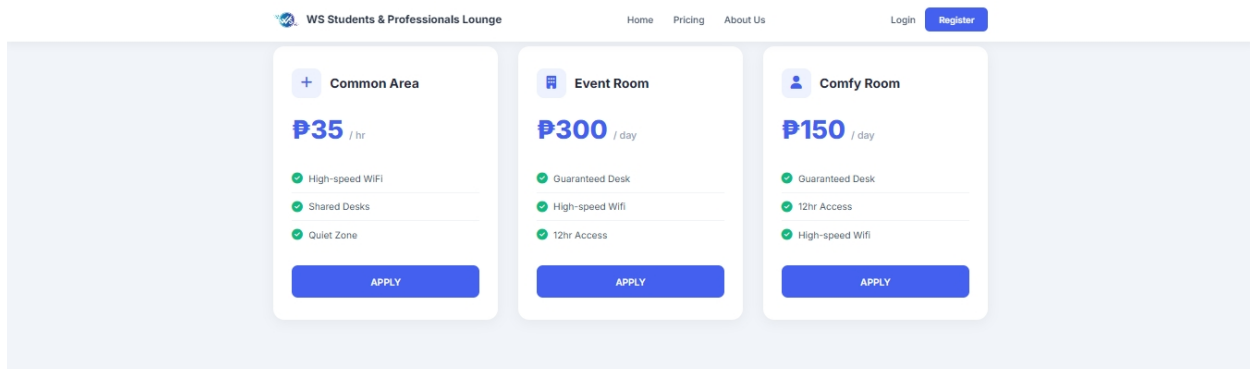
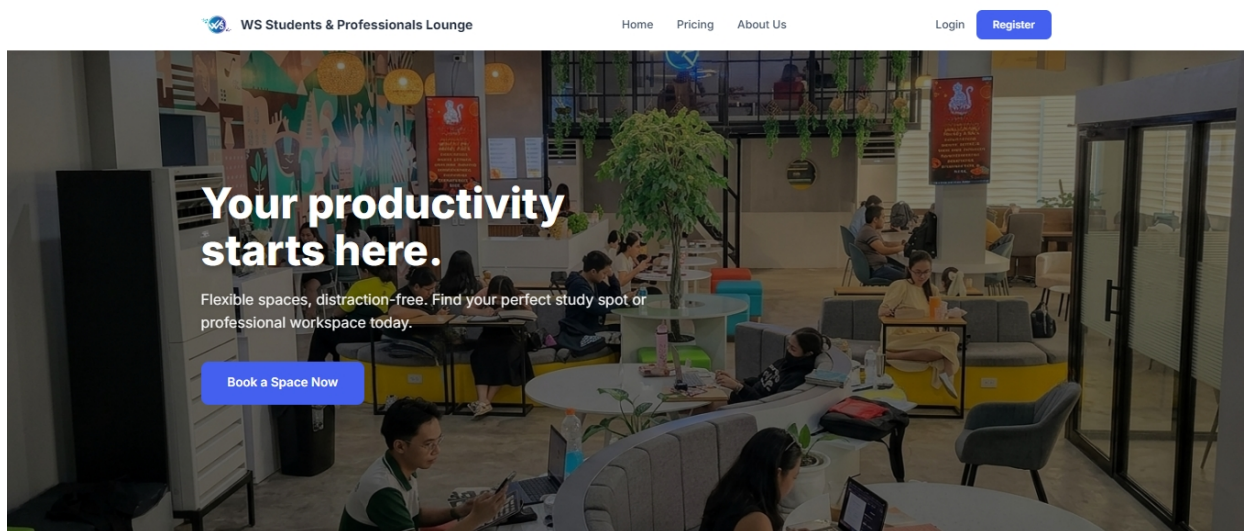
- Registration Page



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- Landing Page



Who We Are

Our mission is to support your focus by providing a quiet, distraction-free environment. Find your perfect study spot or



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- Admin Dashboard



STUDENT & PROFESSIONALS
WS LOUNGE

[Dashboard](#)
[New Reservation](#)
[Reservations](#)
[Members](#)
[Payment Settings](#)
[Reports](#)
[Manage Staff](#)
[Logout \(wslounge\)](#)

Admin Dashboard

Welcome Back, wslounge!

TOTAL MEMBERS
5

ACTIVE PLANS
1

PENDING RESERVATIONS
1

EARNINGS TODAY
₱365.00

Live Room Status

Search customer or room...

WALK-IN

COMMON AREA - OCCUPIED
1

COMMON AREA - AVAILABLE
69

Common Area

OCCUPIED

Occupied
1

Available Slots
69

Common Area

OCCUPIED

Room
Start
End

Common Area
07:53 PM
08:53 PM

Name: Ramil
Customer ID: 13

Add Time End Session

Waiting List - Today

Next: Justine Grace Reyes
(Small Meeting Room 2, 08:55 - 09:55 PM)

Common Area Members

1

Ramil Fernandez
Checked in: 07:53 PM
Time Used: 00:10:30

Occupied
0

Available Slots
70

Info: No active Common Area sessions

Small Meeting Room 1

AVAILABLE

Status: Available

Start: ---
End: ---
Name: ---

Small Meeting Room 2

AVAILABLE

Status: Available

Start: ---
End: ---
Name: ---

Lecture Room

AVAILABLE

Status: Available

Start: ---
End: ---
Name: ---

Conference Room

AVAILABLE

Status: Available

Start: ---
End: ---
Name: ---

Comfy Room

AVAILABLE

Status: Available

Start: ---
End: ---

Event Room 1

AVAILABLE

Status: Available

Start: ---
End: ---

Common Area Members

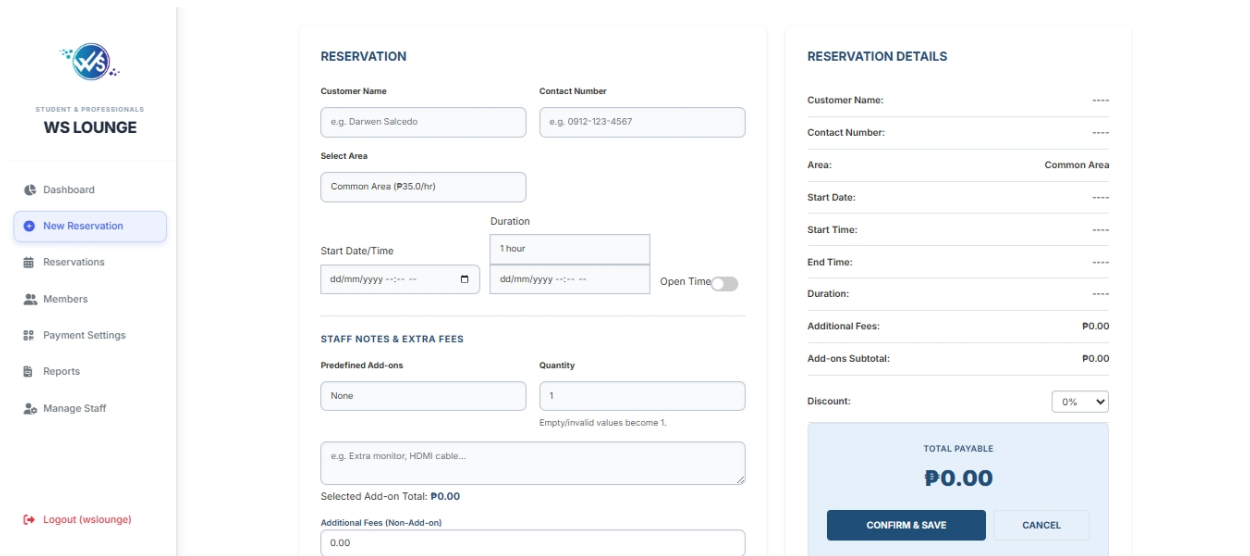
0

No active common area occupants.

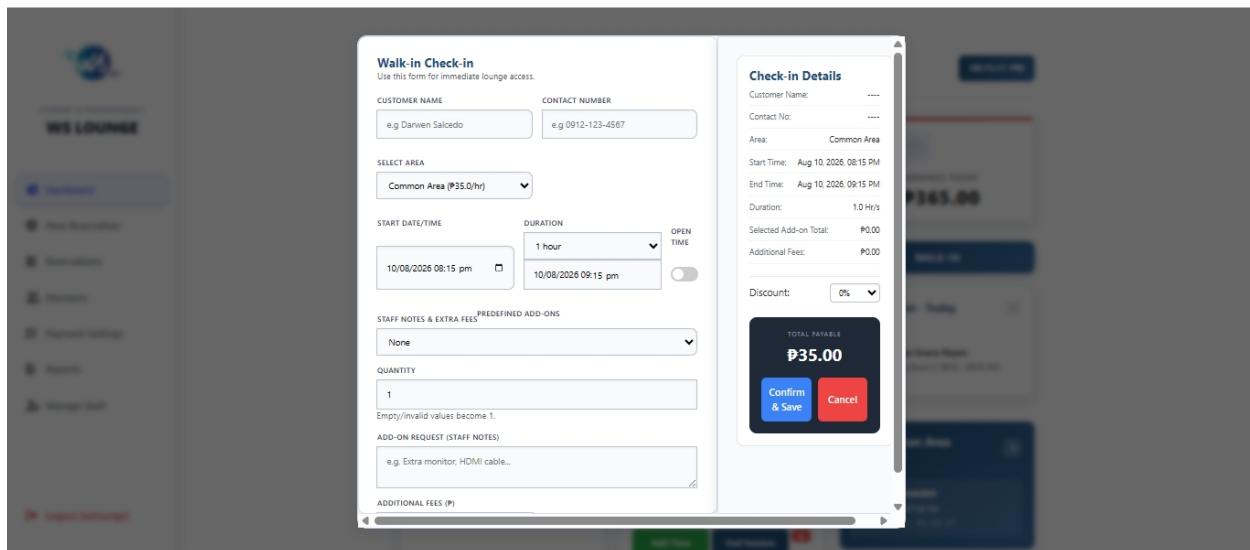
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- Reservation Page



The screenshot displays the 'RESERVATION' page of the WS Lounge system. On the left is a sidebar with the WS Lounge logo and a navigation menu including Dashboard, New Reservation, Reservations, Members, Payment Settings, Reports, and Manage Staff. The main content area is divided into two panels. The left panel, titled 'RESERVATION', contains fields for Customer Name, Contact Number, Select Area (Common Area at P35.0/hr), Start Date/Time, Duration (1 hour), and an Open Time toggle. Below these are 'STAFF NOTES & EXTRA FEES' with predefined add-ons (None) and a quantity of 1. A text area for additional fees is also present. The right panel, titled 'RESERVATION DETAILS', shows a summary of the reservation with fields for Customer Name, Contact Number, Area, Start Date, Start Time, End Time, Duration, Additional Fees (P0.00), Add-ons Subtotal (P0.00), and a Discount (0%). At the bottom, a blue box displays the 'TOTAL PAYABLE' as P0.00, with 'CONFIRM & SAVE' and 'CANCEL' buttons.



The screenshot shows a 'Walk-in Check-in' modal window overlaid on the reservation page. The modal is titled 'Walk-in Check-in' and includes a subtitle 'Use this form for immediate lounge access.' It contains the same reservation fields as the main page, but with pre-filled values: Customer Name (e.g. Darwen Salcedo), Contact Number (e.g. 0912-123-4567), Select Area (Common Area at P35.0/hr), Start Date/Time (10/08/2026 08:15 pm), Duration (1 hour), and Open Time (10/08/2026 09:15 pm). The 'STAFF NOTES & EXTRA FEES' section shows a predefined add-on of 'None' with a quantity of 1. The 'ADD-ON REQUEST (STAFF NOTES)' section has a text area for additional fees. The 'CHECK-IN DETAILS' panel on the right shows the reservation summary with a 'TOTAL PAYABLE' of P35.00 and 'Confirm & Save' and 'Cancel' buttons.

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- Reservations List



STUDENT & PROFESSIONALS
WS LOUNGE

- Dashboard
- New Reservation
- Reservations**
- Members
- Payment Settings
- Reports
- Manage Staff

[Logout \(wslounge\)](#)

Reservations List

Manage confirmed and pending reservations. Use the separate confirmation page for pending approvals.

[Confirm Reservation](#)

ID	CUSTOMER	ROOM	DATE & TIME	STATUS	PAYMENT	RECEIPT	AMOUNT	ADDED BY	ACTIONS
#13	Ramil No Address	Common Area	Aug 10, 2026 07:53 PM - 08:53 PM	Walk-in	N/A	None	P35.00	wslounge (Staff)	Add-ons: None Delete



STUDENT & PROFESSIONALS
WS LOUNGE

- Dashboard
- New Reservation
- Reservations**
- Members
- Payment Settings
- Reports
- Manage Staff

[Logout \(wslounge\)](#)

Confirm Reservation

Review pending reservation requests and confirm, hold, or delete them here.

[Back to Reservation List](#)

ID	CUSTOMER	ROOM	DATE & TIME	PAYMENT	RECEIPT	AMOUNT	ADDED BY	ACTIONS
#150	Justine Grace Reyes No Address	Small Meeting Room 2	Aug 10, 2026 08:55 PM - 09:55 PM	N/A	None	P250.00	wslounge (Staff)	Add-ons: None Confirm Hold Delete

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College of Information Technology Education

- Membership Page



STUDENT & PROFESSIONALS
WS LOUNGE

- Dashboard
- New Reservation
- Reservations
- Members**
- Payment Settings
- Reports
- Manage Staff

[Logout \(wslounge\)](#)

All Members

Monday, August 10, 2026

Membership Requests Member List Deactivated Accounts

Prioritize by: **Date Requested**

Ramil Fernandez Email: ramil@gmail.com Contact Number: 09345678932	Selected Plan INDIVIDUAL RATE Date Requested: 08/10/26 - 07:57 PM	Payment Verification Check	Approve Reject
---	--	--	---



STUDENT & PROFESSIONALS
WS LOUNGE

- Dashboard
- New Reservation
- Reservations
- Members**
- Payment Settings
- Reports
- Manage Staff

[Logout \(wslounge\)](#)

All Members

Monday, August 10, 2026

Membership Requests **Member List** Deactivated Accounts

Official Members
Review and manage current members.

 Active	Ramil Fernandez WS-WLKC-107 Email: ramil@gmail.com Contact: 09345678932	Plan: INDIVIDUAL RATE Status: Checked In
Checked In Check Out Customer View Records Renew Membership Deactivate		

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STUDENT & PROFESSIONALS
WS LOUNGE

Dashboard

New Reservation

Reservations

Members

Payment Settings

Reports

Manage Staff

Logout (wslounge)

All MembersMonday, August 10, 2026

Membership RequestsMember ListDeactivated Accounts

Deactivated Accounts
Soft disabled accounts are preserved for reporting and can be restored.

Deactivated since

**Ramil Fernandez**
WS-WLK-107
Email: ramil@gmail.com
Contact: 09545678932

REACTIVATED ON
Aug 08, 2026

REASON
Manual soft deactivation

Reactivate

Permanent Delete

- Payment/Billing Page



STUDENT & PROFESSIONALS
WS LOUNGE

Dashboard

New Reservation

Reservations

Members

Payment Settings

Reports

Manage Staff

Logout (wslounge)

Payment Settings
Manage GCash and Maya account details for customer downpayments.

GCash
Account Name

Account Number

Instructions

Please make your payment using the details above and upload your payment receipt.

Upload QR Code

Choose FileNo file chosen

Current QR


Save GCash Settings

Maya
Account Name

Account Number

Instructions

Please make your payment using the details above and upload your payment receipt.

Upload QR Code

Choose FileNo file chosen

Current QR


Save Maya Settings

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- Reports History



STUDENT & PROFESSIONALS
WS LOUNGE

- Dashboard
- New Reservation
- Reservations
- Members
- Payment Settings
- Reports**
- Manage Staff

[Logout \(wslounge\)](#)

Daily Reports & Analytics

Start Date
10/08/2026

End Date
10/08/2026

Filter Reports

Completed Sessions

Download Completed Sessions (PDF)

CUSTOMER	CONTACT	EMAIL ADDRESS	ROOM	PAX	STATUS	STAFF	START
Tin	09997672051	wslounge@lounge.com	Small Meeting Room 1	1	Checked-Out	wslounge	Aug 10, 2026 05:47 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Tin	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 05:27 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Justine Reyes	09997672051	ramil@gmail.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 05:25 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Lyza	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 04:00 AM



STUDENT & PROFESSIONALS
WS LOUNGE

- Dashboard
- New Reservation
- Reservations
- Members
- Payment Settings
- Reports**
- Manage Staff

[Logout \(wslounge\)](#)

Ramil	03535346342	ramil@gmail.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 03:07 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Justine Reyes	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 10, 2026 02:23 AM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Flo	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 09, 2026 11:18 PM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							
Justine Reyes	09997672051	wslounge@lounge.com	Common Area	1	Checked-Out	wslounge	Aug 09, 2026 10:59 PM
Notes: No notes Extra Fee: PHP0.00 Add-ons: None							

Daily Analytics Summary

REPORT DATE	TOTAL LOGINS	TOTAL CHECK-INS	MEMBER OF THE DAY	GENERATED AT
August 10, 2026	0	10	N/A	05:48 AM

Data automatically stored upon session end.

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- Manage Staff Accounts



STUDENT & PROFESSIONALS
WS LOUNGE

[Dashboard](#)
[New Reservation](#)
[Reservations](#)
[Members](#)
[Payment Settings](#)
[Reports](#)
[Manage Staff](#)

[Logout \(wslounge\)](#)

Manage Staff Accounts

Create new employee accounts and toggle active status without deleting audit history.

Create Staff / Admin Account

Full Name

Email Address

Phone

Password

Role

Staff

Create Account

Staff Directory

Name	Email	Phone	Role	Status
Admin Kim	kim@gmail.com	09997672053	Staff	Active
Admin Lorie	lorie@gmail.com	09997672054	Staff	Active
Admin Lyn	lyn@gmail.com	09997672052	Staff	Active
Admin Rose	rose@gmail.com	09997672055	Staff	Active
Admin Smile	smile@gmail.com	09997672057	Staff	Inactive
Admin Thyna	thyna@gmail.com	09997672058	Staff	Active
Admin Weyn	weyn@gmail.com	09997672056	Staff	Inactive
admin	admin-f236f8b7@example.com	N/A	Admin	Active
Maam Smile	smilesorongon@gmail.com	N/A	Admin	Active

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- User Dashboard



STUDENT & PROFESSIONALS
WS LOUNGE

[Dashboard](#)
[My Profile](#)
[Use Lounge \(Time In/Out\)](#)
[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

Member Dashboard

Welcome Back, Ramil !



3
Total Reservations



6.01 hrs
Total Solo Hours



5
Solo Plans

✓ **Official Member**

INDIVIDUAL RATE

STARTED ON	EXPIRES ON	COUNTDOWN	REMAINING HOURS	TOTAL HOURS	STATUS
August 10, 2026 08:23 PM	August 10, 2026 09:23 PM	00h 53m 51s		6.01 hrs	ACTIVE ✓ Checked in

Current Session Duration

00:06:00

Hours Used So Far

0.10 hrs

Recent Attendance

DATE	CHECK-IN	CHECK-OUT	HOURS USED



STUDENT & PROFESSIONALS
WS LOUNGE

[Dashboard](#)
[My Profile](#)
[Use Lounge \(Time In/Out\)](#)
[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

ID	Room	Date & Time	Status
#153	Small Meeting Room 2	08/20 08:25 PM	CONFIRMED
#11	Common Area	08/10 05:25 AM	CHECKED-OUT
#8	Common Area	08/10 03:07 AM	CHECKED-OUT

#152
INDIVIDUAL RATE
Expires: 08/10 2026
APPROVED

#151
INDIVIDUAL RATE
Expires: 08/10 2026
CHECKED OUT

#149
INDIVIDUAL RATE
Expires: 08/10 2026
CHECKED OUT

#147
INDIVIDUAL RATE
Expires: 08/10 2026
CHECKED OUT

#100
INDIVIDUAL RATE (4HRS)
Expires: 08/10 2026
CHECKED OUT

[New Reservation](#)[View Rooms](#)[Solo Plans](#)[Timeline](#)

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- Member Reservation Page



STUDENT & PROFESSIONALS

WS LOUNGE

[Dashboard](#)
[My Profile](#)
[Use Lounge \(Time In/Out\)](#)
[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

Rooms & Reservations

Book your preferred space with ease

AVAILABLE ROOMS

Common Area
P35.0/hr (1/70 occupied)

AVAILABLE

Small Meeting Room 1
P60.0/hr

AVAILABLE

Small Meeting Room 2
P60.0/hr

AVAILABLE

Lecture Room
P150.0/hr

AVAILABLE

Conference Room
P250.0/hr

AVAILABLE

Comfy Room
P150.0/hr

AVAILABLE

Event Room 1
P300.0/hr

AVAILABLE

Event Room 2
P300.0/hr

AVAILABLE

RESERVATION

CUSTOMER NAME

CONTACT

Ramil

09345128764

SELECT AREA

Small Meeting Room 2 (P50.0/hr)

START DATE/TIME

Duration

20/08/2026 08:25 pm

1 hour

20/08/2026 09:25 pm

Open Time

STAFF NOTES & EXTRA FEES

PREDEFINED ADD-ONS

QUANTITY

Extra Chair (Free)

2

Empty value will be treated as 1.

SPECIAL INSTRUCTIONS

Enter specific requests here...



STUDENT & PROFESSIONALS

WS LOUNGE

[Dashboard](#)
[My Profile](#)
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[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

[View Payment QR & Details](#)

Grand Total:

P0.00

Downpayment Amount (50%):

P0.00

UPLOAD RECEIPT

Choose File

No file chosen

Verify the amount matches the downpayment (50%) shown above.

RESERVE SPACE

UPCOMING BOOKINGS

dd/mm/yyyy

All Status

EXPORT CSV

ROOMS	START TIME	END TIME	STATUS
Small Meeting Room 2	Aug 20, 2026 08:25 PM	Aug 20, 2026 09:25 PM	CONFIRMED

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- Member Profile Page



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WS LOUNGE

[Dashboard](#)
[My Profile](#)
[Use Lounge \(Time In/Out\)](#)
[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

My Profile

Keep your account details current and update your password securely.

Profile Information

FULL NAME
Ramil Fernandez

EMAIL ADDRESS
ramil@gmail.com

CONTACT NUMBER
09345678932

Member account • Updated through your secure profile settings

Update Your Details

Full Name

Email Address

Contact Number

Save Profile

Change Password

Current Password

New Password

Confirm New Password

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- Membership Plans



STUDENT & PROFESSIONALS
WS LOUNGE

[Dashboard](#)
[My Profile](#)
[Use Lounge \(Time In/Out\)](#)
[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

Solo Plan Rates

Choose your plan, upload your payment receipt, and wait for admin approval. Your current solo plan status is shown below.

SOLO PLAN STATUS

ACTIVE

Plan: INDIVIDUAL RATE
Expiry: 2026-08-10 09:23 PM
Payment Method: GCash
Status: Approved

MEMBERSHIP INFO

ACTIVE MEMBER PLAN

Plan Name:
Started: 2026-08-10 08:23 PM
Expiration: 2026-08-10 09:23 PM
Remaining days: 0.0
Countdown: 00h 59m 41s

RATE COMPARISON

Compare solo rates at a glance. Click a plan card to submit your payment receipt.

PLAN	PRICE	DETAILS
INDIVIDUAL RATE	P35/HR	- Hourly common area usage - WiFi / Charging
INDIVIDUAL RATE (4HRS)	P100	4hrs common area usage - WiFi & Charging
DAY/NIGHT PASS	P200	- Choice of Day (7AM - 10PM) or Night (6PM - 9AM) common area usage - WiFi & Charging



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WS LOUNGE

[Dashboard](#)
[My Profile](#)
[Use Lounge \(Time In/Out\)](#)
[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

- WiFi & Charging
- Gold Card

ACTIVE PLAN IN USE

- WiFi & Charging
- Platinum Card

ACTIVE PLAN IN USE

- WiFi, Charging & Exclusive Locker
- Gold Card

ACTIVE PLAN IN USE

MONTHLY PASS (24HRS)**P2500**

- 24/7 Access to All Branches in Iloilo
- 1 Month (30 days) unlimited common area usage
- WiFi & Charging, Exclusive Locker
- 1hr Sleeping pod & Shower Room use per day (City Proper Branch only)
- Platinum Card

ACTIVE PLAN IN USE

WORKSTATION (24HRS)**P3000**

- 24/7 Access to dedicated Workstation (Lapaz)
- 24/7 Common area access to all Branches
- 1 Dedicated Workstation
- WiFi & Charging & Exclusive Locker
- 1hr Sleeping pod & Shower Room use per day (City Proper Branch only)
- Platinum Card

ACTIVE PLAN IN USE

PLAN SUMMARY

Choose the rate that fits your needs. After selecting a plan, upload your payment receipt and wait for admin approval.

- Plan approvals are reviewed manually by the admin.
- Your uploaded receipt is stored securely under uploads/receipts.
- Active plans remain valid until expiry.

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- Time Logs



STUDENT & PROFESSIONALS
WS LOUNGE

[Dashboard](#)
[My Profile](#)
[Use Lounge \(Time In/Out\)](#)
[Book Space](#)
[Solo Plans](#)

[Logout \(Ramil Fernandez\)](#)

Lounge Usage Logs



5hr 59m
Monthly Total Hours



INDIVIDUAL RATE
Most Used Room



36m
Avg. Session Length



10
Total Completed Sessions

Active Session CountdownAug 10, 2026, 8:27:50 PM

00 : 55 : 31

Status: ACTIVE

Currently using: **INDIVIDUAL RATE**

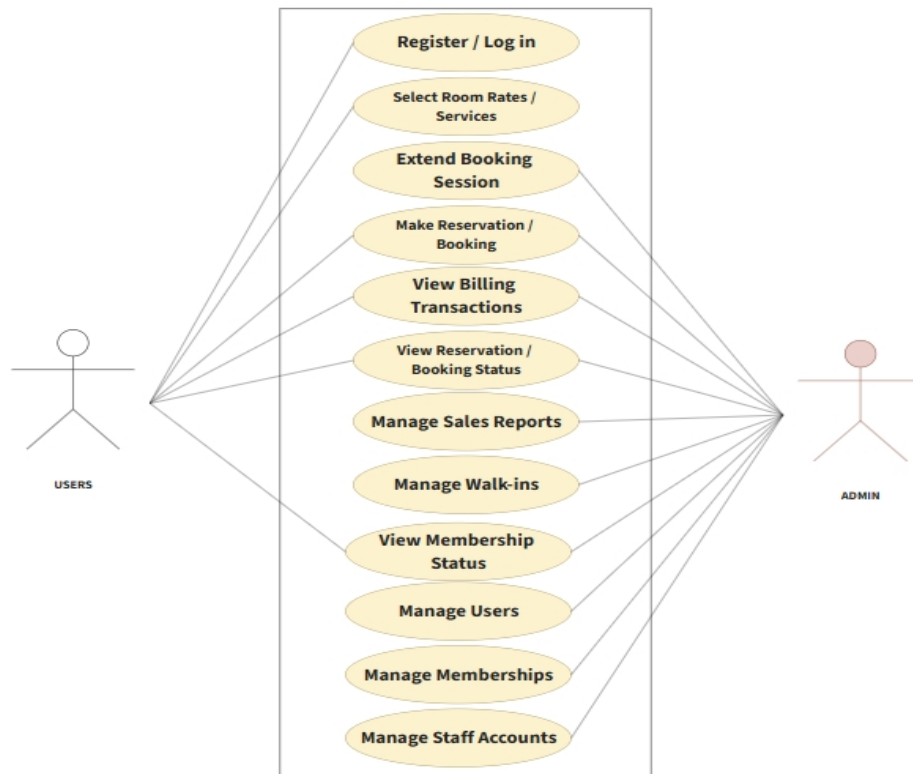
[Check Out Now](#)

Usage History

August 10, 2026 ACTIVE Room: INDIVIDUAL RATE Duration: 08:23 PM - Ongoing Total: 4 mins	August 10, 2026 ENDED Room: INDIVIDUAL RATE Duration: 07:58 PM - 08:15 PM Total: 16 mins	August 10, 2026 ENDED Room: INDIVIDUAL RATE Duration: 05:50 AM - 06:49 AM Total: 59 mins
August 10, 2026 ENDED Room: INDIVIDUAL RATE Duration: 03:04 AM - 05:15 AM Total: 130 mins	August 10, 2026 ENDED Room: INDIVIDUAL RATE Duration: 02:25 AM - 02:26 AM Total: 0 mins	August 09, 2026 ENDED Room: INDIVIDUAL RATE Duration: 10:07 PM - 10:07 PM Total: 0 mins
August 09, 2026 ENDED Room: INDIVIDUAL RATE Duration: 10:07 PM - 10:07 PM Total: 0 mins	August 09, 2026 ENDED Room: INDIVIDUAL RATE Duration: 06:55 PM - 09:25 PM Total: 149 mins	August 09, 2026 ENDED Room: INDIVIDUAL RATE Duration: 12:55 PM - 12:56 PM Total: 0 mins
August 09, 2026 ENDED		

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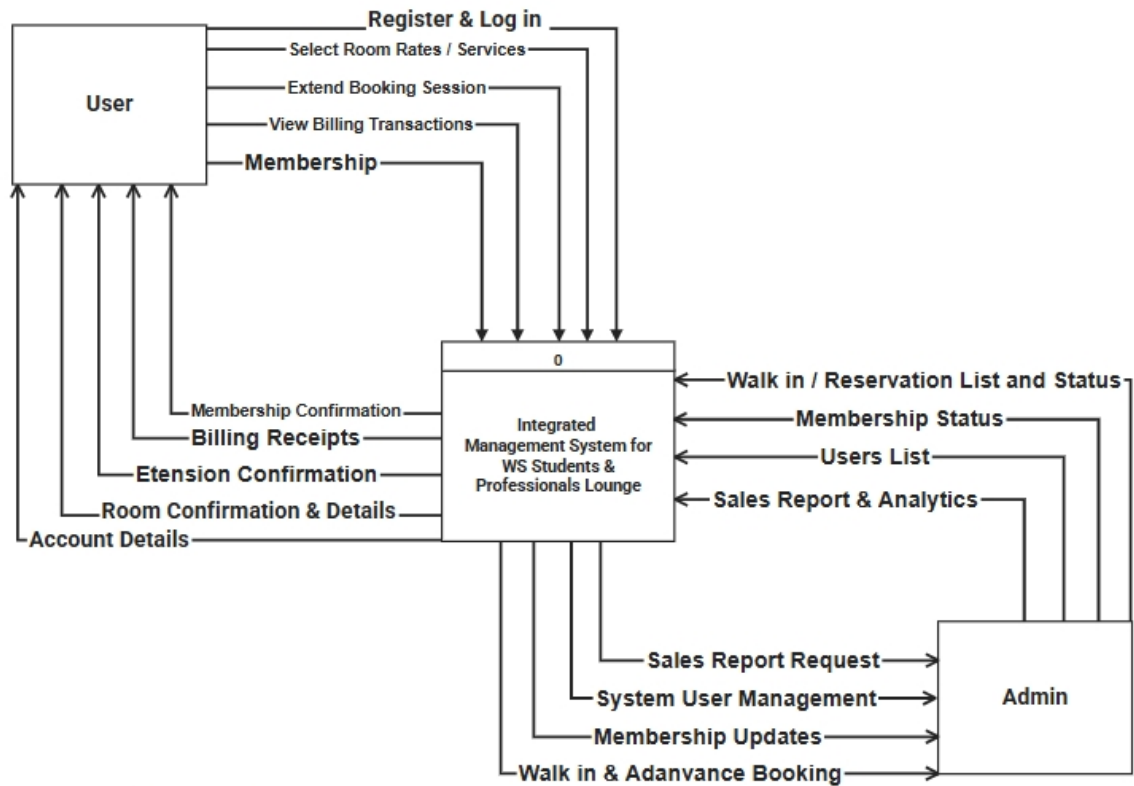
5.4.2 Appendix B: Use-Case Diagram



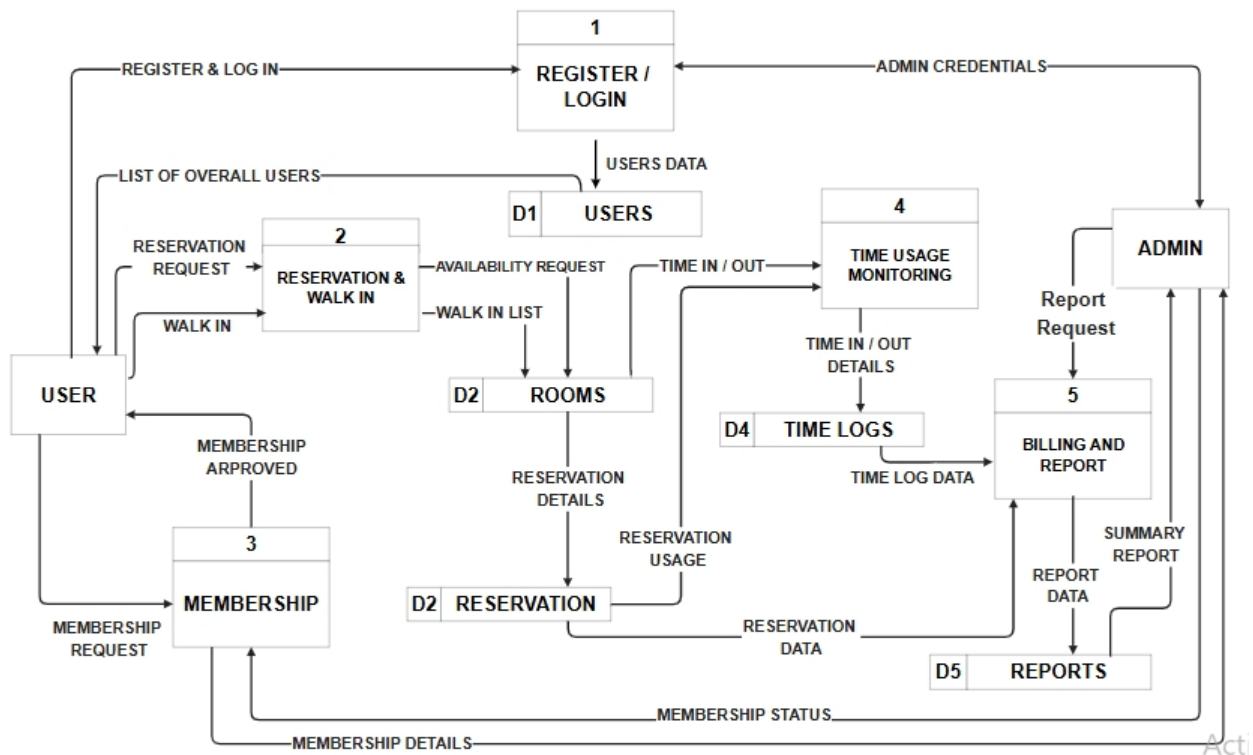
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5.4.3 Appendix C: Data Flow Diagram

Level 0



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5.4.4 Appendix D: User Survey Form and Summary

Demographic Information

What is your role in WS Students & Professionals Lounge?

- ☐ Staff / Owner
- ☐ Regular Customer (Student/Professional)
- ☐ Occasional Visitor

For Students/Professionals/Visitors Survey

1. How long have you been using study hubs or similar lounge facilities?

- ☐ Less than 1 year
- ☐ 1-2 years
- ☐ 3-4 years
- ☐ 5-6 years
- ☐ More than 6 years

2. Which area in the lounge do you prefer to use?

- ☐ Private Study Pods
- ☐ Group Meeting Rooms
- ☐ Common Area
- ☐ Other: _____

3. How many times do you visit the lounge per month?

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1. Less than 5
2. 5-10
3. 11-20
4. 21-30
5. More than 30

4. What type of transaction method have you been using?

- ☐ Email Form
- ☐ Online Booking
- ☐ Phone-based Reservation
- ☐ Walk-in

5. What is your average duration of stay in the lounge?

- ☐ Less than an hour
- ☐ 2-5 hours
- ☐ 6-8 hours
- ☐ Whole day

6. Do you prefer getting a membership?

- ☐ Yes
- ☐ No

7. What would convince you to use a new online booking system instead of making a manual booking? (Select all that apply)

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- ☐ Fast and instant booking confirmation (Email / SMS)
- ☐ Real-time view of available rooms, photos, and exact prices
- ☐ Ability to book anytime (24/7) without waiting for a reply
- ☐ Convenient online payment options (e.g., E-Wallets, Credit Card)
- ☐ Ability to easily modify or cancel my reservation online

8. If this establishment launches a fully automated online reservation platform, how likely are you to use it for your next booking?

- ☐ Extremely Likely
- ☐ Likely
- ☐ Neutral
- ☐ Not Likely
- ☐ Extremely UnLikely

9. How satisfied are you with the current reservation/booking method you are using?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Not Satisfied

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10. How would you rate the support and service provided for your lounge concerns?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

For Staff/Owner

1. How long have you been working in WS Students & Professionals Lounge?

- ☐ Less than a year
- ☐ 1-2 years
- ☐ 3-4 years
- ☐ 5-6 years
- ☐ More than 6 years

2. Have you ever experienced missing or unrecorded transaction logs?

- ☐ Yes, frequently during peak hours
- ☐ Yes, occasionally
- ☐ Rarely
- ☐ No, records are always complete

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3. What issues have you been experienced with current booking or check-in methods? (Select all that apply)

- ☐ Double Bookings
- ☐ Uncertainty if seats/spaces are actually available
- ☐ Delayed responses when inquiring about availability
- ☐ Slow Billing Process
- ☐ None / Satisfied with current method
- ☐ Other: _____

4. How are active customer session times currently monitored?

- ☐ Staff manually check a spreadsheet
- ☐ Relying on customers to keep track of their own time
- ☐ Staff manually check paper logbooks periodically
- ☐ Verbal reminders from staff members

5. How do you currently reserve a room reservation/booking to a customer?

- ☐ Social Media Platform
- ☐ Email/Contact Form
- ☐ Phone Call/Message
- ☐ Walk-in

6. How do staff currently generate and submit daily/weekly/monthly transaction reports to management?

- ☐ Handwritten summary sheets / Physical logbooks

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- ☐ Manually filled-out digital spreadsheets
- ☐ Sending photos of physical receipt logs or logbook pages via chat apps
- ☐ Verbal updates at the end of the shift or day
- ☐ Other: _____

7. How open are you to replacing your current manual booking process with an automated online reservation system?

- ☐ Very Open
- ☐ Somewhat Open
- ☐ Neutral
- ☐ Not Open

8. What is your primary goal for adopting an automated reservation system?

(Select up to 2)

- ☐ Reduce manual errors / double-bookings
- ☐ Save time on administrative tasks
- ☐ Provide 24/7 availability for customers
- ☐ Accept real-time online payments
- ☐ Improve customer tracking and reporting

9. How easy was it to navigate and use the new automated system compared to your previous manual process?

- ☐ Significantly easier
- ☐ Slightly easier
- ☐ About the same

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- ☐ More difficult
- ☐ Significantly more difficult

10. How would you rate the Integrated Study Hub Management System for WS Students & Professionals Lounge?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

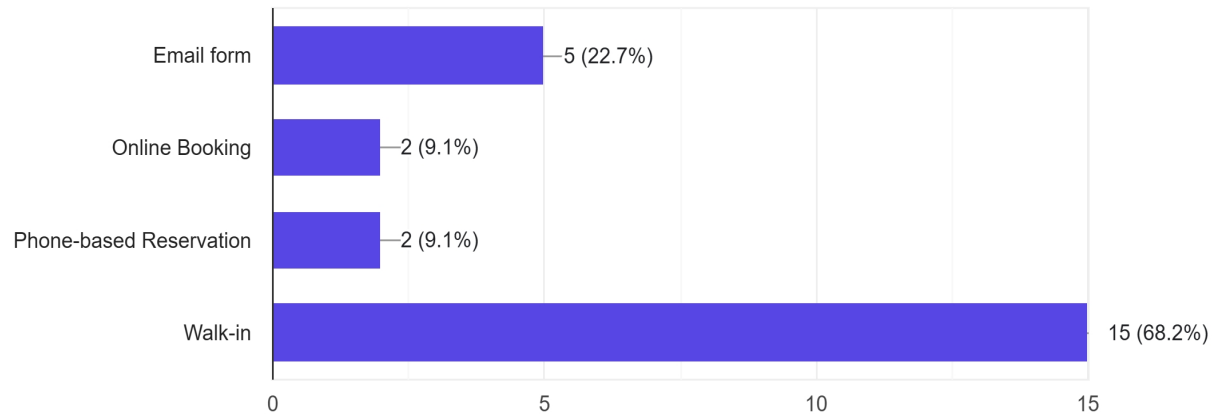
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5.5 Summary of Responses

Users

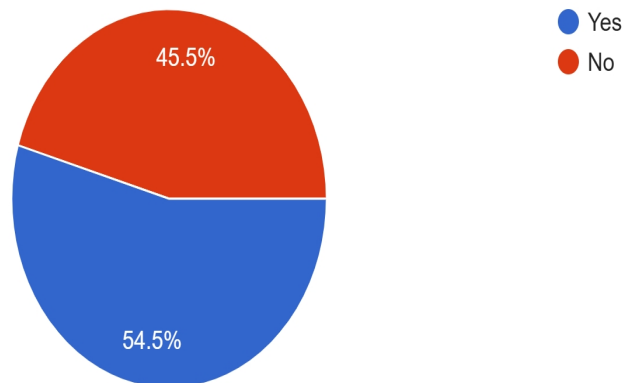
What type of transaction method have you been using?

22 responses



Do you prefer getting a membership?

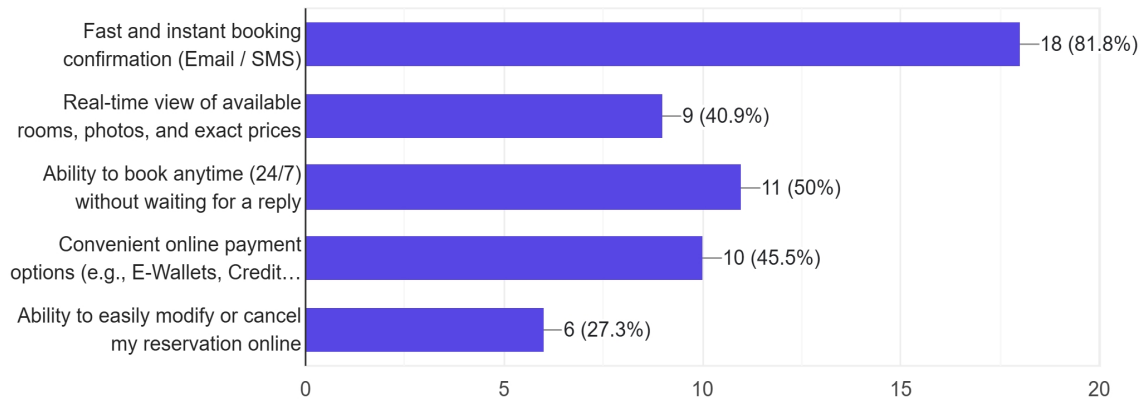
22 responses



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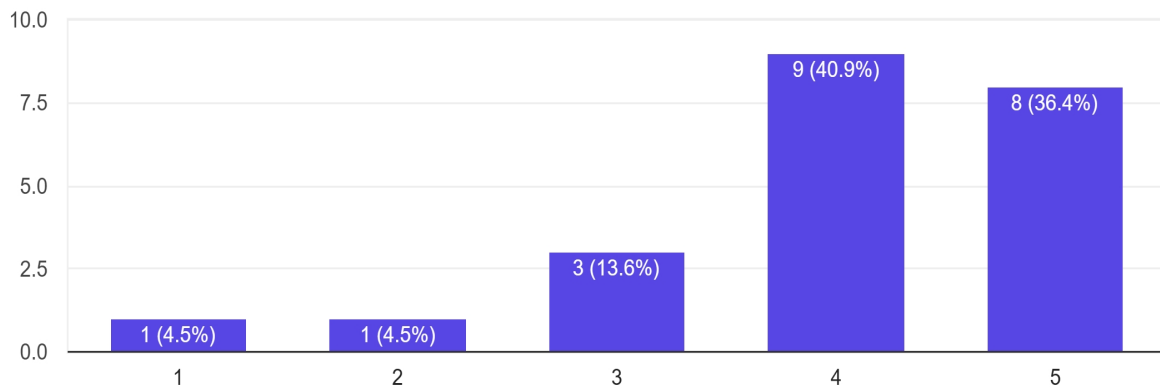
What would convince you to use a new online booking system instead of making a manual booking? (Select all that apply)

22 responses



If this establishment launches a fully automated online reservation platform, how likely are you to use it for your next booking?

22 responses

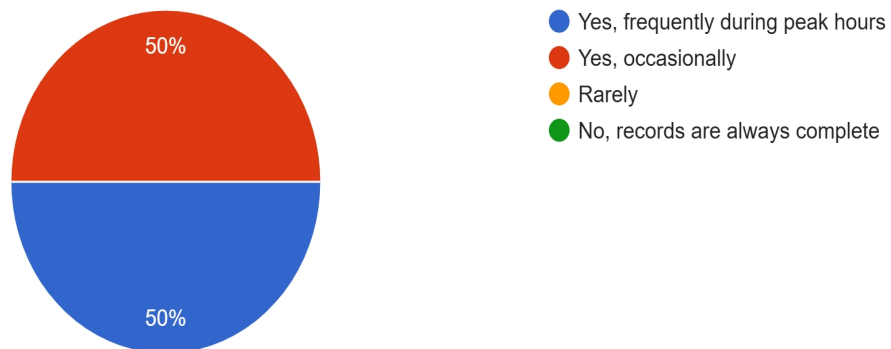


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Staff

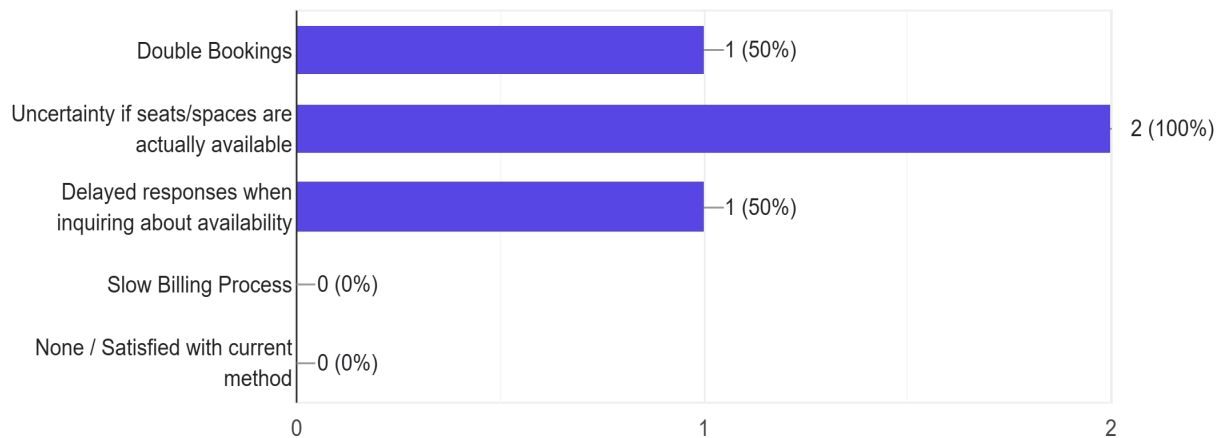
Have you ever experienced missing or unrecorded transaction logs?

2 responses



What issues have you experienced with current booking or check-in methods? (Select all that apply)

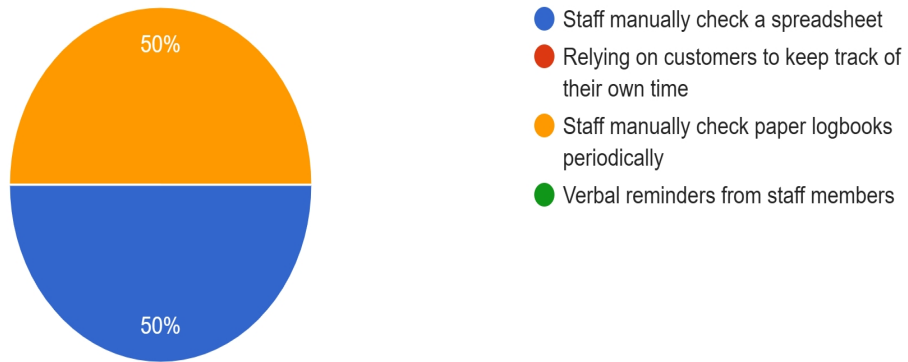
2 responses



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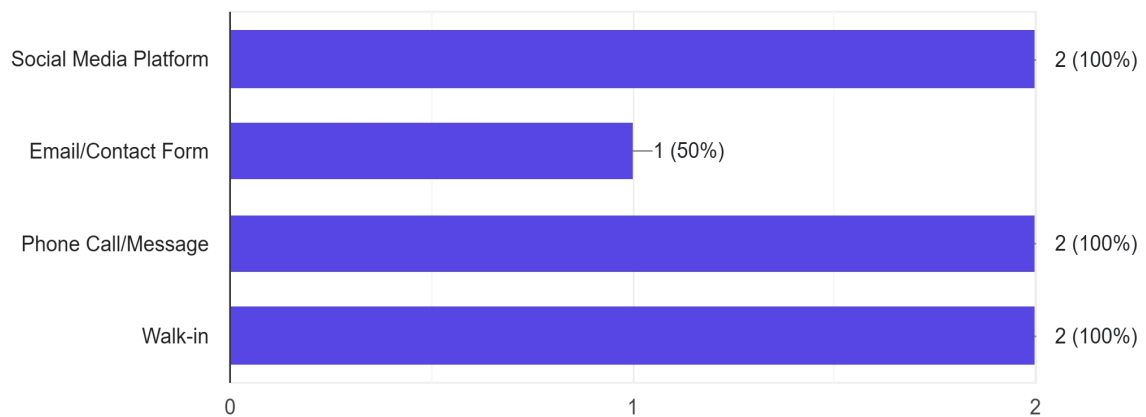
How are active customer session times currently monitored?

2 responses



How do you currently reserve a room reservation/booking to a customer?

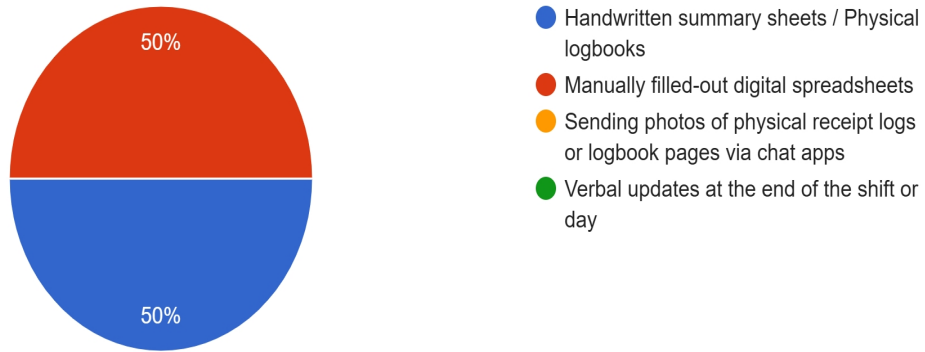
2 responses



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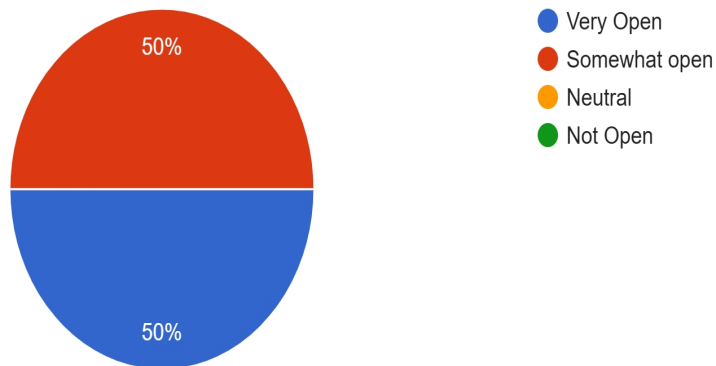
How do staff currently generate and submit daily/weekly/monthly transaction reports to management?

2 responses



How open are you to replacing your current manual booking process with an automated online reservation system?

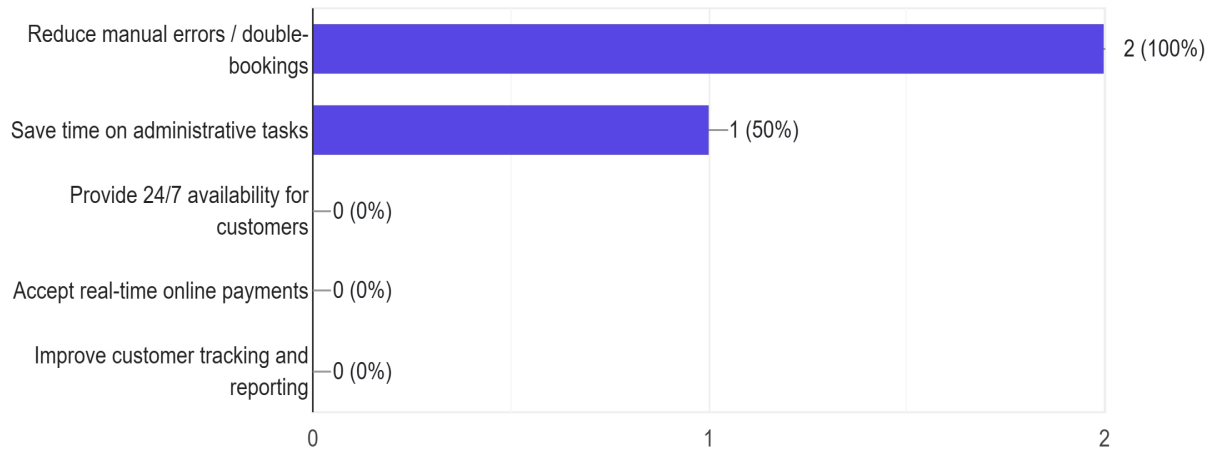
2 responses



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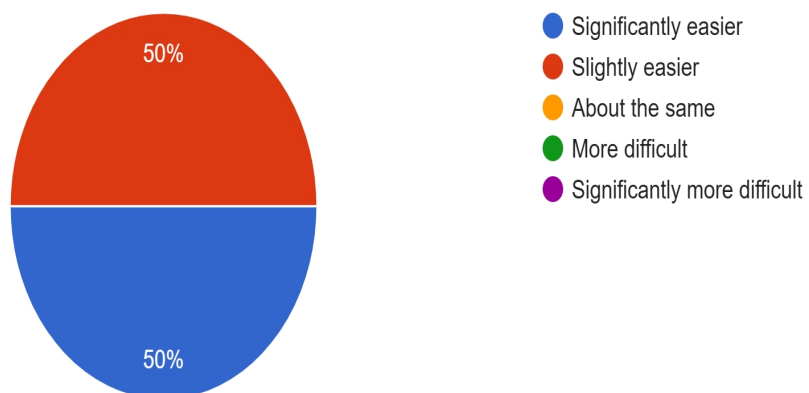
What is your primary goal for adopting an automated reservation system? (Select up to 2)

2 responses



How easy was it to navigate and use the new automated system compared to your previous manual process?

2 responses



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How would you rate the Integrated Study Hub Management System for WS Students & Professionals Lounge?

2 responses

